The Hausdorff Dimension of Kakeya Sets in Four Dimensions

OpenAI

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Abstract

We prove that every subset of $\mathbb{R}^4$ containing a unit line segment in every direction has Hausdorff dimension four, with no compactness or regularity assumption on the set or its witnessing line family. A multiscale reduction turns a hypothetical dimension deficit into a weighted family of lines whose local quadratic approximations require shrinking time intervals. Scalar comparison estimates and geometric arguments for three concentration regimes produce approximations on longer intervals or with greater transverse concentration, contradicting the extremal choice of the weighted family.

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1 Introduction

A *Kakeya set* in $\mathbb{R}^n$ is a set containing a unit line segment in every direction. The Hausdorff-dimension Kakeya conjecture asserts that every such set has Hausdorff dimension n. We prove its four-dimensional case.

Theorem 1.1. *Let $K \subset \mathbb{R}^4$. Suppose that for every $e \in S^3$ there is a point $a_e \in \mathbb{R}^4$ such that*

$$\{a_e + te : 0 \le t \le 1\} \subset K.$$

Then $\dim_H K = 4$.

Theorem 1.1 applies to arbitrary sets K and arbitrary choices of witnessing segments, including nonmeasurable ones. The family of witnessing lines is unrestricted by packing dimension or stickiness.

For arbitrary sets, we use the covering definition

$$\mathcal{H}^s_\delta(K) = \inf\left\{\sum_i (\operatorname{diam} U_i)^s : K \subset \bigcup_i U_i, \quad \operatorname{diam} U_i \le \delta\right\}, \qquad \mathcal{H}^s(K) = \lim_{\delta \downarrow 0} \mathcal{H}^s_\delta(K),$$

where the covers are countable, and $\dim_H K = \inf\{s : \mathcal{H}^s(K) = 0\}$. The reduction in Section 2 uses this covering definition, finite selections of witnessing segments, and outer-measure estimates in direction space.

Related dimension statements and geometric consequences are developed in Section 11. Besides the packing- and bounded-set Minkowski-dimension implications, these include a lower bound in higher dimensions by orthogonal projection, arbitrary direction sets and extension of segments to lines, and Nikodym and curved-Kakeya applications. The latter implications use the transfer results of Keleti and Máthé [20], Gao et al. [11], and Nadjimzadah [23], with their distinct hypotheses stated there.

1.1 Context and input estimates

The Kakeya problem connects the geometry of line segments with estimates for oscillatory operators. The classical needle problem asks how small a planar region can be if a unit segment is to turn continuously within it and return with its endpoints reversed; the area can be arbitrarily small [3, 10]. A central difficulty in the dimension problem is that many segments can overlap substantially even when their directions are well distributed. Besicovitch's constructions show that a Kakeya set can have Lebesgue measure zero, whereas Davies proved full Hausdorff dimension in the plane [3, 10].

For bounded Kakeya sets, the upper Minkowski formulation asks for dimension n, measured by covering numbers at one common scale; the Hausdorff formulation allows covers with varying diameters. For fixed ambient dimension $n \ge 2$, let $T_\delta(a, e)$ be the cylinder centered at $a \in \mathbb{R}^n$,

with axis direction $e \in S^{n-1}$, length one, and transverse radius δ. Define the maximal operator by $K_\delta f(e) = \sup_{a \in \mathbb{R}^n} |T_\delta(a,e)|^{-1} \int_{T_\delta(a,e)} |f(x)|\, dx$. The stronger Kakeya maximal conjecture asks that, for every $\varepsilon > 0$, there be a constant $C_{\varepsilon,n}$ such that

$$\|K_\delta f\|_{L^n(S^{n-1})} \le C_{\varepsilon,n} \delta^{-\varepsilon} \|f\|_{L^n(\mathbb{R}^n)}$$

for all $f \in L^n(\mathbb{R}^n)$ and $0 < \delta < 1$ [17]. For partial maximal estimates, the dimension parameter records the Hausdorff lower bound supplied by that estimate. Theorem 1.1 proves the four-dimensional Hausdorff assertion.

Wolff's bound $\dim_H K \ge (n+2)/2$ gives dimension at least three in $\mathbb{R}^4$. His geometric method studies the hairbrush of a tube: the collection of tubes meeting it [14, 33]. Bourgain connected line incidences in separated slices to sumset and difference-set estimates, using Gowers's quantitative form of the Balog–Szemerédi theorem [5]. Katz and Tao developed sums–differences iterations for higher-dimensional bounds [17]. In three dimensions, Katz, Łaba, and Tao combined multiscale clustering and local planar structure with the arithmetic method to improve the upper Minkowski lower bound beyond $5/2$ [19]. Łaba and Tao then used multiscale geometric structure to obtain upper Minkowski dimension strictly greater than $(n+2)/2$ for $n \ge 4$, in particular greater than three in four dimensions [21].

The later four-dimensional Hausdorff improvements addressed concentration near algebraic sets. For compact Kakeya sets, the polynomial Wolff estimates of Guth and Zahl, together with the algebraic direction bounds of Zahl and Katz and Rogers, give the lower bound $3 + 1/40$ [14, 16, 34]. The polynomial Wolff axioms restrict concentration near low-degree algebraic sets; the accompanying multilinear estimates exploit quantitative linear independence of tube directions [14]. Katz and Zahl's planebrush organizes tubes around a common plane and combines this geometry with Guth and Zahl's trilinear estimate. It gives Hausdorff dimension at least 3.059, while their distinct maximal-function estimate has dimension parameter at least 3.049 [18]. Borges, Chan, Chen, Liu, Xi, and Zhan subsequently improved the maximal parameter to $(159 + \sqrt{145})/56 \approx 3.0543$, combining the planebrush with a planar two-ends Furstenberg estimate [4]. This improves the maximal estimate while leaving the larger Hausdorff bound 3.059 as a separate conclusion.

Rai Choudhuri proved the bound $13/4$ for compact sticky sets, which admit a witnessing line family of packing dimension three [26]. Wang and Zakharov have since obtained a full-dimensional union estimate for almost AD-regular sticky tube families satisfying convex Wolff axioms and a dense-shading condition [32]; those hypotheses do not cover arbitrary Kakeya sets. In three dimensions, Wang and Zahl proved the Kakeya set conjecture [30], and Guth, Wang, and Zahl subsequently streamlined the proof [15].

The three-dimensional estimate used in our reduction is the weighted full-time plank bound of OpenAI [24], recorded as Lemma 2.15. It supplies spatial boxes with enough incidence mass to support the quadratic local fits described below. The further analytic inputs have distinct roles. Determinant multilinear Kakeya [7] controls configurations of independent velocities. The restricted-triple dot-product theorem of Wang and Zahl [31] and the multiplicative-convolution estimate of Orponen et al. [25] provide planar rigidity. The latter is applied in Proposition 7.6 to construct scalar coefficients with small interval probabilities. That step also uses a graph Balog–Szemerédi–Gowers argument and fixed-iterate sumset calculus [28, 29]. The graph argument retains a substantial set of original incidence edges, which are needed to fit the resulting planar structure back to the trajectories.

1.2 The proof route

The proof uses quadratic local fits to capture line concentration that persists across scales. Its basic object is a *chart*: a collection of line pieces in a moving spatial box and a time interval, on which one quadratic polynomial test stays small. Each chart also carries a mass requirement, measured using selected time-line pairs. This prevents a fit to an exceptional collection of negligible weight from counting as progress. Normalizing inside a chart produces another problem of the same kind, with its time interval rescaled to unit length. Charts can therefore be nested.

The two quantities to balance are the thickness of the quadratic relation and the length of the line pieces on which it holds. Longer intervals at a prescribed thickness give better charts. The argument first extracts a weighted line problem from a hypothetical Hausdorff deficit, then uses comparison and geometric structure to produce intervals longer than that problem permits. The steps are as follows.

1. Theorem 2.17 converts a Kakeya counterexample into a weighted family whose density decreases slowly under finer observation, while its terminal charts must use short time intervals. Section 2 defines the model and tracks incidence mass under chart normalization.

2. Lemma 3.3 compares local polynomial tests that agree on many sampled incidences. A derivative discrepancy would force more endpoint volume than the small lists of tests permit. Lemma 3.5 uses this comparison to lengthen charts while keeping their labels recoverable from the original observations.

3. Proposition 4.3 selects nested charts in an extremal weighted problem. Their limiting density and time profiles identify improvements that would contradict the extremal choice. The remaining cases depend on the relative shrinking of the smaller spatial axis, quantified by the optimized narrowness in Section 4.

4. Theorem 6.1 treats isotropic charts. Scalar estimates, multilinear Kakeya and interpolation produce a potential; Proposition 6.8 converts its higher-degree terms into an admissible quadratic test. Corollary 6.9 gives the auxiliary scalar model used in the other cases.

5. Propositions 8.4 and 8.5 treat finite positive narrowness by scalar projection. Wide projected intervals give an improvement directly; otherwise planar rigidity leads to the horizontal model. Section 9 excludes that remaining model by improving its fits or recovering wide intervals.

6. Proposition 10.1 excludes unbounded narrowness. Two nested chart labels first localize trajectories over the full normalized interval, after which scalar projection produces the forbidden improvement.

The local fits must carry enough total incidence mass and have labels recoverable from the original observations. For finite narrowness, Proposition 4.11 obtains these properties by greedy coverage and comparison. The unbounded case recovers them after full-time normalization. Each construction returns to the original weighted problem and contradicts its extremal choice. Section 10 assembles the cases; Theorem 2.17 then proves Theorem 1.1. The comparison, the finite-narrowness conversion, and the conditional higher-jet argument (Proposition 5.6) apply without a low-entropy hypothesis on the original line family.

1.3 Main mechanisms

In more detail, write the lines as $(t, y_0 + tV, w_0 + tW)$, where $y \in \mathbb{R}^2$. An observation retains exact (t, y) and bins the remaining coordinate w; increasing the depth refines this bin. At resolution $N_0 \to \infty$, the selected incidence density in a bin has a common typical size $n(r) = N_0^{-f(r)}$. Here density is measured per $dt\,dy$ and per hidden bin within each component of the weighted mixture. The counterexample gives conditional velocity nonconcentration and

$$f(L) - f(r) \le d(L - r) \qquad (0 \le r \le L), \qquad d < 1,$$

where L is the terminal depth. A chart is called *true* when its incidence mass, divided by its interval length, is at least this density times its spatial volume, up to subpower factors. A chart system's coverage is the selected incidence mass, and mass $N_0^{-o(1)}$ is called substantial. Writing a common interval length as $N_0^{-h+o(1)}$, its *time exponent* is h. The *horizon* $H(E)$ is the infimum of these exponents over true terminal systems of substantial coverage in the weighted problem E, allowing further selections and subsequences. The reduction yields $H(E) > 0$.

The exponent below one makes a comparison principle available. Local graph representations of the polynomial tests give scalar sheets, which are analytic branches of polynomial equations. The density bound and the lower mass bounds of retained entries supply lists of size at most $\delta^{-d+o(1)}$ at accuracy δ, for a fixed $d < 1$. Under the derivative and sampling bounds in Lemma 3.3, two such lists cannot frequently agree in value while keeping their rescaled derivative data apart. Alternating fresh conditional samples of sheets and times turns a persistent derivative discrepancy into a polynomial endpoint map whose image has more volume than the density bounds allow. Successful paths remain unnormalized submeasures, so this volume argument retains their dependence on the preceding choices. The comparison both lengthens terminal charts and joins the local approximations constructed later.

To locate a forbidden improvement, we follow nested charts in a nearly extremal weighted problem. Their relative density and time profiles become linear, with time rate $k > 0$. A further extremal choice controls how much thinner the smaller spatial axis can become than the time scale; the larger width is comparable to that scale up to subpower factors. Call the optimized rate of this *narrowness* ℓ. The choices are arranged so that a definite improvement in time, or an additional gain in narrowness at fixed time when ℓ is finite, contradicts extremality after completion and return to the original problem. Section 4 gives the precise order of these choices and the resulting critical paths.

There are three geometric regimes. When $\ell = 0$, enough separated velocities remain to determine an approximate polynomial field. Conditional scalar estimates first give polynomial fits along individual lines. Multilinear Kakeya and interpolation combine them, and switching between lines makes the field approximately conservative, producing a potential. The potential can initially have degree higher than two. The obstruction has a specific form: the broad directions may lie near a conic. The proof identifies the surviving cubic and quartic terms and uses the existing hidden coordinate to encode them, on the retained trajectories, in a quadratic correction to the chart test. This restores the degree required for an admissible chart (Proposition 6.8). The same argument with one spatial coordinate also gives zero horizon for an auxiliary scalar model; that result supplies the scalar projections used in the other regimes.

When $0 < \ell < \infty$, projection onto the scalar model gives an alternative. Sufficiently wide projected intervals already yield an improvement by comparison. Otherwise planar rigidity forces the trajectories into a horizontal model with equation $Y' = -Z + tv$. Its natural boxes have horizontal size H and transverse size H^2. The noncommuting horizontal directions force the relevant

scalar sheets to be nearly affine. At the critical time rate $2k = 1$, a path with three refreshed line segments fills enough three-dimensional base volume to improve the fit. Away from that rate the aim is to recover wide projected intervals. When $2k < 1$, this first requires repeated scalar estimates that improve the resolution of trajectory parameters over the whole normalized time interval.

When $\ell = \infty$, normalization can introduce large shifts in the density exponents, but these cancel in the relative density ratios used by the argument. A final scalar projection gives charts whose time exponent tends to zero, contradicting the positive horizon inherited from the counterexample.

Two scale conventions are essential when these steps are combined. Section 4 distinguishes the original resolution N_0 from the local tangent resolution N: returned chart systems must satisfy mass and list bounds subpower in N_0, even when the local analysis only supplies bounds subpower in N. Section 2 distinguishes current incidence laws from earlier witness laws. A witness floor used for counting is paired with a density cap for that same law.

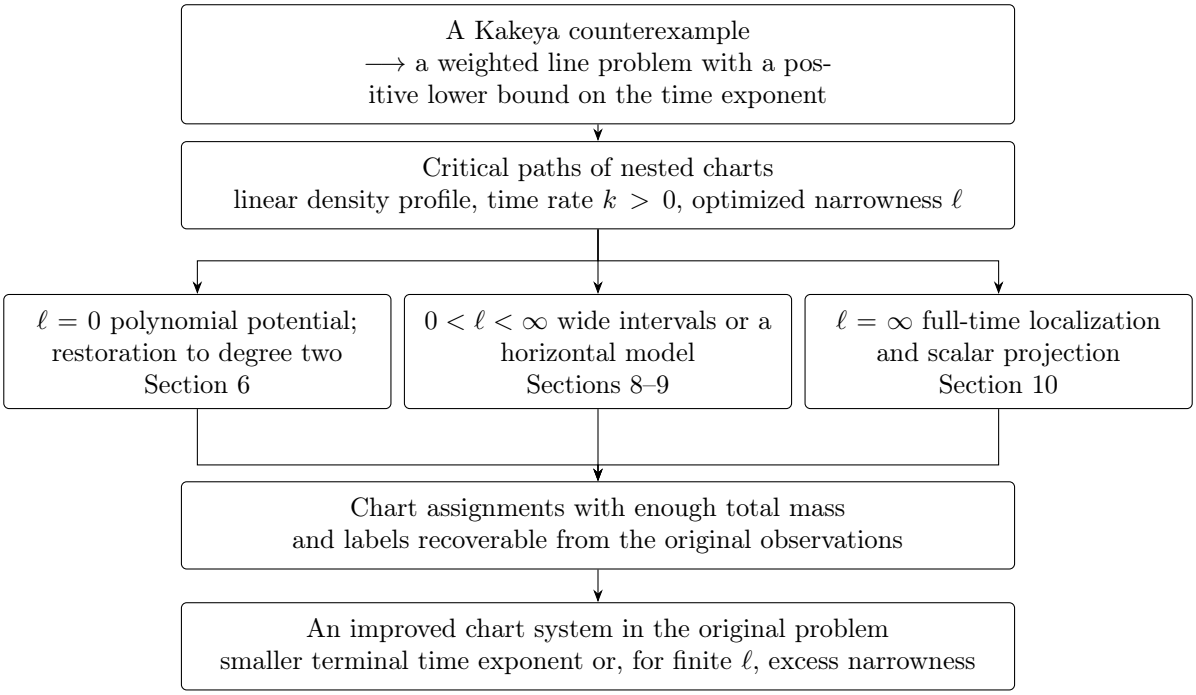


Figure 1: The proof by contradiction, organized by the narrowness of the critical charts. A critical weighted path has one of three narrowness regimes. Each produces improvements on substantial residuals. The finite-narrowness branches use Proposition 4.11 to obtain a chart system with recoverable labels; Section 10 obtains coverage and label recovery after full-time localization. All three return an improved chart system to the original problem. In the middle branch, $2k = 1$ uses the three-leg switch; $2k \neq 1$ returns to wide intervals, with a bootstrap when $2k < 1$. The scalar version of the isotropic argument supplies the projection theorem used in the other two branches.

2 Weighted models and the reduction from arbitrary Kakeya sets

We encode a line family by a probability law on trajectories and a selection of trajectory–time incidences. A local fit confines the base spatial projection of each whole trajectory piece to a moving box and keeps a quadratic polynomial small along that piece. Its mass is measured on the selected incidences. Theorem 2.17 turns a hypothetical Kakeya counterexample into a weighted

problem in which fitting at the final thickness requires a positive power cost in time localization. The infimum of these costs is the horizon.

We first develop the observations and density estimates used to measure these fits. We then define the charts, show how their priors and densities transform under normalization, and prove the reduction from an arbitrary Kakeya set. All subpower factors in this section refer to one exact sequence with resolution tending to infinity; later sequences of exact problems will be introduced separately.

2.1 Exact problems and observations

The model has two versions. Version 2 retains the lines in $\mathbb{R}^4$; version 1 is the scalar model used after projection. In each version a prior samples a trajectory independently of time, and a separate event specifies which trajectory–time incidences are retained.

Definition 2.1 (Exact problems)**.** An exact problem is a sequence indexed by a real parameter $N_0 \to \infty$, with a fixed length $0 < L < \infty$. For $0 \le r \le L$, write $a_r = N_0^{-r}$. A subpower factor $K_0 \ge 1$ satisfies $\log K_0 / \log N_0 \to 0$; its value may be increased from one occurrence to the next. We write $A \sim_{\log} B$ when $K_0^{-1} B \le A \le K_0 B$ for such a factor. Subsequences are permitted.

There are finitely many mixture components γ, of weights $\omega_\gamma \ge 0$ satisfying $\sum_\gamma \omega_\gamma = 1$. In component γ, a trajectory has prior probability law ν_γ, independently of uniform time $t \in [0,1]$. Its base coordinate is

$$y(t) = y_0 + tV \in \mathbb{R}^j, \qquad |y(t)| + |V| \le K_0.$$

We use the following two versions.

1. In version 1, $j = 1$, and $w(t) = w_0 + w_1 t + w_2 t^2$, with $\sup_{[0,1]} |w| \le K_0$. Tests are $P(t, y, w) = w - F(t, y)$, where F is a polynomial of total degree at most two. The derivative scale is $B = 1$, and $X = w$.

2. In version 2, $j = 2$, and $w(t) = w_0 + tW$ is affine. The coefficients of w need not be independent of those of y. Each component has a derivative scale $B > 0$ and a polynomial $P_*(t, y, w)$ of total degree at most two. Writing $X(t) = P_*(t, y(t), w(t))$, the native bounds are

$$\begin{aligned} &\sup_{[0,1]} |X| \le K_0, && \sup_{[0,1]} |\partial_w P_*| \le K_0 B, \\ &|\partial_w P_*| \ge B/K_0 \quad \text{at selected incidences}, && |\partial_{ww} P_*| \le K_0 B^2. \end{aligned} \tag{2.1}$$

   Tests in this version are arbitrary polynomials $P(t, y, w)$ of total degree at most two.

In both versions X is a quadratic polynomial along each trajectory, with subpower bounded coefficients.

An incidence selection is an event E_γ in trajectory–time space, and

$$d\mu_\gamma(l, t) = \mathbf{1}_{E_\gamma}(l, t)\, d\nu_\gamma(l)\, dt, \qquad \mu = \sum_\gamma \omega_\gamma \mu_\gamma, \qquad \mu(\text{all}) \ge K_0^{-1}. \tag{2.2}$$

Thus μ_γ is not normalized to probability. A further selection restricts this event and must retain total mass at least an inverse subpower.

Proof. The coefficient bound for X follows by interpolation on $[0,1]$. □

Here and below polynomial bounds are N_0^M for a finite exponent M fixed throughout the exact sequence. The exponent may be arbitrarily large and may change when a new exact problem is constructed. We impose the following finiteness convention.

Definition 2.2 (Tempered data)**.** The numbers of components, specified parameter ranges and coefficients, and B, B^{-1}, are polynomially bounded. In the full trajectory coefficient coordinates, each prior has piecewise constant Lebesgue density, polynomially bounded on polynomially many semialgebraic pieces. Trajectory and incidence predicates use polynomially many polynomial conditions of bounded individual degree, with arbitrary Boolean combinations. The number of variables in each geometric test is bounded. Real constants in these conditions have no height restriction.

Exact-level selections, labels, cells, and frames obey such uniform bounds. Chart widths and time lengths have polynomially bounded reciprocals. Changes of trajectory coordinates are invertible and have polynomially controlled constant Jacobian. We permit polynomially many copies of coefficient space, with an internal discrete trajectory index. This index is not a component or an observed point feature; all assignments concern indexed trajectories.

Measurable arguments may be used to locate a finite family of boxes or polynomial tests. The output is encoded by those constants and geometric tests, not by a description of the intermediate measurable search.

Components with extremely small success fractions can be removed. After forming chart components, charts with extremely small conditional prior mass can also be removed: there are only polynomially many charts, and the selected measure is bounded by prior times time. The threshold can be a sufficiently small reciprocal polynomial. These observations preserve the convention in Definition 2.2.

Use nested half-open dyadic grids, rounding requested widths upwards within a factor of two. The observation at depth r is

$$p_r = (\gamma, t, y(t), [w(t)]_{a_r/B}). \tag{2.3}$$

Within a component its density is with respect to $dt\,dy$ and counting measure on the hidden bin. Such densities exist: at fixed t, replacing y_0 by $y(t) = y_0 + tV$ has Jacobian one.

A density is typically $n = N_0^{-f}$ if, for every fixed $\tau > 0$, the selected incidences where it is outside $[N_0^{-\tau} n, N_0^{\tau} n]$ have total weighted mass at most $N_0^{-c(\tau)}$, eventually, for some $c(\tau) > 0$. One-sided typical bounds have the corresponding meaning. A problem is homogeneous for a family of observations if each observation has a common exponent, independent of the component and fixed along the exact sequence. Unless otherwise stated the observations are p_r, $0 \le r \le L$, and their typical densities are denoted $n(r) = N_0^{-f(r)}$.

2.2 Density selection and homogeneous profiles

The selected observation densities need not initially have uniform sizes. We will restrict to incidences on which their logarithmic sizes stabilize. Restrictions and finite refinements obey a simple mass estimate; a finite-table approximation will make the subsequent density selections compatible with the tempered data.

Lemma 2.3 (Restriction, refinement, and caps)**.** *Let g be the unnormalized density of an observation and g' the density after restriction. For $0 < \epsilon < 1$,*

$$\int_{\{g' < \epsilon g\}} g' \le \epsilon \int g. \tag{2.4}$$

If an observation is refined by a discrete label with at most D possible values on the retained events, the total refined mass at entries whose density is less than ϵg is at most $\epsilon D \int g$. These statements apply within conditional components and may be summed with the original component weights.

Consequently further inverse-subpower restrictions, and refinements by subpower lists, preserve typical logarithmic densities.

Remark. After the exceptional observations are removed, the restricted measure satisfies the upper density bounds. Integration over arbitrary base regions and summation over discrete entries therefore give the corresponding upper mass bounds.

Proof. Restriction gives $0 \le g' \le g$, and integrating the defining inequality on $\{g' < \epsilon g\}$ proves (2.4). For a refinement with densities g'_c, summing the inequalities $g'_c < \epsilon g$ over at most D labels at each old observation and integrating gives the refinement bound. With $\epsilon = N_0^{-\tau}$, a subpower D or a subpower normalization loss is absorbed into a smaller positive power.

To track exceptional mass, write the pushed-forward measure as a good part plus an omitted part, with densities $g_{\mathrm{good}}, g_{\mathrm{bad}}$. On the new states where $g_{\mathrm{bad}} > g_{\mathrm{good}}$, their total mass is at most $2 \int g_{\mathrm{bad}}$. On the other states the total density is at most twice the good density. This transfers a ceiling proved for the restricted good measure to a typical ceiling for the full resulting measure. The restricted good measure supplies the ceiling used in integrals; the omitted mass is bounded separately by the preceding total-mass comparison. □

A datum is known by p_s if, after permitted pruning, it belongs to a deterministic list of size at most K_0 depending only on p_s. List knowledge, rather than a choice of one representative, is the convention throughout. The exact base coordinates in an observation vary continuously. To choose labels or density ranges while preserving temperedness, we approximate their joint law on a sufficiently fine finite base mesh. The next lemma controls the error uniformly over subsequent choices.

Lemma 2.4 (Joint base flattening)**.** *Fix tempered discrete labels and predicates of polynomial total count. Given a fixed $T > 0$, there is a fixed finite D such that uniformly redistributing the base (t, y) inside its mesh cells of width $\rho \sim N_0^{-D}$, while retaining the discrete entries, changes the joint measure in total variation by $O(N_0^{-T})$. Thus a posterior list selector for these entries can be replaced, with that error in success, by a cell-constant tempered list table.*

Proof. Use coordinates (t, y, V) and the remaining trajectory coefficients, extending all densities by zero outside their domains. Holding the other variables fixed, each polynomial condition has a bounded number of changes on a base-coordinate axis, unless it is identically zero there; the latter condition has no changes. There are polynomially many conditions. The weighted density and the discrete labels therefore change across only polynomially many boundaries on each such fiber. Their heights and the ranges of the remaining variables are polynomially bounded. Integration first on the fiber and then in the other variables shows that a base translation of size at most ρ changes the joint measure by at most $N_0^C \rho$, with C fixed by the tempered bounds. Summation over the finitely many base axes and internal index copies gives a bound of the same form.

Within-cell averaging has the same bound, up to a dimensional constant: write its L^1 difference from the original density as the integral of pairwise differences of density values in the same cell, then compare these with translations of length at most a constant times ρ. Choosing $D > C + T + 1$ proves the assertion. All labels are included in this argument jointly.

A measurable selector has the same success up to total variation error on the flattened law. On that law the posterior of the discrete entries is constant inside each cell with its conditioning bins fixed; a constant maximizing list can be chosen there. There are polynomially many such cells and

entries. The resulting lookup table is tempered, and its success transfers back by the same total variation estimate. □

When two grids overlap with at most K_0 members at fixed exact base and other conditioning data, their common refinement has the same typical exponent by Lemma 2.3. For a one-way upper bound it suffices to sum the good measure over the corresponding covering entries.

Lemma 2.5 (Homogenization)**.** *After a tempered further selection of inverse-subpower mass and a subsequence, an exact problem has a homogeneous profile $f(r)$ on every prescribed compact depth interval. It is nondecreasing and 1-Lipschitz. Jointly, one may homogenize the observation $(p_r, [V]_{a_u})$ on compact nonnegative (r,u)-ranges, obtaining a function $f(r,u)$ that is nondecreasing in each variable and has Lipschitz constants 1 in r and j in u. Moreover $f(r,0) = f(r)$.*

Proof. Choose a countable dense depth grid containing all required endpoints. At stage N_0, test a finite initial part of this grid, increasing its size sufficiently slowly. By Lemma 2.4, use a common fixed polynomial base mesh fine enough for all the observations and for a power-small total variation error. The tested hidden and velocity grids are nested and lie in fixed compact depth ranges, so their joint entries are determined by the finest tested bins. Their total number, including base cells, is polynomial.

Polynomial bounds give a polynomial ceiling for each flattened density. Densities below N_0^{-C}, for a sufficiently large fixed C, have power-small total mass, by summing over the polynomial ranges and entries. Quantize the remaining logarithmic densities in intervals of width $\epsilon_{N_0} \to 0$. If m_{N_0} is the number of tested observations, choose the rates so slowly that the number of density-type lists is

$$\big(O(\epsilon_{N_0}^{-1})\big)^{m_{N_0}} = N_0^{o(1)}.$$

One list carries inverse-subpower mass. Restriction to that list is a cell table testing the joint discrete observations, hence is tempered. Its upper bounds persist, and Lemma 2.3 shows that its densities lose a fixed power relative to the preselection values only on power-small mass.

The total variation estimate transfers these conclusions to the exact base densities. Indeed, where two densities differ by a large multiplicative factor, their L^1 difference controls the mass under the larger density there. Pass to a subsequence on which the type exponents converge at every fixed tested depth.

Refinement from depth r to $r' \geq r$ has at most $K_0 N_0^{r'-r}$ hidden-bin choices, and refinement from velocity depth u to $u' \geq u$ has at most $K_0 N_0^{j(u'-u)}$ choices. Monotonicity of densities under refinement and Lemma 2.3 imply

$$0 \leq f(r',u) - f(r,u) \leq r' - r, \qquad 0 \leq f(r,u') - f(r,u) \leq j(u'-u).$$

These inequalities extend the profiles uniquely to all intermediate depths; nesting squeezes their typical densities between nearby tested ones. Since $|V| \leq K_0$, appending the unit velocity bin adds only K_0 choices, proving $f(r,0) = f(r)$. □

Other fixed logarithmic quantities with polynomial upper and reciprocal bounds may be homogenized by the same argument. In subsequent cap estimates, fixed tolerances on fixed finite grids are sent to zero sufficiently slowly, and the grids increase sufficiently slowly, that the exceptional mass is negligible compared with the particular inverse-subpower selection in use. This convention concerns the present exact N_0-sequence.

The class used in the contradiction imposes two further bounds. The first limits how rapidly the typical density can decrease as the hidden observation is refined toward the terminal depth. The second limits velocity concentration within a terminal observation.

Definition 2.6 (The cap class)**.** Fix $S, \eta > 0$. A homogeneous exact problem belongs to $C(d)$, $d < 1$, if

$$f(L) - f(r) \le d(L - r) \qquad (0 \le r \le L), \tag{2.5}$$

and, for $0 < u \le \eta L$, the joint density of $(p_L, [V]_{a_u})$ has typical upper bound

$$n(L) N_0^{-Su+o(1)}. \tag{2.6}$$

The $o(1)$ is interpreted by the fixed-tolerance convention above. The parameters S, η are held fixed when the class is used.

Lemma 2.7 (Angular caps under coarsening)**.** *Suppose* (2.6) *holds. For* $0 \le b < L$ *and* $0 < u \le \eta L$*, the conditional probability*

$$\mathbb{P}\{[V]_{a_u} = v \mid p_b\}, \qquad v = [V]_{a_u},$$

under the selected incidence law has typical upper bound $N_0^{-Su+o(1)}$*, evaluated at the sampled angular entry. Normalizing the total incidence mass does not change this conditional distribution. Good restricted measures with subpower slack can be prepared simultaneously on slowly increasing finite depth grids, losing negligibly relative to a specified inverse-subpower selection.*

Proof. Fix the exponents and a tolerance. Remove the fine entries where the joint angular upper bound or the pure p_L lower bound fails. For each remaining fine entry and a specified angular bin, its restricted numerator is bounded by $N_0^{-Su+O(\tau)}$ times the old pure density. Summing these numerators over the fine hidden bins inside a coarse bin, then dividing by the corresponding sum of old pure densities, proves the coarse bound for the good submeasure relative to the old denominator. The omitted-mass comparison in Lemma 2.3 gives the typical assertion, and (2.4) handles a later inverse-subpower restriction of the denominator.

For finitely many tests their power-small exceptions can be summed. Choose the tolerance and number of tests slowly to obtain the final assertion. These exclusions may be purely analytical; if an output selection needs them, Lemma 2.4 implements the corresponding cell tests. The typical bound is evaluated at the sampled angular entry; simultaneous sums use the good restricted measure. □

2.3 Time charts and prior-mass bounds

We now describe the local fits whose time exponent defines the horizon. A chart system's coverage is its selected incidence mass. In the present exact sequence, mass $N_0^{-o(1)}$ is called substantial. A chart controls each assigned trajectory throughout its time interval. Its incidence selection specifies where the lower derivative bound is required and supplies the mass entering the true floor. This selection need not occupy the whole interval.

Definition 2.8 (Charts, true packets, and known atlases)**.** At depth $r \in (0, L]$, a chart system has a common dyadic time partition into intervals I of length $q = N_0^{-h+o(1)}$, $h \ge 0$. Within each component and each interval, it assigns disjoint trajectory sets A_c to chart labels c, and selects incidences from the given event on these assignments. The total selected mass is at least K_0^{-1}.

1. A chart has an affine spatial center $y_c(t)$ and an invertible matrix D_c, with

$$\|D_c\| \le qK_0, \qquad J_c = |\det D_c|, \qquad \sup_{t \in I} |D_c^{-1}(y(t) - y_c(t))| \le K_0 \quad (l \in A_c). \tag{2.7}$$

2. The polynomial test belongs to the appropriate model class and satisfies

$$\begin{aligned} \sup_{t\in I} |P(t, y(t), w(t))| &\le K_0 a_r, \\ \sup_{t\in I} |\partial_w P(t, y(t), w(t))| &\le K_0 B, \\ |\partial_w P| &\ge B/K_0 \quad \text{on its selected incidences}, \\ |\partial_{ww} P| &\le K_0 B^2/a_r. \end{aligned} \tag{2.8}$$

   All chart data, assignments, selections, inverse coordinate changes, and reciprocal widths are tempered.

3. The trajectory and incidence masses are

$$\nu_c = \nu_{\gamma(c)}(A_c), \qquad m_c = \mu_{\gamma(c)}(\text{selected incidences assigned to } c).$$

   A packet is an individual chart together with its label, time interval, geometric and test data, full trajectory assignment A_c, and current incidence selection. It carries the assigned trajectory measure $\nu_{\gamma(c)}|_{A_c}$, of mass ν_c, and the unnormalized selected incidence measure, of mass m_c. An incidence-only restriction shrinks the latter without conditioning the assigned trajectory measure on success. A packet is true when

$$\frac{m_c}{q} \ge K_0^{-1} n(r) J_c. \tag{2.9}$$

   A system is true if every used packet is true.

A known atlas at depth r is a chart system whose label is known by p_L. It has all the assignment, coverage, geometric, test, and temperedness properties above, but need not satisfy (2.9). We also allow known atlases at $r = 0$, with $a_0 = 1$, when the same displayed bounds hold.

Since $y - y_c$ is affine on I, (2.7) also bounds $qD_c^{-1}(V - y_c')$ by K_0. Disjointness of assignments in each interval gives the basic budget

$$m_c \le q\nu_c, \qquad \sum_c \omega_{\gamma(c)} q\nu_c \le 1. \tag{2.10}$$

The sum includes all components, intervals, and labels of one system.

The following elementary estimate converts the small-value and lower-derivative tests into a bound on the hidden-coordinate bins.

Lemma 2.9 (Quadratic sublevel sets)**.** *Let Q be a real polynomial of degree at most two, and let $\varepsilon, b, \delta > 0$. The set*

$$\{w \in \mathbb{R} : |Q(w)| \le \varepsilon,\ |Q'(w)| \ge b\}$$

has total length at most $4\varepsilon/b$ and meets at most $C(1 + \varepsilon/(b\delta))$ bins of any grid of width δ, for an absolute constant C.

Proof. The condition $|Q'| \ge b$ is the union of at most two intervals on each of which Q is monotone. On either interval, the inverse derivative bound gives length at most $2\varepsilon/b$ for the indicated sublevel set. Each resulting interval meets at most two more grid bins than its length divided by δ. □

The defining true floor compares incidence mass with observation density and spatial volume. The next lemma combines that floor with the assignment budget to obtain comparability with the prior mass ν_c, and also confines the chart label to a subpower list determined by the observation.

Lemma 2.10 (Saturation and label knowledge)**.** *A true system may be restricted, retaining inverse-subpower total mass and enlarging subpower slacks, so that every used chart satisfies*

$$\frac{m_c}{q} \sim_{\log} \nu_c \sim_{\log} n(r)J_c, \tag{2.11}$$

and its label is known by p_r. The true floor persists on the resulting selected incidences.

Proof. For a large subpower H, deleting charts with $q\nu_c > Hm_c$ loses at most

$$\sum_{q\nu_c > Hm_c} \omega_{\gamma(c)} m_c \le H^{-1} \sum_c \omega_{\gamma(c)} q\nu_c \le H^{-1}.$$

Choose H^{-1} negligible relative to the system mass.

Analytically remove the exceptional p_r-entries above the ceiling $K_0 n(r)$, choosing the slack so the removed mass is negligible relative to that same mass. At a fixed base in chart c, the conditions $|P| \le K_0 a_r$ and $|P_w| \ge B/K_0$ admit only K_0 hidden bins of width a_r/B. Indeed, Lemma 2.9 gives this count with $\varepsilon = K_0 a_r$, $b = B/K_0$, and $\delta = a_r/B$, after enlarging the subpower factor. The chart base volume is at most $K_0 q J_c$. Thus the good portion of m_c is at most $K_0 n(r) q J_c$. Discard whole charts whose full mass is more than twice this bound. Their total mass is controlled by twice the globally omitted mass. Together with the first deletion and the true floor, this proves (2.11) before a final enlargement of the slack.

To obtain lists, let a chart cross an observation if its time, spatial, value, and event-derivative tests are compatible with that observation. If $T(p_r)$ is the number of crossings, the same volume calculation, now integrating the capped original measure rather than only assigned incidences, gives

$$\begin{aligned} \int T(p_r)\, d\mu_{\mathrm{cap}} &\le K_0 \sum_c \omega_{\gamma(c)} n(r) q J_c \\ &\le K_0 \sum_c \omega_{\gamma(c)} m_c. \end{aligned}$$

Markov's inequality, with a sufficiently large subpower threshold, leaves subpower lists and loses negligible system mass. The joint flattening lemma implements these lists as tempered tables if necessary. Finally discard charts that lost more than half their selected mass in these operations. Their total loss is at most twice the removed incidence mass, and (2.9) survives with enlarged slack. □

Later restrictions can reduce the selected mass of an individual chart. The next lemma gives two ways to continue a count: restore a floor for the current entries by pruning, or charge earlier witnesses using a cap for their own frozen measure.

Lemma 2.11 (Restoring current floors and charging witnesses)**.**

1. *Fix a chart layer with the assignment budget* (2.10)*. Suppose a later selected measure has inverse-subpower mass in the same normalization. For each chart let at most $D \le K_0$ additional entries be used, and let $\widetilde{m}_{c,e}$ be their current masses. For any $0 < \epsilon < 1$, deleting entries with*

$$\widetilde{m}_{c,e} < \frac{\epsilon q \nu_c}{D} \tag{2.12}$$

loses at most ϵ total weighted mass. In particular, ϵ may be an inverse-subpower negligible relative to the current total mass, giving current floors against the full assignment budget. For

finitely many prescribed layers and positive tolerances ϵ_i with $\sum_i \epsilon_i \le \epsilon$, the corresponding floors hold on one final retained measure, with total weighted loss at most ϵ.

If time is further partitioned, the corresponding threshold uses $|I'|\nu_c$ for a subinterval $I' \subset I$; summing these budgets over the subintervals gives $q\nu_c$.

2. *For a collection of disjoint witness submeasures of a fixed measure λ, all supported in a domain U, the total witness mass is at most $\lambda(U)$. With multiplicity at most D, the bound is $D\lambda(U)$.*

Remark. The witness estimate uses a cap for the same fixed measure λ. Floors for a later restricted measure are obtained separately from the first part.

Proof. Summing (2.12) over at most D entries per chart and using (2.10) bounds the deleted mass by ϵ. For finitely many layers, choose positive tolerances ϵ_i with $\sum_i \epsilon_i \le \epsilon$, and use ϵ_i in the threshold for layer i. Keep its assignment budgets and entry counts fixed. Repeatedly delete all remaining incidences of any entry whose current mass is positive and below its fixed threshold, reconsidering every layer after each deletion. An entry becomes empty permanently when deleted, so it is charged at most once. The assignment budget bounds the weighted charges from layer i by ϵ_i, including deletions caused by earlier losses in other layers. There are finitely many entries, so the process terminates with every surviving entry meeting its threshold in the final measure and total loss at most ϵ. The removed entries form a polynomially sized table of original predicates, preserving temperedness. For a slowly increasing collection satisfying the standing tempered bounds, choose its size and tolerance allocation slowly enough that the floor losses remain subpower. The time-subinterval assertion follows from $\sum_{I' \subset I} |I'|\nu_c = q\nu_c$.

For the last assertion, if the witnesses of the counted labels are disjoint submeasures of a frozen measure λ and all lie in a domain U, the sum of their masses is at most $\lambda(U)$. A multiplicity bound D replaces this by $D\lambda(U)$. The cap and the witnesses must therefore refer to the same measure λ, after every restriction required for that cap has been imposed. □

Lemma 2.12 (Fullness of the largest spatial scale)**.** *In a cap-class problem, a chart system known by p_L, whether true or not, has*

$$\|D_c\| \sim_{\log} q$$

on an inverse-subpower selection, after homogenizing the ratio if necessary.

Proof. The upper bound is part of (2.7). If the ratio were at most $N_0^{-\alpha}$ for a fixed $\alpha > 0$ on inverse-subpower mass, the endpoint bounds in (2.7) would confine V, for each chart, to a ball of radius $K_0 N_0^{-\alpha}$ about y'_c. Choose a fixed $u > 0$ with $u < \alpha/2$ and $u \le \eta L$. Such a ball meets only K_0 velocity bins of width a_u. Since chart labels are known by p_L, at each retained terminal observation only K_0 such velocity bins are needed. On the good restricted measure each has numerator at most $N_0^{-Su+o(1)}$ times the old pure denominator, by (2.6). Summing these bounds and then integrating shows that the alleged mass is power-small. The exceptional entries were removed before this sum. This contradiction excludes every fixed-power deficiency. □

2.4 Normalization, composition, and horizon

To iterate the construction, we regard each chart as a new component with its conditional trajectory prior and its own unit time interval. The density transformation must be tracked within that component before child charts can be lifted back to the original problem. The following lemma records both operations and the label-knowledge condition needed for intermediate true floors.

Lemma 2.13 (Chart normalization and lifting)**.** *Let a homogeneous exact problem of length L have a terminal-known atlas at depth $0 \le r < L$. After discarding extremely light charts and homogenizing their Jacobians and prior masses, it defines a new homogeneous exact problem of length $L' = L - r$. For its typical densities,*

$$n'(L') \sim_{\log} n(L)\frac{J_c}{\nu_c}, \tag{2.13}$$

$$n'(s) \le N_0^{o(1)} n(r+s)\frac{J_c}{\nu_c} \qquad (0 \le s < L'). \tag{2.14}$$

If the parent label is already known by p_r, there is equality up to subpower factors in (2.14)*. The class $C(d)$, with the same S, η, is preserved.*

A true terminal system in the new problem lifts to a true terminal system in the old problem. Intermediate true systems also lift through a parent with the stated intermediate density equality, in particular through a true parent prepared by Lemma 2.10.

Proof. The chart assignments determine the new product priors. We first compute their density transformation, then check the cap bounds and lift child charts.

The normalized priors. Put

$$Z = \sum_c \omega_{\gamma(c)} q \nu_c, \qquad \omega_c' = \frac{\omega_{\gamma(c)} q \nu_c}{Z}.$$

If the atlas success is $M \ge K_0^{-1}$, then $M \le Z \le 1$. Use $c = (\gamma, I, A_c)$ as a new component, with trajectory law $\nu_\gamma(\cdot \mid A_c)$ and independent uniform time

$$t = t_c + q\tau, \qquad y = y_c(t) + D_c z, \qquad 0 \le \tau \le 1.$$

The new selected measure is the pushforward of the old assigned selected measure divided by $q\nu_c$. Its weighted total is the old atlas selected mass divided by Z. In particular it is still a restriction of the new product priors and has inverse-subpower mass.

In version 1, if the parent test is $w - F(t, y)$, set

$$w' = \frac{w - F(t, y)}{a_r}.$$

This is quadratic along trajectories and bounded by K_0. In version 2, retain w and use the native test $P_*' = P/a_r$, with derivative scale $B' = B/a_r$. The parent test bounds give

$$\sup |P_*'| \le K_0, \quad \sup |\partial_w P_*'| \le K_0 B', \quad |\partial_w P_*'| \ge B'/K_0, \quad |\partial_{ww} P_*'| \le K_0 (B')^2$$

in the required places. All base and velocity bounds follow from (2.7). At $r = 0$, if the existing native test already has these bounds after the spatial normalization, it may instead be retained unchanged; this is useful for full-time spatial atlases.

These coordinate changes preserve the permitted models. Their maps on coefficient space have constant Jacobian: the base changes are affine, and in version 1 the subtraction of the quadratic $F(t, y)$ is triangular with respect to the w-coefficients. They and their inverses have polynomial control. Charts whose conditional prior is below a sufficiently small reciprocal polynomial have negligible total success, by polynomial chart count. On the remaining charts conditioning preserves temperedness.

Density and cap bounds. At fixed old base and chart label, new hidden-bin width at depth s is the old width a_{r+s}/B, with a chart- and base-dependent offset in version 1. The two grids have bounded overlap. For transported bins, the exact within-component density transformation is

$$\rho'_c = \frac{qJ_c}{q\nu_c}\rho_c = \frac{J_c}{\nu_c}\rho_c. \tag{2.15}$$

The numerator qJ_c is the base Jacobian; the denominator $q\nu_c$ normalizes the new product prior. Neither the component weight nor Z belongs in this within-component factor.

Refining p_L by the known chart label does not change its typical density exponent. The within-component factor in (2.15), together with bounded bin overlap and Lemma 2.3, therefore gives (2.13). Global exceptional masses are multiplied by $Z^{-1} \le K_0$, so their power-small character is preserved. At an earlier depth, refinement by an arbitrary label can only decrease the unnormalized density, giving (2.14). Knowledge by p_r, and hence by finer hidden observations, gives the reverse typical inequality. For a true parent saturated by Lemma 2.10, the formulas simplify to

$$n'(s) \sim_{\log} \frac{n(r+s)}{n(r)}, \qquad n'(0) \sim_{\log} 1.$$

For the angular bound, a new velocity bin of width N_0^{-u} maps by $V = y'_c + D_c V'/q$ into an old velocity ball of radius $K_0 N_0^{-u}$. At each exact chart and base it requires only K_0 old terminal/angular bins. Summation of the good old joint numerators precedes the density transformation. Lemma 2.3 controls both the old exceptional mass and the new states where it dominates, giving

$$\rho'_{\text{joint}} \le N_0^{o(1)} n(L) \frac{J_c}{\nu_c} N_0^{-Su} \sim_{\log} n'(L') N_0^{-Su}$$

for $u \le \eta L'$. Finally,

$$\log_{N_0} \frac{n'(s)}{n'(L')} \le f(L) - f(r+s) + o(1) \le d(L'-s) + o(1),$$

which proves the endpoint cap.

Lifting child charts. Let a child have relative time length q_d, spatial Jacobian J_d, and selected mass m'_d in the new component. Its old quantities are

$$q_{cd} = qq_d, \quad J_{cd} = J_c J_d, \quad m_{cd} = q\nu_c m'_d, \quad \frac{m_{cd}}{q_{cd}} = \nu_c \frac{m'_d}{q_d}. \tag{2.16}$$

At the terminal depth, a child true floor and (2.13) give the old floor $K_0^{-1} n(L) J_c J_d$. At an intermediate depth the same conclusion uses the intermediate equality; an upper bound alone would not suffice.

The tests lift by multiplication by a_r. In version 1, a child test $w' - G(\tau, z)$ becomes

$$w - F(t, y) - a_r G(\tau, z),$$

which is still of the special quadratic form. In version 2, $a_r Q(\tau, z, w)$ remains quadratic, has old derivative scale B, and satisfies

$$|\partial_{ww}(a_r Q)| \le K_0 a_r \frac{(B/a_r)^2}{a_s} = K_0 \frac{B^2}{a_{r+s}}.$$

The small-value and first-derivative bounds transform in the same way. Spatial matrices and times compose, whole-block fitting is preserved, and assignments may record the entire path with nested time partitions. All operations retain tempered bounds. □

In version 2, largest-axis fullness gives a geometric interpretation to

$$\log_{N_0} \frac{q^2}{J_c}, \tag{2.17}$$

which we call the narrowness in the specified depth units: it is the additional thinness of the smaller spatial axis, up to subpower factors. Under composition,

$$\frac{q_{cd}^2}{J_{cd}} = \frac{q^2}{J_c}\frac{q_d^2}{J_d}.$$

Thus narrowness exponents add exactly, without any assumption that the spatial axes are parallel. Statements about a common value of this exponent use a homogeneous subfamily.

The exponent h in the common interval length $N_0^{-h+o(1)}$ is its *time exponent*.

Definition 2.14 (Minimum horizon)**.** For a homogeneous exact problem E of length L, let $H(E)$ be the infimum of the exponents $h \geq 0$ for which a true system at terminal depth L exists. The infimum includes systems on arbitrary subsequences and further permitted incidence selections, in the coordinates of E. Such selections do not change the terminal density exponent.

The rest of the section proves two complementary bounds. Every homogeneous exact problem satisfies $H(E) \leq L$: terminal fits can be built on intervals of length comparable to a_L. A hypothetical Kakeya counterexample, however, yields a cap-class problem with $H(E) > 0$. Both arguments use the weighted plank estimate below.

2.5 The weighted plank input and the initial horizon bound

We first prove that true terminal systems always exist with horizon at most L. At time scale a_L, the native quadratic signal already supplies the polynomial test. A weighted plank estimate will supply spatial boxes whose incidence mass meets the true floor.

Lemma 2.15 (Weighted full-time plank estimate)**.** *Fix a bounded affine-line chart in $\mathbb{R}^2$, and a fixed permitted enlargement of mesh cells. For every $\varepsilon > 0$, there are $\eta_\varepsilon > 0$ and $C_\varepsilon < \infty$ with the following property. Let*

$$M_i(t) = b_i + tu_i, \qquad 0 \leq t \leq 1,$$

be finitely many indexed traces, with b_i, u_i in the fixed bounded chart, and arbitrary positive weights ω_i. Repeated traces are allowed. Let $D \geq 1$, $\rho = D^{-1}$, and suppose that every full-time plank

$$\mathcal{P} = \left\{ (t,x) : \begin{array}{l} 0 \leq t \leq 1, \\ |\langle x - c - tv, e_1\rangle| \leq a\rho, \\ |\langle x - c - tv, e_2\rangle| \leq b\rho \end{array} \right\}, \qquad 1 \leq a \leq b \leq D,$$

where (e_1, e_2) is an arbitrary orthonormal spatial frame and $c, v \in \mathbb{R}^2$, satisfies

$$\sum_{i : M_i \subset \mathcal{P}} \omega_i \leq Aab. \tag{2.18}$$

Fixed dilations of these plank tests are permitted. Mark time bins J_i, each of length ρ, with $\#J_i \geq D^{1-\eta_\varepsilon}$. If E_{vis} is the union of the spatial–time mesh cells visited at their bin centers, or fixed enlargements of those cells, then

$$\sum_i \omega_i \#J_i \leq C_\varepsilon A D^\varepsilon \#E_{\mathrm{vis}}. \tag{2.19}$$

Proof. This is the indexed weighted formulation of OpenAI [24, Lemma 2.3]. The plank condition concerns containment of the entire trace, not merely its marked portions. The cited statement permits arbitrary positive weights and repeated indices, and has precisely the long-mark hypothesis above. It is proved there by weighted convex clustering from the union estimate of Guth et al. [15, Theorem 1.1]. □

We record explicitly two extensions of use below. A continuous weighted law of bounded affine traces can be rounded in coefficient space at a mesh much smaller than ρ. Group traces with the same rounded coefficients and the same finite mark set, and assign each group its total prior mass. This gives a finite indexed family for Lemma 2.15. Fixed enlargements of cells and planks recover the original traces, and do not alter the estimates beyond constant factors. These finite approximations are used for the estimate; their mark sets need not be output labels of a chart construction.

Also suppose the chart parameters are bounded by a subpower K_0, the mesh is a fixed power of N_0^{-1}, and the available labeled lengths are at least K_0^{-1}. A common spatial rescaling by a subpower places the traces in a fixed bounded chart. Refining meshes and enlarging cells by subpower factors changes all mark and cell counts by subpower factors. The long-mark condition is then satisfied for every fixed $\varepsilon > 0$ at sufficiently large N_0. Choose $\varepsilon \downarrow 0$ only after these fixed-ε requirements, by a slow diagonal. The factors $C_\varepsilon D^\varepsilon$ are thereby subpower. In physical widths u, v, the resulting conclusion bounds raw weighted average multiplicity by a subpower times the supremum of $\rho^2 \sum_{M_i \subset \mathcal{P}} \omega_i/(uv)$. This is the form used when a plank is inferred from a small visited-cell set.

Proposition 2.16 (Initial horizon bound)**.** *Every homogeneous exact problem satisfies $H(E) \le L$.*

Proof. At the terminal time scale, the native test already controls the quadratic signal. We choose spatial planks greedily and show that an uncovered residual would produce another admissible plank through Lemma 2.15.

Initial labels and tests. Choose a dyadic $q \sim a_L$. In each component and each q-interval I, group trajectories by y at the midpoint of I, to width q, and by X there, to width a_L. Call the joint label G. Along the block, both y and X move by at most $K_0 q$: for X, use its quadratic coefficients and the uniform bound in the exact problem. At an observed incidence, the hidden bin determines X to $K_0 a_L$. In version 2, Taylor's formula in w uses the native first-derivative and curvature bounds; the curvature contribution is $O(K_0 a_L^2)$. Thus G is known by p_L up to subpower lists.

Let b_G be the X-bin center. Use the test $w - b_G$ in version 1, or $P_* - b_G$ in version 2. It has the required terminal small-value and derivative bounds throughout the block and at selected incidences. It remains to find spatial planks carrying the true mass floor.

Translation by the spatial bin center, spatial rescaling by Rq, and time rescaling to $[0, 1]$ place all affine traces in one fixed bounded chart, provided R is a sufficiently large subpower. Put $J_0 = (Rq)^j$. In these coordinates use the old trajectory measure inside G, without normalizing it to a probability; only time is normalized to the block. The typical exact-base density in this group is at least $n(L)J_0$ up to subpower factors: refine by the known label G, change coordinates, and sum over compatible hidden bins.

Choose a common mesh $\rho = D_m^{-1} \sim N_0^{-D}$, with D_m dyadic. Here D is fixed sufficiently large for the tempered bounds and total variation errors, before the loss parameters below are chosen. Joint flattening transfers the preceding lower bound to the cell averages, apart from power-small exceptional mass, weighted over all components, groups, and blocks in original time. Coordinate changes and block normalizations introduce only polynomial factors.

Greedy spatial planks. Fix $0 < \xi < 1$. In each group greedily select spatial planks with arbitrary time-independent orthonormal spatial frames, affine centers, and normalized half-widths $\rho \le r_i \lesssim 1$,

$1 \le i \le j$. A plank may be selected if the remaining incidence mass on trajectories wholly contained in it, with block time normalized, is at least

$$N_0^{-\xi} n(L) J_0 \prod_{i=1}^{j} r_i. \tag{2.20}$$

Assign all such available trajectories to that chart and remove them from the group. Fixed constants in the width cutoffs are chosen to allow subsequent enlargements.

This procedure has polynomially many steps, uniformly for $0 < \xi < 1$: the mass removed in any step is bounded below by a reciprocal polynomial, and the total within-component block mass is at most one. Full containment is tested at endpoints in each spatial axis, so assignments and successive exclusions are tempered. A nonempty plank can have its center and speed controlled by choosing a contained bounded trace as center and enlarging the widths by a fixed factor. Its old spatial Jacobian is $J_0 \prod_i r_i$, and the test remains the one fixed by G.

The residual and its visited cells. Suppose the algorithm covers less than half the initial selected mass along a subsequence. From the residual we remove cells that fail the original group floor with tolerance $\xi/4$, followed by trajectory–block entries whose remaining normalized labeled duration is less than the initial total selected mass divided by a sufficiently large constant. The first loss is power-small. The second is bounded by that duration threshold, because the trajectory assignments and component–block weights have total prior budget at most one. There remains inverse-subpower mass, and every used trajectory has labeled normalized duration at least K_0^{-1}.

Write m_G for the original normalized incidence mass of a group. Weighted averaging gives a group with surviving normalized mass at least $m_G/K_0 > 0$. Every residual visited good cell has original mass at least

$$N_0^{-\xi/4} n(L) J_0 \rho^{j+1}.$$

The number of such cells is consequently at most

$$\frac{m_G}{N_0^{-\xi/4} n(L) J_0 \rho^{j+1}}. \tag{2.21}$$

The mass used in this count is the original whole-cell witness. A residual cell need not retain that floor.

Mark all time bins with positive surviving labeled duration on each used trace. There are at least D_m/K_0 marks per trace, so Lemma 2.15 applies for any fixed $\varepsilon > 0$, eventually. Moving a point to the bin center requires only a fixed cell enlargement because normalized speeds are bounded. Each trace keeps its old trajectory weight.

For $j = 1$, add an independent static coordinate uniform on $[0, 1]$; this increases the visited-cell count by at most $O(\rho^{-1})$ and preserves total weighted mark count. In both cases, using (2.21),

$$\frac{\sum_i \omega_i \# J_i}{\# E_{\mathrm{vis}}} \gtrsim K_0^{-1} N_0^{-\xi/4} n(L) J_0 \rho^2, \tag{2.22}$$

since the numerator is at least $m_G D_m / K_0$.

A new plank from the residual. Taking the supremum plank constant in (2.18), the weighted estimate therefore supplies a full-time normalized plank of physical spatial area uv with

$$\frac{\text{used trajectory weight in the plank}}{uv} \gtrsim_\varepsilon K_0^{-1} N_0^{-\xi/4} D_m^{-\varepsilon} n(L) J_0. \tag{2.23}$$

The permitted fixed enlargement also contains the unrounded traces.

In the padded case, project the rectangle onto the true spatial coordinate, and denote its projected width by W. The maximal vertical section length of a rectangle of area comparable to uv is at most Cuv/W. For example, writing its side half-widths as u, v and its angle as θ, its projected half-width is $u|\cos\theta| + v|\sin\theta|$, while a vertical chord has length at most

$$2\min\left(\frac{u}{|\sin\theta|}, \frac{v}{|\cos\theta|}\right),$$

with the evident interpretation at zero denominators. Their product is at most a fixed multiple of uv. For each original trace the accepted fraction of the static dummy coordinate is no larger than this section bound, already at one time. Thus projection turns (2.23) into the same lower bound for original trace weight divided by W.

The contradiction and limiting selection. Every counted original residual trace has labeled duration at least K_0^{-1}. In either dimension, the resulting original incidence mass in the projected or unpadded plank is therefore larger than the greedy threshold, for large N_0, if

$$D\varepsilon < \xi/4.$$

The two fixed losses then total less than $\xi/2$; all other losses are subpower. This contradicts the stopping rule.

Consequently at least half the selected mass is covered. Let $\xi \downarrow 0$ sufficiently slowly, after each fixed-ξ, ε estimate is valid. The polynomial complexity bounds above were uniform for $\xi < 1$, so the output remains tempered. The threshold (2.20) is now the true floor with a subpower loss, and $q \sim a_L$ gives time exponent $h = L$. □

2.6 From an arbitrary Kakeya counterexample to positive horizon

Theorem 2.17 (Reduction to a bad weighted problem)**.** *If a set $K \subset \mathbb{R}^4$, with no measurability or compactness assumption, contains a unit segment in every direction and satisfies* $\dim_H K < 4$*, then there are $0 < d < 1$, $S, \eta > 0$, and a tempered homogeneous exact problem E of version 2 in $C(d)$ such that $H(E) > 0$.*

Proof. We give the selection from K first, then construct the cap-class problem, and finally prove the positive horizon bound.

A finite selection from outer measure. Fix a bounded open graph-slope chart of directions, with slopes $s \in \mathbb{R}^3$, writing spatial lines as

$$x(t) = b + ts, \qquad x = (y, w).$$

A unit segment in one of these directions has a nondegenerate time projection. Every such segment contains the graph over some interval with rational endpoints, and its intercept belongs to a bounded box with integer endpoints. These are countably many choices. Countable outer subadditivity therefore supplies one interval and one intercept box for which the set S_0 of eligible slopes has positive outer Lebesgue measure, say $a > 0$. After a fixed affine coordinate change, the common time interval is $[0, 1]$, and all these slopes and intercepts lie in fixed bounded boxes. This coordinate change preserves Hausdorff dimension.

Choose $0 < \sigma < 1$ with $\dim_H K < 4 - \sigma$. There are covers by balls of arbitrarily small maximum radius with

$$\sum_i r_i^{4-\sigma} \le 1.$$

Only balls meeting the fixed bounded region containing the eligible graph segments are needed for the following argument; their centers are bounded as well. For $k \geq 1$, let U_k be the union of balls with $2^{-k-1} < r_i \leq 2^{-k}$. There are at most $C2^{k(4-\sigma)}$ such balls.

Choose a fixed $c > 0$ so that $\sum_{k\geq 1} ck^{-2} < 1$. Every eligible witness has some k for which

$$|\{t \in [0,1] : b + ts \in U_k\}| \geq ck^{-2},$$

because the sum of these time measures is at least one. Define $S_k \subset S_0$ by existence of a bounded witness with this property. The sets S_k need not be measurable, but $S_0 \subset \bigcup_k S_k$. Choose $c' > 0$ with $c' \sum_{k\geq 1} k^{-2} < 1$. Outer subadditivity implies that for some k

$$|S_k|^* \geq c' a k^{-2}. \tag{2.24}$$

Indeed the reverse strict inequality for every k would imply $|S_0|^* < a$. The same scale thus has both the time-fraction condition defining S_k and (2.24). As the maximum cover radius tends to zero, every possible such k tends to infinity.

Put $\delta = 2^{-k}$. A maximal δ-separated subset of the bounded set S_k is finite and covers S_k by δ-balls. Its cardinality M therefore satisfies

$$ck^{-2}\delta^{-3} \leq M \leq C\delta^{-3}. \tag{2.25}$$

Choose one successful witness for each of these finitely many slopes. This is the only witness choice needed, and uses no measurable selection.

Smoothing and three density bounds. Let $N_0 = \delta^{-1}$, initially $L = 1$, $P_* = w$, and $B = 1$. Use the uniform mixture of the selected lines and perturb all three intercept and all three slope coordinates by independent uniform boxes of radius a fixed small multiple of δ. Select incidences in fixed enlargements of the balls in U_k. Each original covered time stays in those enlargements under every allowed perturbation, so selected mass is at least ck^{-2}.

This is a tempered version-2 problem. The mixture may be represented by the internal indexed copies of Definition 2.2; its count and density heights are polynomial, its supports are boxes, and the finite union of enlarged balls is a bounded-degree polynomial predicate of polynomial size. The separated slope centers and (2.25) give the full three-slope density ceiling

$$\frac{d\nu_s}{ds} \leq Ck^2 \leq K_0. \tag{2.26}$$

Conditional on the full slope, the y-intercept density is at most $C\delta^{-2}$: within each mixture component this follows from independent intercept smoothing, and conditional mixture weights sum to one.

Homogenize jointly for $0 \leq r \leq 1$, $0 \leq u \leq 4$, where u is the depth of the two-dimensional V-bin. Write

$$n(r,u) = N_0^{-f(r,u)}, \quad m(r,u) = N_0^r n(r,u), \quad g(r,u) = r - f(r,u).$$

The resulting limiting exponents obey

$$g(1,0) \geq \sigma, \qquad g(0,0) \leq 0, \qquad g(r,u) \leq r + 2 - 2u. \tag{2.27}$$

For the first inequality, at width δ each enlarged covering ball uses $O(1)$ hidden w-bins over a base (t,y)-volume $O(\delta^3)$. The total observation-space measure of these states is at most $C\delta^{\sigma-1}$. The selected mass is inverse-subpower and the typical density is $n(1,0)$, so $n(1,0) \geq K_0^{-1}\delta^{1-\sigma}$. This gives $g(1,0) \geq \sigma$.

For the third inequality, the V-marginal has bounded region and density at most K_0, by integrating (2.26) over the bounded remaining slope. A V-bin has probability at most $K_0 N_0^{-2u}$. At fixed t, the conditional $y(t)$-density, given the full slope, is at most $C\delta^{-2}$. Ignoring the hidden w-bin gives the pointwise joint ceiling $K_0 N_0^{2-2u}$, hence the claimed bound for $g(r,u)$.

For the second inequality, the projected y-lines have a full-time plank cap at every fine mesh $\rho = D_m^{-1}$:

$$\nu\{l : y_l([0,1]) \subset \mathcal{P}\} \le K_0 ab\rho^2 \tag{2.28}$$

for spatial radii $a\rho, b\rho$. Containment at $t = 0, 1$ confines the velocity to the corresponding rectangle of radii $2a\rho, 2b\rho$, and its density is at most K_0.

Suppose instead that $n(0,0)$ grows by a fixed positive power. Choose $\alpha > 0$ smaller than that power, and flatten at a sufficiently fine fixed polynomial mesh $\rho \sim N_0^{-D}$. Retain typical cells. Their projected base cell masses have the lower bound $N_0^{\alpha}\rho^3$, after decreasing α if needed; the bounded w-range meets only subpower many unit bins. There are therefore at most $K_0 N_0^{-\alpha}\rho^{-3}$ visited base cells. Prune short labeled durations using the product prior, leaving inverse-subpower total mass and duration at least K_0^{-1} per used trace. Write $m_{\mathrm{ret}} \ge K_0^{-1}$ for the retained incidence mass, and mark the visited time bins of each trace. Each mark covers at most ρ units of time. In the finite weighted approximation used for Lemma 2.15, this gives

$$\sum_i \omega_i \# J_i \ge \frac{m_{\mathrm{ret}}}{\rho}.$$

Dividing by the visited-cell bound, and enlarging the subpower factor, yields the raw weighted average multiplicity

$$\frac{\sum_i \omega_i \# J_i}{\# E_{\mathrm{vis}}} \ge \frac{m_{\mathrm{ret}}/\rho}{K_0 N_0^{-\alpha}\rho^{-3}} \ge K_0^{-1} N_0^{\alpha}\rho^2.$$

On the other hand, Lemma 2.15 and (2.28) bound it by $C_\varepsilon K_0 D_m^{\varepsilon}\rho^2$. Choose $D\varepsilon < \alpha/2$. This is impossible for large N_0, proving $g(0,0) \le 0$.

Selecting an endpoint and an angular scale. The bounds on g force positive growth between hidden depths zero and one and give an upper bound at large velocity depth. We use both facts to select an endpoint with the one-sided comparisons required by the cap class. Choose small constants $R, S > 0$, for example with $R + 4S < \sigma/2$ and $R, S < 1/10$, and maximize

$$g(r,u) - Rr + Su \quad \text{on } [0,1] \times [0,4]. \tag{2.29}$$

Continuity follows from homogenization. At a maximizer (r_0, u_0), we have

$$r_0 > 0, \qquad u_0 < 4, \qquad g(r_0, u_0) \ge \sigma/2. \tag{2.30}$$

Indeed the value at $(1,0)$ is at least $\sigma - R$. At $r = 0$, monotonicity in u and $g(0,0) \le 0$ bound the value by $4S$. At $u = 4$, the last inequality in (2.27) bounds it by $r - 6 - Rr + 4S < 0$. Finally maximality gives $g(r_0, u_0) \ge \sigma - R - 4S$.

Partition the y-intercept and V coordinates into full-time boxes B of side a_{u_0}, with dyadic rounding. Make them components with conditional trajectory priors and weights proportional to their original probabilities $\nu(B)$. After extremely light boxes are discarded and their weights are homogenized, subtracting the central affine y-line $y_B(t)$ and dividing y by a_{u_0} normalizes the base coordinates; w is retained. At fixed exact (t, y), the old V-bin determines the intercept bin up to boundedly many possibilities, since $y_0 = y - tV$. Thus appending the crop label B to $(p_s, [V]_{a_{u_0}})$ is a bounded-list refinement at every depth s.

Let $\rho_B(s)$ be the density of this refined observation in the old, unnormalized law. Time is unchanged, the base change $y = y_B(t) + a_{u_0} y_{\mathrm{new}}$ has Jacobian $a_{u_0}^2$, and conditioning the trajectory prior divides by $\nu(B)$. For transported bins the within-component density is therefore exactly

$$\rho_{\mathrm{new},B}(s) = \frac{a_{u_0}^2}{\nu(B)} \rho_B(s).$$

This is the specialization of (2.15) to a full-time crop. The actual rounded grids have bounded overlap. Combining that overlap and the bounded crop-label ambiguity with Lemma 2.3 gives the typical equivalence, for $0 \le s \le r_0$,

$$n_{\mathrm{new}}(s) \sim_{\log} n(s, u_0) \frac{a_{u_0}^2}{\nu(B)}. \tag{2.31}$$

A new angular bin of depth u corresponds, with the same bounded-overlap convention, to the old joint angular refinement at depth $u_0 + u$.

Maximality in (2.29) gives

$$\begin{aligned} f(r_0, u_0) - f(s, u_0) &\le (1 - R)(r_0 - s), \quad 0 \le s \le r_0, \\ f(r_0, u_0 + u) - f(r_0, u_0) &\ge Su, \qquad\qquad 0 \le u \le 4 - u_0. \end{aligned} \tag{2.32}$$

Consequently the new exact problem, with length $L = r_0$, belongs to $C(1 - R)$ for the chosen S and, for example, $\eta = (4 - u_0)/(2r_0) > 0$.

It remains to prove positive horizon. We return a true terminal chart's mass floor to the original prior, then bound the prior mass of all trajectories that can satisfy its geometric tests. Comparing these bounds will force its time interval to be short.

The original-prior floor of a terminal chart. Consider any true terminal chart in this new problem, including one obtained after a further permitted selection or on a subsequence. Let its time length be q, its spatial Jacobian in the cropped coordinates be J_{new}, and set

$$J_{\mathrm{phys}} = a_{u_0}^2 J_{\mathrm{new}}.$$

Multiplying its true floor by $\nu(B)$ and using (2.31), its selected mass in the original, unnormalized prior satisfies

$$\frac{m_{\mathrm{old},c}}{q} \ge K_0^{-1} n(r_0, u_0) J_{\mathrm{phys}} = K_0^{-1} m(r_0, u_0) a_{r_0} J_{\mathrm{phys}}. \tag{2.33}$$

This calculation cancels $\nu(B)$ even when the crop has a very small polynomial mass. Both priors retain their normalization before the analytical incidence restriction.

Every trajectory contributing positive selected time to this floor satisfies the whole-block spatial, small-value, and upper-derivative tests in the original coordinates, and satisfies $|P_w| \ge K_0^{-1}$ at some time. The original prior mass of these trajectories is at least $m_{\mathrm{old},c}/q$.

The slope-volume bound. Fix this chart. Let $\Sigma \subset \mathbb{R}^3$ consist of the bounded slopes s for which some bounded intercept b satisfies

$$\sup_{t \in I} |D_{\mathrm{phys}}^{-1}(y_{b,s}(t) - y_c(t))| \le K_0, \qquad \sup_{t \in I} |P(t, b + ts)| \le K_0 a_{r_0},$$

and

$$\sup_{t \in I} |P_w(t, b + ts)| \le K_0, \qquad \sup_{t \in I} |P_w(t, b + ts)| \ge K_0^{-1}.$$

Here $P(t, b+ts)$ means that the three spatial coordinates are split as (y, w), and D_{phys} is the chart's original spatial y-matrix. Use fixed bounded slope and intercept boxes containing the original smoothed prior support.

The set Σ is defined by these geometric tests alone. We impose neither crop membership, finite witness identity, density selection, nor selected-time conditions. Thus Σ contains the slopes of all trajectories contributing to (2.33).

We claim that

$$|\Sigma| \le K_0 \frac{a_{r_0} J_{\text{phys}}}{q^3}. \tag{2.34}$$

The original full-slope density ceiling will then convert this geometric bound into an upper bound for the prior mass of an eligible chart. To prove the claim, we select one eligible intercept for each slope and compare the volume swept out by these lines with the volume allowed by the chart tests.

The definition of Σ uses a fixed number of real variables and fixed-degree polynomial conditions, with quantification over b and t. All chart coefficients, bounds, and interval endpoints are parameters. Quantifier elimination, semialgebraic choice, and C^1 cell decomposition therefore provide an eligible choice $b = b(s)$ whose graph has uniformly bounded formula complexity; its C^1 full-dimensional open pieces cover Σ up to a null set. We use here the parameter-uniform forms of these foundational results from real algebraic geometry; see [1]. Their bounds depend on the formula format, not on the numerical magnitudes of the chart coefficients.

For each t, put $F_t(s) = b(s) + ts$. On a C^1 piece,

$$\det DF_t(s) = \det(Db(s) + t\,\mathrm{Id}), \tag{2.35}$$

a monic cubic in t. There is also a uniform bound on the number of regular preimages of any F_t. Indeed the fiber has fixed semialgebraic complexity, uniformly in t, the target point, and the chart parameters, so has a uniformly bounded number of connected components. A preimage with nonzero Jacobian is isolated in its fiber by local inversion inside its open C^1 piece, and hence is a separate component. This proves the bound without any bound on Db, and excludes a multiplicity growing with N_0.

Times with a large Jacobian and derivative. For a fixed slope s, remove from I intervals of radius cq around the real parts of the three complex roots of the monic cubic in (2.35). The total removed length is at most $6cq$. Outside them each factor has modulus at least cq, so

$$|\det DF_t(s)| \ge c^3 q^3.$$

Also, $g_s(t) = P_w(t, b(s) + ts)$ is real affine and has $\sup_I |g_s| \ge K_0^{-1}$. For small fixed $\epsilon > 0$,

$$|\{t \in I : |g_s(t)| < \epsilon/K_0\}| \le 4\epsilon q. \tag{2.36}$$

To verify this, let $M = \sup_I |g_s|$. If the variation of g_s on I is less than $M/2$, then $|g_s| \ge M/2$ throughout. Otherwise its slope has modulus at least $M/(2q)$, and the inverse image of $(-\epsilon/K_0, \epsilon/K_0)$ has length at most $4\epsilon q$.

Choose c, ϵ so that each bad set has length at most $q/4$. Let Σ_t be the slopes in the open C^1 pieces passing both tests. Then

$$\int_I |\Sigma_t|\, dt \ge \frac{q}{2}|\Sigma|. \tag{2.37}$$

The two good-time conditions therefore overlap uniformly for each slope by the bound on the sum of the two exceptional lengths.

The area formula and the uniform regular-fiber bound give

$$q^4|\Sigma| \lesssim q^3 \int_I |\Sigma_t|\, dt \lesssim \int_I |F_t(\Sigma_t)|\, dt \tag{2.38}$$

$$\leq K_0 q J_{\mathrm{phys}} a_{r_0}. \tag{2.39}$$

For the last inequality, the allowed y-region has area at most $K_0 J_{\mathrm{phys}}$. At fixed t, y, the conditions

$$|P(t,y,w)| \leq K_0 a_{r_0}, \qquad |P_w(t,y,w)| \geq \epsilon/K_0$$

confine w to total length at most $K_0 a_{r_0}$, after enlarging the subpower factor. This is Lemma 2.9, applied at fixed (t,y). Integration over I gives (2.39); division by q^4 proves (2.34).

Comparison with the mass floor. The original full-slope density ceiling (2.26) bounds the original prior mass of all geometrically eligible trajectories by $K_0|\Sigma|$. Combining this with (2.33) and (2.34) and cancelling $a_{r_0} J_{\mathrm{phys}} > 0$ yields

$$m(r_0, u_0) \leq K_0 q^{-3}.$$

By (2.30), $m(r_0, u_0) \geq N_0^{\sigma/2}$. Thus any true terminal system with $q = N_0^{-h+o(1)}$ has $h \geq \sigma/6$. The same estimate holds on every subsequence and further allowed selection, so $H(E) \geq \sigma/6 > 0$. This proves the theorem. □

We call a cap-class problem with $H(E) > 0$ bad. The remaining sections exclude bad problems for every fixed $S, \eta > 0$, first in version 1 and then in version 2. By Theorem 2.17, their exclusion proves the theorem for arbitrary Kakeya sets.

3 Comparison of algebraic sheets and extension of terminal time

We prove a comparison principle for two small lists of scalar functions and apply it to extend the terminal charts of an exact problem in time. The comparison forces sheets that agree on many sampled incidences to agree on a whole neighborhood. Algebraic continuation then carries these local comparisons to a common complex point, where finite jets provide labels for representative equations. This is how a fit on a longer time block can remain known from the old terminal observation.

The comparison uses fresh conditional samplings. Successful paths are kept as unnormalized submeasures; the sampling kernels are never conditioned on a later step succeeding.

We use real quantifier elimination and semialgebraic cell decomposition with parameters in fixed dimension and degree: formulas of fixed format define semialgebraic sets, and the numbers of connected components and isolated points of their fibers have bounds uniform in their real coefficients. A one-variable semialgebraic function which is finite and bounded on compact subintervals of $[2, \infty)$ has at most polynomial growth at infinity. These foundational forms, including their parameter-uniform versions, may be found in Basu et al. [1, Chapters 3, 5, 7, and 14].

3.1 Uniform norms and continuation for algebraic sheets

A holomorphic function will be called an *algebraic sheet of degree at most D* if it satisfies

$$P(z, f(z)) = 0$$

for a nonzero polynomial P of total degree at most D. There is no lower bound on a nonzero coefficient of P. All balls and polydiscs in the next lemma have fixed geometry. In particular, the

real normalization ball has nonempty real interior and stays a fixed distance inside the complex domain.

The next lemma belongs to the Bernstein–Remez theory of algebraic functions; compare Roytwarf and Yomdin [27, Sections 3.3–3.5]. We include a proof of the formulation used here.

Lemma 3.1 (Algebraic analytic norm comparison)**.** *Let $\Omega \subset \mathbb{C}^n$ be a fixed ball or polydisc, and let $B \subset \Omega \cap \mathbb{R}^n$ be a fixed closed ball with nonempty interior. Fix a compact set $K \Subset \Omega$. For algebraic sheets of degree at most D, normalized by $\|f\|_B = 1$, the following bounds are uniform:*

1. *for every fixed integer q, $\max_{|\alpha|\le q} \|\partial^\alpha f\|_K \le C_q$;*
2. *at every $z \in K$,*
$$\max_{|\alpha|\le D} |\partial^\alpha f(z)| \ge c_K > 0.$$

Consequently, let $B_1 \subset \Omega \cap \mathbb{R}^n$ be another fixed closed real ball with nonempty interior. For every measurable $E \subset B$ with $\theta = |E|/|B| > 0$,

$$\|f\|_{B_1} + \max_{|\alpha|\le q} \|\partial^\alpha f\|_{B_1} \le C_q \theta^{-C} \|f\|_E. \tag{3.1}$$

The statements are homogeneous without normalization. In particular, the second bound has the equivalent form

$$\|f\|_B \le c_K^{-1} \max_{|\alpha|\le D} |\partial^\alpha f(z)| \qquad (z \in K),$$

with the same degree D, constant c_K, and geometric data. The bounds are unchanged, with the corresponding scaled derivatives, by affine translation and rescaling of the base, including translation to a parallel real slice through a complex point. A sufficiently long finite jet at the center of an interior ball controls the norm on that ball.

Proof. We first prove local boundedness of the normalized family, allowing the degree in the function variable to drop in a limit. Suppose f_ν is such a family. Normalize the coefficient vector of a relation $P_\nu(z, W)$ to have norm one and pass to a subsequence on which it converges to a nonzero polynomial P. This limiting polynomial must involve W. Indeed, if $P = P(z)$, then for every $x \in B$, boundedness of $f_\nu(x)$ gives

$$P(x) = \lim_{\nu\to\infty} P_\nu(x, f_\nu(x)) = 0.$$

A polynomial vanishing on the real interior of B is zero, a contradiction.

Write $a(z)$ for the leading nonzero coefficient of P as a polynomial in W, and set $Z = \{a = 0\}$. On a compact subset of $\Omega \setminus Z$, all roots of $P(z, \cdot)$ lie in a fixed disk. A sufficiently large circle in the W-plane is therefore root-free, uniformly over that compact set. The same circle remains root-free for P_ν for large ν. Although some roots of P_ν may escape to infinity, a continuous chosen root cannot cross this circle.

Choose a point in the interior of $B \setminus Z$. The chosen value f_ν there is bounded. It follows, by continuation along a compact path and a neighborhood of that path in $\Omega \setminus Z$, that f_ν is uniformly bounded there. The complement $\Omega \setminus Z$ is path connected: one can join two points by complex line segments and make small detours around the finitely many zeros met on generic complex lines. The preceding bound therefore holds locally throughout that complement.

It remains to cross Z. At a point of Z, choose a complex line on which a is not identically zero, and a small disk in that line whose boundary avoids Z. After translating the disk slightly, all of

its boundary circles still avoid Z. These circles form a compact subset of $\Omega \setminus Z$, where the bounds have already been proved. The maximum principle on the disks now bounds f_ν in a neighborhood of the original point. A finite cover gives local uniform boundedness on every compact subset of Ω.

Cauchy's estimates and compactness yield a subsequence converging holomorphically on compacta, together with all fixed derivatives, to a function f. Uniform convergence on B preserves $\|f\|_B = 1$; in particular f is not zero. The relation $P(z, f(z)) = 0$ persists. Divide P by its largest polynomial factor which is a power of W. The resulting relation has a nonzero constant coefficient $a_0(z)$:

$$a_0(z) = -\sum_{k \geq 1} a_k(z) f(z)^k.$$

If f vanishes to order r at a point, then a_0 vanishes to order at least r there. Since a_0 is a nonzero polynomial of degree at most D, we have $r \leq D$. Compactness, also for a sequence of possible violating points in K, proves both asserted uniform bounds.

For completeness, these jet bounds imply a quantitative sublevel estimate. At each point of a fixed real interior ball, some directional derivative of order at most D of either $\Re f$ or $\Im f$ has absolute value at least a fixed positive constant. One may use a fixed finite set of real directions, by finite-dimensional norm equivalence for symmetric tensors. Bounds on one additional derivative make this lower bound persist on a neighborhood of fixed radius. If the order is zero, that neighborhood contains no sufficiently small sublevel values. If the order is $p \geq 1$, restrict to segments parallel to the selected direction. From a subset of length m choose $p + 1$ points separated by at least a constant times m. The divided difference formula and the mean value theorem imply

$$|\{s : |f(s)| \leq u\}| \leq C u^{1/p}.$$

Integrating over transverse slices and using a finite cover gives, for some fixed $c > 0$,

$$|\{x \in B_1 : |f(x)| \leq u \|f\|_{B_1}\}| \leq C u^c |B_1|, \qquad 0 < u < 1. \tag{3.2}$$

The same compactness argument compares the norms on the fixed interior balls involved here. Applying (3.2) with B in place of B_1 to $E \subset B$ bounds $\|f\|_B$ by $C\theta^{-C}\|f\|_E$; norm comparison and the derivative bounds then give (3.1) on B_1. Finally, the uniform lower bound on the finite jet, after normalization on the comparison ball, proves the last assertion. □

We record closure properties needed when using this lemma. Sums, differences, and fixed-degree polynomial evaluations of a fixed number of algebraic sheets have polynomial relations of bounded degree. Over the rational function field in z, form the product of the linear factors corresponding to all combinations of roots and clear denominators. Symmetry in each collection of roots bounds the degrees of the resulting coefficients. Restrictions to affine complex lines also have bounded-degree relations. If direct specialization makes a relation identically zero, perturb the line on an interior disk, normalize the coefficients of nontrivial specialized relations, and take a limit. Derivatives of the simple linear or quadratic sheets used below have this property as well: implicit differentiation gives rational expressions in the base variables and the same radicals, with nonzero denominators on the comparison domain. Eliminating the radicals and clearing these denominators gives the required relations. Thus Lemma 3.1 applies to all the differences, evaluations, and fixed derivatives used here.

Local comparisons will eventually be recorded at a common complex point. The next lemma bounds the change in norm when a sheet is continued along a path that avoids its branch points, uniformly even when the equation coefficients degenerate.

Lemma 3.2 (Polynomial bounds for continuation)**.** *Fix the base dimension, polynomial degrees, number of sheets, and a bound on the number of pieces of a broken straight path. Work in a fixed bounded complex base region. The sheets may be affine functions, roots of linear equations, or simple roots of quadratic equations in the function variable whose quadratic coefficient is constant. Differences and fixed-degree polynomial evaluations of these sheets are allowed.*

Suppose the linear coefficients needed for linear roots and the discriminants needed for quadratic roots are nonzero throughout an A^{-1}-neighborhood of the path, where $A \geq 2$. On endpoint comparison balls of radius c/A, with fixed margins inside that neighborhood, the norms of a continued function compare in both directions with loss A^C. Fixed derivatives are controlled by these norms, and sufficiently many endpoint jets control the norms, with losses of the same form. The exponent C is independent of the values of the equation coefficients.

Proof. There are two separate uniformity arguments. For a fixed A, cover the path by $O(A)$ overlapping balls of radius comparable to A^{-1}, with fixed enlargements in the prescribed neighborhood. Each branch is simple and holomorphic on these enlargements. Its polynomial evaluations have bounded-degree relations, by the preceding closure properties. Applying Lemma 3.1 on successive overlapping balls gives a finite bound $C^{O(A)}$ for the endpoint norm ratio. It is uniform in all coefficient values and is bounded for A in compact subintervals of $[2, \infty)$. A function with zero starting norm vanishes under continuation, so it need not enter a ratio.

We next show that the best such bound has a fixed semialgebraic description. Split all complex parameters into real and imaginary parts. The equation coefficients, the boundedly many path vertices, and the endpoint branch choices form a fixed-dimensional parameter space. Nonvanishing on the tube is expressed by real quantification over a segment parameter and a displacement of size less than A^{-1}.

For a square root of a nonvanishing discriminant, continuation of the endpoint signs can be encoded with finitely many crossings. Choose a generic ray from zero, avoiding the discriminant values at the path vertices. Along each straight piece the discriminant is a polynomial in a real segment parameter. Except when its argument is constant, there are only boundedly many critical directions: differentiate the argument, whose numerator is a real polynomial of bounded degree. Choose the ray away from those directions; a constant-argument piece can also be avoided. Crossings of the ray are then transversal and their number is bounded by the degree. Relative to a square root of the ray direction, label an endpoint square root by the sign of its imaginary part. Each crossing changes this sign, and the parity of the bounded list of crossings specifies exactly the continued endpoint root. Crossing parameters, their order, transversality, and the parity condition can all be described by polynomial conditions and real quantifiers of fixed complexity.

To evaluate a norm on an endpoint ball, append one straight segment from its center to a variable point of that ball and use the same description of continuation. Quantification over that point and its selected roots describes the norm inequalities. It follows from quantifier elimination that the supremum of the endpoint ratio over all admissible data is a one-variable semialgebraic function of A. The first argument proves that this supremum is finite and locally bounded. The polynomial growth theorem for one-variable semialgebraic functions therefore bounds it by A^C; these are the fixed-format semialgebraic facts recalled above. The $O(A)$ comparison balls were used only to prove finiteness and are not part of this semialgebraic description.

Reversing the path proves the reverse comparison. Applying Lemma 3.1 on the endpoint balls, and accounting for their radius in derivatives, proves the jet assertions. □

In particular, a tiny constant quadratic coefficient causes no loss outside the stated bound. For example, the roots of $\varepsilon W^2 + zW + 1$ can have very different sizes. When its two branch points are

closer than the prescribed tube radius, an admissible path cannot separate them and change the relevant continuation parity. In general the coefficient-independent fixed-A bound and the fixed-format continuation description perform precisely this separation of a bounded chosen branch from unused escaping roots.

3.2 Two-color matching

The next lemma compares close values with separated derivative data. The density bound and lower mass bounds used in the chart application supply lists of size at most $\delta^{-d+o(1)}$ at accuracy δ, for a fixed $d < 1$. Under the lemma's derivative and sampling bounds, two such lists cannot frequently agree in value while their rescaled derivative data stay apart. The functions need not be algebraic; that restriction enters only when we pass from derivative data to graph norms.

Lemma 3.3 (Two-color matching)**.** *Fix $D \geq 1$, an integer $J > D$, and $d_0 < 1$. There is no sequence $N \to \infty$ with the following properties. There are two indexed lists of real C^{J+1} functions on a unit interval, each of cardinality at most $N^{d+o(1)}$, where $d \leq d_0$, and all derivatives through order $J+1$ are bounded by $K = N^{o(1)}$. There is a probability law on events (i_1, i_2, t) such that, for each color $\kappa = 1, 2$, its (i_κ, t)-marginal is bounded by*

$$K\pi_{\kappa,i_\kappa}\,dt, \qquad \sum_i \pi_{\kappa,i} = 1, \tag{3.3}$$

and every retained event satisfies

$$|f_{1,i_1}(t) - f_{2,i_2}(t)| \leq KN^{-1}, \tag{3.4}$$

$$\max_{1\leq j\leq D} |f^{(j)}_{1,i_1}(t) - f^{(j)}_{2,i_2}(t)| \geq K^{-1}. \tag{3.5}$$

The impossibility is uniform with sufficiently small fixed power losses in place of all displayed subpower losses, for these fixed orders and $d \leq d_0$.

Proof. Suppose such a sequence exists. We record the functions' jets at successive time and value scales. The list size bounds the number of finest records. On most nearby scales, an alternating path along functions from the two lists has a polynomial endpoint map with a nonzero Jacobian. Its image forces more records than this bound allows. The resampling argument below controls the law of these paths while retaining their full dependence on the preceding choices. This use of alternating paths and an endpoint Jacobian is analogous to the method of refinements for incidence operators; compare Christ [8, Sections 3–4].

All probability restrictions in the proof have subpower mass, until the bounded-length switching experiment is introduced. Normalizing such a restriction changes (3.3) only by a subpower factor. We enlarge K as necessary.

Jet scales and nested states. Logarithmically pigeonhole the derivative gaps through order J. After a subsequence their exponents are β_j, meaning that the j-th gap has size $N^{-\beta_j+o(1)}$ on the retained events. Exponents above a fixed large constant greater than one are cut off; at a cutoff we use only the corresponding upper bound. We have

$$\beta_j \geq 0, \qquad \beta_0 \geq 1, \qquad \beta_j = 0 \text{ for some } 1 \leq j \leq D.$$

At value scale N^{-a}, choose a time radius $N^{-r(a)}$ on which every Taylor term of the difference is at most N^{-a}, up to subpower factors. The j-th term requires $r(a) \geq (a - \beta_j)/j$. The corresponding

accuracy for an individual j-th derivative is $N^{-(a-jr(a))}$. This exponent need not increase with a, so we retain its largest preceding value. Thus, for $0 \le a \le 1$, define

$$r(a) = \max_{1\le j\le J} \frac{a-\beta_j}{j}, \qquad l_j(a) = \max_{0\le x\le a} \{x - jr(x)\}. \tag{3.6}$$

Then $r(0) = 0$, and r is increasing, convex, and piecewise linear. Moreover,

$$l_0(a) = a, \qquad 0 \le l_j(a) \le \beta_j, \qquad l_j(a) - l_{j+i}(a) \le ir(a). \tag{3.7}$$

The last inequality follows by evaluating l_{j+i} at a point where the maximum for l_j is attained. Since $r(x) \ge x/D$, for $a > 0$ we also have

$$(J+1-j)r(a) > l_j(a). \tag{3.8}$$

Indeed, if a maximizing point x for $l_j(a)$ is positive, use $(J+1)r(x) > x$ and monotonicity of r; if $x = 0$, then $l_j(a) = 0$ and $r(a) > 0$.

On an open piece where the unique active term in r is m, convexity gives

$$r(a) = (a-\beta_m)/m, \tag{3.9}$$

$$l_j(a) = a - jr(a) \qquad (0 \le j \le m), \tag{3.10}$$

$$l_j(a) > a - jr(a) \qquad (j > m). \tag{3.11}$$

For $j \le m$, the function $x - jr(x)$ is nondecreasing up to the active piece. For $j > m$, moving slightly to the left within that piece strictly increases it. The active β_m has not been cut off, because the cutoff terms cannot maximize r on $[0,1]$.

We choose the finite grid before constructing its states or selecting their atom masses. Fix an integer $C > 5J$. Trim a small fixed amount from the ends of every active piece, also staying away from zero, and omit pieces shorter than the trim. On the remaining compact intervals, the gaps

$$l_{m+i}(a) + ir(a) - \beta_m > 0 \qquad (1 \le i \le J-m)$$

and all needed remainder gaps are uniformly positive. Choose a common small step b, after these trims, and subdivide each remaining interval. Write A_0 for its first subdivision point. We will use steps $[a, a+b]$ only when $a + Cb$ still lies in that interval. Take a finite increasing grid containing 1, all these subdivision points, and every required $a+b$ and $a+Cb$. The construction that follows is performed anew for each such grid. All its selections and limits in N precede any decrease of b or of the trims.

For a level a in this fixed grid, the state F_a records the dyadic interval I_a containing the event time, of length comparable to $N^{-r(a)}$, the bins of both arms' derivatives of orders $0, \dots, J$ at the center of I_a, with widths $N^{-l_j(a)}$, and all earlier grid states. The dyadic partitions are nested. Taylor translation over I_a has uncertainty at derivative order j bounded by

$$K \sum_{i=0}^{J-j} N^{-l_{j+i}(a)-ir(a)} + KN^{-(J+1-j)r(a)} \le KN^{-l_j(a)}. \tag{3.12}$$

Here (3.7) and (3.8) handle, respectively, the binned terms and the remainder. Applying the same calculation to the two arms, starting from the event gaps and $l_j \le \beta_j$, shows that their center bins differ by only K neighbors, at every recorded level.

An index from one color and I_1 determine that arm's entire history. The other arm's history then has only subpowerly many neighbor choices. Therefore

$$|\operatorname{supp} F_1| \le N^{d+r(1)+o(1)}. \tag{3.13}$$

For this fixed grid, make a further subpower selection on which the masses of occurring atoms, measured *before this selection*, are $N^{-u_a+o(1)}$, uniformly at each grid level. Extremely light atoms can first be omitted using (3.13). The exponents are nonnegative and nondecreasing, and

$$u_1 \le d + r(1). \tag{3.14}$$

If a retained coarse atom at a has retained descendants at $c > a$, their number is at most

$$N^{u_c-u_a+o(1)}. \tag{3.15}$$

To see this without any assertion about later conditional probabilities, denote the pre-selection law by p_{pre}. Each retained child has p_{pre}-mass at least $N^{-u_c-o(1)}$, and the disjoint children lie in a parent of mass at most $N^{-u_a+o(1)}$. Summing their pre-selection masses proves (3.15). Write $\mathbb{P}$ for the final normalized event law.

The upper bound (3.14) will contradict the growth forced by matching. More precisely, on almost all small steps in an active m-piece we will prove

$$u_{a+b} - u_a \ge b\left(1 + \frac{1}{m} - o_b(1)\right).$$

Summing these increments, then decreasing b and removing the trims, will force the uniform upper bound $d + r(1)$ to be at least $1 + r(1)$. The next two constructions establish this increment on most steps of the already fixed grid.

Derivative bins with little entropy growth. We use the entropy chain rule and conditional mutual information in their standard forms; see Cover and Thomas [9, Chapter 2]. Fix $D_1 > 5$. The variable Z records the pair of m-th derivative bins at the center of I_{A_0}, now at width $N^{-\beta_m - D_1 b}$. Since $l_m(A_0) = \beta_m$,

$$H(Z \mid F_{A_0}) \le 2D_1 b \log N + O(1).$$

At a step retained in the grid construction, let

$$L_a = \log \frac{\mathbb{P}(Z \mid F_{a+Cb})}{\mathbb{P}(Z \mid F_a)}$$

at the sampled derivative-bin pair. Conditional mutual information and the entropy chain rule give

$$\sum_a \mathbb{E} L_a \le C H(Z \mid F_{A_0}).$$

The negative part has bounded expectation. Indeed, for probability vectors p, q,

$$\sum_{p(z)<q(z)} p(z) \log \frac{q(z)}{p(z)} \le \frac{1}{e} \sum_z q(z) \le \frac{1}{e}.$$

Conditional on the fine state, this inequality applies with p its law of derivative bins and q the coarse law of derivative bins. Since there are order b^{-1} steps, the average of $\mathbb{E}(L_a)_+$ is $O(b^2 \log N + 1)$. Markov's inequality gives an average probability $O(b^{3/4}) + o_N(1)$ for $L_a > b^{5/4} \log N$. A second

application shows that, except for a proportion $O(b^{1/2})+o_N(1)$ of the steps, this failure probability is at most $b^{1/4}$. Call these steps good. Thus at a good step,

$$\mathbb{P}(Z=z \mid F_{a+Cb}) \le N^{b^{5/4}}\mathbb{P}(Z=z \mid F_a) \tag{3.16}$$

at the sampled value, apart from probability tending to zero with b.

The switching experiment. Fix a good step. We will make $m+1$ traversals, whose m internal arrival times supply the variables of a full-rank endpoint map. Its endpoint must lie in the small union of targets supplied by (3.15); comparing those volumes will force the required increment. Start with $E^0 \sim \mathbb{P}$, keep every time in the same I_a, and proceed as follows. At the beginning of traversal $p+1$, draw a fresh event $\widetilde{E}^p$ from $\mathbb{P}$ conditional only on

$$F_{a+Cb}(\widetilde{E}^p) = F_{a+Cb}(E^p).$$

Write $t_p = t(E^p)$ for the running time and $\widetilde{t}_p = t(\widetilde{E}^p)$ for the refreshed event's witness time. Both lie in the same fine interval I_{a+Cb}, but the refresh leaves the running time at t_p. Choose a color κ_p by the rule below and set

$$\varphi_p = f_{\kappa_p, i_{\kappa_p}(\widetilde{E}^p)}.$$

Draw E^{p+1}, including its time t_{p+1}, freshly from $\mathbb{P}$ conditional only on this arm's color, index, and I_a. Traverse φ_p from t_p to t_{p+1}. The increments can have either sign.

Before any failures are removed, E^p and $\widetilde{E}^p$ have marginals bounded by $2^p\mathbb{P}$. Refreshing on a state preserves a marginal bound of this form. For a traversal, dominate the adaptively selected arm kernel by the sum of the two color kernels; each color kernel preserves $\mathbb{P}$ after averaging over its index and interval. This proves the assertion inductively.

The possible coarse states stay in a subpower neighbor set of the start. A refresh preserves F_a, since the finer state includes it. A traversal preserves the chosen arm's bins at a and at all earlier levels: its index and I_a determine these data. At either endpoint the other arm is within K-neighbor bins. There are only finitely many coordinates at the fixed grid and only $m+1$ traversals. Thus all possible coarse states lie in a start-determined set of size K, after increasing K.

Let t_a be the center of I_a, and subtract the order J Taylor polynomial T of one starting arm there. On traversal $p+1$, put $\varphi = \varphi_p$ and set

$$s = (t-t_a)N^{r(a)}, \qquad Y_j = N^{l_j(a)}\partial_t^j(\varphi - T)(t), \quad 0 \le j < m.$$

Along a traversal,

$$\frac{dY_j}{ds} = Y_{j+1} \quad (j < m-1), \qquad \frac{dY_{m-1}}{ds} = U + O(N^{-2b}), \tag{3.17}$$

where U is a constant of size at most K. Indeed the factor in the last derivative is N^{β_m}. Taylor expansion of this residual m-th derivative at t_a shows that its variation is bounded by

$$K\sum_{i=1}^{J-m} N^{\beta_m - l_{m+i}(a) - ir(a)} + KN^{\beta_m-(J+1-m)r(a)}.$$

The strict trimmed gaps make this smaller than N^{-2b}, for b sufficiently small.

Choose U from the fresh derivative-bin pair, the higher jet bins at A_0 recorded in the coarse history, and the fixed frame T. Taylor translation from the center of I_{A_0} to t_a gives an error $O(N^{-D_1 b})$ in its m-th term, while the higher-bin errors have exponents

$$l_{m+i}(A_0) + ir(A_0) - \beta_m > 0.$$

The remainder has the same strict gap as before. This discretizes U to the precision required in (3.17) using only the start, the coarse state, the derivative-bin pair, and the color. The two controls available in a fresh pair differ by at least K^{-1}: the m-th derivative gap at its event has exponent β_m, and all the approximation errors are smaller by a fixed power. Choose either color on the first leg. Thereafter at least one of the two available controls is separated from the preceding control by $1/(2K)$; choose such a color and enlarge K.

At a switch, the old and new arms have K-neighbor fine bins at $a + Cb$. The running time t_p and the witness time $\widetilde{t}_p$ lie in that finer interval, so the bins control the jump at the actual switching time t_p. Applying (3.12) there gives a jump in Y_j of size at most

$$KN^{l_j(a)-l_j(a+Cb)} = KN^{-(1-j/m)Cb} \le N^{-2b}, \qquad j < m.$$

Here $C > 5J$ absorbs the subpower factors into the stated N^{-2b} bound. This also accounts for the comparison to the starting arm at the first switch.

A step fails when (3.16) fails at its refresh or when its chosen color-index/interval pair has $\mathbb{P}$-mass less than $\pi_{\kappa,i}|I_a|/\log N$, with the weights from (3.3). For each color the sum of these latter thresholds over all indices and intervals is $1/\log N$. The marginal domination just proved therefore makes the total failure probability small. On a retained pair, the new-time conditional density is bounded by

$$\frac{K\pi_{\kappa,i}}{\pi_{\kappa,i}|I_a|/\log N} \le \frac{K}{|I_a|}.$$

Consequently the m internal arrival times can also be required to be pairwise separated by $|I_a|/K$, with negligible additional loss: test each new internal time against the preceding ones and choose the final reciprocal-subpower separation threshold smaller than the reciprocal of the preceding density bound. The experiment has successful probability bounded below when averaged over starts, first for a sufficiently small fixed b and then for large N.

Domination conditional on the entire past. Fix the exact start E^0. For every coarse state in its start-determined neighbor set, push its conditional law of derivative bins to each of the two possible discretized controls, using that state's earlier data and the fixed frame T. The sum of these pushforwards is a measure Λ on controls, with $\Lambda(\mathbb{R}) \le K$.

Let $\mathcal{H}_p$ contain the entire sampling history before the next refresh, including all hidden event data. This history fixes the current finer and coarse states. The next refresh has exactly the finer-state conditional law, independently of the rest of $\mathcal{H}_p$. On its good derivative-bin pairs, (3.16) dominates that law by the corresponding coarse law of derivative bins. The chosen arm can depend on the full history and the full refreshed event, but its point mass is bounded by the sum of the two candidate-control point masses. Finally, conditional on that complete refreshed event and its actual chosen index, the retained new-time density is at most $K\,ds$. Therefore the *unnormalized* successful step kernel satisfies

$$Q_{\mathcal{H}_p}(dU\,ds) \le KN^{b^{5/4}}\Lambda(dU)\,ds \quad \text{for every admissible } \mathcal{H}_p. \tag{3.18}$$

The same Λ works for every such history with this start.

For a nonnegative necessary test depending only on the start and the successive controls and times, (3.18) may be iterated backwards. Integration of the final step with its full previous history fixed leaves an upper integral depending only on the preceding displayed controls and times. Repeating this operation backwards gives a product bound for the unnormalized successful kernels, without conditioning any kernel on future survival. This is why possible hidden dependence of future events on the refreshed line identity does not change the product upper bound.

The endpoint map and its target. Write $U_1, \dots, U_{m+1}$ and $s_1, \dots, s_{m+1}$ for the controls and arrival times, and keep the starting time s_0 fixed. Solve (3.17) with exact constant controls and no switching errors. Its endpoint differs from the actual low jets by $O(N^{-2b})$. At fixed final time s_{m+1}, the control contribution to its j-th coordinate is

$$\frac{1}{(m-j)!}\left[U_1(s_{m+1}-s_0)^{m-j}+\sum_{p=1}^{m}(U_{p+1}-U_p)(s_{m+1}-s_p)^{m-j}\right]. \tag{3.19}$$

The omitted part depends only on the initial jets and s_{m+1}. The formula follows by telescoping the integrals of constant controls; it remains valid for signed increments.

For fixed controls the map from $(s_1, \dots, s_m, s_{m+1})$ to the final time and low jets is polynomial of fixed degree. Differentiating (3.19) with respect to s_p gives

$$-\frac{U_{p+1}-U_p}{(m-j-1)!}(s_{m+1}-s_p)^{m-j-1}.$$

Thus the absolute Jacobian determinant is exactly

$$\frac{\displaystyle\prod_{p=1}^{m}|U_{p+1}-U_p|\prod_{1\le p<q\le m}|s_p-s_q|}{\displaystyle\prod_{k=0}^{m-1}k!}. \tag{3.20}$$

It is at least K^{-1} on successful paths. The number of preimages on this regular locus is bounded in terms of the degree and dimension: each is an isolated solution of a fixed-degree polynomial system, and the fixed-format semialgebraic fiber bound applies uniformly in the coefficients.

Set $c = a + b$. A possible final child state F_c records its own interval I_c, and the final time lies in that interval. At each such time, translate each final-arm jet bin from the center of I_c, subtract T, and apply the coarse normalization. The uncertainty in derivative order $j < m$ is bounded by

$$\begin{aligned} KN^{l_j(a)}\left(\sum_{k=j}^{J}N^{-l_k(c)-(k-j)r(c)}+N^{-(J+1-j)r(c)}\right) \\ \le KN^{l_j(a)-l_j(c)}=KN^{-(1-j/m)b}. \end{aligned} \tag{3.21}$$

This uses the child's interval, not the whole coarse interval. It includes every higher-jet uncertainty and the Taylor remainder. The simulation error N^{-2b} fits these widths. The normalized final-time interval has length $O(N^{-b/m})$. Hence, for either choice of final color, the target associated with one child has volume at most

$$KN^{-b(\frac{1}{m}+\sum_{j=0}^{m-1}(1-j/m))}=KN^{-b(\frac{1}{m}+\frac{m+1}{2})}. \tag{3.22}$$

Its center may move rapidly with the final time; Fubini's theorem uses only the widths of the fibers in (3.21).

Take the union of these targets over both colors and all retained children of every possible coarse neighbor of the start. By (3.15), it has at most $N^{u_{a+b}-u_a+o(1)}$ constituent targets. Membership in this union, together with the internal-time and consecutive-control separations, is a necessary test depending only on the start, controls, and times. Other hidden success conditions can be discarded for the upper bound.

For each fixed separated control tuple, the change-of-variables formula, (3.20), and the bounded regular-fiber multiplicity bound the time integral by K times the target volume. This estimate is uniform in the controls. We may therefore integrate it against Λ^{m+1}, even when Λ is atomic, and use (3.18). The conditional successful probability is at most

$$N^{u_{a+b}-u_a-b(\frac{1}{m}+\frac{m+1}{2})+(m+1)b^{5/4}+o_N(1)}.$$

The averaged lower bound for success gives, on every good step,

$$u_{a+b} - u_a \geq b\left(1 + \frac{1}{m} - o_b(1)\right), \tag{3.23}$$

since $(m+1)/2 \geq 1$.

For this fixed trimmed grid, first let $N \to \infty$, passing to a subsequence that fixes the set of good steps. Summing (3.23) gives a lower bound on the total growth, since all omitted increments are nonnegative. After this fixed-grid limit, we let $b \to 0$, so that the proportion of excluded steps tends to zero, and then let the trims tend to zero. The sum of the b/m terms tends to $\int_0^1 r'(a)\,da = r(1)$, and the sum of the b terms tends to one. Thus (3.14) implies $d \geq 1$, a contradiction. The homogeneous exponents can depend on the fixed grid; only their uniform upper bound is used when passing to finer grids.

Finally, if no uniform small fixed-power tolerance existed, choose counterexamples with loss exponents tending to zero and N tending to infinity. Their lists can be bounded using d_0, and they would form the subpower counterexample sequence just excluded. This proves the last assertion. □

Corollary 3.4 (Local matching of scalar graphs)**.** *Fix the base dimension, algebraic degrees, real comparison balls with fixed holomorphic margins, and $d_0 < 1$. There is $\varepsilon_* > 0$ with the following property. Let $M, \Delta > 0$, and put $R = M/\Delta$. For sufficiently large R, two lists of real-on-real algebraic sheets cannot satisfy all of the following, with $0 < \varepsilon \leq \varepsilon_*$ and $d \leq d_0$:*

1. *each list has at most $R^{d+\varepsilon}$ members, and each member is within MR^{ε} in norm of one common algebraic reference sheet;*
2. *a subprobability law on pairs of indices and a common real base point in the comparison domain has total mass at least $R^{-\varepsilon}$, with each index-base marginal bounded by $R^{\varepsilon}\pi_{\kappa,i}\,dz$, where the $\pi_{\kappa,i}$ are probability weights, separately in each color;*
3. *on its events, the two values differ by at most ΔR^{ε}, whereas the norm of their difference on the comparison ball is at least $MR^{-\varepsilon}$.*

Remark. Fixed multiplicative constants may be absorbed by decreasing ε_* or taking R larger.

Proof. The positive pair mass makes both lists nonempty. Choose a member f_* of one list as reference. By the common-reference bound and the triangle inequality, every list member satisfies $\|f_i - f_*\| \leq 2MR^{\varepsilon}$. The reference has the same algebraic degree bound and common holomorphic domain as the list members. After subtracting f_* and dividing by M, the closure properties and Lemma 3.1 give derivative upper bounds through any required fixed order with loss $R^{C\varepsilon}$. At every incident point, the difference has a jet of bounded order of size at least $R^{-C\varepsilon}$. The zeroth-order value cannot supply this lower bound when R is large and ε is sufficiently small, because its upper bound is $R^{-1+\varepsilon}$. A finite set of real directions detects a positive-order directional derivative. Select one order, direction, and interior product box carrying a fixed fraction of the pair mass.

After slicing parallel to that direction, Fubini's theorem supplies a fiber interval with sliced pair mass at least $R^{-C\varepsilon}$. Before normalizing this slice, each of its two index-time marginals is still bounded by $R^{C\varepsilon}$ times the original probability index weights. Normalization introduces only another factor $R^{C\varepsilon}$. After rescaling the fiber interval, whose geometry and holomorphic margins are fixed, the restricted sheets meet Lemma 3.3 with $N = R$, with only fixed multiples of ε in all loss exponents. Choose ε_* below that lemma's tolerance divided by these constants. □

3.3 Trading terminal thickness for time

We now apply scalar-sheet matching to a true terminal chart system. The aim is to enlarge its time intervals by a factor N_0^e, while losing at most Ce in thickness depth. The new chart labels must still be known from the old terminal observation. This last requirement determines the construction: one old incidence will choose a new label, and a second incidence on the same trajectory will provide a short list containing it.

Both versions of the tempered exact model in Definitions 2.1 and 2.2 are allowed. The cap class is Definition 2.6, and the charts and atlas assignments are those of Definition 2.8: their fits hold on full trajectory pieces, while the mass floor is measured on selected incidences. The observation p_L is (2.3) at the terminal depth; thickness is $a_r = N_0^{-r}$, and the time exponent is measured in the original sequence as in Definition 2.14.

Lemma 3.5 (Terminal time extension)**.** *Fix $d_0 < 1$ and angular parameters $S, \eta > 0$. There are constants $C < \infty$ and $c > 0$, depending only on these fixed parameters and the fixed model dimensions and test degrees, with the following property. Suppose a tempered exact problem of either version in Definitions 2.1–2.2 belongs to $C(d)$ as defined in Definition 2.6, with $d \le d_0$ and length L, and has a true terminal system in the sense of Definition 2.8 at time exponent $h > 0$. For every*

$$0 < e < \min(h, cL)$$

there is, on a retained subsequence, a known atlas at time exponent $h - e$, whose labels are known at the old observation p_L, and whose fitting thickness is $a_{L'}$, where

$$0 < L' < L, \qquad L - L' \le Ce.$$

The resulting charts, trajectory assignments, tests, and selected incidences are tempered in the original exact sequence. The constant C is independent of the particular exact problem, its length, and its tempered complexity exponent.

We recall the quadratic identities used to replace implicit tests by graphs. Write z for the base variables of a test $P(z, w)$ of total degree at most two, and put $c = P_{ww}$, which is constant. Then

$$\mathcal{D} = P_w^2 - 2cP$$

is independent of w. If P, P_w, c are real, $P_w \neq 0$, and $2|cP| \le \frac{1}{2}|P_w|^2$, then $\mathcal{D} > 0$, and the real root f for which $P_w(z, f)$ has the sign of $P_w(z, w)$ satisfies

$$|w - f| = \frac{2|P(w)|}{|P_w(w)| + \sqrt{\mathcal{D}}} \le \frac{2|P(w)|}{|P_w(w)|}. \tag{3.24}$$

For $c = 0$, use the corresponding linear formula. For $c \neq 0$, the affine critical center F_{ctr} satisfies

$$w - F_{\mathrm{ctr}}(z) = \frac{P_w(z, w)}{P_{ww}}. \tag{3.25}$$

These formulas follow by completing the square. A nonvanishing discriminant on a simply connected complex ball supplies a simple holomorphic branch there; a linear equation with nonzero coefficient supplies its branch by division.

Proof. All subpower factors in this proof refer to the original exact sequence at resolution N_0. Put $\delta = N_0^{-L}$. In each component replace w by Bw; the derivative unit is then one, and terminal hidden bins have width δ, up to dyadic rounding. We undo this change at the end. Every constant implicit in $N_0^{O(e)}$ is independent of the exact problem and of L, h, e. For fixed e, subpower factors can be absorbed into such losses. The construction constants will be fixed before the final restriction on e/L.

The proof has three parts. We first compare the old equations at two incidences in a longer time block. We then record their agreement by finitely many derivatives at a common complex point and choose one real representative equation for each derivative bin. Finally, we fix one source time per block, use the source incidence to assign trajectories to these bins, and normalize the representative equations to obtain the new charts.

The paired incidence law. Prepare the true terminal system by Lemma 2.10, so its labels are known by p_L and its true floors persist. Let $q_0 = N_0^{-h+o(1)}$ be its dyadic time length. Select whole labels and homogenize their spatial singular values. By largest-axis fullness these are, up to subpower factors, $q_0, q_0 g$ in version two, where $g = N_0^{-\beta}$ and $\beta \geq 0$. In version one put $g = 1$ and use the single spatial radius q_0. Set

$$n_Q = n(L) q_0^j g. \tag{3.26}$$

Each used old label has selected incidence mass at least n_Q/K_0 in normalized q_0-block time. We use these floors to count old labels; subsequent restrictions of paired incidences need not retain a floor for each label.

Choose a coarser dyadic partition into blocks I of length q_1, with q_1/q_0 comparable to N_0^e. This is possible because $e < h$. In a component γ and block I, sample a trajectory from its original prior ν_γ, and sample t_0, t independently and uniformly in I. Retain the pair when both times are selected incidences of the old terminal system. Weight this experiment by $\omega_\gamma q_1$.

Let $s_I(\ell)$ be the selected time fraction of trajectory ℓ in I. The total pair mass is

$$\sum_{\gamma, I} \omega_\gamma q_1 \int s_I(\ell)^2 \, d\nu_\gamma(\ell).$$

By Jensen's inequality it is at least the square of the old selected mass, and hence is bounded below by an inverse-subpower factor. Since $0 \leq s_I \leq 1$, either one-time marginal of this unnormalized law is dominated by the old selected incidence law.

Write A and A' for the old labels at t_0 and t, including their q_0-subblocks. Their tests, in the normalized hidden coordinate, are P_A and $P_{A'}$. We call A the source label. Its own incidence supplies

$$|P_{A,w}(t_0, y(t_0), w(t_0))| \geq K_0^{-1}.$$

The label A' is known from the observation at t; its own derivative floor is at t.

Here is the form of the labels we will construct. From A we will obtain a coarse moving spatial box D and a simple graph near the source point. For each D, a common complex point α_D will serve to compare these graphs: after continuation, we bin a fixed finite tuple of their derivatives at α_D. Call this bin b. The pair $c_0 = (D, b)$ will determine a representative real equation, independently of the old source identity. On the retained paired incidences, A' will determine a subpower list of possibilities for c_0. Only a logarithmic derivative-size refinement will later be added to the output label.

At this stage t_0 remains averaged. In particular the estimates used to choose a representative equation will hold before one fixes the source time. Once that time and the complex comparison points are fixed in a block, the source construction will assign at most one c_0 to each trajectory throughout the block.

In one component and new block the old label count is at most

$$\frac{K_0 N_0^e}{n_Q}. \tag{3.27}$$

Indeed the new block contains $O(N_0^e)$ old subblocks, and within each the old floors and disjoint assignments give at most K_0/n_Q labels. All pair masses below within this component and block use ν_γ and the two normalized uniform time priors; restrictions are left unnormalized.

Polynomial extrapolation from an old subblock to the new block gives, for either old test on its assigned trajectory,

$$\sup_I |P| \le \delta N_0^{C_0 e}, \qquad \sup_I |P_w| \le N_0^{C_0 e}, \tag{3.28}$$

after increasing a fixed C_0. The polynomials on a line have degrees at most two and one, respectively, in both model versions. Spatial deviations from the old moving chart center also enlarge by at most $K_0 O(N_0^e)$.

For comparison at the source, we also need to evaluate the witness test away from its own incidence. The derivative of each test is at least N_0^{-e} in absolute value at the opposite sampled time, except on power-small absolute pair mass. To prove this, fix the trajectory and its own selected incidence first. The derivative is affine in time and has maximum at least K_0^{-1} on I. An affine function of maximum M is below εM in absolute value on a subset of relative length $O(\varepsilon)$. The other time is still drawn from the uniform prior; imposing its selection can only reduce this bad mass. Taking $\varepsilon = K_0 N_0^{-e}$ and treating both colors proves the claim.

Cap the one-time densities at both vertices on a fixed thickness-depth grid of spacing at most e, including $0, L$, by

$$N_0^e n(r).$$

Also cap the joint terminal/angular density at angular depth $u = Re$ by

$$n(L) N_0^{e - SRe},$$

where R is a fixed constant to be chosen shortly and $Re \le \eta L$. The typical bounds and marginal domination make these deletions power-small. They are analytical masks used for estimates of unnormalized measures.

A common spatial frame. In version two, consider a pair whose old narrow normals differ in sine angle by more than $g N_0^{Re}$; there is something to prove only when $g N_0^{Re} < 1$. The velocity lies in two strips of widths $K_0 g$, because the old spatial chart bounds hold throughout their respective subblocks. Their intersection has diameter at most $K_0 N_0^{-Re}$.

We bound the exceptional pair mass using the angular cap before any conditioning on that exception. Fix A. At the opposite time, the spatial point lies in its extrapolated rectangle of area at most

$$q_0^2 g N_0^{C_0 e}.$$

At each exact base, apply Lemma 2.9 with the value width from (3.28), the opposite-time derivative floor N_0^{-e}, and hidden-bin width δ. This gives at most $N_0^{O(e)}$ terminal hidden bins. Enlarge C_0, independently of R, so that the count is at most $N_0^{C_0 e}$. Given the terminal observation, the old knowledge of A' leaves only K_0 choices. Each disagreeing pair confines the velocity to K_0 angular

bins at depth Re. For the opposite-time marginal of the unnormalized pair law, integrate the capped density over the spatial rectangle and these hidden and angular bins, then sum over A. The bound is

$$\frac{K_0 N_0^e}{n_Q}\big(q_0^2 g N_0^{C_0 e}\big)\big(N_0^{C_0 e}\big)\big(n(L) N_0^{(1-SR)e}\big).$$

The four factors are, respectively, the old-label count, spatial area, hidden-bin count, and joint density cap; the partner-label and angular-bin multiplicities are absorbed in K_0. The normalized time integral is $\int_I dt/q_1 = 1$, and $n_Q = n(L) q_0^2 g$ in version two. Thus the exceptional mass is at most

$$K_0 N_0^{(2+2C_0-SR)e}. \tag{3.29}$$

Choose R large enough that this is power-small. In version one no normal comparison is needed.

Put $g^{\#} = \min(1, g N_0^{Re})$, with $g^{\#} = 1$ in version one. We now choose a coarse moving box D, deterministically from A, of spatial radii $q_1, q_1 g^{\#}$ in version two, and q_1 in version one. When $g^{\#} < 1$, grid the unoriented normal at spacing $g^{\#}$. In that rounded frame grid the normal component of the old center velocity at spacing $g^{\#}$, and grid its position at the new block midpoint at spacings $q_1, q_1 g^{\#}$. Use the resulting midpoint and normal velocity, and take zero tangential center velocity. When $g^{\#} = 1$, use fixed axes and a static midpoint grid of spacing q_1. The entire spatial trace of the line on I lies in this coarse moving box up to subpower factors.

Moreover, A' determines a list of only K_0 possible D's on retained pairs. Enumerate the boundedly many possible rounded orientations of A neighboring the normal of A'. For each, project the midpoint and center velocity of A' in that same rounded frame and enumerate their subpower neighborhoods in the stated grids. To justify these neighborhoods, compare both old centers to the common line. Their midpoint errors have norm at most $K_0 q_1$ and old normal components at most $K_0 q_1 g$; their velocity errors are at most K_0 and $K_0 g$, respectively. A rotation error of size $g^{\#}$ acts on these differences, giving $K_0 q_1 g^{\#}$ and $K_0 g^{\#}$. It does not act on an uncontrolled absolute midpoint. This proves the stated compatibility lists.

Use coordinates $z = (\tau, \xi)$ for D: rescale time in I to a unit interval, subtract its moving center, and divide the spatial coordinates by the displayed radii. A common subpower dilation makes the relevant traces lie in a fixed box, with bounded speeds in normalized time. The spatial Jacobian back to physical coordinates is at most

$$N_0^{C_1 e} q_0^j g. \tag{3.30}$$

Here C_1 can depend on the already fixed R. Write z_0 for the base point at t_0.

Let J_D be the spatial Jacobian of these coordinates, including the common dilation. For a fixed entry (A, D), or (A', D), push the restricted unnormalized pair law to $z_0 = (\tau_0, \xi_0)$, integrating out the other sampled time. Its density with respect to $d\tau_0\, d\xi_0$ is bounded by

$$N_0^{C_1 e} n_Q. \tag{3.31}$$

Indeed, the source's own test or the witness's extrapolated test at t_0, together with the respective derivative lower bound, allows at most $N_0^{C_0 e}$ hidden terminal bins by Lemma 2.9, with the same value width and derivative floor as in the preceding count. The capped one-time density is $N_0^e n(L)$, and the change of base variables satisfies

$$\frac{dt_0\, dy}{q_1} = J_D\, d\tau_0\, d\xi_0.$$

Consequently the density is at most $J_D N_0^e n(L) N_0^{C_0 e} \le N_0^{C_1 e} n_Q$, after increasing C_1 in (3.30).

We will repeatedly use the following counting consequence. Suppose a group construction gives at most $N_0^{C_1 e}$ options per old label. Omitting entries of mass less than

$$n_Q N_0^{-C' e}$$

removes a power-small total mass if C' is a sufficiently large fixed constant. This follows by multiplying the threshold by the option count and (3.27), in each component/block, and then summing with the original weights. For the entries (A, D) and (A', D) this gives lower mass bounds before deletion; those are the masses used in the subsequent counts.

Stable simple graphs and comparison mass. We now construct the graphs whose agreement will later be recorded at the anchor. Write $P_{A,D}$ for the source equation expressed in D's real affine base coordinates, after the initial normalization $w \mapsto Bw$, and use $P_{A',D}$ for the witness equation in the same coordinates. These equations have total degree at most two and constant second w-derivative. For either equation write P temporarily, and choose an early constant C_A larger than the value and derivative losses in (3.28). If

$$|P_{ww}| \geq \delta^{-1} N_0^{-C_A e},$$

replace this test for comparison by $\widetilde{P} = w - F_{\mathrm{ctr}}(z)$. The relation (3.25) and the derivative upper bound give

$$|w - F_{\mathrm{ctr}}(z)| \leq \delta N_0^{O(e)}$$

throughout the new block. Otherwise let $\widetilde{P} = P$. Denote the resulting equations by $\widetilde{P}_{A,D}$ and $\widetilde{P}_{A',D}$. The conversion, including the curvature test and any affine replacement, depends only on the old label and D, before a source time or a trajectory is fixed. For every unreplaced test, the choice of C_A makes $|P_{ww}P|$ much smaller than $|P_w|^2$ at the tested point $z_0, w(t_0)$, where $|P_w| \geq N_0^{-e}$. Thus

$$N_0^{-O(e)} \leq \mathcal{D}(z_0) := \widetilde{P}_w^2 - 2\widetilde{P}_{ww}\widetilde{P} \leq N_0^{O(e)}. \tag{3.32}$$

For replaced tests the discriminant is one.

For each converted equation, the coefficients of its discriminant polynomial in z are bounded by $N_0^{O(e)}$, uniformly over the retained entries. Indeed, the entry's pre-deletion mass is at least $n_Q N_0^{-C' e}$, whereas its source-base density is at most $n_Q N_0^{C_1 e}$. Its projection therefore has Lebesgue measure at least $N_0^{-(C'+C_1)e}$. On that projection the upper bound in (3.32) holds. Polynomial Remez therefore bounds the coefficients and hence the discriminant and its derivatives on a fixed complex enlargement. This argument uses the original heavy entry, before later joint deletions.

Fix a net of real balls U of radius $N_0^{-C_U e}$ covering the z_0-box, with mesh at most half that radius. Choose C_U using the uniform discriminant derivative bounds and the lower bounds in (3.32). For each pair, a net center within half a radius of z_0 then gives a ball whose fixed complex enlargement $8U$ has both of that pair's discriminants nonzero. Assign the first such ball in a fixed ordering. Admissibility is a condition on the pair and the ball; no ball is required to avoid the zeros of every equation. The pair's two converted equations have simple holomorphic branches on $8U$, real on the real ball. Choose the nearest real branch to $w(t_0)$. For an unreplaced test (3.24) applies; for a replaced test use its affine graph. In either case,

$$|w(t_0) - f(z_0)| \leq \delta N_0^{O(e)}. \tag{3.33}$$

We claim that, apart from power-small pair mass, the two chosen sheets differ in norm on $2U$ by at most

$$\delta N_0^{C_* e} \tag{3.34}$$

for a fixed sufficiently large C_*. To prove this, consider discrepancies with norm between M and $2M$, for dyadic $M \geq \delta N_0^{C_* e}$. Group by M, D, U, component and new block, and by the two center jet cubes at spacing M, in the scaled coordinates of U, through a fixed sufficiently high order. Paired cubes are bounded neighbors by Lemma 3.1. The same lemma shows that all members of such a group are within $O(M)$ of a common reference sheet in norm.

Each old label has at most $N_0^{O(e)}$ group options, with an exponent independent of C_*. The D-compatibility list is subpower, the fine ball grid has $N_0^{O(e)}$ members, and there are at most two branches per ball and boundedly many neighboring jet cubes. The range of M is polynomial in N_0, so it contributes only $O(\log N_0)$ choices. For this last assertion, the chosen branch value near z_0 is polynomially bounded by the trajectory coefficient bounds and (3.33). Implicit differentiation, using the derivative lower bound from (3.32), bounds its fixed jets polynomially. Affine replacements use an inverse curvature only above their cutoff. Finite-jet norm control bounds the whole comparison norm. The polynomial exponent here may depend on the exact problem; its logarithmic contribution is absorbed in N_0^e for large N_0. An unused far quadratic root is irrelevant to this argument.

Let m_G be the mass of one group in its component/block. At a given base point, all its values $w(t_0)$ lie in a moving band of width $O(M)$, using (3.33) and choosing C_* after its loss constants. If $M \leq 1$, choose a capped grid depth with $M \leq a_r \leq M N_0^e$. Only boundedly many hidden bins of that width meet the band. The cap-class inequality gives

$$n(r) \leq n(L)(a_r/\delta)^d.$$

With (3.30), this proves

$$m_G \leq N_0^{O(e)} n_Q (M/\delta)^d. \tag{3.35}$$

When $M > 1$, use depth zero. The native value and incidence derivative tests in version two allow only a subpower number of unit hidden bins at each exact base, by Lemma 2.9; in version one w is subpower bounded. Since $d \geq 0$ for a nonempty cap class, the depth-zero cap again gives (3.35).

In each color omit group indices, consisting of a label and branch, whose marginal mass before the joint deletion is less than $n_Q N_0^{-C' e}$. Choose C' so that the total removed mass, summed using the preceding group-option count and the old label count, is less than N_0^{-2e}. The remaining lists have cardinality

$$\leq N_0^{O(e)} (M/\delta)^d$$

by (3.35). If the total mass of these large-discrepancy groups were at least N_0^{-e}, some group would retain mass $m'_G \geq m_G/2$ after both color deletions. We normalize that group's surviving law to probability. In each color the initial index weights n_Q/m'_G sum to at most $2N_0^{C' e}$, because each retained index had its stated pre-deletion floor. After these weights are normalized to probability and the ball is rescaled, (3.31) bounds each index-base density by $N_0^{O(e)}$ times its normalized index weight.

The lists now agree in value to $\delta N_0^{O(e)}$, have paired norm discrepancies comparable to M, and satisfy all other hypotheses of Corollary 3.4. Every displayed loss $N_0^{O(e)}$ is a sufficiently small fixed power of M/δ if C_* is chosen large enough. The constants determining these losses and the option count were fixed before C_*. The corollary gives a contradiction. This proves (3.34) outside power-small mass.

Continuation to common complex points. The graphs now agree near their sampled source point, but those comparison balls vary with the point. Recording this agreement at one common complex point for each frame D gives a bin that the witness label can list. The source time remains averaged throughout this continuation and the subsequent choice of representative equations.

For each D, including its component and new-block context, choose a random anchor α_D uniformly in a fixed bounded complex box with interior. Also choose independently a random σ_D in a fixed bounded complex disk. For a source point z_0, use the complex line

$$z = z_0 + s(\alpha_D - z_0)$$

and the broken path $0 \longrightarrow \sigma_D \longrightarrow 1$ in its s-plane.

For each retained pair, with probability $1 - N_0^{-\Omega(e)}$ over these choices, both converted discriminants have lower bounds $N_0^{-O(e)}$ on a tube of radius $N_0^{-O(e)}$ around this common path. Here are the quantitative details. The coefficient bounds and the lower bound at z_0 imply a polynomial norm lower bound on the fixed anchor domain. Polynomial sublevel estimates therefore make both values at α_D at least N_0^{-Be}, except with power-small probability, for a fixed sufficiently large B. Restrict a discriminant to the complex line; its coefficients in s are bounded by $N_0^{O(e)}$. Its nonzero values at $0, 1$, together with these derivative bounds, place those endpoints at distance at least $N_0^{-O(e)}$ from all zeros in the bounded path region.

Remove disks of a much smaller radius N_0^{-Be}, with B increased, around the boundedly many such zeros. A segment from either endpoint to a uniformly chosen σ_D meets one disk only on a set of power-small probability: the disk subtends an angle bounded by its radius divided by its distance from that endpoint, and the σ_D-density is bounded. On a path avoiding these disks, factor the one-variable polynomial and compare each factor with its value at zero. Nearby zeros reduce the ratio by at most fixed powers of N_0^{-e}; far zeros reduce it by at most fixed constant factors. This gives a lower bound $N_0^{-O(e)}$ all along the path. The coefficient bounds in the full z-variables extend it to the stated smaller tube. These exceptional probabilities may be averaged against the original pair law, so the absolute mass they remove is power-small.

Choose common endpoint comparison balls of radius $N_0^{-O(e)}$, with fixed margins inside these tubes and, at the source, inside the original stable ball. We continue the two chosen branches along these paths. Lemma 3.1 first extends (3.34) to the interior complex source domain, and Lemma 3.2 then transfers the comparison to the anchor. There the difference and all required fixed jets in D's z-coordinates are at most $\delta N_0^{O(e)}$.

For each D, bin a sufficiently long jet of the source branch at α_D, separating real and imaginary parts, at one common spacing

$$\delta N_0^{C_b e}, \tag{3.36}$$

where C_b is fixed after the propagation constants. On retained pairs these anchor bins are bounded neighbors. The reached jets have polynomial range: their source norms have polynomial bounds by the preceding near-branch argument, and Lemma 3.2 propagates the norm of the branch itself. Thus we can prescribe polynomial ranges for all bins used on successes, even when an unused far root has much larger coordinates.

Given A', the source D belongs to a subpower compatibility list. For each such D, enumerate the simple roots at the anchor of the converted equation $\widetilde{P}_{A',D}$, including its affine replacement when the early curvature test required one. There are at most two, and their jet tuples are determined by those roots. These are the same converted sheets used in the local comparison. Different continuation paths can interchange these two tuples but cannot create further ones. Taking the neighboring bins of both tuples gives a list of only subpowerly many possible source bins. Thus the old-p_L list has subpower size. The intermediate $N_0^{O(e)}$ option counts have been reduced to boundedly many jet choices per compatible frame.

A representative test for each bin. The source label A now supplies the branch to be fitted on the new block. The opposite-time label A' supplies only the list just constructed, which makes the

eventual output label visible at p_L. For each D and used bin choose a representative equation $\widehat{P}$ among the converted real equations $\widetilde{P}_{A,D}$ from all old source subblocks. Require the fixed discriminant coefficient bounds, its prescribed $N_0^{-O(e)}$ lower absolute bound at α_D, and a branch whose jet lies in that bin. These are fixed equation/anchor eligibility tests, allowing affine replacements. Choose the first eligible finite option in a predetermined order. This table depends on the anchor, D, and bin, and not on t_0 or the sampled trajectory. Every source on a retained success is itself eligible.

On a common anchor ball with an inverse-power radius, the representative branch $\widehat{f}$ is simple and holomorphic. For the source branch f in the same bin, the finite-jet part of Lemma 3.1 gives

$$\|f - \widehat{f}\| \le \delta N_0^{O(e)}.$$

The derivative of $\widehat{P}$ on $\widehat{f}$ is bounded by the square root of its discriminant, hence by $N_0^{O(e)}$; also $|\widehat{P}_{ww}| \le \delta^{-1}$. Taylor expansion in w therefore yields on an interior anchor ball

$$|\widehat{P}(z, f(z))| \le \delta N_0^{O(e)}, \qquad |\widehat{P}_w(z, f(z))| \le N_0^{O(e)}. \tag{3.37}$$

For example, the quadratic error in the first estimate is bounded by $\delta^{-1}(\delta N_0^{O(e)})^2 = \delta N_0^{O(e)}$.

Lemma 3.2 returns these evaluations to the source along the source-safe path: each is a fixed-degree polynomial evaluation of the continued branch of $\widetilde{P}_{A,D}$. Only the source discriminant is required to stay nonzero on this path. The representative equation is evaluated on that source branch, without continuing the representative branch or requiring it to have a real root near every source point.

On a real segment near t_0 of relative new-block length at least $N_0^{-O(e)}$, the actual line remains in the source endpoint ball and

$$|w - f(z)| \le \delta N_0^{O(e)}.$$

For a replaced source this is its affine-center estimate. For an unreplaced source, $P_{A,D,w}$ is affine along the line, has the upper bound in (3.28), and has its own-time lower bound

$$|P_{A,D,w}(z_0, w(t_0))| \ge K_0^{-1}.$$

This is the original selected-incidence floor for the source label A. Shrink the segment by a fixed power of N_0^{-e} so that this lower bound persists within a constant factor; shrink it again to fit the source endpoint ball, using the bounded z-speed. The early cutoff C_A still makes $|P_{ww}P| \ll |P_w|^2$ there. The identity (3.24) thus applies continuously to the same source branch chosen at t_0. If t_0 is near the edge of the new block, use the one-sided portion of this segment; it has the same lower order of length.

The bound $|\widehat{P}_{ww}| \le \delta^{-1}$ permits replacement of $f(z)$ by the actual w in the returned estimates (3.37). Polynomial extrapolation along the line then gives, on the entire new block,

$$\sup_I |\widehat{P}(z, w)| \le \delta N_0^{C_2 e}, \qquad \sup_I |\widehat{P}_w(z, w)| \le N_0^{C_2 e} \tag{3.38}$$

for a fixed C_2. In version two all trajectory coordinates are affine. In version one the special form $w - F(t, y)$ still has degree at most two along a trajectory. Thus the extrapolation has fixed degree in both cases and introduces only the indicated fixed powers.

We also need a lower bound for the representative discriminant at the real source:

$$|\widehat{\mathcal{D}}(z_0)| = |(\widehat{P}_w)^2 - 2\widehat{P}_{ww}\widehat{P}| \ge N_0^{-C_3 e}. \tag{3.39}$$

This fails only on power-small pair mass, with an exponent C_3 chosen before the later curvature split. To prove this, fix the anchors and the resulting representative table. Each representative has the prescribed inverse-power discriminant lower bound at its anchor. Polynomial norm comparison transfers it to the fixed real z-box B_{real}: for an already fixed exponent b_{anc}, uniformly over the table,

$$\|\widehat{\mathcal{D}}\|_{B_{\mathrm{real}}} \geq N_0^{-b_{\mathrm{anc}}e}.$$

Let $\theta > 0$ be a fixed polynomial sublevel exponent for these degrees and this box. Then

$$\big|\{z \in B_{\mathrm{real}} : |\widehat{\mathcal{D}}(z)| < N_0^{-C_3 e}\}\big| \leq C N_0^{-\theta(C_3 - b_{\mathrm{anc}})e}.$$

For summation, fix a source label A. Its D is fixed. At the fixed anchor its converted equation $\widetilde{P}_{A,D}$ has at most two root/jet tuples, hence at most two source bins and a bounded number of associated representatives. The number of possible homotopy classes of a varying path does not enter this count. Charge the exceptional volumes against (3.31) for this source, then sum using (3.27). In each component and block the bad pair mass is at most

$$\frac{K_0 N_0^e}{n_Q}\big(n_Q N_0^{C_1 e}\big) C N_0^{-\theta(C_3 - b_{\mathrm{anc}})e} \leq K_0 N_0^{[C_1 + 1 - \theta(C_3 - b_{\mathrm{anc}})]e}.$$

The bounded representative count is absorbed in K_0. Summing with $\omega_\gamma q_1$ gives the same global bound. Choose

$$C_3 > b_{\mathrm{anc}} + \frac{C_1 + 2}{\theta}.$$

This makes the exception power-small using only constants already fixed. In particular it is independent of the later curvature threshold. The charge is made while t_0 is still averaged, using the representative table's independence of t_0.

Fixing the source and assigning comparison bins. The representative equations now satisfy the whole-block bounds (3.38). Outside power-small pair mass, their discriminants also satisfy (3.39) at the source. These bounds permit trajectory assignments on the longer blocks. The source label determines an assignment but is excluded from its output label.

Let $\mathbf{a}$ denote the anchor and detour table, and let $\mathcal{S}_{\gamma,I}(\ell, t_0, t; \mathbf{a})$ be the event that the pair passes all restrictions through (3.39). For $\vartheta = (\mathbf{a}, t_0)$, define

$$s_{\gamma,I}(\vartheta) = \iint_I 1_{\mathcal{S}_{\gamma,I}}(\ell, t_0, t; \mathbf{a})\, \frac{dt}{q_1}\, d\nu_\gamma(\ell).$$

Sample ϑ using the anchor and detour laws above and independent uniform $t_0 \in I$. For each component and block, choose $\vartheta_{\gamma,I} = (\mathbf{a}_{\gamma,I}, t^{\mathrm{src}}_{\gamma,I})$ with mass at least its average. The preceding estimates give

$$M_{\mathrm{fr}} := \sum_{\gamma,I} \omega_\gamma q_1 s_{\gamma,I}(\vartheta_{\gamma,I}) \geq \sum_{\gamma,I} \omega_\gamma q_1 \mathbb{E}_\vartheta s_{\gamma,I}(\vartheta) = N_0^{-o(1)}.$$

The lower bound concerns this weighted sum; individual blocks may have much smaller mass. In a fixed component and block, write $t_0 = t^{\mathrm{src}}_{\gamma,I}$. The frozen successful measure is

$$d\mu^{\mathrm{fr}}_{\gamma,I}(\ell, t) = 1_{\mathcal{S}_{\gamma,I}}(\ell, t_0, t; \mathbf{a}_{\gamma,I})\, d\nu_\gamma(\ell) \frac{dt}{q_1}.$$

It remains an unnormalized restriction of the original product prior. We now define geometric assignments containing its successful trajectories.

At the fixed source time, require old selected source-chart membership. This determines the old label A, its coarse frame D, and the nearest simple branch of $\widetilde{P}_{A,D}$. Require its discriminant to be nonzero on the fixed broken path, with the prescribed simple-branch conditions at its endpoints, and continue the source branch to the anchor. Require its value and jets to lie in the stored polynomial ranges. Let b be its anchor-jet bin, and require that this bin have a representative table entry. Retain only trajectories satisfying the whole-block spatial bound, both bounds in (3.38), and (3.39) at t_0, for the table entry $\widehat{P}_{D,b}$. Every frozen success satisfies these conditions.

Assign each such trajectory the preliminary label

$$c_0 = (D, b),$$

merging all source labels that give the same pair. The fixed source time and disjoint old source assignments give at most one label per trajectory in each new block. In particular this assignment is independent of the later time t. Its representative equation is the fixed table entry $\widehat{P}_{D,b}$; the old source identity is no longer recorded.

For these assignments retain the original later incidences whose old terminal label A' supplies D and b in its compatibility lists. This selection contains the frozen successes, has weighted mass at least M_{fr}, and is still a restriction of $d\nu_\gamma\, dt/q_1$. It is specified by the old incidence predicates, the fixed tables, and geometric tests; the analytical density masks are not part of the selection. At p_L, the old label A' belongs to a subpower list. Each such label gives a subpower list of frames, and for each frame at most two anchor-jet tuples with boundedly many neighboring bins; intersect this list with the stored polynomial bin range. Thus the preliminary label c_0 is known by p_L.

It remains to turn each representative equation into an admissible real test. The two curvature cases below preserve these assignments; only the low-curvature case refines their labels.

Normalizing the representative tests. The bounds (3.38) and the choice of C_3 are already fixed. Choose $C_4 > C_3/2 + 2$, then choose

$$C_s > C_2 + C_4 + 1,$$

and finally choose

$$T > \max(C_2 + C_3 + 1, C_s + C_4 + 1). \tag{3.40}$$

First consider a representative with

$$|\widehat{P}_{ww}| \le \delta^{-1} N_0^{-Te}.$$

On every assigned trajectory, the product of curvature and value in (3.38) is at most $N_0^{(C_2-T)e}$, smaller than the discriminant lower bound in (3.39). At the real source,

$$(\widehat{P}_w)^2 = \widehat{\mathcal{D}} + 2\widehat{P}_{ww}\widehat{P}.$$

It follows that the discriminant is positive and $|\widehat{P}_w(z_0, w(t_0))| \ge N_0^{-C_4 e}$. Consequently the trajectory's block maximum satisfies

$$N_0^{-C_4 e} \le M_\partial := \sup_I |\widehat{P}_w(z, w)| \le N_0^{C_2 e}.$$

Let v be the lower endpoint of the dyadic bin containing M_∂, refine the label to $c = (D, b, v)$, and use the test $\widehat{P}_{D,b}/v$. These refinements remain disjoint trajectory assignments throughout the block. The full range of v contains only $O(\log N_0)$ values, so the refined label is still known by p_L. In version one the derivative is identically one, and we take $v = 1$.

The normalized derivative has block maximum between fixed positive constants. The value bound is at most $\delta N_0^{(C_2+C_4)e}$, and the curvature bound is

$$\delta^{-1} N_0^{(-T+C_4)e}.$$

By (3.40), these fit the thickness

$$\delta_s = \delta N_0^{C_s e}.$$

The derivative upper bound is of order one. For its incidence lower bound, omit later times where its absolute value is below an inverse-subpower number κ. On each assigned trajectory this derivative is affine and has order-one block maximum, so the omitted set has relative length $O(\kappa)$. Disjointness of the assignments gives total weighted loss at most $O(\kappa)$ against the original trajectory and time priors. The same bound holds for the selected submeasure. Choose $\kappa = N_0^{-o(1)}$ with $\kappa = o(M_{\mathrm{fr}})$; the loss is then negligible compared with the available mass.

For the remaining representatives,

$$|\widehat{P}_{ww}| > \delta^{-1} N_0^{-Te},$$

use the real affine critical center and the test $w - F_{\mathrm{ctr}}(z)$, keeping the label $c = c_0$. By (3.25) and (3.38),

$$|w - F_{\mathrm{ctr}}(z)| \le \delta N_0^{(T+C_2)e}$$

on the entire new block. This fits thickness $\delta_h = \delta N_0^{(T+C_2+1)e}$, with derivative identically one and zero curvature. A real representative root near the source is not needed in this case.

Choose a curvature case carrying at least half the remaining mass, and pass to a subsequence if necessary to obtain a common thickness exponent. All constants were chosen in the order C_A, the stable-ball and comparison constants, the continuation and binning constants giving C_2, then C_3, C_4, C_s, T. In particular C_3 is independent of T: its sublevel estimate was proved for the representative table while the source time was still averaged.

Let C exceed both final thickness exponents and all needed smallness constants. Restrict $e < cL$, with fixed $c > 0$ small enough that $L - Ce > 0$ and $Re \le \eta L$. The chosen thickness is $a_{L'}$, with $0 < L' < L$ and $L - L' \le Ce$. The block length q_1 has exponent $h - e$.

Finite encoding and return to the original coordinates. The assignments just constructed use polynomially many old labels, coarse boxes, anchor bins, and representative table entries. The chosen source branches and their propagated jets have polynomial ranges, as established above; unused far roots need not be stored. The continuation predicates have bounded semialgebraic complexity by Lemma 3.2, and fixed-order implicit differentiation gives the same property for the endpoint jets. Nearest-root choices, geometric inequalities, and affine block maxima also have fixed-degree descriptions. A whole-block bound quantifies only over one time variable for each individual test. Combining these conditions with the old predicates and finite tables therefore preserves temperedness.

All equation and coordinate coefficients have polynomial bounds. The affine-center constructions invert curvatures only above their specified cutoffs, and the low-curvature normalization divides only by numbers at least a constant times $N_0^{-C_4 e}$. Thus the final tests and inverse coordinate changes remain polynomially controlled. The measurable choices made during the proof enter only through the fixed source times, anchors, equations, and tables, not through analytical masks or their searches.

Finally compose the tests with the real affine coordinate map $z = z(t, y)$ and undo $w \mapsto Bw$. In version one this preserves the form $w - F(t, y)$; in version two it preserves total degree at most two and restores the derivative unit B. The moving spatial frames contain the assigned full traces and

have largest singular value at most q_1 times a subpower factor. We have constructed tempered disjoint trajectory assignments with the required whole-block fits, inverse-subpower selected incidence mass, and labels known by the old p_L. They form the asserted known atlas. □

4 Critical paths and improvements of known atlases

The preceding extension theorem permits a choice of exact problems in which the relative density exponent and the optimal time exponent evolve linearly. We construct these paths first. We then pass to short portions of them, keeping an explicit exclusion in the original resolution. The last part of the section shows how local candidates return to that original exclusion with sufficient mass, geometric bounds and terminal label knowledge.

4.1 Extremal exact problems

Fix the model version and j, S, η, and suppose bad problems exist. We optimize in three stages: approach the least density cap that permits positive horizon, maximize the horizon relative to length there, and, in version 2, optimize the narrowness available on the resulting sequences. The first two stages are expressed by

$$d_* = \inf\{d < 1 : \exists E \in C(d),\ H(E) > 0\}, \qquad k(d) = \sup_{E \in C(d)} H(E)/L, \qquad k = \lim_{d \downarrow d_*} k(d). \tag{4.1}$$

Here $0 \le d_* < 1$; in the right-hand limit the classes are nonempty and nested by monotonicity.

The supremum defining $k(d)$ uses all lengths, equivalently length one. Replacing the base N_0 by N_0^L divides depth, horizon and log-density exponents by L, preserving $C(d)$, including its angular parameters, and the chart rules. This scaling takes place within one exact problem, with its positive length L fixed.

Lemma 4.1 (Composition of horizon costs)**.** *Let $E \in C(d)$ have length L. If a known atlas at depth $r < L$ has time exponent h, and F is a homogeneous remaining problem in the original exponent units, then*

$$H(E) \le h + H(F). \tag{4.2}$$

If H_{out} is the horizon infimum for the coarsening of E to endpoint $r > 0$, then

$$H(E) \le H_{\mathrm{out}} + k(d)(L - r). \tag{4.3}$$

These statements allow all the subsequences and incidence restrictions in the definition of the respective infima.

Proof. Choose a true terminal system in F with horizon within an arbitrary positive tolerance of $H(F)$, on a subsequence where that system exists. By the terminal equality in Lemma 2.13, lifting through the known atlas adds the time exponents and gives a system true at the original terminal depth. Letting the tolerance tend to zero proves (4.2).

Coarsening to r keeps the trajectory and incidence data and the profile on $[0, r]$. The angular coarsening in Lemma 2.7 gives the required angular cap at p_r. Thus only the terminal profile-slope condition at r has to be checked when a cap class is asserted for the coarsening. A true terminal system of that coarsening can be prepared with its label known by p_r; it is then a true intermediate system for E, and its label is also known by the finer p_L, with the permitted list loss.

For outer true systems with costs tending to H_{out}, normalization produces remaining problems in $C(d)$ of length $L - r$, whose horizons are at most $k(d)(L - r)$. The first inequality therefore gives

(4.3) as the outer tolerance tends to zero. A restriction to subpower incidence mass leaves the homogeneous density exponents unchanged. Moreover, systems on such restrictions and subsequences are already among the competitors defining $H(E)$. This proves the inequalities with their stated quantifiers. □

Lemma 4.2 (A positive critical time rate)**.** *The rates in* (4.1) *satisfy* $0 < k \le 1$.

Proof. The upper bound follows from Proposition 2.16. For a bad problem E, choose $e > 0$ small compared with $H(E) > 0$ and its length, and choose a true terminal system with horizon $h < H(E) + e/4$. By Lemma 3.5, it gives a known atlas with horizon $h - e$ and positive remaining length at most Ce. Normalization through this atlas gives a remaining problem in $C(d)$ whose horizon, by (4.2), is at least

$$H(E) - (h - e) > 3e/4. \tag{4.4}$$

Its horizon-to-length ratio is therefore at least $3/(4C)$. For d near d_*, choose one $d_0 < 1$ with $d \le d_0$; the extension constant is uniform for this range and for the fixed j, S, η. Bad problems exist arbitrarily close to the defining dimension infimum. The lower bound for k follows. □

Scale units to $L = 1$ when discussing *maximizing sequences*. These are sequences of exact problems $E_i \in C(d_i)$ with $d_i \to d_*$ and $H(E_i) \to k$; they exist by the definition. We have $\limsup_i k(d_i) \le k$, by comparison with dimensions slightly larger than d_*. For each fixed i, all exponents are first taken along its exact N_0-sequence; the outer limit in i comes afterward. A single (i, N_0)-diagonal does not replace these inner limits.

In version 2, fix one maximizing sequence. Consider true terminal systems whose time exponents tend to k. For each such system within E_i, pass to a homogeneous subfamily of charts to obtain a narrowness exponent, as in (2.17), in length-one units. Define ω to be the supremum of the attainable lim sup values of these exponents. The choices include systems on any subsequence of the outer indices i, and all permitted subsequences and selections within the individual exact problems. Choices near the horizon infimum with some homogeneous narrowness exponent exist, and narrowness is nonnegative.

Put $\ell = \inf \omega$, where the infimum ranges over all maximizing sequences and may be infinite. In version 1 we use $\ell = 0$ formally, without optimizing width.

Proposition 4.3 (Critical paths)**.** *Fix the model version and its parameters j, S, η, and suppose bad problems exist. Let d_*, k be as in* (4.1)*, and use the maximizing-sequence and narrowness conventions of this subsection. The following conclusions hold.*

1. *Every maximizing sequence of length-one exact problems satisfies*

$$\sup_{0 \le r \le 1} \left| f_i(r) - f_i(0) - d_* r \right| \longrightarrow 0. \tag{4.5}$$

2. *For every fixed finite ordered grid of target depths ending at one, there are nested true systems at actual depths tending to those targets. Their cumulative horizons tend to kr, and their segment horizons, in their own length units, tend to k. Both the coarsened and remaining problems used at each step are maximizing after their individual length normalizations. Full layer assignments, priors and incidence witnesses can be retained in addition to the final path incidences.*

3. *In version 2 with $\ell < \infty$, for every $\zeta > 0$ one may choose the base maximizing sequence and these paths so that cumulative narrowness at a target r, and narrowness increments between targets, equal the corresponding linear values with slope ℓ, up to $O(\zeta) + o_i(1)$. The base sequence has terminal optimal narrowness supremum at most $\ell + \zeta$.*

4. *If $\ell = \infty$, every newly obtained maximizing sequence admits, on subsequences, true terminal systems with relative horizons tending to k and relative narrowness tending to infinity. Thus arbitrarily large narrowness increments can be required on each of finitely many successive positive-length segments.*

Remark. Only the relative profiles occur in (4.5); no bound on $f_i(0)$ is required.

Proof. The proof follows the three optimizations above. The critical density and horizon composition rule out a lower profile secant. Once this forces linear profiles, composition of horizon costs produces nested maximizing segments. In the finite-narrowness case, the terminal upper bound then controls each segment.

Relative profiles. We first establish the relative-profile assertion,

$$f_i(r) - f_i(0) \longrightarrow d_* r \tag{4.6}$$

uniformly.

Take any uniform subsequential limit of the relative profiles by their increment bounds. If it has a secant on an interior $[a, b]$ of slope strictly below d_*, fix $0 < d'' < d_*$ still larger than the secant slope. Minimize $f_i(x) - d''x$ on $[a, b]$, obtaining r_i bounded away from a for large i by the strict secant comparison and Lipschitz bound. At the coarsened endpoint r_i we have the slope cap d'' back to a; there is also a global terminal cap strictly less than one by the ambient Lipschitz bound. Indeed from $x < a$,

$$f_i(r_i) - f_i(x) \le r_i - x - (1 - d'')(r_i - a). \tag{4.7}$$

Let G_i be this coarsened problem. We claim that $H(G_i) = 0$. Otherwise fix this i, and let C_i, c_i be the constants in Lemma 3.5 for its global terminal cap. Choose

$$0 < e < \min\big(H(G_i), c_i r_i, (r_i - a)/C_i\big)$$

and a true terminal system with horizon $h < H(G_i) + e/4$. Since $h \ge H(G_i) > e$, terminal extension applies. It gives a known atlas with horizon $h - e$ at a depth t_i satisfying

$$0 < r_i - t_i \le C_i e < r_i - a,$$

so $t_i > a$.

Normalization through this atlas gives a homogeneous remaining problem F of length $L_F = r_i - t_i$. The terminal density equality and intermediate density upper bound in Lemma 2.13 give, for its profile f_F,

$$f_F(L_F) - f_F(s) \le f_i(r_i) - f_i(t_i + s) \le d''(L_F - s), \qquad 0 \le s \le L_F.$$

The second inequality uses only the slope cap on $[a, r_i]$, since the entire remaining interval lies there. The angular cap also passes through normalization, so $F \in C(d'')$. By (4.2),

$$H(F) \ge H(G_i) - (h - e) > 3e/4 > 0,$$

contradicting the definition of d_*. All choices in this argument are within the fixed exact problem G_i; neither e nor its extension constants need be uniform in i.

Using these zero outer costs back in the original problems gives $H(E_i) \le k(d_i)(1 - r_i)$, contradicting maximization since $k > 0$. If $d_* = 0$, a lower secant is already excluded by monotonicity. Thus in every case no lower interior secant exists. Together with the endpoint cap (in the limit) and continuity this forces the linear profile, proving the assertion.

Nested true systems. Fix a target $0 < r < 1$ on a maximizing sequence. Choose $\epsilon_i > 0$ tending to zero sufficiently slowly relative to the uniform profile errors, and minimize $f_i(x) - (d_* + \epsilon_i)x$ on $[0, r]$. The linear approximation forces a minimizer $r_i \to r$. At this actual endpoint the terminal profile cap back to the start is $d_* + \epsilon_i$.

The coarsened problems at r_i, each normalized to length one, form a maximizing sequence. Indeed, their outer costs before rescaling satisfy

$$H_{\text{out}} \le k(d_* + \epsilon_i)r_i, \qquad H(E_i) \le H_{\text{out}} + k(d_i)(1 - r_i), \tag{4.8}$$

so $H_{\text{out}}/r_i \to k$.

Take true systems on these outer problems with time exponents tending to k in relative units.

Normalization of the original finer problems through these packets gives homogeneous remaining problems that are again maximizing when normalized to length one, with base $N_0^{1-r_i}$: their costs in old units lie between $H(E_i) - (k + o_i(1))r_i$ and $k(d_i)(1 - r_i)$. These conclusions persist on further subsequences with the same choices.

We can therefore iterate over any fixed finite ordered grid of target depths ending at one. At each step use the next target fraction of the remaining length, choosing that fixed fraction from the limiting remaining length. The maximizing property just proved permits the next application; at fraction one use the full remaining maximizing sequence. This gives nested true systems whose actual depths tend to the targets and whose segment horizons have the required limits, with the final node at the true endpoint.

Each normalization divides by its individual length, and returning to old units multiplies by that length, not merely its limit. Only finitely many subsequences of i, and of the individual exact sequences, are needed for a fixed grid.

Intermediate true packets lift through true parents by the equality case of Lemma 2.13. At each stage, homogenization and the preparation following Definition 2.8 retain the full layer assignments, priors and incidence witnesses, together with the final path incidences common to all layer selections. The final path alone need not saturate an earlier chart. In a parent, for example, the trajectory prior remains conditioned on its full assignment after further incidence selections. Child trajectory sets use this fixed prior and lift to subsets of the parent assignment.

Labels may include their whole histories; list knowledge composes via the parent's knowledge and the bounded overlaps of the corresponding hidden grids once chart and exact base are specified. Time lengths and spatial Jacobians multiply. All these paths for a fixed finite grid and i use actual exact subsequences and their subpower slacks.

Narrowness optimization. In version 2 with $\ell < \infty$, fix a small tolerance $\zeta > 0$ and choose a whole base maximizing sequence with $\omega \le \ell + \zeta$. At each step of the fixed finite-grid construction, including the final segment, choose a true system whose relative time exponent tends to k and whose relative narrowness has $\liminf$ at least $\ell - \zeta$. To make this choice, apply the definition of ℓ to the newly coarsened maximizing sequence in its length-one units: choose an attainable $\limsup$ large enough, then pass to a subsequence. The new remaining problems continue to maximize by the preceding composition bounds.

Composition through the final depth and true lifting give a terminal true system with cumulative time exponent tending to k, so its cumulative narrowness has $\limsup \le \omega$ for the chosen base sequence. Narrowness exponents add, with each relative exponent multiplied by its actual segment length in original units. The segment lower bounds therefore give the cumulative lower bound at a node of target r; subtracting the lower bounds for the subsequent segments from the terminal upper bound $\ell + \zeta + o_i(1)$ gives the upper bound. Thus the cumulative narrowness is $\ell r + O(\zeta) + o_i(1)$, and the corresponding increments have the same linear behavior. Cumulative time exponents tend to kr. All preceding bounds concern the constructed systems on the indicated exact subsequences.

If instead $\ell = \infty$, every newly obtained maximizing sequence admits, on a subsequence, true terminal systems with relative horizons tending to k and relative narrowness tending to infinity. Choose arbitrarily high narrowness thresholds and diagonalize over i in the definition of ω. □

Corollary 4.4 (Depth-zero normalization and vanishing-depth nodes)**.** *In the setting of Proposition 4.3, a sequence of depth-zero, horizon-zero known normalizations of a maximizing sequence is again maximizing, provided the normalizations have the actual subpower success and terminal knowledge required in Lemma 2.13. This assertion does not require bounded absolute density exponents.*

On any maximizing sequence one may choose true nodes at positive depths $b_i \to 0$ whose time exponents tend to zero, retaining the relative-profile estimates at those nodes in the slowly diagonalized sense.

Proof. Let F_i be a depth-zero normalized problem of length one. Normalization preserves $C(d_i)$, and lifting gives

$$H(E_i) \le H(F_i) \le k(d_i). \tag{4.9}$$

Both outer bounds tend to k. Thus F_i is maximizing, and (4.5) applies to $f_{F_i}(r) - f_{F_i}(0)$, without extracting a limit of either absolute term.

For every fixed $b \in (0,1)$, the construction in Proposition 4.3 gives actual depths $b_i \to b$, relative profile errors tending to zero, and horizons $kb + o_i(1)$. Apply this statement successively with $b = 1/m$. For each fixed m first take the outer index far enough and retain the required exact subsequences. Then increase m sufficiently slowly that the depth error and horizon error are at most $1/m$, and that the profile errors are as small relative to b as subsequently required. Each choice still uses actual exact problems and their true systems. It gives $b_i > 0$, $b_i \to 0$, and horizon tending to zero. Additional finite requirements, such as the narrowness choices in Proposition 4.3, can be included before each increase of m. □

Lemma 4.5 (Excluded improvements)**.** *On the maximizing sequence and fixed finite grid of Proposition 4.3, there are two forbidden outputs of known atlases at an interior path node (including the parent where a short piece starts) as $i \to \infty$. Exponents refer cumulatively to the original problem; one can homogenize within exact problems. Write r_i, h_i for that node's depth and time exponent. Knowledge is required by the original p_L. Path incidences alone may be used, with subpower coverage and slacks in each exact problem. The gains $\Delta_i > 0$ have a uniform positive lower bound along the outer sequence.*

1. *A thickness-depth improvement by $\Delta_i > 0$ beyond r_i, still leaving positive length, with cumulative time exponent at most $h_i + k\Delta_i/2$.*
2. *In the finite-ℓ width optimization, extra narrowness by $\Delta_i \gg \zeta > 0$ over the node's, keeping exactly the node depth and horizon.*

Proof. The first on an infinite subsequence would give by normalization and lifting

$$H(E_i) \le h_i + k\Delta_i/2 + k(d_i)(1 - r_i - \Delta_i) \le k - k\Delta_i/2 + o_i(1), \tag{4.10}$$

impossible. In the second the remaining problems after the output atlas still maximize, by the same composition reasoning (they have remaining cost at least $H(E_i) - h_i$ before length normalization). Take their true terminal packets with relative narrowness $\ge \ell - \zeta$ in $\liminf$ and optimal limiting horizon along a further subsequence. Lifting now gives terminal horizon tending to k and cumulative narrowness beating $\ell + \zeta$, by the previously built segment lower bounds and the extra gain (e.g. with lower bound $> 4\zeta$). This violates the base sequence's upper bound. The contradiction uses the true terminal systems and their full-label knowledge when lifting the remaining problem. □

4.2 A tangent diagonal with an original-resolution exclusion

We next extract short portions of these paths. The local estimates will use tangent units, while their final outputs must still violate an exclusion in the original exact problem.

Take stages $n \to \infty$ and a tangent depth unit $\epsilon_n \to 0$. For fixed n put disjoint successive windows of length $D_n\epsilon_n$ in an interior interval bounded away from depths zero and one, using $\gtrsim 1/(D_n\epsilon_n)$ windows. Their starts are potential parent nodes. Subdivide each with a finite mesh of spacing $o_n(1)$ in relative units $s = (\text{offset from start})/\epsilon_n$; form true paths for this whole grid, including the terminal depth.

Use a finest indicated coarse velocity precision of depth $\epsilon_n U_n$ in the original problem coordinates, where $U_n, D_n \to \infty$, $U_n D_n \epsilon_n \to 0$.

Choose a forbidden gain threshold $\gamma_n\epsilon_n$, $\gamma_n \to 0$ positive, and path optimization tolerance $\zeta \ll \gamma_n\epsilon_n$ as above. Take i sufficiently large in the resulting sequence, then take a sufficiently late sample N_0 of its exact path subsequence and use

$$N = N_0^{\epsilon_n} \to \infty, \qquad K = N^{o(1)} \tag{4.11}$$

along the stages. All node depths, time exponents and profiles needed on the finite grids can use errors $o_n(\epsilon_n)$ in original exponent units by the preceding limiting results and this order; cumulative narrowness in version 2 also uses the $O(\zeta)$ tolerance.

Thus later labels at specified s mean rounded path nodes with error tending to zero in tangent exponent units. Additional analytical grids in fixed finite exponent ranges can increase sufficiently slowly. K denotes changeable subpower factors *on the diagonal*, possibly much too large as slacks for an output.

At a fixed stage n, choose the outer index i large enough that Lemma 4.5 excludes the buffered gain $\gamma_n\epsilon_n$ for this finite grid. Fix its exact path subsequence. Let

$$K_{\text{path}} = N_0^{o_{N_0|i}(1)} \tag{4.12}$$

include its true-path geometric and list bounds, cumulative products, preparation costs and reciprocal selected path mass. Retain an original tempered final path submeasure of mass $\mathfrak{p}_{\text{path}} \geq K_{\text{path}}^{-1}$ as a reference, together with all earlier layer assignments and witnesses.

Lemma 4.6 (The finite-sample exclusion)**.** *Fix the stage n, outer index i, and exact path subsequence in the construction of Section 4.2, with tangent resolution* (4.11)*, reference-path bound* (4.12)*, and the exclusion of Lemma 4.5. Let $M^{\#}$ be any finite original-coordinate complexity exponent. For all sufficiently late N_0 there is no output known atlas satisfying all of the following:*

1. *One of the improvements in Lemma 4.5, with its buffered gain at least $\gamma_n\epsilon_n$, with thickness depth bounded away from the terminal endpoint. A width improvement keeps the actual node depth and time length and holds uniformly on its output labels.*

2. *Counts, coefficient magnitudes, reciprocal widths and lengths, piece and condition counts are at most $N_0^{M^{\#}}$, with degree per condition at most $M^{\#}$.*

3. *All geometric slacks and original p_L-list sizes are at most $(K_{\text{path}} \log N_0)^n$, and the original incidence mass is at least $\mathfrak{p}_{\text{path}}/\log N_0$.*

Proof. Otherwise there are infinitely many such outputs along this one exact sequence. The fixed complexity bound makes them uniformly tempered. Their slacks, list sizes and reciprocal masses are subpower in the original resolution N_0, since n, i are fixed. Homogenize the output geometry

in logarithmic order and extract limiting horizon and thickness-depth exponents. The strictly buffered improvement persists. For a width improvement its uniformity on labels, relative to the homogeneous old widths, ensures that it persists as well. Replacing a thickness exponent by its limit changes thickness by an $N_0^{o(1)}$ factor; both the value and curvature requirements are preserved after enlarging the actual subpower slack. These outputs therefore form an actual known atlas on an exact subsequence, excluded by Lemma 4.5. □

Let $M_{\mathrm{path}} \geq 1$ bound all original and path complexity exponents, including law complexity, degrees and coordinate changes. It is fixed within this exact problem before the sample is chosen. Declare, for example,

$$M^{\#} = 2^{(M_{\mathrm{path}}+n)^2}. \tag{4.13}$$

The order of the choices is

$$\text{finite stage and grid} \longrightarrow \text{outer exact problem and its path} \longrightarrow (M_{\mathrm{path}}, M^{\#}) \longrightarrow N_0. \tag{4.14}$$

Choose N_0 beyond the cutoff of Lemma 4.6 for this already declared $M^{\#}$. It may be taken arbitrarily late there. If G_n denotes the total number of relevant grid nodes, impose also

$$\frac{G_n}{\log N_0} \leq \frac{1}{n}, \qquad N_0^{-1} K_{\mathrm{path}} \log N_0 \leq \frac{1}{n}, \tag{4.15}$$

$$\frac{\log\big((K_{\mathrm{path}} \log N_0)^n\big)}{\epsilon_n \log N_0} \leq \frac{1}{n}. \tag{4.16}$$

Each condition is possible in the fixed exact sequence. All other finite-stage approximation requirements are enforced at the same time. In particular the required original profile and horizon errors are $o_n(\epsilon_n)$, and the named path factors and logarithms are $K = N^{o(1)}$ on the resulting tangent diagonal. Analytical grids in fixed finite exponent ranges can be increased sufficiently slowly.

There will be a countable family of fixed candidate scenarios. A scenario will be identified only after the diagonal and a successful subsequence have been chosen. The implementation below proves that, for every fixed scenario $\mathcal{S}$, the original-coordinate complexity of every possible output is bounded by a polynomial $P_{\mathcal{S}}(M_{\mathrm{path}})$, uniformly over the selected tests, masks and posterior rules. This eventually satisfies the predeclared complexity bound. Indeed, if $P_{\mathcal{S}}(M) \leq C_{\mathcal{S}}(1+M)^{d_{\mathcal{S}}}$, then

$$\sup_{M \geq 1} \{d_{\mathcal{S}} \log_2(1+M) - M^2\} < \infty \tag{4.17}$$

implies $P_{\mathcal{S}}(M) \leq 2^{(M+n)^2}$ for all $M \geq 1$ once n is large enough depending only on $\mathcal{S}$. Likewise an actual slack or list power fixed by $\mathcal{S}$ is eventually less than n. Discarding finitely many stages leaves the previously chosen complexity bound and samples in force. The contradiction will therefore use the original-resolution exclusion of Lemma 4.6, with the horizon H evaluated in the original exact problem.

4.3 Preparing the tangent laws

The path scales determine the local geometry. The remaining preparation concerns the density and velocity bounds that will survive later restrictions of the incidence law.

Write $d = d_*$, $N = N_0^{\epsilon_n}$, and use $K = N^{o(1)}$ for analytical estimates. For a path probability μ_{prep} and a selected interior parent window, write

$$\mu_{\mathrm{prep}} = \sum_A p_A \lambda_A^0, \qquad p_A = \mu_{\mathrm{prep}}\{A\},$$

where λ_A^0 is its conditional probability in parent A. Let ν_A denote the trajectory prior conditioned on the full assignment of that parent, and let dt denote uniform parent time. Set $v = V$ in version 1 or when $\ell = 0$; when $0 < \ell < \infty$, let Z, Y be the major and minor parent coordinates and set $v = Z'$. Denote the velocity marginal of λ_A^0 by π_A. These conventions apply to each law and window supplied by the proposition.

Proposition 4.7 (Prepared tangent data)**.** *Assume $\ell < \infty$, and use the diagonal of Section 4.2 with the choice order* (4.14) *and sample requirements* (4.15)–(4.16)*. One can retain all but $o(1)$ of the relative mass of the original final path and choose an interior parent window and a prepared path probability μ_{prep} with the following properties. On the retained parents, $\lambda_A^0 \le K\nu_A \times dt$. Subsequent omissions restrict this mixture and leave its reference weights and probabilities fixed.*

1. *At every required bounded tangent depth s, the true label Q_s has thickness $a_s = N^{-s}$, time length $q_s \sim_{\log,N} N^{-ks}$, and spatial determinant $J_s \sim_{\log,N} q_s^j N^{-\ell s}$ in version 2; omit the last factor in version 1. Its largest width is comparable to q_s in logarithmic order, and its full assigned trajectory mass satisfies $\nu_Q \sim_{\log,N} a_s^d J_s$.*

2. *For the restricted incidence laws, pure hidden and native-signal bins of width p have the instantaneous joint base-density ceiling Kp^d on the required finite grids. In unit coordinates of a Q_s-box, the base marginal has density at most $Kq_s\nu_Q$, with masses still measured in parent time.*

3. *The fixed conditional parent velocity law satisfies $\pi_A(B_p) \le Kp^S$. If E denotes the end history, then before likelihood deletion the required velocity bin satisfies*

$$\sum_A p_A I_{\lambda_A^0}([v]; E) = o(\log N).$$

4. *Required finite-mesh point states and path labels have lists at the original p_L bounded by fixed powers of actual path factors and logarithms. Sufficiently deep histories leave only K options for these point states. Each fitted test has at most K local graph choices approximating w to Ka_s on retained incidences in their cores. After composition with the native test, their fixed-order derivatives are bounded by K on the used normalized interiors. These graph bounds and capped full-layer incidence witnesses can be prepared before arbitrary later inverse-subpower selections.*

When $0 < \ell < \infty$, major-coordinate fullness and the coefficients of the tilted-box rows can be prepared with actual N_0-subpower bounds.

Remark. The information bound is averaged with the fixed parent weights. Lemma 4.8 gives the joint and conditional consequences of these fixed laws.

Proof. The preparations must precede the choices that use them. We first remove density and angular exceptions at all potential parents. One normalization then fixes the probability law used to choose the entropy window. Graph and list preparations are subsequent restrictions of that law.

Coordinates and masks at every potential parent. For now retain all potential parent nodes in the finite grid. If A is a label of cumulative depth b, let q_A be its original time length, J_A its spatial determinant, and $\nu_{\mathrm{old}}(A)$ its full assigned trajectory mass in its old component. Saturation gives

$$\nu_{\mathrm{old}}(A) \sim_{\log} n_{\mathrm{old}}(b) J_A.$$

After the base normalization of Lemma 2.13, the parent prior ν_A is the old trajectory prior conditioned on this full assignment, and parent time is uniform on $[0,1]$. In version 2 rescale the hidden coordinate so that its parent derivative unit is $B = 1$. Its value need not be bounded in tangent units; the native signal $X = P_*$ has the native bounds.

At width $p = N^{-r}$, a parent hidden bin corresponds, at fixed base and label, to boundedly many old bins at depth $b + \epsilon_n r$. After the original high-density exceptions at these depths have been removed for all required parents and finite grids, the within-component density factor under parent normalization is $J_A/\nu_{\mathrm{old}}(A)$. Consequently the remaining unnormalized density is at most

$$K\frac{J_A}{\nu_{\mathrm{old}}(A)}n_{\mathrm{old}}(b+\epsilon_n r) \le K\frac{n_{\mathrm{old}}(b+\epsilon_n r)}{n_{\mathrm{old}}(b)} \le Kp^d.$$

The last inequality uses the relative-profile errors $o_n(\epsilon_n)$ fixed when choosing i. The native derivative and curvature bounds give the same ceiling for X-bins. These are ceilings for restricted measures, so they remain valid under all subsequent deletions.

Angular ceilings can be prepared on the same finite collection of parents. Refinement of the original terminal observation by a known parent label has typical density at least $n(L)/K_{\mathrm{path}}$ under the unnormalized final path measure. With the exceptional denominators and old joint angular exceptions removed before velocity projection, a normalized velocity cell of side p pulls back to an old velocity ball of radius Kp. In the positive-narrowness case the omitted minor component is already much smaller than this radius, since the parent is at an interior original depth. For every tested cell the remaining angular numerator is therefore bounded by Kp^S times the original refined terminal denominator. Integrating gives the corresponding bound relative to the parent masses before these masks. The numerator sum includes only the retained incidences.

We also prepare the spatial fullness that will be needed in original units. For $0 < \ell < \infty$, use the cumulative principal axes of each potential parent. Its minor spatial scale per unit time is smaller than the major scale by an actual power of N_0. By Lemma 2.12, cumulative child largest widths per unit time are full up to actual subpower factors. These prunings are made simultaneously for the finitely many required ancestor–child pairs. For a descendant at a specified relative path depth, let D_s be its actual relative spatial matrix and q_s its actual relative time length. Then

$$\rho_s = \|D_s^T e_Z\| \ge q_s/K_{\mathrm{path}}.$$

Indeed $\|D_s\|/q_s$ is actually subpower bounded. If its major row were deficient by a fixed power, composition with the parent's matrix, whose minor scale already has a fixed-power deficiency, would make the cumulative child largest scale deficient as well. This contradicts the fullness just prepared.

All density exceptions have power-small mass at fixed tolerances in each exact problem. The original order n, i, N_0 allows the finite grids, tolerances, and fullness prunings to be chosen with total loss $o(1)$ relative to the reference path. The entropy-window choice follows these simultaneous preparations.

The fixed probability law and the entropy window. Normalize the surviving path once and call its probability law μ_{prep}. For a potential parent A, let s_A be its surviving success fraction in $\nu_A \times dt$. For $s_A > 0$, set

$$\lambda_A^0 = \frac{(\nu_A \times dt)|_{\text{prepared path in } A}}{s_A}, \qquad p_A = \mu_{\mathrm{prep}}\{A\}.$$

At any one parent node, $\mu_{\mathrm{prep}} = \sum_A p_A \lambda_A^0$. The weight p_A is proportional to its old component weight times $q_A \nu_{\mathrm{old}}(A) s_A$.

Parents with too small a success fraction, or too small a surviving denominator relative to their pre-mask mass, have negligible total weight: sum the former losses with $\sum_A \omega_{\mathrm{old}} q_A \nu_{\mathrm{old}}(A) \le 1$, and

the latter with the pre-mask parent masses. These parents may be omitted when using the chosen window. On the parents retained for estimates,

$$\lambda_A^0 \le K\nu_A \times dt, \qquad \pi_A(B_p) \le Kp^S, \tag{4.18}$$

where π_A is the velocity marginal of λ_A^0. The division by s_A increases the bound by at most K, so the pure ceilings above also hold for these conditional probabilities.

Choose the window using the single law μ_{prep}, before any further graph or likelihood deletion. Let V_* be the indicated original velocity bin, of depth $\epsilon_n U_n$. For disjoint successive windows, the increments

$$I(V_*; \text{history through the window end} \mid \text{history through its start})$$

are nonnegative and sum to at most $H(V_*)$, by the chain rule for nested histories. The original velocity range gives $H(V_*) \le O(U_n \log N) + o(\log N)$. There are $\gtrsim (D_n\epsilon_n)^{-1}$ windows, so one has increment $o(\log N)$, since $U_n D_n \epsilon_n \to 0$.

Use the normalized coordinates of this window's parent. In version 1 and for $\ell = 0$, put $v = V$; for positive finite narrowness, put $v = Z'$. Given the parent, the required v-bin has at most K possibilities from V_*, and hence its conditional mutual information with the end history is larger by at most $\log K = o(\log N)$. For $\ell = 0$, both parent scales per unit time are full to tangent factors K, which suffices for this ambiguity estimate; positive narrowness uses the major-coordinate fullness prepared above. The requested velocity grid is no finer than N^{-U_n} in its own units.

Fix p_A, λ_A^0, and π_A from now on. Omitting the exceptional parents and imposing graph masks restricts this mixture without redefining its reference laws or its entropy estimate. Substantial coverage will mean positive probability relative to the original reference path.

Path scales and the two time conventions. For a descendant $Q = Q_s$, put $a = N^{-s}$, $h = N^{-ks}$, and, in version 2, $g = N^{-\ell s}$. The relative-profile and path conclusions give

$$a_s = N^{-s}, \qquad q_s \sim_{\log,N} h, \qquad J_s \sim_{\log,N} h^j g, \qquad \nu_Q \sim_{\log,N} a^d J_s. \tag{4.19}$$

In version 1 omit g. Write P_s for the fitted true test. Its actual node thickness a_Q is comparable to a in logarithmic order. The true value, derivative, and curvature bounds continue to hold in the actual units of that node. The full trajectory assignments remain disjoint within each time block and nested in the fixed parent prior.

At an exact base, the true value and incidence-derivative bounds permit only K hidden bins of width a_Q. The map to unit base coordinates u of Q_s has Jacobian $q_s J_s$; denote this map by base_Q. The pure cap therefore gives

$$d(\text{base}_Q)_*(\lambda_A^0|_Q)(u) \le Ka^d q_s J_s \, du \le Kq_s\nu_Q \, du. \tag{4.20}$$

This is a cap in parent-time mass. Dividing by the block length q_s gives the cap $K\nu_Q$ in normalized Q_s-time. If the trajectory prior is normalized on the full assignment of Q, the pure ceiling is Kp^d/ν_Q in parent base coordinates. After changing to unit Q-base coordinates and unit block time, it is $Kp^d J_s/\nu_Q$. Here p remains the hidden-bin width in parent units; the denominator is the full assignment mass, not the current selected incidence mass. The assignment mass bounds are

$$\sum_{Q \text{ in one block}} \nu_Q \le 1, \qquad \sum_{Q \text{ at one node}} q_s\nu_Q \le 1.$$

They allow light labels to be removed whenever current incidence floors are needed, with thresholds chosen below the current selected mass. Spatial enlargements by subpower factors only change K.

For positive finite narrowness, subtract a multiple of the Z-row of D_s from its Y-row to make the rows orthogonal. The subtraction coefficient is at most $\|D_s^T e_Y\|/\rho_s$, and the new transverse row has length $|\det D_s|/\rho_s$. Thus the child is a tilted box of widths h, hg, and its assigned velocities satisfy

$$Y' = A_s \cdot (1, v) + O(Kg).$$

The row A_s is determined by Q_s and has actual subpower coefficients. Its affine center velocity is controlled by the whole-block position bounds on occupied trajectories. In actual units the transverse width per time is comparable, up to actual subpower factors, to J_s/q_s^2.

Stable graphs and point-state lists. Later arguments need velocity bounds after recording finite-mesh point states. We first construct the finite list of possible states; its size will be the input to the appended-entry floor below. Use normalized base coordinates u in Q_s, keeping the hidden variable w in parent units with $B = 1$. Let $K_g = N^{o(1)}$ contain the present test bounds. In version 2 they give

$$|P_s| \le K_g a_Q, \qquad K_g^{-1} \le |P_{s,w}| \le K_g, \qquad |P_{s,ww}| \le K_g/a_Q$$

on the counted incidences. First omit labels whose current mass is too small relative to $q_s \nu_Q$. The assignment mass bound controls the total loss, and (4.20) then gives a base projection of inverse-subpower volume in every used label.

The polynomial discriminant

$$\mathcal{D}(u) = P_{s,w}^2 - 2P_{s,ww}P_s$$

is independent of w. Its values on this projection are bounded by K; polynomial Remez therefore bounds $\mathcal{D}$ and its first derivatives by K on the needed complex enlargements of the normalized packet box. Choose a subpower H with $H/K_g^3 \to \infty$, and split the tests at $|P_{s,ww}| = (a_Q H)^{-1}$. Above this cutoff the affine quadratic center satisfies

$$|w - F_{\mathrm{ctr}}(u)| = \frac{|P_{s,w}(u,w)|}{|P_{s,ww}|} \le K_g H a_Q \le Ka.$$

Below the cutoff,

$$2|P_{s,ww}P_s| \le \frac{2K_g}{H} = o(K_g^{-2}) = o(|P_{s,w}|^2), \qquad \mathcal{D} \ge \tfrac{1}{2}K_g^{-2} \tag{4.21}$$

at those incidences. Partition the normalized box into core subboxes of inverse-subpower radius, each with a fixed-factor complex enlargement and a further margin. Choose the radius so that the variation of $\mathcal{D}$ on each occupied enlargement is much smaller than this lower bound. For a quadratic test, each such enlargement has two stable simple holomorphic branches, real on its real core. The branch with the incidence derivative sign approximates w to Ka, by (3.24). For a linear equation the same argument keeps its coefficient nonzero.

There are K core/branch entries per label, including affine center entries. The same $q_s\nu_Q$ mass bounds permit light entries to be removed at a smaller reciprocal-subpower threshold. For each retained entry, (4.20) then gives inverse-subpower real base volume. Composing its graph with the native test P_* gives an error Ka in X at the incidences by the native derivative and curvature bounds, and $|X| \le K$ there. The composition is an algebraic holomorphic sheet; Lemma 3.1 bounds it and its required derivatives by K on the used interiors. In version 1 use the explicit graph directly. These bounds now belong to the fixed graphs and persist under later incidence restrictions.

An end history refining Q_s localizes its normalized base coordinates to uncertainty at most Kq_{end}/q_s. Indeed, nested relative frames have largest width bounded by relative time length times path slacks, and occupied chart centers move at bounded speed up to those slacks. For a prescribed

mesh in (t, y, X), first choose a much finer than that mesh and then choose the end depth so that the graph derivative bound times Kq_{end}/q_s is also much finer. Each graph then gives only K possible mesh states. The finite core/branch list proves the required ambiguity bound before any lower bound for velocity-bin weights is used. Cores lie inside the derivative-controlled interiors; bounded overlapping covers handle their boundaries.

These preparations use arbitrary fixed mesh exponents first, with sufficient window depth, and then slowly increasing grids. They may be made after the entropy-window choice with negligible total loss, without redefining λ_A^0, and before arbitrary later sparse selections.

Original observation lists and full-layer witnesses. Finite-mesh point states in (t, y, X), together with the path labels in the window, have actual-list accuracy given the original p_L. Given the actual path charts in these lists, the base is exact, and the much finer original terminal hidden bin, together with the native derivative and curvature bounds at the incidence, controls X. This is actual knowledge in the original problem; the graph lists separately supply the K-ambiguity needed in tangent estimates.

For the stored full-layer incidence witnesses, the same original pure-cap masks are imposed on the fixed density grids needed in the coordinate transfers. Their mass losses are power-small at every fixed tolerance in the exact problem. If a layer label loses at least half its incidence mass, saturation bounds its old block length times its full trajectory mass by an actual path factor times the lost incidence mass. Summing with the old component weights shows that dropping all such labels from the final path loses negligible mass when N_0 is sufficiently late. This prepares capped witnesses in each needed layer. Their earlier floors belong to these full-layer witnesses. The velocity-law estimates remain one-time estimates on the stated laws.

All these preparations use the already fixed diagonal. Their initial cap masks retain $1 - o(1)$ of the reference path, and their graph and witness prunings have negligible additional relative loss. The parent weights and conditional probabilities remain the reference laws fixed at the entropy-window step. □

4.4 Conditioning finite states and taking time slices

The prepared law has little information about the velocity in its end history. The next lemma converts that information bound to a pointwise joint ceiling. It also records how that ceiling passes to a point state and how much mass must be deleted before it becomes a conditional ceiling under a later restriction.

Lemma 4.8 (Likelihood truncation and finite-state conditioning)**.** *As $N \to \infty$, let $p_A \geq 0$, $\sum_A p_A = 1$, be mixture weights and let λ_A^0 be probability laws carrying two finite labels E, b. Write*

$$R_A^0(E,b) = \lambda_A^0\{E,b\}, \qquad \lambda_{A,\mathrm{pre}}(E) = \sum_b R_A^0(E,b), \qquad \pi_A(b) = \sum_E R_A^0(E,b).$$

Suppose

$$\sum_A p_A \sum_{E,b} R_A^0(E,b) \log \frac{R_A^0(E,b)}{\lambda_{A,\mathrm{pre}}(E)\pi_A(b)} = o(\log N).$$

Then deleting $o(1)$ mass from the whole mixture gives joint submeasures with

$$R_{A,\mathrm{good}}(E,b) \leq K\lambda_{A,\mathrm{pre}}(E)\pi_A(b), \qquad K = N^{o(1)}. \tag{4.22}$$

Let $R_{A,2}$ be any later submeasure, extended by a label d' with at most $D \le K$ possibilities per E, whose (E,b) marginal is dominated by $R_{A,\mathrm{good}}$. Define its pre-omission entry masses by

$$m_{A,\mathrm{preomit}}(E,d') = \sum_b R_{A,2}(E,d',b).$$

For a subpower K_1, deleting at most D/K_1 mixture mass gives retained joint measures $R_{A,3}$ satisfying

$$R_{A,3}(E,d',b) \le KK_1\pi_A(b)m_{A,\mathrm{preomit}}(E,d').$$

For any coarser state $C = C(E,d')$, put

$$w_A(C) = \sum_{(E,d'):C(E,d')=C} m_{A,\mathrm{preomit}}(E,d').$$

Then $R_{A,3}(C,b) \le KK_1\pi_A(b)w_A(C)$. For a further restriction T_A of $R_{A,3}$, deleting states whose current mass is below $w_A(C)/K_2$ removes at most $1/K_2$ mixture mass and gives, on surviving states of positive mass,

$$T_A(b \mid C) \le KK_1K_2\pi_A(b).$$

The subpowers K_1, K_2 may be chosen so each deletion is negligible relative to a specified current inverse-subpower mixture mass.

Proof. Put $R_{A,\mathrm{prod}}(E,b) = \lambda_{A,\mathrm{pre}}(E)\pi_A(b)$. The negative part of the log ratio has expectation at most one in each parent, since

$$\sum_{R_A^0 < R_{A,\mathrm{prod}}} R_A^0 \log(R_{A,\mathrm{prod}}/R_A^0) \le 1$$

by $x\log(1/x) \le 1/e$. Consequently its positive part has mixture expectation $o(\log N)$. Choose $\delta_N \downarrow 0$ slowly enough that this expectation is $o(\delta_N \log N)$. By Markov's inequality, removing the sampled pairs where the log ratio exceeds $\delta_N \log N$ loses $o(1)$ mass in the whole mixture. The retained joint measure satisfies

$$R_{A,\mathrm{good}}(E,b) \le N^{\delta_N}\lambda_{A,\mathrm{pre}}(E)\pi_A(b) \le K\lambda_{A,\mathrm{pre}}(E)\pi_A(b). \tag{4.23}$$

The inequality holds in each retained parent; the small-loss assertion is only for their weighted mixture. Any further restrictions of this numerator leave its reference denominator unchanged.

Let $R_{A,2}$ be a later submeasure of the retained incidence law, carrying a datum d' with at most $D \le K$ possibilities per value of E. Its marginal on (E,b) is dominated by $R_{A,\mathrm{good}}$. With the pre-omission masses defined in the statement, retain the entries satisfying the floor

$$m_{A,\mathrm{preomit}}(E,d') \ge \lambda_{A,\mathrm{pre}}(E)/K_1. \tag{4.24}$$

The omitted mass is at most $DK_1^{-1}\sum_E \lambda_{A,\mathrm{pre}}(E) = D/K_1$ in that parent's probability units. Choose the subpower K_1 so this mass loss is negligible relative to the selected mixture mass. For the remaining measure $R_{A,3}$, the numerator and floor give

$$\begin{aligned} R_{A,3}(E,d',b) &\le R_{A,\mathrm{good}}(E,b) \le K\pi_A(b)\lambda_{A,\mathrm{pre}}(E) \\ &\le KK_1\pi_A(b)m_{A,\mathrm{preomit}}(E,d'). \end{aligned} \tag{4.25}$$

Since whole entries were omitted, their retained conditional probabilities of b are bounded by $KK_1\pi_A(b)$.

Summing (4.25) over the entries giving $C = C(E, d')$ yields $R_{A,3}(C, b) \le KK_1\pi_A(b)w_A(C)$. The numerator uses $R_{A,3}$, while $w_A(C)$ retains all the pre-omission weights from $R_{A,2}$.

A further sparse restriction $T_A \le R_{A,3}$ still has this numerator bound, but its current denominator can shrink. Before using a conditional cap under T_A, choose a new subpower K_2 and discard states with

$$T_A(C) < w_A(C)/K_2.$$

Their total mass is at most $K_2^{-1}\sum_C w_A(C)$ in that parent, and hence at most K_2^{-1} in the mixture. Taking this below the current selected mass gives, on the surviving states, $T_A(b \mid C) \le KK_1K_2\pi_A(b)$. □

Remark (Finite-state velocity bounds). Under λ_A^0, Lemma 4.8 applies with E the deep end history and $b = [v]$. The graph preparation supplies at most K point states per sufficiently deep history, so these states may serve as d'. A single finer dyadic (t, y, X) mesh simultaneously refines any finite collection of the rough grids; summing its joint ceilings gives the ceilings for their states. The reference weights $w_A(C)$ are marginals of the preceding restricted measure and retain its pure cell ceilings. Each new sparse conditional law uses new denominator pruning, whereas the unnormalized numerator ceiling persists under restriction. All initial likelihood and graph-ambiguity masks precede unrestricted sparse selections. The resulting joint and conditional bounds concern one time at a time.

The resulting bounds are integrated in time. The following elementary estimate gives fixed-time bounds for the trajectory prior while keeping that prior unchanged.

Lemma 4.9 (Retained duration and fixed-time prior caps)**.** *Let ν be a trajectory probability and $d\lambda(l, t) = G(l, t)\, d\nu(l)\, dt$, where $0 \le G \le K$ on $[0, 1]$. Fix $K_2 \ge 1$, partition time into dyadic intervals I, and set*

$$L_I = \left\{ l : \int_I G(l, \tau)\, d\tau \ge |I|/K_2 \right\}.$$

Deleting the complementary line–interval pairs removes at most $1/K_2$ incidence mass. Fix $t \in I$. If trajectory constraints C_t and C_τ^+ satisfy $C_t(l) \Rightarrow C_\tau^+(l)$ for every $\tau \in I$, then

$$\nu(L_I \cap C_t) \le \frac{K_2}{|I|} \int_I \int \mathbf{1}_{C_\tau^+}(l) G(l, \tau)\, d\nu(l)\, d\tau. \tag{4.26}$$

For almost every t, the instantaneous restricted incidence mass on C_t is at most $K\nu(L_I \cap C_t)$.

Proof. On a deleted line–interval pair the incidence duration is less than $|I|/K_2$. Integrating over the probability ν and summing $\sum_I |I| = 1$ proves the deletion bound. On $L_I \cap C_t$, the implication in the hypothesis and the duration floor give

$$\int_I \mathbf{1}_{C_\tau^+}(l) G(l, \tau)\, d\tau \ge |I|/K_2.$$

Integration over these trajectories proves the displayed estimate. The last assertion follows from $G \le K$. □

Remark (Fixed-time caps). In the prepared tangent problem, a time partition much finer than the finite collection of spatial meshes gives fixed-time estimates through Lemma 4.9. Bounded motion of (t, y, X) and of affine trajectory quantities gives the implication $C_t \Rightarrow C_\tau^+$ for fixed enlargements of mesh constraints, while a velocity bin stays fixed on a trajectory. Every integrated pure or joint cell

ceiling for these quantities therefore yields the corresponding fixed-time active-prior cap by (4.26). This transfer uses bounded motion of the displayed quantities.

After a selection, small conditional successes are omitted using the fixed mixture weights p_A; integrated exceptional numerators are removed before further sparse conditioning. Choosing K_2 sufficiently large makes its deletion loss negligible relative to the current selected mass. The construction first uses fixed finite mesh exponents and sufficiently fine time meshes, then increases the ranges slowly, retaining the single diagonal already chosen. In the return to actual outputs, substantial coverage continues to mean positive probability on the original reference path, and the final pigeonholes may divide by stage constants much smaller than $\log N_0$.

4.5 From local fits to improved known atlases

The later geometric arguments will find a single improved fit inside any substantial residual of the reference path. Such a fit need not carry enough original mass, and its label need not yet be known from the original observation. We first specify the local fit and then prove a conversion criterion that supplies both properties.

Definition 4.10 (Improvement candidates)**.** Work in a parent of the prepared path probability supplied by Proposition 4.7, and write $d = d_*$. Fix a path packet $Q = Q_s$, with actual time length q_s, thickness $a = N^{-s}$, and assigned trajectory mass ν_Q. The tangent depth s remains bounded. In both definitions below, incidence mass means residual path mass in the parent, divided by q_s.

A *time-improvement candidate of gain* $c > 0$ is a test of the proper quadratic model class, at width $a' = aN^{-c}$, on trajectory pieces assigned to Q. Its value and hidden-derivative upper bounds hold throughout the Q-block:

$$|P| \le Ka', \qquad |P_w| \le K, \qquad |P_{ww}| \le K/a'.$$

On counted incidences it also satisfies $|P_w| \ge K^{-1}$, and these incidences have mass at least

$$K^{-1}N^{-dc}\nu_Q.$$

When $0 < \ell < \infty$, put $g = N^{-\ell s}$. A *strip candidate of gain* $c > 0$ is a fixed row R such that

$$|Y' - R \cdot (1, v)| \le KgN^{-c}$$

on trajectory pieces carrying incidence mass at least $K^{-1}\nu_Q$, in the same convention. Actual rounded path nodes are understood in both definitions.

These mass tests may use restrictions of λ_A^0 without renormalization, or the parent trajectory-times-uniform-time prior restricted to the incidences. On each used parent the two measures differ by the fixed factor $s_A^{-1} \le K$.

For the whole-block upper estimates, extrapolating a polynomial along a trajectory from relative duration K^{-1} loses only a factor K. The derivative lower bound is required only on the counted incidences.

A time candidate improves the hidden-coordinate fit without changing the packet's time block; a strip candidate improves its transverse velocity prediction. The mass in the definition is relative to the packet's full trajectory assignment. The next proposition converts candidates in every substantial residual into one original known atlas.

Proposition 4.11 (Conversion of candidates to actual improvements)**.** *Consider the tangent diagonal of Section 4.2, with the prepared law supplied by Proposition 4.7. Suppose that, on a further subsequence of every subsequence carrying any substantial residual of the reference path, that residual supplies a time-improvement or strip candidate in the sense of Definition 4.10. Its depth remains in a bounded tangent range and its gain has a fixed positive finite limit, or is a fixed positive number.*

Then one of the original-resolution outputs excluded by Lemma 4.6 exists. Its original success is at least $\mathfrak{p}_{\mathrm{path}}/\log N_0$*, and its lists and geometric slacks are bounded by a fixed scenario-dependent power of* $K_{\mathrm{path}} \log N_0$*. Its complete original-coordinate complexity, including list tables, is bounded by a polynomial in* M_{path} *depending only on that fixed scenario. Consequently the candidate hypothesis is impossible on this diagonal.*

The proof first converts candidate existence into substantial coverage by one fixed scenario. For strip candidates, the covered trajectories give narrower spatial charts. For time candidates, comparison of their local graphs produces short lists and improved chart tests. In both cases, Lemma 4.12 below recovers the lists on the original law; the final complexity and mass estimates then allow Lemma 4.6 to apply. Lemma 4.12 is also used in the unbounded-narrowness argument.

Lemma 4.12 (Original-law recovery of finite lists)**.** *Let* μ *be an original unnormalized tempered path measure, and let* $\nu \le \mu$ *be an arbitrary measurable submeasure. Fix a finite geometric output-label map* $\mathcal{A}$*, defined on the original law and agreeing on* ν *with the labels used in the analytical construction. A failure label may be used off the eligible pieces and is never included in a list. Write the original observation as* $O = (x, o)$*, where* $x = (t, y)$ *is exact base and* o *contains its original observed discrete entries, including the hidden-coordinate bin and component.*

Suppose a measurable rule $L(x, o)$*, with* $|L(x, o)| \le D_{\mathrm{list}}$*, satisfies*

$$\nu\{\mathcal{A} \in L(O)\} \ge m. \tag{4.27}$$

Retain all original observation tags, path tags and output labels in the joint pushforward law λ *of* μ*, without making unobserved tags available to the list rule. If* $F_\rho\lambda$ *is uniform base-cell averaging at mesh* ρ *and* $\|\lambda - F_\rho\lambda\|_{\mathrm{TV}} \le e$*, there is a list table* $L_C(o)$*, indexed only by the base cell and original observed entries, with the same list size and*

$$\mu\{\mathcal{A} \in L_{C(x)}(o)\} \ge m - 2e. \tag{4.28}$$

If the density, ranges, copies and geometric predicates of this original joint law have complexity bounded by a fixed polynomial in M_{path} *and fixed construction constants, the mesh for* $e = O(N_0^{-1})$ *and the resulting table have complexity bounded by another such polynomial. Neither bound depends on the analytical complexity of* ν *or of* L*.*

Proof. For the fixed geometric map $\mathcal{A}$, submeasure domination gives

$$\mu\{\mathcal{A} \in L(O)\} \ge \nu\{\mathcal{A} \in L(O)\} \ge m. \tag{4.29}$$

Use total variation in the convention of supremum over measurable events. In each base cell C and observed discrete entry o, choose the D_{list} output labels with greatest total $\lambda(C, o, \cdot)$-mass; ties are broken in a fixed ordering. On $F_\rho\lambda$ this label-weight vector is independent of the position in C. Thus, writing E_L for the list success event,

$$\begin{aligned}\lambda(E_{L_C}) &\ge (F_\rho\lambda)(E_{L_C}) - e\\ &\ge (F_\rho\lambda)(E_L) - e\\ &\ge \lambda(E_L) - 2e \ge m - 2e.\end{aligned}$$

All auxiliary tags remain joint data in this calculation. The maximization uses only the observed pair C, o. The original bin is kept as a discrete tag during averaging, so the table can use its actual value on the original law even when the hidden-bin boundaries cross C.

Averaging the original law. We next bound the total-variation error in terms of the original geometric predicates, including after the hidden trajectory coefficients are marginalized. In original coefficient coordinates replace the spatial intercept by $y(t)$. The resulting variables are (x, ξ), with $x = (t, y(t))$ and ξ consisting of V and the remaining coefficients; this change has Jacobian one. Include the finite internal copies. Before integrating out ξ, regard the density as a vector indexed by all retained discrete tags, extending it by zero outside its support.

The original tempered density is piecewise constant on its defining semialgebraic pieces. Fix the other variables and translate one coordinate of x by h. A nonzero polynomial predicate of degree D has at most D roots on that axis; an identically zero restriction has no sign changes. The weighted density and its tags therefore have at most the sum of these degrees many possible changes. Boolean priority rules and finite table assignments add no new boundary locations. If H bounds the density, T the number of predicates, D their degrees, R the coordinate ranges and J the copy count, integration over the other variables gives

$$\|\tau_h f - f\|_{L^1} \le CH(TD+1)J(1+R)^q|h|, \tag{4.30}$$

where q is fixed by the ambient coefficient dimension. Marginalizing ξ contracts L^1. Averaging over pairs of points in a base cell, and changing one coordinate at a time, now gives

$$\|\lambda - F_\rho \lambda\|_{\mathrm{TV}} \le N_0^{B_{\mathcal{S}}(M_{\mathrm{path}})}\rho, \tag{4.31}$$

after absorbing fixed constants. Here $B_{\mathcal{S}}$ is a polynomial in the old complexity bound for every fixed construction scenario $\mathcal{S}$. This argument applies to the original geometric predicates and path law.

Choose $\rho = N_0^{-D_{\mathrm{mesh}}}$ with $D_{\mathrm{mesh}} \ge B_{\mathcal{S}}(M_{\mathrm{path}}) + 2$. The error is then $O(N_0^{-1})$, with room to spare. If the original base ranges are bounded by $N_0^{P_0(M_{\mathrm{path}})}$, the number of base cells is at most

$$N_0^{(j+1)(D_{\mathrm{mesh}}+P_0(M_{\mathrm{path}}))+O(1)}. \tag{4.32}$$

The number of observed bin entries, components and possible labels is also polynomial. Multiplying these counts gives the asserted polynomial table exponent. Arbitrary changes of a maximizing posterior only change the contents of this table. The table-size bound depends on the counts of cells, observed entries and labels. □

Proof of Proposition 4.11. Candidate existence means existence of a parent, node, chart and test; the auxiliary construction proving it may use much sparser configurations. Only the residual is used to certify the candidate's mass floor.

The preparations of Proposition 4.7 and Lemma 4.8 retain $1-o(1)$ of reference mass and supply the pure caps and joint end-history bounds with π_A. Bounded varying exponents use neighboring widths in the increasing grids, at a K loss. Aggregation uses the fixed parent weights p_A and unnormalized submeasures of λ_A^0. These reference laws remain fixed under subsequent deletions, so substantial mixture coverage is substantial coverage on the reference path.

Coverage by one fixed candidate scenario. First fix a gain $c > 0$, a candidate type, and a bounded range of path nodes in tangent units. Use a fixed power slack N^σ in the tests and mass threshold, where $\sigma > 0$ will satisfy the bounds below. In each chart Q, choose tests successively that carry the threshold mass on lines not yet assigned.

Each chosen test is assigned all still available Q-trajectories satisfying its line inequalities. For time candidates the value and derivative upper bounds hold throughout the block, and the derivative lower bound holds somewhere in it. Only occurrences where that lower bound holds count toward the mass threshold. Eligibility may be measured under the unnormalized parent prior restricted to prepared path incidences.

Disjointness makes this priority assignment stop after at most $N^{dc+\sigma}$ choices per Q for time candidates, or N^{σ} for strips, with a K loss if path normalization is used. The assignment is made separately for each node and scenario. The retained geometric line predicates record the chosen tests. Their eligibility was certified separately by the analytical mass calculation.

Choose a countable family of scenarios. Each fixes a type, a bounded tangent-depth range, a positive gain c, and a sufficiently small power slack σ; include arbitrarily fine choices of gains and slacks. The smallness conditions below are uniform on compact ranges of positive gains. If a residual candidate has gain $c_n \to c_* > 0$, choose a gain from this family sufficiently close to c_* and a slack σ for which the difference of gains, multiplied by every fixed exponent in the tests and mass floor, is less than $\sigma/4$. The candidate's subpower losses and the depth-grid rounding use the remaining slack for all sufficiently late stages. This choice also includes the change in the factor N^{-dc}. Every candidate with positive limiting gain is thus eligible for some fixed scenario eventually.

For each available scenario and node, make the finite greedy assignment maximal. The selections for different scenarios are made separately on the same prepared path. Suppose that the union of every fixed finite family has reference coverage tending to zero. Choose $J_n \to \infty$ sufficiently slowly that the union of the first J_n scenarios still has coverage tending to zero. After *all* path incidences on those assigned trajectory pieces throughout the respective blocks have been removed, the remaining path has substantial mass. The candidate hypothesis gives, on a further subsequence, a candidate of positive limiting gain in a bounded depth range. The preceding choice from this family supplies a fixed compatible scenario, eventually among the first J_n. Its witnessing trajectories are still free for that scenario in the relevant chart, since every incidence on every earlier assigned piece was removed. This contradicts maximality.

Hence a fixed finite family, of size J, covers a fixed reference fraction $\eta_0 > 0$ on a subsequence. Pigeonholing in that finite family fixes one scenario on a further subsequence. Choosing one available node per sample, among at most G_n, leaves coverage

$$\beta_n \geq \frac{\eta_0}{JG_n}. \tag{4.33}$$

The node can vary with the sample; its depth remains in the fixed scenario's bounded range. By (4.15), $\beta_n \log N_0 \to \infty$. Moreover $\beta_n^{-1} = N^{o(1)}$ by the diagonal preparation. These are distinct facts: the latter permits the analytical comparison, while the former supplies the actual output mass.

The following comparisons lose only negligible fractions of this coverage or bounded factors. In the time case retain the derivative lower bound $N^{-2\sigma}$ at the covered incidences. The derivative on each assigned line is constant or affine, with maximum absolute value at least $N^{-\sigma}$, so the excluded times have relative length $O(N^{-\sigma})$. Disjointness of assignments and (4.18) bound their total mass by a fixed negative power, hence by $o(\beta_n)$.

Conversion of strip coverage. Take a true label deeper within the same parent, at a fixed target depth large enough that its velocity-strip thickness is $\ll gN^{-c}$; here $\ell > 0$. Let A_{deep} be its row, and put $D = R - A_{\mathrm{deep}} = (D_0, D_1)$. On a counted incidence,

$$|D_0 + D_1 v| \leq CgN^{-c+\sigma}.$$

If $\|D\| > gN^{-c/2}$, boundedness $|v| \le K$ implies either that no such incidence exists or that $|D_1| \ge gN^{-c/2}/K$, after enlarging K. The possible velocities then lie in an interval of length

$$\frac{2CgN^{-c+\sigma}}{|D_1|} \le KN^{-c/2+\sigma} \le N^{-c/3}. \tag{4.34}$$

For each deep history E, its ancestor Q_s has at most N^σ assigned rows. The joint numerator bound (4.23) and the angular ceiling therefore charge all these misaligned rows by at most

$$KN^{\sigma-Sc/3}\lambda_{A,\mathrm{pre}}(E).$$

Fine velocity bins with bounded enlargements cover the intervals. Summing the resulting bounds over histories, labels, and parents with weights p_A gives total mass at most $KN^{\sigma-Sc/3}$. For example, choose $\sigma < \min(c/12, Sc/12)$. This is a fixed-power saving and hence is $o(\beta_n)$, since $\beta_n = N^{-o(1)}$. Thus all but a negligible fraction of counted coverage satisfies

$$\|R - A_{\mathrm{deep}}\| \le gN^{-c/2}. \tag{4.35}$$

The unbinned rows stored in the priority assignment also have coefficients bounded by a fixed power of N. A row with much larger norm and any eligible lines would confine bounded velocities to an interval of arbitrarily small fixed-power length. The same joint bound, summed over the preceding histories in Q, charges that interval by its velocity probability times $Kq_s\nu_Q$. Here the preceding history mass is at most $Kq_s\nu_Q$ by (4.18). For a sufficiently large fixed power this contradicts the greedy eligibility threshold.

Set $\tau = (J_s/q_s^2)N^{-c/4}$, using the actual relative J_s, q_s at Q, and bin the two coefficients of R at width τ, with values R_0^b, R_1^b. On aligned events the deeper path label gives an actual list for this bin at original p_L, since $gN^{-c/2} \ll \tau$. The row R itself then has coefficients bounded by $K_{\mathrm{path}}^{O(1)}$, using the actual deep-row bounds from Z-projection fullness. The output retains only predictors satisfying these bounds. On assigned lines,

$$|Y' - R_0^b - R_1^b Z'| \le K_{\mathrm{path}}^{O(1)}\tau \tag{4.36}$$

by the actual bound on $v = Z'$ and the positive power buffer. Also bin $Y - R_1^b Z$ at the block midpoint at width $q_s\tau$. Given the observed positions and velocity-row bin, this position bin has actually subpower ambiguity. The output label consists of the old label Q, its binned row, and this position bin. Merging earlier priority assignments with the same three entries preserves each trajectory's block-invariant assignment. The row and position bins determine the output labels to be recovered from the original observation.

The old Z-center is retained. The center for $Y - R_1^b Z$ is the selected partition value moving with derivative R_0^b, and the spatial columns are $q_s(1, R_1^b)$ and $q_s(0, \tau)$ in parent (Z, Y) coordinates. This geometry has actual path-slack bounds throughout the block. Its determinant in parent coordinates is $q_s^2\tau = J_s N^{-c/4}$, so composition through the parent multiplies the old cumulative determinant at Q by $N^{-c/4}$. The true test, depth, and time remain fixed. The resulting atlas has the forbidden narrowness improvement and the required aggregate list success on the original path; its tempered implementation is given below.

Comparison of the time candidates. We compare candidate graphs compatible with the same original observation. They agree at its base point to the candidate fitting accuracy. We will prove that, except on a negligible fraction of compatible pairs, they are also close on a whole short comparison box. Their finite jets can then replace the candidate indices as listable chart labels.

Work in each assigned Q_s, using its actual normalized base coordinates u (unit-block time and spatial coordinates via its moving affine center and frame); w remains in parent units. Write a_Q for the actual node thickness in these parent units, $a_Q/a \sim_{\log,N} 1$. Any subpower factors in the following preparatory comparisons can be absorbed using N^σ, for fixed σ and sufficiently late stages. The constants in exponents $O(\sigma)$ below can be fixed independently of σ small. Write $m_Q = p_A q_s \nu_Q$, so $\sum_Q m_Q \le 1$ summing over parents and blocks at the chosen node.

For each assigned test, the base marginal on the counted occurrences is bounded by

$$m_Q N^{-dc+O(\sigma)}\, du. \tag{4.37}$$

Indeed the candidate value and lower derivative tests require only $O(N^{O(\sigma)})$ hidden bins of width a' at each exact base (use monotone intervals of the quadratic); use the $K(a')^d$ cap and Jacobian $q_s J_s$, with $\nu_Q \sim_{\log,N} a^d J_s$. This bound holds also on each test's original matched incidence set witnessing the greedy mass threshold. That set's projection in u has volume $\ge N^{-O(\sigma)}$, by the threshold and (4.37) (likewise if using the unnormalized parent prior for eligibility).

Local graphs in w are obtained for each assigned test and the true Q_s test by the following curvature distinction. Write the test as P, with $a_{\mathrm{fit}} = a'$ for a candidate and $a_{\mathrm{fit}} = a$ for the true test. At late stages its whole-block value bound is $a_{\mathrm{fit}} N^\sigma$, its whole-block derivative upper bound is N^σ, and its derivative lower bound at counted points is $N^{-2\sigma}$, up to harmless fixed constants. If $|P_{ww}| \ge N^{-C\sigma}/a_{\mathrm{fit}}$, for a sufficiently large fixed C, say 20, use the affine quadratic center. It stays within $a_{\mathrm{fit}} N^{O(\sigma)}$ of the entire assigned line piece; if an equation is needed, use w minus this center.

For the remaining tests the discriminant

$$\mathcal{D} = P_w^2 - 2P_{ww}P \tag{4.38}$$

is independent of w and at counted points positive and comparable to P_w^2. On complex enlargements of the entire normalized chart box (K_{path}-sized ranges in u allowed) this polynomial and its first derivatives are bounded by $N^{O(\sigma)}$. Indeed its upper bound on the matched projection just discussed follows immediately from the tests there, for both the assigned candidate and true test; extend by polynomial Remez. The projection bound here comes from each assigned candidate's initial matching before greedy enlargement.

The time of Q_s is subdivided dyadically at relative size ρ comparable to $N^{-C'\sigma}$, with C' sufficiently large and fixed. In each short block, the midpoint normalized spatial position of each line determines a ρ-box. There are at most $N^{O(\sigma)}$ resulting base subboxes, including time, per Q. Given Q and the exact base position, these subboxes have lists at the original p_L of size bounded by actual path-slack powers: motion in normalized coordinates over a short block is at most $K_{\mathrm{path}}^{O(1)}\rho$.

Around each subbox center, take comparison balls or polydiscs with radius a fixed multiple of $K_{\mathrm{path}}^{O(1)}\rho$, large enough to contain the corresponding real line pieces with a fixed extra holomorphic margin. Keep only test–subbox choices certified by a counted occurrence, which supplies the derivative lower bound at one point. The chart derivative bound for $\mathcal{D}$, together with the choice of C', makes its variation throughout this comparison domain much smaller than $N^{-4\sigma}$. For each nonreplaced quadratic test, the discriminant therefore stays nonzero and gives simple holomorphic branches throughout the domain, real on the real box. For a linear equation in w, the same check keeps its coefficient nonzero.

For such nonreplaced tests, on any assigned line for these choices the low-curvature cutoff guarantees $2|P_{ww}P| \ll \mathcal{D}$ over the whole real short piece, so the sign of P_w is constant there, picking one branch f. At every time w has distance at most $O(a_{\mathrm{fit}} N^{O(\sigma)})$ from that branch: P_w has the same sign there and at w, comparable in absolute value to $\sqrt{\mathcal{D}}$, so the derivative lower bound works on the intervening interval. Replaced tests simply use their affine centers. Thus branch signs for true

and candidate tests can be assigned invariantly over the short block. Denote the true comparison graph in one such choice by f_0.

Comparison at a common observation. The geometric preparation lets us compare the two candidate graphs associated with independent occurrences at one original observation. Let μ_{cov} be the unnormalized law of counted covered occurrences, including the parent weights. Set $O = (p_L, Q_s, \mathrm{subbox}, \mathrm{true\ branch})$, with history included in Q_s, and let κ_O be the conditional probability of an occurrence under this law. Sample O with its covered-law pushforward, then sample two independent posteriors:

$$\mu_{\mathrm{pair}} = \int \kappa_O \otimes \kappa_O \, d(O_* \mu_{\mathrm{cov}})(O). \tag{4.39}$$

Each marginal is μ_{cov}, and the pair law retains its unnormalized total mass, at least $N^{-o(1)}$. The candidate graphs agree at the sampled common base point to $a' N^{O(\sigma)}$, because original p_L, given the parent, determines w much more finely. In version 1 the subtracted parent graph is fixed by the chart and exact base, so the same conclusion holds.

Fix $d_0 = (1+d)/2 < 1$. Let ε_* be the tolerance in Corollary 3.4 for these fixed comparison domains and degrees. Decrease it to at most one, and set

$$\varepsilon = \varepsilon_*/2, \qquad \theta = \varepsilon_*/8. \tag{4.40}$$

We will choose σ after these constants. We claim that, outside an $o(1)$ relative fraction of the pair law, the candidate graphs agree in norm on the comparison real box, with its fixed margin around the used pieces, to $aN^{-2\theta c}$.

First discard entries (index, subbox, candidate and true branch signs) with marginal mass below $m_Q N^{-dc - C''\sigma}$. The index and subbox counts make their total mass negligible on either marginal when C'' is sufficiently large. For each remaining entry, the candidate graph differs from f_0 by at most $aN^{O(\sigma)}$ on its incidence base projection. By (4.37), this projection has relative measure at least $N^{-O(\sigma)}$ in the comparison box. Lemma 3.1 on the interiors with fixed margins then gives the same norm bound, with an adjusted $O(\sigma)$ exponent.

Normalizing a group of bad pairs. Suppose a nonvanishing relative fraction of the pair law were bad. The light-entry omissions leave a group $(Q, \mathrm{subbox}, \mathrm{true\ branch})$ of bad pairs of mass at least $m_Q N^{-O(\sigma)}$; the weighted count justifying this choice is given below. After normalizing that group's bad pairs to probability, Corollary 3.4 compares its two candidate lists, relative to the common true graph f_0, with $M = a$ and $\Delta = a'$. After rescaling the base, (4.37) gives index-base ceilings with weights N^{-dc} and losses $N^{O(\sigma)}$. To check that they remain valid for probability weights in each color, we also account for the bad-pair normalization. Indeed, if W_c is the sum of N^{-dc} over that color's retained indices and B is the unnormalized bad-pair mass in the selected group, the index density bound becomes

$$\frac{f_i}{B} \le N^{O(\sigma)} \frac{m_Q W_c}{B} \frac{N^{-dc}}{W_c}. \tag{4.41}$$

The list count gives $W_c \le N^{O(\sigma)}$, and the group choice gives $B \ge m_Q N^{-O(\sigma)}$. Thus the multiplier of the probability weight N^{-dc}/W_c is $N^{O(\sigma)}$, as required. The existence of such a group uses the sum $\sum_Q m_Q \le 1$ and the polynomial-in-N^σ subbox count. If a nonvanishing relative fraction of the covered pair law were bad, its absolute mass would be at least a fixed multiple of $\beta_n = N^{-o(1)}$. Light-entry errors are power-small and hence $o(\beta_n)$; the factor β_n^{-1} is absorbed into the fixed-power comparison allowance.

The normalization just proved supplies the marginal bounds required by Corollary 3.4. To check its remaining quantitative hypotheses, put $R = M/\Delta = N^c$, with $M = a$ and $\Delta = a'$. All functions

are holomorphic sheets with bounded-degree relations. A bad pair has norm gap at least

$$aN^{-2\theta c} = MR^{-2\theta} > MR^{-\varepsilon}.$$

The list counts are at most $R^{d+C\sigma/c+o(1)}$. The norm upper bounds relative to the true graph f_0, the pointwise matching precision, and the normalized marginal multipliers just computed have losses at most $R^{C\sigma/c+o(1)}$, for finitely many fixed constants C. Choose σ so that every $C\sigma/c < \varepsilon_*/8$. At late stages the remaining $o(1)$ losses, including β_n^{-1}, fit within another $\varepsilon_*/8$. The normalized bad-pair law has mass one, so it also satisfies the required lower mass bound. Since $d < d_0$, all hypotheses of Corollary 3.4 hold with $\varepsilon = \varepsilon_*/2$, a contradiction. This proves the claim.

The improved time charts. Bin sufficiently many candidate branch jets at the subbox center (in comparison-ball scaled coordinates) to steps $aN^{-\theta c}$. Pairs with the proved closeness give bounded neighbor bins by the norm rule. For clarity, if π_O is the posterior law of the actual jet bin at a conditioning observation O, and $\mathcal{N}(b)$ is the bounded neighbor list of a bin b, then

$$\sum_b \pi_O(b)\pi_O(\mathcal{N}(b)) = \mathbb{P}\{B_2 \in \mathcal{N}(B_1) \mid O\}. \tag{4.42}$$

Choose a maximizing b separately for each O. Its neighbor list has success at least the displayed average. Integrating preserves $1 - o(1)$ of the covered mass once the bad-pair fraction is $o(1)$; in particular it preserves a fixed positive fraction. The neighbor lists retain this aggregate covered mass even when each individual candidate has only inverse-subpower mass.

Define the preliminary geometric label

$$c_{\mathrm{geom}} = (Q_s, \text{short block}, \text{spatial subbox}, \text{true branch sign}, \text{candidate jet bin}).$$

Merging the earlier priority assignments with the same c_{geom}, while retaining their invariant branch signs, gives a trajectory assignment fixed throughout the short block. The original candidate index is not part of c_{geom}. The conditioning entries preceding the jet bin have only actual path-slack ambiguity at the original p_L, so the posterior-neighbor lists give lists for this full preliminary label with the same type of bound.

For each nonempty jet bin in its group, choose one representative equation $\widehat{P}$ and its assigned branch $\hat{f}$. The representative is a fixed function of c_{geom}. Every member branch in this bin differs from $\hat{f}$ by $O(aN^{-\theta c})$ on the real box: sufficiently many center jets control the norm of their holomorphic algebraic difference. Thus

$$|w - \hat{f}| \lesssim aN^{-\theta c}$$

on each member's entire short line piece, after choosing σ small enough in the graph approximations. We now turn the representative into an admissible chart test.

- If $|\widehat{P}_{ww}| \le (aN^{-\theta c/2})^{-1}$, normalize by the absolute w-derivative on the branch at the box center (this equals 1 for explicit tests, notably in version 1). In the simple-root case this factor is $\ge N^{-O(\sigma)}$ and comparable uniformly, to constants, to the absolute w-derivative of the test on its branch throughout, by discriminant stability. Passing from branch to member w changes the derivative by $O(N^{-\theta c/2})$. Thus the normalized equation has actual order-one derivative bounds, value $\lesssim aN^{-\theta c}$ and curvature $\le a^{-1}N^{\theta c/2+O(\sigma)}$ on the required data. These fit at thickness $a_Q N^{-2\theta c/3}$.

- For larger curvature, the representative branch is within $aN^{-\theta c/2+O(\sigma)}$ of its affine center by the test's derivative upper bound on that branch. Use w minus that affine center as test with thickness $a_Q N^{-\theta c/3}$.

Choose σ after θ to meet the preceding matching tolerances and graph-approximation bounds. Also impose $C'\sigma < k\theta c/12$. The strict buffers absorb the subpower discrepancy a_Q/a. Select a representative type carrying a fixed fraction of the list success, so the output has a common depth.

The true spatial frame is scaled by ρ, with its center adjusted to the subbox's midpoint spatial bin in moving chart coordinates. Its bounds hold throughout with actual path slacks. Under the return to original coordinates in Lemma 2.13, thickness and tests acquire the parent-depth factor and the derivative units are restored. In version 2, w uses the resulting parent B-scale; in version 1 the parent graph subtraction preserves the special test form. The path labels, subbox, true branch sign, and deterministic bin lists together increase the list size by only actual path-slack and logarithmic powers.

The preliminary label c_{geom} is retained: its chosen representative determines the curvature type and final test. The assignment is restricted to trajectories satisfying the resulting whole-block geometric, value, and derivative upper bounds, with the derivative lower bound imposed on selected incidences. The comparisons above show that the retained list successes satisfy these tests. We have therefore constructed geometric charts and full-label lists on the covered analytical law.

The depth gain is at least $(\theta c/3)\epsilon_n$, whereas the additional time exponent is $C'\sigma\epsilon_n + o(\epsilon_n)$. Our choice of σ makes this less than half k times the depth gain. Either representative type therefore gives the forbidden improved-depth atlas.

Encoding the geometric predicates. Both constructions have supplied chart geometry, tests, full labels and block-invariant trajectory assignments. We first bound the coefficients and complexity of their geometric predicates.

The stored data are the ordered quadratic tests or row predictors at the single chosen node. The assignments use membership in the old chart's trajectory set, whole-block line inequalities, and finite priority rules. Subbox certification flags, branch signs, merger tables, and representative tests are fixed data for each chart and test; a simple branch is selected by its midpoint derivative sign. Nonempty jet bins may be reindexed by discrete identifiers; predicates then use those identifiers. The geometric assignments are defined by the stored tests and finite rules just described, while masks and posterior search serve the analytical list construction.

For every time candidate, the coefficients can be bounded explicitly. In normalized chart coordinates write $P(u,w) = Aw^2 + B(u)w + C(u)$, with B affine and C quadratic. Curvature gives $|A| \le N^{s+c+O(\sigma)}$. The original threshold witness, before greedy enlargement, has base projection of volume $\delta \ge N^{-C_{\mathcal{S}}\sigma}$ by (4.37); its chart weight cancels when the floor is divided by the ceiling. On that witness $|w| \le W \le N_0^{P_0(M_{\mathrm{path}})}$ by original problem and path bounds. The derivative and value estimates yield

$$|B(u)| \le N^{\sigma} + 2|A|W, \qquad |C(u)| \le a'N^{\sigma} + |A|W^2 + |B(u)|W. \tag{4.43}$$

Fixed-degree polynomial Remez from this projection loses at most a factor δ^{-C}; it bounds all coefficients of B, C by $N_0^{P_{\mathcal{S}}(M_{\mathrm{path}})}$. The bound used for w is in original exponent units, including its affine part. Inverse curvatures for affine centers are needed only above cutoffs; rescaling by a branch derivative uses the stability lower bound. Raw strip coefficients were bounded above. Coordinate compositions back to the original component are all polynomially controlled. Counts of new choices per path chart are bounded by fixed tangent powers. The whole-block tests have fixed-degree semialgebraic complexity, with block time quantified separately for each individual test.

Lists on the original law. On the masked successes the full output labels have lists at the original p_L of size at most $(K_{\mathrm{path}} \log N_0)^{r_{\mathcal{S}}}$, with $r_{\mathcal{S}}$ fixed by the scenario. For the original unnormalized reference path and its masked covered submeasure, Lemma 4.12 applied to this geometric assignment produces a tempered cell-list selection with the same list size and only $O(N_0^{-1})$ original mass loss.

The averaging applies to the original joint law. The final selector observes only the original exact base and original discrete observation entries.

Complexity and mass. The coefficient and predicate bounds above, followed by Lemma 4.12, bound every original-coordinate complexity exponent by a polynomial in M_{path} depending only on the fixed scenario. This includes the flattening depth and list-table count. Only the chosen scenario is encoded. Its polynomial bound fits $M^{\#}$ at all sufficiently large stages by (4.13); its fixed power of actual path factors and logarithms likewise fits the allowance n.

The comparison and representative choice retain some fixed fraction $c_0 > 0$ of β_n. After original-law list recovery, the original final success is therefore at least

$$\frac{c_0\eta_0}{JG_n}\mathfrak{p}_{\mathrm{path}} - O(N_0^{-1}) > \frac{\mathfrak{p}_{\mathrm{path}}}{\log N_0} \tag{4.44}$$

for large stages, by (4.15). The original mass loss is negligible since

$$N_0^{-1}\log N_0/\mathfrak{p}_{\mathrm{path}} \le N_0^{-1}K_{\mathrm{path}}\log N_0 \to 0.$$

The fixed positive tangent gain exceeds γ_n, and its time exponent is less than half k times its gain. All requirements of Lemma 4.6 are now satisfied, giving the contradiction. □

4.6 Velocity breadth in isotropic parents

The isotropic argument will use finite point states to control gradients of packet graphs. A large gradient would predict the velocity in a power-thin affine strip. We now rule out such concentration on a substantial family by constructing a narrower parent atlas. The original-law list recovery and finite encoding just established realize this width improvement while keeping the parent depth and horizon fixed.

Corollary 4.13 (Breadth before sparse selections)**.** *In version 2 with $\ell = 0$, let a substantial path residual carry a measurable discrete state, including its tangent parent. Suppose the state is known on these events at the original p_L with list size bounded by a fixed power of actual path factors and logarithms, and has at most K possible values given the used deep end histories. For every fixed $\kappa, u > 0$, the total residual probability of states whose conditional velocity law puts mass at least κ within distance N^{-u} of an affine line tends to zero.*

In version 1 the corresponding assertion holds for intervals. These statements can be prepared at reciprocal-subpower separation thresholds before arbitrary sparse selections. They persist under selections of whole conditioning states, or controlled fixed-fraction selections within them.

Remark. The version-2 breadth conclusion is restricted to these permitted selections; it makes no assertion for every velocity-dependent inverse-subpower subfamily.

Proof. If the version-2 assertion fails, on a subsequence there are state-predicted strips of the indicated width carrying a fixed positive amount of residual mass, still on substantial events of the prepared good law.

A finite family of heavy strips. Use unit normal and offset pairs (n_v, b'_v) for those predicted lines, with Euclidean parameter distance modulo simultaneous sign. Parent velocities are bounded by $T \ge 1$ of order actual path-slack powers; the required offsets then have size $O(T)$. On almost all the narrow events the predicted strip, say enlarged to $2N^{-u}$ using finer velocity bins, has parent velocity law π-weight at least $\kappa_N = 1/K$ for sufficiently small reciprocal-subpower threshold. Indeed those of smaller mass under the velocity law have negligible total probability by summing the unnormalized

joint bound of velocity with the end history using the subpower list there (choose κ_N small relative to all those losses). State predictions need only be specified on occurring states; their choices are finite here.

Per parent take maximal parameter-separated representatives at spacing $r_1 = N^{-u/2}$ for such heavy strips. Their number M_1 is $\le K$. For two separated ones with common narrow-band points in $|v| \le T$, the angle sine is at least a constant times $r_1/(1+T)$: take the sign with closer normals; offset difference there is at most $4N^{-u} + T|n_{v,1} - n_{v,2}|$. Thus the intersection is contained in a ball of diameter $O((1+T)N^{-u}/r_1) \le N^{-u/3}$, hence a fixed angular power in π. By Cauchy–Schwarz on the sum of strip indicators,

$$(M_1\kappa_N)^2 \le M_1 + KN^{-Su/3}M_1^2, \tag{4.45}$$

giving the claim. Moreover for representatives separated by $> r_2 = N^{-u/4}$, their enlarged strips of widths $10(1+T)r_1$ still have intersections of negligible mass under the velocity law even summed over all pairs: the same reasoning gives diameter $\lesssim (1+T)^2 r_1/r_2$ or no common point in the used range.

Trajectory assignments and the narrower atlas. On narrow events the velocity therefore lies in an enlarged representative strip, and any choice of such containing strip agrees to $O(r_2)$ in parameter (modulo sign) with the predicted pair outside negligible mass. Indeed there is one within r_1 of the prediction by maximality, and the excluded overlaps depend just on parent and velocity and have small probability by the bound for the velocity law. The first containing representative is assigned using only the parent trajectory velocity, so the assignment is invariant over the block. Bin its parameters (with one orientation each) to rectangular r_2-bins; this gives only boundedly many options given the state prediction on successful events. Fix one unit normal and offset pair $(\hat{n}, \hat{b})$ per occurring bin consistent with it. Then

$$|\hat{n} \cdot y' - \hat{b}| \lesssim (1+T)r_2 \tag{4.46}$$

on assigned pieces. Bin also the corresponding $\hat{n}$-position at parent-block midpoint to width r_2 in parent units, known with actual slack up to lists via observed base and this velocity prediction. Merge assignments using those two bin labels. Use a frame with unit tangential column and width r_2 along $\hat{n}$ in parent spatial coordinates (move the transverse bin with derivative $\hat{b}$); compose with the parent frame, keeping parent test, depth and horizon. All slacks are thus bounded by fixed powers of actual path bounds, and the determinant has gained the new narrowness factor.

On the successful incidences, the assumed state knowledge supplies actual lists for the full output labels at the original p_L. The finite representative and bin assignments are stored together with the original path predicates. Lemma 4.12, applied to this geometric label map on the original path and its successful submeasure, realizes the analytical list success by tempered tables with the same list size and only $O(N_0^{-1})$ original mass loss; the stored data are the geometric assignments and original path predicates. As in the preceding construction, the original mass and fixed-complexity bounds fit Lemma 4.6. This contradicts the width improvement excluded at the parent.

In version 1 the corresponding avoidance of narrow velocity intervals on substantial families already follows from the angular cap and the deep-history joint bound (summing the subpower state ambiguities there). These almost-all breadth/separation conclusions can use reciprocal-subpower accuracies by taking fixed-power thresholds to zero sufficiently slowly. In applications requiring general position, one can prepare them before sparse selections and keep them through selections of whole conditioning states (or with small fixed-fraction losses controlled there). The permitted version-2 selections are those stated in the remark after the corollary. □

5 Scalar concentration and polynomial fitting

We prove two consequences of scalar concentration. First, a small spacetime support forces a heavy bin of quadratic trajectory coefficients. Second, when a common row subtracts the tangential coordinates from a quadratic signal, concentration of the residual scalar forces polynomial fits to that row along labeled trajectories. The second argument begins with entropy estimates under simultaneous time tests. A planar dot-product theorem then gives affine row fits and first-jet information; divided differences raise the jet order, and a separate time test produces the polynomial fits.

The arguments in this section take place in tangent units. Thus $K = N^{o(1)}$ denotes a changeable subpower factor, and a measure is dense if its mass is at least K^{-1}. All coefficient ranges and all fixed-dimensional norms below are bounded by K. The time interval is $[0, 1]$. A trajectory index may include auxiliary data; a bin of its coefficients never bins those additional indices.

We use measurable time labels $E_l \subset [0, 1]$. An instantaneous labeled mass is

$$\nu\{l : t \in E_l, \text{ the indicated conditions hold}\};$$

it is not divided by the success probability at time t. This convention also covers an incidence density $w(l, t) \le K_{\mathrm{pre}}$ with $K_{\mathrm{pre}} = N^{o(1)}$, relative to $\nu \otimes dt$. Indeed, if its mass is $\delta \ge K_{\mathrm{pre}}^{-1}$, deleting $w < \delta/2$ loses at most $\delta/2$. On the remaining indicator label set, the two measures compare by subpower factors, and its product measure is at least $\delta/(2K_{\mathrm{pre}})$. The same argument applies in typical conditional components. Restrictions made in this section change the labels and leave the indicated trajectory prior fixed unless a new normalization is explicitly stated.

5.1 Concentration of quadratic coefficients

Lemma 5.1 (Scalar coefficient clustering)**.** *Let*

$$x_l(t) = x_{0,l} + t x_{1,l} + t^2 x_{2,l},$$

and let the trajectory prior ν have total mass at most K. Fix $0 \le d < 1$ and a resolution $p = N^{-c+o(1)}$, where $c > 0$ is fixed. Suppose that a dense labeled incidence measure, dominated by $K\, d\nu\, dt$, occupies at most Kp^{-1-d} squares of side p in the (t, x)-plane. Suppose also that

$$\nu\{l : (x_{0,l}, x_{1,l}, x_{2,l}) \in B\} \le Kr^d \tag{5.1}$$

for every coefficient ball B of radius $r \ge p$ in the bounded coefficient range. Then some coefficient bin of side p has trajectory mass at least p^d/K. The conclusion can be obtained using only trajectories with labeled length at least K^{-1}, and it applies anew to every dense residual satisfying the same upper bounds.

Proof. First discard trajectories of labeled length below a sufficiently small reciprocal subpower. The remaining trajectory mass is at least K^{-1}, and each retained trajectory meets at least p^{-1}/K labeled time bins. Adjoin an independent uniform variable $D \in [0, 1]$ and lift the signal to the affine planar motion

$$z_{l,D}(t) = \big(x_{0,l} + t(x_{1,l} - D),\ D + t x_{2,l}\big). \tag{5.2}$$

The identity

$$(1, t) \cdot z_{l,D}(t) = x_l(t)$$

shows that the marked lifted motions occupy at most Kp^{-2-d} spacetime cubes. In fact, above one occupied (t,x)-square there are at most K/p second-coordinate bins, and then K possible first-coordinate bins. Motion within one time bin changes this count only by a subpower factor.

Apply Lemma 2.15, namely the weighted full-trace form of OpenAI [24, Lemma 2.3]. Rescaling a subpower bounded spatial chart into a fixed bounded chart and rounding line parameters much more finely than p are allowed by that lemma. The total weighted number of marks is at least p^{-1}/K, whereas the occupied-cell count is at most Kp^{-2-d}. Their ratio is at least p^{1+d}/K. Consequently there is a full-time plank with fixed orthogonal spatial axes, affine center, and physical half-widths

$$p/K \le u \le v \le K$$

whose lifted trajectory mass is at least

$$K^{-1}p^{d-1}uv. \tag{5.3}$$

Here the conversion from widths in cell units contributes p^{-2}; the fixed positive loss in the quoted estimate is sent to zero slowly after its scale is fixed.

All scalar quadratics arising from this plank lie in one coefficient ball of radius Kv. To see this, take the difference of two lifted motions in the plank. Its scalar projection by $(1,t)$ has size $O(v)$ throughout unit time, and evaluation at three fixed separated times bounds its three coefficients by $O(v)$.

For fixed scalar coefficients, varying D changes the lifted motion by $D(-t,1)$. If $n=(n_1,n_2)$ is the unit narrow normal, then

$$\max\{|n_2|,|n_2-n_1|\} \ge c$$

for an absolute $c>0$. The narrow constraint at one of $t=0,1$ therefore accepts a set of D's of length at most Cu. The coefficient cap implies that the lifted plank mass is at most Kuv^d. When $v<p$, use the cap at p and $v \ge p/K$, absorbing the resulting subpower factor into K. Comparing with (5.3) gives

$$v^{1-d} \le Kp^{1-d}, \qquad v \le Kp.$$

Dividing (5.3) by the dummy-coordinate acceptance bound gives coefficient mass at least

$$K^{-1}p^{d-1}v \ge p^d/K.$$

The coefficient ball has radius Kp and meets only K ordinary p-bins; one of them has the asserted mass. All pruning and estimates used only the upper hypotheses and a dense lower mass, proving the residual assertion. □

5.2 Scalar-normal data and conditioning

The next definition separates the fixed trajectory prior from the time labels that we will repeatedly restrict. Its three inputs control scalar supports, comparisons of the row and residual slope, and spatial mass conditional on a coefficient bin.

Definition 5.2 (Scalar-normal data)**.** Fix $j \in \{1,2\}$, $0 \le d < 1$, and a floor ζ. Let ν be a trajectory probability, let x_l be quadratic and $y_l : [0,1] \to \mathbb{R}^j$ affine, and suppose their coefficients are bounded by K. Let E_l have dense mass under $\nu \otimes dt$, and let $F(t) \in (\mathbb{R}^j)^*$ be a measurable, time-only row of norm at most K. Write

$$S_l(t) = x_l(t) - F(t)y_l(t).$$

For a bounded set A, let $N_q(A)$ count the cells of a fixed half-open mesh of side q that meet A, with the usual dyadic rounding. The following bounds are required at the available resolutions:

1. For every active time, every interval J of length r, and $\zeta \le q \le r \le 1$,
$$N_q\big(J \cap \{S_l(t) : l \text{ with } t \in E_l\}\big) \le K(r/q)^d. \tag{5.4}$$
For every spatial cube $Q \subset \mathbb{R}^{1+j}$ of side q,
$$\nu\{l : t \in E_l,\ (x_l(t), y_l(t)) \in Q\} \le Kq^{j+d}. \tag{5.5}$$

2. At active times,
$$|F(t) - F(u)| \le K(|t-u| + \zeta). \tag{5.6}$$
Writing $R_l(t) = x_l'(t) - F(t)y_l'$, labeled spacetime points satisfy
$$|R_l(t) - R_{\tilde{l}}(u)| \le K\big(\mathrm{dist}((t, x_l(t), y_l(t)), (u, x_{\tilde{l}}(u), y_{\tilde{l}}(u))) + \zeta\big). \tag{5.7}$$

3. Let l_r bin all the displayed polynomial coefficients at side r, and let B be a participating bin of positive prior mass. For every spatial q-cube, $\zeta \le q \le r$,
$$\frac{\nu\{l \in B : t \in E_l,\ (x_l(t), y_l(t)) \in Q\}}{\nu(B)} \le K(q/r)^{j+d}. \tag{5.8}$$

These are absolute labeled masses relative to the stated prior. A common null set of times may be removed. A finite component parameter is permitted, with the hypotheses holding in each retained conditional component; every conditioning variable below then includes that component.

We use fixed dyadic grids, allowing bounded enlargements. All exponents of resolutions used in one application range over a fixed finite interval. Slowly densifying exponent grids give intermediate covering bounds by subdivision, since consecutive scale ratios are subpower. The conditional cap (5.8) is needed on compatible parameter grids at the entropy scales. A restriction of labels preserves every upper hypothesis in Definition 5.2 with the same prior.

For these data we will approximate F by polynomials on labeled portions of individual trajectories within short time blocks. Each portion will have inverse-subpower relative length and contain the incidence being fitted; the polynomial may depend on that incidence. The next entropy estimates express the obstruction behind this fit: a scalar with support exponent $d < 1$ cannot determine a tangential projection whose entropy has exponent one. Later discrepancies will be tested on events involving several times, so we prove these estimates anew on every dense tested event.

For clarity, write H and I for Shannon entropy and conditional mutual information, and set $o_* = o(\log N)$. Exact data are conditioned upon before only the displayed outputs are discretized.

Lemma 5.3 (Entropy on a dense tested event)**.** *Assume Definition 5.2. Presample a finite time tuple independently of the trajectory in its component, with absolutely continuous marginal laws for its tested times. Let T contain the component and the entire tuple. Let $\mathcal{A}$ be any dense event on which all tested labels hold, and normalize its restriction to a probability Q. At every tested time t and available dyadic scales,*
$$\mathsf{H}_Q(S(t)_r \mid T) = d\log(1/r) + o_*, \tag{5.9}$$
$$\mathsf{H}_Q(y(t)_r \mid T, S(t)_r) = j\log(1/r) + o_*, \tag{5.10}$$
$$\mathsf{I}_Q(l_r; S(t)_q \mid T, S(t)_r) = o_*, \qquad q < r. \tag{5.11}$$
Moreover, for every T-measurable unit row e and $r \ge q$,
$$\mathsf{H}_Q((ey(t))_r \mid T, S(t)_q) \ge \log(1/r) - o_*. \tag{5.12}$$
The same assertions hold anew on every further dense event; the event may depend on all the times and on the trajectory.

Proof. Let P be the law before restriction, $p_{\mathcal{A}} = P(\mathcal{A})$, and $s(T) = P(\mathcal{A} \mid T)$. The following elementary denominator bound will also be used with additional coefficient-bin conditioning:

$$\mathbb{E}_Q \frac{1}{s(T)} \le \frac{1}{p_{\mathcal{A}}}, \qquad \mathbb{E}_Q \log \frac{1}{s(T)} \le \log \frac{1}{p_{\mathcal{A}}} = o_*. \tag{5.13}$$

The first inequality follows by integrating $s(T)/p_{\mathcal{A}}$ against the law of T on $\{s > 0\}$; the second is Jensen's inequality. The averaged denominator bound is sufficient for the entropy estimates below.

The shear $(x, y) \mapsto (S, y)$ is T-known and has subpower distortion. Put $Z = (S_l(t)_r, y_l(t)_r)$. Since $\mathcal{A}$ implies the tested label at t, every joint bin z satisfies

$$P(\mathcal{A}, Z = z \mid T) \le P(t \in E_l, Z = z \mid T) \le K r^{j+d}.$$

For the last inequality, the trajectory law at fixed T is the original within-component prior, and the sheared bin is covered by K spatial r-cubes to which (5.5) applies. Thus, whenever $s(T) > 0$,

$$Q(Z = z \mid T) = \frac{P(\mathcal{A}, Z = z \mid T)}{s(T)} \le \frac{K r^{j+d}}{s(T)}.$$

The entropy is therefore at least

$$(j + d) \log(1/r) - O(\log K) - \mathbb{E}_Q \log(1/s(T)).$$

The support bounds are $K r^{-d}$ for S_r and $K r^{-j}$ for y_r. Subtracting either support entropy proves (5.9) and (5.10), including their matching upper bounds.

For (5.11), put $B = l_r$ and $s(T, B) = P(\mathcal{A} \mid T, B)$. Before restriction the conditional trajectory law is exactly the within-bin prior in (5.8), because the tuple was trajectory-independent. In B, there are at most $K(r/q)^j$ relevant y_q-bins for a fixed S_q-bin. Hence

$$P(\mathcal{A}, S_q = z \mid T, B) \le K(q/r)^d.$$

The calculation in (5.13) gives

$$\mathbb{E}_Q \log(1/s(T, B)) \le \log(1/p_{\mathcal{A}}),$$

with no dependence on the number or individual prior masses of B. Consequently

$$\mathsf{H}_Q(S_q \mid T, B) \ge d \log(r/q) - o_*.$$

Since the coefficients vary by $O(r)$ within B, the time-known shear permits only K values of S_r there. Chain rule gives

$$\mathsf{H}_Q(S_q \mid T, B, S_r) \ge \mathsf{H}_Q(S_q \mid T, B) - \mathsf{H}_Q(S_r \mid T, B) \ge d \log(r/q) - o_*.$$

On the other hand, (5.4) bounds $\mathsf{H}_Q(S_q \mid T, S_r)$ above by $d \log(r/q) + o_*$. Their difference proves (5.11).

For (5.12), fix T and a projection bin of width r. The corresponding slab in the bounded y-range meets at most $K r q^{-j}$ cubes of side q. Subtract its logarithm from (5.10) at scale q. All estimates used the new event's actual probability $p_{\mathcal{A}}$; they can therefore be repeated on any dense event. Absolutely continuous time marginals suffice to avoid the common null set, and a density ceiling for the tuple law is unnecessary. □

We shall repeatedly use the following exact entropy inequality, valid for any discrete Z, A, B, C, with arbitrary further conditioning T:

$$\mathsf{H}(Z \mid T, C) \le \mathsf{H}(Z \mid T, B) + \mathsf{H}(Z \mid T, A) + \mathsf{I}(A; B \mid T, C). \tag{5.14}$$

Indeed, expand the left side as $\mathsf{H}(Z \mid T, B, C) + \mathsf{I}(Z; B \mid T, C)$, enlarge Z to (A, Z) in the mutual information, and use chain rule. In our applications, Z has negligible entropy both given (T, S_q) and given (T, l_r). Taking $A = S_q$, $B = l_r$, and $C = S_r$, the mutual-information estimate (5.11) then gives negligible entropy conditional on (T, S_r) alone. Boundary ambiguities caused by bounded enlargements contribute at most $O(\log K)$ to the entropy.

5.3 The first jet

Proposition 5.4 (Scalar-normal first jet)**.** *Assume Definition 5.2. Fix a finite nested hierarchy of dyadic time lengths of fixed positive power, with finest length H_*. If the available floor is sufficiently fine—$\zeta \le H_*^3/K$ suffices after enlarging K—then a dense restriction of the single-time labels admits a time-only row $A(t)$, bounded by K, such that*

$$|F(v) - F(u) - (v - u)A(u)| \le KH^{11/10} \tag{5.15}$$

whenever the retained active times u, v lie in the same designated interval of length H. Put

$$J_1(l, t) = x_l'(t) - F(t)y_l' - A(t)y_l(t).$$

For every finite presampled tuple and every dense tested event as in Lemma 5.3, on which all tested times carry the retained labels,

$$\mathsf{H}(J_1(l, t)_r \mid T, S_l(t)_r) = o_* \quad \text{if } r \ge H_*^{(1-d)/64} \tag{5.16}$$

at the available resolutions. The row A and the single-time restriction are fixed before the tuple and the later event are chosen.

The first step is geometric: we fit each component of F by an affine function on selected times in each hierarchy block. After local normalization, the residual comparison expresses the residual slope as an approximately Lipschitz function of one tangential coordinate. Collisions of scalar values relate this graph to the graph of the corresponding component of F through dot products. The scalar support has exponent $d < 1$, which will exclude the expansion alternative in the following theorem. Its line-concentration alternative then supplies the affine fit.

Comparing these fits across lengths will produce the row A and its Taylor estimate. We will then concentrate the residual slopes J_1. This last step gives fixed scalar and coefficient-bin covers, so that the entropy estimate can be proved on every later dense tested event.

Theorem 5.5 (Planar dot-product alternative)**.** *For every $\varepsilon > 0$ there are $\eta, \delta_0 > 0$ such that the following holds for $0 < p \le \delta_0$. Let $\mathcal{A}, \mathcal{B} \subset [0, 1]^2$, and suppose that*

$$N_s(E \cap B(z, w)) \le p^{-\eta} w/s \qquad \text{for } E \in \{\mathcal{A}, \mathcal{B}\}, \quad p \le s \le w. \tag{5.17}$$

At least one of the following alternatives holds:

1. *There are affine lines $\ell, \ell^\perp$ with perpendicular directions such that both*

$$N_p(\mathcal{A} \cap \{z : \operatorname{dist}(z, \ell) \le p\}), \qquad N_p(\mathcal{B} \cap \{z : \operatorname{dist}(z, \ell^\perp) \le p\})$$

are at least $p^{\varepsilon - 1}$.

2. *Every restricted triple set $\mathcal{H} \subset \mathcal{A} \times \mathcal{B} \times \mathcal{B}$ with $N_p(\mathcal{H}) \geq p^{\eta-3}$ admits a scale $s \geq p$ and an interval J, $|J| \geq p^{-\eta}s$, for which*

$$N_s\big(J \cap \{a \cdot (b - b') : (a, b, b') \in \mathcal{H}\}\big) \geq (|J|/s)^{1-\varepsilon}.$$

We use the restatement of Theorem 5.2 at the start of Section 8 of Wang and Zahl [31], together with their Definition 4.2, with theorem numbering fixed to arXiv:2210.09581v2 (2025). In particular, (5.17) is only an upper covering condition, and the second alternative concerns a restricted triple set. We use the first alternative only for $\mathcal{A}$.

Proof of Proposition 5.4. Before any label selection, insert finitely many intermediate dyadic lengths so that adjacent lengths $H' \leq H$ satisfy $H' \geq H^{33/32}/2$. They will permit comparison of the affine slopes after the fits have been constructed.

We construct those fits one row component and one hierarchy length at a time. The row F is common to all trajectories. A temporary group of trajectories can therefore locate fitting times, after which we may retain all current labels at those times. The next preparation ensures that this recovery retains dense incidence mass.

Preparing the current labels. We perform the following preparation before fitting each row component at each hierarchy length H. Denote the current labels by E_l^{cur}, and write

$$m(t) = \nu\{l : t \in E_l^{\mathrm{cur}}\}, \qquad \delta = \int_0^1 m(t)\, dt \geq K^{-1}.$$

Choose a reciprocal-subpower threshold $\tau = o(\delta)$. First remove times with $m(t) < \tau$. Next remove each dyadic H-block whose remaining incidence mass is less than τH. The two removals discard mass at most τ each, so the resulting labels E_l^{pre} have mass at least $\delta/2$. Every surviving block has mass at least $\tau H \geq H/K$, and at every surviving active time the instantaneous mass is still $m(t) \geq \tau$: both removals kept or discarded all current labels at a time. Since ν is a probability, each block has mass at most H. Thus at least $\delta/(2H)$ blocks survive. We now run the local construction separately in each of these blocks.

Local normalization. Fix a surviving block $I = [t_I, t_I + H]$, put $u = (t - t_I)/H$, and define

$$U_l = \{u \in [0, 1] : t_I + Hu \in E_l^{\mathrm{pre}} \cap I\}.$$

Averaging and Cauchy–Schwarz provide u_* such that

$$\int \mathbf{1}_{\{u_* \in U_l\}} |U_l|\, d\nu(l) \geq K^{-1}.$$

During the local construction retain trajectories labeled at $t_* = t_I + Hu_*$, and put

$$F_* = F(t_*), \qquad B_l = x_l(t_I) - F_* y_l(t_I).$$

The difference $B_l - S_l(t_*)$ is $O(KH)$. Thus the occupied H-bins C of B_l number at most KH^{-d}, and each has prior mass at most KH^d. For the latter statement, sum (5.5) at time t_* over the KH^{-j} tangential H-bins compatible with $|B_l - S_l(t_*)| \leq KH$. In each C, divide the restricted prior by H^d and call the result ν_C. Deleting groups with $\int |U_l|\, d\nu_C < K_1^{-1}$ loses at most K/K_1 of the original normalized incidence mass. Choose a sufficiently large subpower K_1, so that a dense family of good groups remains.

For a lower endpoint b_C, define

$$Y_l = y_l(t_I), \qquad z_l = (B_l - b_C)/H, \qquad q_l = x_l'(t_I) - F_* y_l',$$
$$\phi(u) = (F(t_I + Hu) - F_*)/H, \quad L_l(u) = z_l + uq_l - \phi(u)Y_l.$$

The normalized row ϕ measures the variation that we must fit on this block. The scalar model L_l separates that row from the frozen position Y_l and residual slope q_l. These quantities are bounded by K on the labels, and ϕ obeys a Lipschitz comparison with slack ζ/H. Writing $x_{2,l}$ for the quadratic coefficient, direct expansion gives the exact identity

$$\frac{S_l(t_I + Hu) - b_C}{H} = L_l(u) + \mathsf{H}\big(u^2 x_{2,l} - u\phi(u)y_l'\big). \tag{5.18}$$

The final term is $O(KH)$. Consequently at $p = H^{1/4}$, and also at $p_0 = H^{1/8}$, the $L_l(u)$-supports retain the upper d-count. Moreover,

$$\nu_C\{l : u \in U_l,\ (Y_l, L_l(u)) \text{ in given } p\text{-bins}\} \le Kp^{j+d}. \tag{5.19}$$

Indeed, in the original variables the corresponding region has tangential widths Kp and normal width KHp. It is covered by KH^{-j} cubes of side Hp, and division by H^d converts their total cap to Kp^{j+d}. The same proof gives (5.19) with p_0. Finally, the residual comparison at t_* gives

$$|q_l - q_{\tilde{l}}| \le K(|Y_l - Y_{\tilde{l}}| + H). \tag{5.20}$$

Here $y_l(t_*) = Y_l + O(KH)$, the relevant scalar values differ by $O(KH)$, and replacing $x_l'(t_*)$ by $x_l'(t_I)$ introduces an error $O(KH)$. Thus the quadratic part of x_l contributes only the recorded $O(KH)$ errors. At the coarser scale $p = H^{1/4}$, freezing the other coordinates of Y_l makes q_l an approximately Lipschitz function of the remaining coordinate. We now use the planar theorem to fit the corresponding component of ϕ.

Fitting one row component. Slice the other $j-1$ coordinates of Y_l into p-bins; no slice is needed when $j = 1$. A group consisting of C and one such slice has prior mass at most $KH^d p^{j-1}$, by the spatial cap at t_*. After division by $H^d p^{j-1}$ and deletion of light groups, choose a sliced measure ν_s with

$$\nu_s(\text{all trajectories}) \le K, \qquad \int |U_l|\, d\nu_s(l) \ge K^{-1}.$$

Let v_l be the remaining coordinate of Y_l. Choose representatives of q_l on occupied v-bins and interpolate between their centers. The relation (5.20) gives a function G satisfying

$$|G(v) - G(v')| \le K(|v - v'| + p).$$

Translate $L_l(u)$ by the other components of $\phi(u)$ times the slice centers. The resulting labeled values satisfy

$$L_l^a(u) = z_l + uG(v_l) - \phi_1(u)v_l + O(Kp). \tag{5.21}$$

They have the local upper d-count, and

$$\nu_s\{l : u \in U_l,\ (v_l, L_l^a(u)) \text{ in given } p\text{-bins}\} \le Kp^{1+d}. \tag{5.22}$$

Scalar collisions and dot products. The local scalar support forces many pairs with matching outputs. Choose u_0 with $\int \mathbf{1}_{\{u_0 \in U_l\}} |U_l|\, d\nu_s \ge K^{-1}$. For pairs of trajectories labeled at both u_0 and u, count collisions of their $L^a(u)$-bins using the unnormalized measure $d\nu_s(l)\, d\nu_s(l')\, du$. Cauchy–Schwarz and

the Kp^{-d} output-bin count give collision mass at least p^d/K. Fixing the first trajectory's starting scalar bin at u_0 leaves at least p^{2d}/K of this mass. On the retained pairs, (5.21) implies

$$\begin{aligned}(u-u_0)(G(v_l)-G(v_{l'}))-(\phi_1(u)-\phi_1(u_0))(v_l-v_{l'})\\ =L^a_{l'}(u_0)-L^a_l(u_0)+O(Kp).\end{aligned}\tag{5.23}$$

At fixed u and fixed p-bins of $v_l, v_{l'}$, the first trajectory has weight at most Kp^{1+d} by its fixed starting scalar bin. The equation confines the second starting scalar value to K bins, so its weight has the same bound. Each $(u, v_l, v_{l'})$-cell therefore carries at most Kp^{3+2d}. The total pair weight at a fixed u is at most Kp^{2d}: sum (5.22) over the tangential bins for the first starting value and the second matching value.

For each time p-bin J, let $T_J \subset J$ be its active pair times for the retained collisions, before any quadrant restriction. Discard bins with $|T_J| < p/K_1$, and keep these pre-quadrant sets T_J attached to all surviving bins. There are $O(p^{-1})$ time bins, so the discarded pair mass is at most Kp^{2d}/K_1. Take K_1 large enough to preserve the lower bound. The represented triples

$$a=(u-u_0,-\phi_1(u)+\phi_1(u_0)),\quad b=(G(v_l),v_l),\quad b'=(G(v_{l'}),v_{l'})$$

then have p-covering number at least p^{-3}/K.

Retain a quadrant for a containing a fixed fraction of the pair mass. Let $\mathcal{H}$ be the image of these retained collision triples, let $\mathcal{A}$ be its first-coordinate projection, and let $\mathcal{B}$ be the union of its two trajectory-coordinate projections. Thus $\mathcal{H} \subset \mathcal{A}\times\mathcal{B}\times\mathcal{B}$, with the same lower covering count up to a fixed factor. Reflect both planar sets by the same coordinate reflection, dilate each by a positive scalar between K^{-1} and 1, and translate only the second set to place the sets in $[0,1]^2$. Translation of the second set cancels in $b-b'$; no translation of the first set is made. Their graph comparisons give the upper covering condition (5.17) with a subpower constant: covering a horizontal interval for the first graph, or a vertical interval for the second, by s-intervals gives at most Kw/s graph cells. Normalization changes this constant and the triple lower count only by subpowers.

Fix $0<\varepsilon<1-d$. For sufficiently large N, the $p^{\pm\eta}$ allowances in Theorem 5.5 absorb these subpower losses. Its expansion alternative is impossible: by (5.23), all dot products are within Kp of a translate of the starting scalar support, so they use at most $K(|J|/s)^d$ s-bins in J. This would imply

$$(|J|/s)^{1-\varepsilon-d}\le K,\qquad |J|/s\ge p^{-\eta},$$

a contradiction. The additive error and inverse dilations change the covering bounds by only subpower factors at $s\ge p$; intervals larger than the unit cutoff can be split into subpower many intervals in the bounded range.

We therefore obtain at least $p^{\varepsilon-1}/K$ time bins near one line in the graph of ϕ_1. The line is a graph over time with slope at most K: two represented points have horizontal separation at least p^ε/K, larger than their Kp proximity errors, and the graph comparison bounds their slope. Let $\mathcal{J}_I$ be the resulting family of represented time bins. In each $J\in\mathcal{J}_I$, the graph comparison extends the affine fit from a represented near-line point to every time in the stored set T_J, with error Kp. Indeed, their time separation is at most p, and both the graph comparison and the fitted line have slope bounded by K. Use precisely the time set

$$T_I=t_I+H\bigcup_{J\in\mathcal{J}_I}T_J.$$

Every T_J has length at least p/K_1; the disjoint bins and their number therefore give

$$|T_I|\ge Hp^\varepsilon/K.$$

In particular, the fitting set consists of prepared active times, not the whole selected bins. Let ε tend to zero sufficiently slowly, after the corresponding η, δ_0 and large-N thresholds have been fixed. Then $|T_I| \geq H/K$, while the normalized fit error remains Kp.

Recovering the time labels. In the original variables this is an affine component fit to F with error $KH^{5/4}$ on T_I. At these times retain all prepared labels E_l^{pre}, including those outside the temporary sliced group. Their total incidence mass satisfies

$$\sum_{I \text{ surviving}} \int_{T_I} m(t)\, dt \geq \tau \sum_I |T_I| \geq \tau \frac{\delta}{2H} \frac{H}{K} = \frac{\delta \tau}{2K}. \tag{5.24}$$

This is inverse-subpower. It uses both the instantaneous floor and the number of surviving blocks, each established before the local search. For $j = 2$, repeat with the second component on these new labels, recomputing m, δ, τ first. Do the same at every length in the fixed finite hierarchy.

Subsequent label restrictions preserve the earlier fits and all upper inputs. We obtain affine rows $P_{\mathfrak{l}}(t)$, with slopes A_I of size at most K, satisfying

$$|F(t) - P_{\mathfrak{l}}(t)| \leq KH^{5/4} \tag{5.25}$$

on every retained active time in the corresponding interval. Let $I_*(t)$ be the finest dyadic block containing t, using a fixed half-open endpoint convention. Define

$$A(t) = A_{I_*(t)}$$

on the finest blocks with retained labels, and set $A(t) = 0$ elsewhere. This is a fixed time-only row bounded by K. Subsequent pruning restricts labels without changing this row.

Comparison across lengths. We now compare the affine fits through the intermediate lengths fixed before any selections. Let μ_{fit} be the current labeled measure. At every hierarchy length H, delete the labels in blocks I with $\mu_{\mathrm{fit}}(I) < H/K_1$, testing all blocks against this same measure. Each length removes mass at most $1/K_1$, so choose K_1 large enough to make the total loss negligible. Every child block that still contains a retained incidence had μ_{fit}-mass at least H'/K_1. Since ν is a probability, its fitting times have length at least H'/K_1. Both its own affine fit and its parent's hold on those times, whether or not all of them survive the deletion. Choose two separated by H'/K. Comparing the fits there gives

$$|A_{I'} - A_I| \leq K \frac{H^{5/4}}{H'} \leq KH^{7/32}.$$

Summing through the fixed finite chain compares the finest slope $A(u)$ with A_I by $KH^{7/32}$. Together with (5.25), this gives

$$|F(v) - F(u) - (v-u)A(u)| \leq K\big(H^{5/4} + H^{1+7/32}\big) \leq KH^{11/10}.$$

This proves (5.15), which persists under the label restrictions below.

Concentrating the residual slopes. The affine fits now hold on a dense set of incidences. We next restrict the trajectory labels so that their residual slopes admit short lists once the time and scalar value are known. At the finest length $H = H_*$, repeat the unsliced C-group normalization on the retained labels. With $p_0 = H^{1/8}$, the affine approximation to ϕ gives an affine approximation to L_l with slope

$$k_l = q_l - A_I Y_l$$

and error at most Kp_0. The slopes vary by at most Kp_0 in each Y-cell of side p_0, by (5.20). Put $\gamma = (1-d)/4$. We claim that some interval of width p_0^γ contains slopes of labeled incidence mass at least K^{-1} in every good group, with uniform subpower constants.

If this failed, there would be a sequence of groups and a fixed $c > 0$ on which every such interval has incidence mass at most N^{-c}. Let $\tau = N^{-c/4}$. Delete trajectories with labeled length below τ, and then Y-cells whose remaining trajectory mass is below $p_0^j\tau$. The two losses are at most $K\tau$. The remaining output collisions, integrated in time, still have mass at least p_0^d/K, by Cauchy–Schwarz.

For a fixed first trajectory, second trajectories with slope within p_0^γ occupy at most

$$KN^{-c}\tau^{-2}p_0^{-j}$$

retained Y-cells. Indeed, their slopes, including the Kp_0 variation within each cell, lie in a bounded number of intervals of the assumed small incidence mass. Their labeled lengths are at least τ, so their prior mass is at most KN^{-c}/τ; each retained cell has prior mass at least $p_0^j\tau$. For each such cell the matching output weight at a time is at most Kp_0^{j+d}, by (5.19). Thus the close-slope collision contribution is at most

$$Kp_0^dN^{-c}\tau^{-2} = Kp_0^dN^{-c/2}.$$

For the other pairs, the two affine approximations can agree to Kp_0 only on times of length at most $Kp_0^{1-\gamma}$. Their total contribution is at most that quantity times a subpower bound for the product prior mass. Both contributions are $o(p_0^d/K)$, because $1-\gamma > d$. This contradicts the collision lower bound and proves the claim. If the claim's subpower constants were not uniform across groups, an offending sequence of groups would give exactly the contradiction just proved.

Keep one such slope interval in each good group. Since

$$J_1(l,t) - k_l = \mathsf{H}\big(2ux_{2,l} - \phi(u)y_l' - uA_Iy_l'\big) = O(KH), \tag{5.26}$$

this gives a deterministic interval cover for J_1. For fixed $t, S_l(t)_H$, only K base bins C are possible: $|B_l - S_l(t)| \le KH$. Hence J_1 lies in K intervals of width

$$Kp_0^\gamma = KH^{(1-d)/32}.$$

At any reading scale $r \ge H^{(1-d)/64}$, these are K r-bins. The cover survives appending all tuple times to T and imposing any later event. The formula for J_1, whose rows are now fixed and bounded, also gives K r-bins when T, l_r is fixed. Apply (5.14) and (5.11) anew on each dense tested event, with fine scalar S_H and coarse scalar S_r. This proves (5.16). The argument rederives the entropy estimate on the specified event, using covers that persist under restriction. Together with the Taylor estimate already proved, this completes the proposition. □

5.4 Higher jets and polynomial fits

Fix $0 \le d < 1$, an integer $M \ge 2$, and a dyadic block length $h = N^{-c+o(1)}$, where $c > 0$ is fixed. We seek a degree-M polynomial fitting the row on a labeled portion of a trajectory of length at least h/K. The first-jet information from Proposition 5.4 will be differentiated along a finite tree of nearby times. Its row and single-time labels are fixed before this tree is sampled. The additional time used to test the final polynomial is sampled separately, so the polynomial coefficients can be chosen without that time. At each pair scale H, the child jets must already be readable at width $H^{1.01}$, so that division by the pair gap still leaves a positive power of accuracy. The finest scale supplies the first jets; the root accuracy must be finer than the error used in the final polynomial test. We fix all these scales before selecting labels.

Define

$$c_0 = .1, \qquad r_i = \min(.009, c_{i-1}/2), \qquad c_i = r_i/2 \quad (1 \le i \le M-1),$$

and put

$$r_{\min} = \min_{1 \le i \le M-1} r_i = .009\, 4^{-(M-2)}, \qquad \alpha_0 = (1-d)/64.$$

Both numbers are positive and depend only on M, d. Starting with h, choose nested dyadic lengths

$$h \gg H_1 \gg H_2 \gg \cdots \gg H_{M-1} \gg H_*,$$

by taking at each step the largest dyadic length satisfying the corresponding inequality

$$\begin{aligned} H_1^{r_{\min}} &\le h^{M+11}, \\ H_i^{r_{\min}} &\le H_{i-1}^{1.02} \qquad (2 \le i \le M-1), \\ H_*^{\alpha_0} &\le H_{M-1}^{1.02}. \end{aligned} \tag{5.27}$$

Choose the available floor to satisfy

$$\zeta \le K^{-1} \min\{H_*^3, h^{M+1/2}\}. \tag{5.28}$$

These choices use finite fixed power ratios. We will verify the required entropy readings in the proof. The strict exponent margins absorb dyadic rounding and subpower factors. When specifying the floor by a fixed power, one may place it a further fixed power of h below the displayed minimum to allow every later subpower enlargement of K.

Proposition 5.6 (Higher-jet polynomial fitting)**.** *For the parameters M, d, h of Section 5.4 and the hierarchy* (5.27)*, assume Definition 5.2, including the parameter-bin cap, and the floor condition* (5.28)*. Then a dense set of labeled incidences (l, t) has the following property. There is a row-valued polynomial P of degree at most M, with coefficients bounded by K, such that in the h-block containing t,*

$$|F(v) - P(v)| \le K h^{M+1/4}$$

on labeled times $v \in E_l$ of length at least h/K, including $v = t$. The polynomial may depend on the incidence. The conclusion applies anew to any dense restriction of the input labels with the same upper hypotheses.

Proof. The proof constructs higher jets by divided differences on the time tree. The entropy lower bound for tangential projections forces the successive rows to be compatible. The same bound at a separate time then forces the polynomial fit, and a final fiber argument recovers the required labeled duration.

With the hierarchy and floor fixed in (5.27) and (5.28), Proposition 5.4 applies after its finite intermediate Taylor scales have been inserted, before any label selections. We retain the resulting single-time labels and the row $A(t)$.

An independent time tree. Choose u uniformly in $[0, 1]$. Add a second time uniformly in its H_1-interval. At the next level, split each of these times into itself and a new independent uniform draw in its H_2-interval. Continue for $M-1$ pair levels. Separately draw t' uniformly in the h-block of u, independently of all the other draws conditional on that block. This entire experiment is independent of the trajectory before any labels are imposed. Let T contain the component, the whole tree, and t'.

All tested times are labeled with dense probability. Here is a direct verification. For a fixed trajectory and a fixed containing interval, write a for the fraction of labeled times. At a split,

conditional on the smaller partition interval, the two child times are independent uniform points of that interval. The mean of the product of their subtree success probabilities is the square of the mean subtree probability. Jensen's inequality, repeated up the finite tree, gives a lower bound by $a^{2^{M-1}}$ in each h-block. The additional independent t' multiplies this by a. Averaging over blocks, trajectories and components, and using Jensen once more, bounds total success below by the $(2^{M-1}+1)$-st power of the single-time label mass. This is inverse-subpower.

We may also require that every pair gap at scale H_i be at least H_i/K. Before label restriction, each excluded gap has probability at most $2/K$. Since the number of pairs is fixed, take this K large enough compared with the reciprocal all-label probability. The loss is then negligible relative to that probability. These restrictions are performed on the presampled tuple law. Lemma 5.3 is applied anew on each further dense event of this presampled experiment.

Inductive information statement. Put $F_0 = F$ and $F_1 = A$. Starting at the leaves, a subtree rooted at time t will carry rows $F_1(t), \dots, F_m(t)$ and jets

$$J_i(t) = x_l^{(i)}(t) - F_i(t)y_l(t) - iF_{i-1}(t)y_l', \qquad 1 \le i \le m. \tag{5.29}$$

Every row is a function of its subtree times and the component only. In particular it is independent of the trajectory and does not use the separate time t'. The rows are bounded by K on the retained tuples. At the readings required below, we maintain

$$\mathsf{H}(J_i(t)_r \mid T, S_l(t)_r) = o_*\tag{5.30}$$

on each further dense event in the original trajectory–tuple experiment. At the leaves this is (5.16) for the fixed row A. Its proof remains valid with the whole tree and t' appended to T: the deterministic fine-scalar and coefficient-bin covers survive, and Lemma 5.3 is proved anew on the chosen event.

Consider a pair t, w at scale H, whose two subtrees have jets through order m. Set $q = H^{1.01}$, with dyadic rounding. Their internal scales will have been chosen fine enough for the child assertions at q. The Taylor expansions of x_l, y_l, together with (5.15), give

$$S_l(w) = S_l(t) + (w-t)J_1(t) + O(KH^{1.1}).\tag{5.31}$$

The error is smaller than q. Thus $S_l(w)_q$, and then all the q-jets at both endpoints, have negligible entropy given $T, S_l(t)_q$. This uses the child information statements on the current event and chain rule. Write

$$\nabla Z = \frac{Z(w) - Z(t)}{w - t},$$

where endpoint rows use their own subtree data. Since $|w-t| \ge H/K$, the ∇J_i are determined to accuracy $KH^{.01}$, with entropy cost o_*, by $T, S_l(t)_q$.

For $i \ge 1$, quadraticity of x_l and affinity of y_l give the exact identity

$$\nabla J_i = x_l^{(i+1)}(t) - (\nabla F_i)y_l(t) - \big(F_i(w) + i\nabla F_{i-1}\big)y_l'.\tag{5.32}$$

For $i = 1$, the difference quotient of x_l' equals x_l''; for $i \ge 2$, both sides of the corresponding derivative identity are zero. The remaining terms in (5.32) are simply the exact product difference for an affine y_l.

Proceed through $i = 1, \dots, m$, using the exponents fixed above and maintaining

$$\nabla F_{i-1} = F_i(t) + O(KH^{c_{i-1}}), \qquad c_{i-1} > 0.\tag{5.33}$$

Initially (5.15), divided by the pair gap, supplies the chosen exponent $c_0 = .1$.

First, ∇F_i is bounded by a subpower after negligible loss. If a fixed-power tail $|\nabla F_i| \geq N^{\alpha}$, $\alpha > 0$, had dense mass on a subsequence, divide (5.32) by that norm. The other displayed coefficients are bounded by K, using (5.33), and $x_l^{(i+1)}, y_l'$ have size K. The equation would therefore encode the projection of $y_l(t)$ in the T-known direction $(\nabla F_i)/|\nabla F_i|$, with negligible entropy, at some fixed positive power accuracy coarser than q. For example, if $H = N^{-b+o(1)}$, take a reading $N^{-\beta}$ with $0 < \beta < \min(\alpha, b)$. All nuisance terms after division are below this reading, as is the divided q-bin error. This contradicts (5.12) on that very tail event. It follows that every fixed-power tail is negligible relative to the current dense tuple mass, and a slow choice of cutoffs yields $|\nabla F_i| \leq K$.

We now also have $F_i(w) - F_i(t) = O(KH)$. Recall that

$$r_i = \min(.009, c_{i-1}/2) > 0.$$

The identity (5.32) and the compatibility estimate (5.33) encode

$$U_{i+1} = x_l^{(i+1)}(t) - (\nabla F_i) y_l(t) - (i+1) F_i(t) y_l' \tag{5.34}$$

to width H^{r_i}, at entropy cost o_*, given $T, S_l(t)_q$. Indeed, the error $KH^{.01}$, the $O(KH)$ row difference, and the $O(KH^{c_{i-1}})$ compatibility error are all smaller than this width.

If $i < m$, subtract the already known jet $J_{i+1}(t)$. This encodes $(\nabla F_i - F_{i+1}(t)) y_l(t)$ to the same width. On an event where the row norm is at least $H^{r_i/2}$, division would encode its unit projection at width $H^{r_i/3}$, again contradicting (5.12) on any dense such event. After a negligible deletion,

$$\nabla F_i - F_{i+1}(t) = O(KH^{r_i/2}).$$

This proves the compatibility assertion with the chosen $c_i = r_i/2$, allowing us to proceed to the next value of i.

For $i = m$, define $F_{m+1}(t) = \nabla F_m$ and keep the older rows at t. This row is computed only from the two child rows and their times. The new jet equals U_{m+1}. Its bin at $r = H^{r_m}$, and at coarser required powers, has negligible entropy given $T, S_l(t)_q$. Given T, l_r, the bounded rows and coefficient-bin diameter allow only K bins for that jet. Applying (5.14) with

$$Z = J_{m+1}(t)_r, \quad A = S_l(t)_q, \quad B = l_r, \quad C = S_l(t)_r$$

and invoking (5.11) on the current event gives (5.30) for the new jet. The older jets retain their hereditary assertions.

There are finitely many rows and discrepancy tests for fixed M. For each fixed positive threshold exponent, a nonnegligible relative bad mass would itself yield a dense event on a subsequence, contradicting the projection estimate just used. Choose threshold exponents decreasing to zero so slowly that the union of the finitely many bad sets has relative mass tending to zero. The resulting row bounds are deterministic on the retained tuples. Every later dense event inherits those deterministic bounds; its entropy assertions are then derived anew from the child assertions, (5.14), and Lemma 5.3. These eventwise applications establish the hereditary entropy assertion. These cutoff tests use only the rows already defined from the time tree.

Verification of the reading scales. The induction is complete once each child jet is available at the accuracy used by its parent and the root is accurate enough for the final test. A jet formed at a pair scale H_i is available at width $H_i^{r_m}$ for its relevant order m, and at all coarser required readings. Since $r_m \geq r_{\min}$, the root widths satisfy

$$KH_1^{r_m} \leq KH_1^{r_{\min}} \leq Kh^{M+11} \ll p := h^{M+10}.$$

At a child scale H_i, $i \geq 2$, the corresponding bound is

$$KH_i^{r_m} \leq KH_{i-1}^{1.02} \ll H_{i-1}^{1.01},$$

which supplies the reading q requested by its parent. Older jets inherited from finer subtrees obey the same requirements. For the leaf jets, (5.16) applies at every reading at least $H_*^{\alpha_0}$; by (5.27), this is finer than $H_{M-1}^{1.01}$, with the same fixed margin. Finally, $\zeta \leq H_*^3/K$ covers the first-jet construction, its fine scalar scale H_*, and all the intermediate readings just used. The projection readings in the tail and compatibility tests are coarser than their scalar conditioning scale q. Thus the entire induction uses available scales.

Testing the polynomial at a separate time. At the root u, set

$$P_F(v) = \sum_{i=0}^{M} \frac{F_i(u)}{i!}(v-u)^i, \qquad E = F(t') - P_F(t'), \qquad J_0(u) = S_l(u).$$

The coefficients of P_F are bounded by K, including when expanded in ordinary powers of v, because M is fixed and $u \in [0,1]$. The coefficients are functions of the component and the tree excluding t'. Exact evolution of the quadratic x_l and affine y_l gives

$$S_l(t') = \sum_{i=0}^{M} \frac{J_i(u)}{i!}(t'-u)^i - Ey_l(t') + O(Kh^{M+1}). \tag{5.35}$$

The last term is the single extra degree produced when the degree M polynomial P_F multiplies the affine y_l.

Suppose $|E| \geq h^{M+1/4}$ on a dense event. Work on that event and condition on $T, S_l(u)_p$. The root jet bins at p have total additional entropy o_*; after specifying them, (5.35) places $S_l(t')$ in an interval of length $K|E|$. Put $r = h^{1/4}$. The local support count (5.4) allows at most Kr^{-d} values at width $|E|r$ in that interval. Both the width and interval length are T-known. The variable width satisfies

$$|E|r \geq h^{M+1/4}h^{1/4} = h^{M+1/2} \geq K\zeta,$$

by (5.28); its containing interval is larger still. Thus both are within the available scale range; dyadic enlargement or the unit cutoff loses only a factor K.

After division by $|E|$, the Taylor remainder is $O(Kh^{3/4})$; the root-jet bin error is smaller, and the scalar-bin error is $O(r)$. Also $y_l(t') = y_l(u) + O(Kh)$. Write J for the tuple of root-jet bins at width p. Given $T, S_l(u)_p, J$, the expansion therefore confines the width-r bin of $(E/|E|)y_l(u)$ to at most Kr^{-d} possibilities. Since $\mathsf{H}(J \mid T, S_l(u)_p) = o_*$, chain rule gives

$$\mathsf{H}\big(((E/|E|)y_l(u))_r \mid T, S_l(u)_p\big) \leq d\log(1/r) + o_*.$$

The bound in (5.12), applied on this same event, gives the lower bound $\log(1/r) - o_*$. Since $d < 1$ and h is a fixed positive power, this is impossible. We retain dense tuple mass with $|F(t') - P_F(t')| \leq Kh^{M+1/4}$.

Recovering labeled time fibers. Let z consist of the component, the trajectory, and the rest of the time tree, excluding t'. The polynomial P_F is fixed at z, and before restrictions the conditional law of t' is uniform in its h-block. If the good tuple event has mass $p_g \geq K^{-1}$, write $g(z)$ for the fraction of successful t''s in that block. Then $\int g(z)\,dP(z) = p_g$. For any positive δ,

$$\int_{\{g < p_g\delta\}} g(z)\,dP(z) \leq p_g\delta.$$

Choose $\delta \to 0$ sufficiently slowly and retain only the other fibers. They still carry dense tuple mass, and every retained fiber has successful labeled length at least $hp_g\delta \geq h/K$.

The pre-restriction marginal law of (l, t') is the original prior times uniform time: uniform u selects h-blocks with their length weights, and t' is uniform inside the chosen block. Projecting the good event therefore gives a dense set of incidences (l, t'). Each has a witnessing polynomial fitting on a successful fiber of length h/K, including that incidence. If desired, round the bounded polynomial coefficients on a sufficiently fine finite grid, adding error much smaller than $h^{M+1/4}$, and test its successful length on the original labels. This makes the existential qualification test measurable and preserves the asserted bounds.

Every construction began with upper inputs and dense mass, so it may be performed anew on a dense residual. This proves the proposition. □

6 The isotropic improvement

In the one-spatial-coordinate model and the isotropic two-spatial-coordinate model, incidence information along trajectories forces a quadratic test with smaller thickness on the same time interval. The argument first constructs a polynomial field and an approximate potential for it. The final task is to recover the degree allowed for a chart. In two spatial coordinates, broad directions can still lie on a conic; the higher-order terms they fail to detect are expressed through the hidden coordinate to obtain a quadratic test.

Theorem 6.1 (Isotropic improvement)**.** *Consider the tangent data of Proposition 4.7, with $0 < k \leq 1$ and $0 \leq d < 1$, either in version 1, where $j = 1$, or in version 2 with $j = 2$ and optimized narrowness $\ell = 0$. In every substantial residual of the reference path one can, after passing to a subsequence, find a packet $Q = Q_s$, of thickness $a = N^{-s}$ and time length $h \sim_{\log,N} a^k$, and a time-improvement candidate of thickness*

$$a_2 = ah^{e_2}, \qquad e_2 > 0.$$

The candidate occupies residual incidences of unnormalized parent-time mass at least

$$\frac{h\nu_Q}{K}\left(\frac{a_2}{a}\right)^d, \qquad K = N^{o(1)}.$$

The block-normalized incidence mass in Definition 4.10 is this mass divided by h. Writing the candidate as P_{new}, its bounds $|P_{\mathrm{new}}| \leq Ka_2$ and $|\partial_w P_{\mathrm{new}}| \leq K$ hold on its assigned trajectories throughout the same h-length block. The hidden derivative is bounded below by K^{-1} on the counted incidences, and its hidden curvature is at most K/a_2. In version 1 the test has the special form $w - F(t, y)$, with F quadratic. In version 2 it is a quadratic in (t, y, w).

Here *substantial* refers to positive probability in the original reference path, whereas a dense discovery family may have only inverse-subpower probability. The distinction is needed when Proposition 4.11 converts candidates into a known atlas with actual N_0-subpower slacks in the original resolution.

The deduction of the theorem uses four stages. The support estimates and Proposition 6.2 produce a velocity graph. Proposition 6.3 gives polynomial fits on its reference trajectories, which Cartesian interpolation combines into a field satisfying (6.22). Proposition 6.5 makes that field approximately conservative; its potential then gives the heavy intercept groups in (6.28) and (6.29). Finally, Proposition 6.8 turns those fits into quadratic candidates. The full assembly appears after Proposition 6.8. For version 1, this also gives zero horizon for every exact problem in the stated class, as recorded in Corollary 6.9.

6.1 Incidence measures and packet estimates

Write

$$z = (t, y), \qquad m = 1 + j, \qquad \mathbf{v} = (1, V), \qquad W = (z, X).$$

A parent component and its label are included in every state below. The parent prior is ν, and restrictions of the incidence measure are denoted by λ. They are generally left unnormalized:

$$d\lambda(l, t) = G_{\mathrm{wt}}(l, t)\, d\nu(l)\, dt, \qquad 0 \le G_{\mathrm{wt}} \le K. \tag{6.1}$$

All initial losses are summed in the path mixture of parents. If a line portion has weighted duration at least H/K_1, its preceding incidence mass bounds its prior weight times H/K_1; its ordinary duration is at least $H/(KK_1)$. This duration estimate is used before further restrictions, with the preceding incidence mass as its denominator.

All exponent ranges are fixed first, with room for finer comparisons. A finite list of pure and joint caps is prepared before sparse selection. Lists of exponents subsequently increase slowly enough that products of their subpower losses remain subpower. Allowed deletion losses are fixed for the whole finite list before it is enlarged. We use successively slower grids for physical counts, parameter estimates, field comparisons, strip filling, and residual comparisons. Interpolation between neighboring dyadic widths whose ratio is at most K is performed only after the endpoint estimates. Its losses are not repeatedly multiplied within the grid they complete.

We use the following consequences of Proposition 4.7 and Lemma 4.8. For every required fixed power width p, the scalar cap in X is Kp^d per unit z-volume. The velocity law π has a positive angular exponent. Joint upper bounds have the same pure cap multiplied by $K\pi(J)$, for a velocity bin J. They remain valid for later submeasures relative to the earlier state weights. Conditional domination by the velocity law on a later dense family follows after deleting states with insufficient denominators. The joint and conditional estimates here concern one time at a time.

The true packets Q_u, at thickness $p = N^{-u}$ and time length $H \sim_{\log,N} p^k$, have

$$\nu_Q \sim_{\log,N} p^d H^j. \tag{6.2}$$

Their assigned trajectory sets are disjoint in each time block, and assignments are invariant throughout that block. All spatial axes have length H up to K, and center velocities are bounded by K. In normalized packet coordinates the base incidence density is at most $KH\nu_Q$. The stable graph lists in the tangent preparation have at most K ball/sign entries per packet. Their branches are holomorphic algebraic functions of bounded relation degree on complex enlargements of their real comparison balls. We use Lemma 3.1 only on interiors with these margins.

6.2 Physical support and parameter masses

The physical support will have exponent $m + d$, strictly below its ambient dimension $m + 1$. This deficit will force the lifted velocities $(\mathbf{v}, X')$ near an m-dimensional graph. We also need bounds for the masses of trajectory coefficient bins. Their lower bounds provide the denominators needed to retain controlled spatial caps after conditioning and rescaling.

We first prove, after negligible deletion on the required grids,

$$\#W_p \le Kp^{-(m+d)}, \qquad \lambda(W_p) \le Kp^{m+d}, \qquad \lambda(z_p) \le Kp^m. \tag{6.3}$$

Here $\#W_p$ counts occupied cubes in a parent. The two mass ceilings follow from the pure caps. The support argument requires a gradient bound for the packet graphs.

Gradients of the packet graphs. At scale p, each packet has at most K local predictions

$$X = f(z) + O(Kp).$$

In version 2 these are obtained by composing the native quadratic test with a simple root of the packet equation, or with its affine quadratic center in the large-curvature case. The root error is $O(Kp)$ in the hidden coordinate; the native derivative and curvature bounds make the error in X also $O(Kp)$. Comparison balls have inverse-subpower radii in coordinates scaled by H, and can be assigned from the exact base by a lattice, with at most two root signs. Removing entries of incidence mass below $H\nu_Q/K_1$ removes mass at most K/K_1, since $\sum_Q H\nu_Q \le 1$. The remaining real projections have relative volume at least K^{-1} by the packet base ceiling. Thus Lemma 3.1 bounds all fixed derivatives of these predictions by K in the scaled coordinates.

Prune line/packet/ball/sign portions of weighted duration below H/K_1, with total mass loss at most K/K_1. On a surviving line, $f(z_l(t)) - X_l(t)$ is holomorphic algebraic of bounded relation degree, and bounded by Kp on a time set of length at least H/K. Its chord lies in the complex comparison enlargement with a fixed relative margin. The one-variable form of Lemma 3.1 gives

$$|D_{\mathbf{v}} f - X'| \le Kp/H. \tag{6.4}$$

As $k \le 1$ and $|X'| \le K$, the directional derivative is bounded by K.

If $|\nabla_z f|$ were power-large on a substantial subfamily, (6.4) and $|V| \le K$ would make its spatial part power-large, and

$$\partial_t f + \nabla_y f \cdot V = O(K)$$

would predict V in a power-thin affine strip (an interval for $j = 1$). The state consisting of packet, ball, sign and a fine base bin has actual list knowledge at p^L and only K ambiguity from a deep end history. The Hessian is bounded by K/H^2, so the base bin can freeze the gradient to error one. This contradicts Corollary 4.13 in version 2, or the angular conclusion of Proposition 4.7 in version 1. Sending fixed-power tolerances slowly to zero allows $|\nabla_z f| \le K$ outside negligible mass. Remove light graph entries again. Their remaining base volume and Lemma 3.1, applied to physical derivative components, extend this bound throughout the used interiors.

Physical support counts. The gradient bound now controls the number of physical cells met by a packet. A packet uses at most $K(1 + H/p)^m$ physical W_p-cubes. Since $H \ge p/K$, summing with (6.2) over packets and H^{-1} blocks gives

$$\frac{K}{H}\frac{1}{p^d H^j}\left(\frac{H}{p}\right)^m \le Kp^{-(m+d)}.$$

This proves (6.3). Remove globally light W_p-cubes of mass below p^{m+d}/K_1; a sufficiently large subpower K_1 makes the loss negligible. The old ceilings then give the hereditary support bounds

$$\#\{W_p \subset W_{w_0}\} \le K(w_0/p)^{m+d} \quad (w_0 \ge p), \qquad \#\{W_p \text{ over a fixed } z_p\} \le Kp^{-d}, \tag{6.5}$$

with bounded enlargements at boundaries.

Masses of coefficient bins. These support counts also bound scalar supports after the affine trajectory coefficients have been fixed. Let B_p be an aligned dyadic cube in all coefficients of

$$X = x_0 + tx_1 + t^2 x_2, \qquad y = y_0 + tV.$$

Additional trajectory tags stay in the prior but are not part of this coefficient bin. There is one fixed, unnormalized, preceding long-line prior ν° for which every participating block satisfies

$$K^{-1}p^{j+d}\pi([V]_p) \le \nu^\circ(B_p) \le Kp^{j+d}\pi([V]_p). \tag{6.6}$$

A physical cell W_p and velocity bin $[V]_p$ admit at most K participating B_p's. We now prove these assertions, keeping prior and incidence denominators separate.

The joint caps imply, for a time interval I, a y-box of volume J_y, an X-interval of width w_0, and a velocity bin J,

$$\lambda(\text{these constraints}) \le K|I|J_y w_0^d \pi(J). \tag{6.7}$$

For the current finite grid of spatial meshes, take one common time partition much finer than every mesh on that grid. Deleting line/bin pairs of weighted duration below $|I|/K_1$ removes mass at most $1/K_1$. Bounded speed makes a constraint at one time hold with a mesh enlargement throughout I. Here a line is *active at* $t \in I$ when its whole line–I mark survives this duration test. Activity does not condition the trajectory prior on incidence success at the particular time t. Dividing (6.7) by the retained duration, as in (4.26), gives the fixed-time active-prior estimate

$$\nu\{l : \text{ active at } t,\ (y_l(t), X_l(t)) \in B,\ [V_l]_\beta = J\} \le K p^{j+d} \pi(J) \tag{6.8}$$

for a spatial cube B of side p; the rectangular version holds too. Remove null exceptional times and lines with tiny total weighted occupation. Restrict ν, without normalization, to lines of occupation at least $1/K$; this defines ν°. Freeze this prior after the common finite-grid preparation. Later labels satisfy $\lambda \le K\nu^\circ\, dt$; restricting them never renormalizes ν°.

For an affine bin $\alpha = [y_0, V]_p$, let $\Pi_\alpha = \pi([V]_p)$. Integrating on the preceding long occupation gives

$$\nu^\circ(\alpha) \le K p^j \Pi_\alpha, \qquad \nu^\circ\{\alpha, (x_0, x_1, x_2) \in B_r\} \le K r^d p^j \Pi_\alpha \quad (r \ge p).$$

At each time these coefficient restrictions put the position in the corresponding moving boxes, so (6.7) applies. This proves the upper half of (6.6).

If the lower half failed by a fixed power on a substantial residual, a parent and an α would have bad incidence mass at least $p^j \Pi_\alpha / K$. This follows by summation, because the sum of $p^j \Pi_\alpha$ over affine bins is at most K; ignore velocity bins of zero probability. Restrict ν° to the bad coefficient blocks in this α, and divide it by $p^j \Pi_\alpha$. The scalar coefficient-ball caps are $K r^d$, the bad labels have dense mass, and their (t, X)-support has at most $K p^{-1-d}$ cells: α fixes y to K options per time cell, and (6.5) counts scalar options. Lemma 5.1 gives a scalar coefficient p-bin of mass at least p^d/K. Splitting among boundedly many aligned bins if needed contradicts the fixed-power deficiency of every bad B_p. Slowly relaxing the deficiencies proves the lower bound. For $p^{-1} \le K$, delete light bins directly by their subpower count.

Coefficient lists at a physical cell. The lower mass bound allows us to count how many coefficient bins can meet one physical cell with the same velocity bin. A block B_p uses at most K/p physical cells W_p. Discard entries below $p\nu^\circ(B_p)/K_1$, with total mass loss K/K_1. For a fixed cell and velocity bin their summed mass is at most $K p^{m+d} \pi([V]_p)$, by the joint cap, and every surviving entry has mass at least $p^{m+d}\pi([V]_p)/K$. There are at most K such blocks. In particular,

$$X' \text{ to width } p \text{ has at most } K \text{ options given } (W_p, [V]_p). \tag{6.9}$$

Between grid widths $p' < p''$ with $p''/p' \le K$, discard child velocity bins whose mass under the velocity law is too small relative to the containing bin. Each parent has at most K children, so domination by the velocity law makes the loss negligible. The finer lower bound implies intermediate lower bounds by inclusion; intermediate upper lists use finer lists and at most K refinements. For upper prior and fixed-time estimates, sum the finer velocity weights. All of (6.6) concerns the same ν°; later restrictions do not assert new incidence saturation.

6.3 A velocity graph and its coarse lists

Fix the scales of the intended improvement before constructing its graph. Choose $a = N^{-s}$, $s > 0$, its actual packet time $h \sim_{\log,N} a^k$, and an integer $M \geq 2$ with $h^M \ll a^2$ by a fixed power. Choose the scalar-normal floor ζ sufficiently fine for Proposition 5.6 at h, M, and then choose

$$\rho \ll \zeta, \qquad \rho \ll h^{M+1/4}.$$

The exponent $\kappa > 0$ in the next proposition depends only on m, d. Take b so fine that b/ρ and b^κ/ρ are negligible at every required comparison scale. The finite interpolation meshes and descendant graph scales are included in this choice, with fixed power margins. These choices depend on the fixed exponents and M, before any test of failure for the graph; none will be changed inside such a failure set.

On any substantial residual, renew breadth at the fine physical states W_b. Outside negligible mass, each conditional velocity law puts as small a fixed probability as desired in any fixed-power thin affine strip, or interval if $j = 1$. Letting that power tend slowly to zero gives reciprocal-subpower separation. Thus m independent draws from a state span with determinant at least K^{-1}, with probability bounded below. For $j = 2$, choose separated points and then avoid their joining lines; for $j = 1$, choose a separated pair. The same assertion holds for unit directions. The probability remains bounded below after sufficiently small fixed fractional losses in each state. Conditional domination by $K\pi$ is prepared at the same time by deleting light states, retaining the unnormalized bounds against the earlier state weights.

We use the determinant multilinear Kakeya estimate in the following form. If $\mathcal{T}_i$, $1 \leq i \leq n$, are weighted families of radius-r neighborhoods of full affine lines in $\mathbb{R}^n$, with unit directions e_T, nonnegative weights w_T, and $M_i = \sum_{T \in \mathcal{T}_i} w_T$, then

$$\int_{\mathbb{R}^n} \left(\sum_{\substack{T_i \in \mathcal{T}_i \\ 1 \leq i \leq n}} |\det(e_{T_1}, \ldots, e_{T_n})| \prod_{i=1}^n w_{T_i} \mathbf{1}_{T_i}(x) \right)^{1/(n-1)} dx \leq C_n r^n \prod_{i=1}^n M_i^{1/(n-1)}. \tag{6.10}$$

This is the equal ambient and wedge dimension case of Carbery and Valdimarsson [7, Theorem 1], after scaling unit-radius tubes by r. The near-endpoint multilinear theorem is due to Bennett et al. [2]; Guth [13] proved the endpoint, and Bourgain and Guth [6, Section 7] obtained a wedge-weighted formulation. Only $2 \leq n \leq 4$ is used. The statement for line measures follows by rounding line parameters arbitrarily finely on a bounded region, summing their weights, enlarging tubes by a bounded factor, and using slightly stronger determinant thresholds before passage to the limit. The estimate depends on their total weights.

Proposition 6.2 (Physical velocity graph)**.** *In the tangent setting of Theorem 6.1, use the physical states and measures of Section 6.1 and the scale choices of Section 6.3. For a sufficiently fine fixed power b, there is a row $G = (G_0, G_y)$, constant on each W_b and bounded by K, such that on a substantial reference family*

$$|X'(t) - G(W_b) \cdot \mathbf{v}| \leq b^\kappa, \tag{6.11}$$

where $\kappa > 0$ depends only on m and $1 - d$. For every required dyadic $s_0 \geq b^\kappa$, the values of G to width s_0 have at most K possibilities in a fixed coarse physical cell W_{s_0}.

Proof. The support deficit first forces lifted directions near a plane. Breadth then makes that plane a graph, and the coefficient-bin lists control the number of possible graph rows in a coarse cell.

Put $n' = m + 1$, choose

$$0 < \theta < (1-d)/3, \qquad 0 < \tau < \theta/(n'-1).$$

If no m-plane contains, say, 0.9 of the unit lifted directions $(\mathbf{v}, X')$ within distance b^τ, successive independent samples have distances at least b^τ from the preceding spans with probability bounded below. Enlarge a smaller span to an m-plane when applying the hypothesis. The resulting n'-fold determinant is at least a constant times $b^{\tau(n'-1)}$, and hence is at least b^θ for large N.

Suppose these bad cells carry substantial mass. Group them into dyadic W-cubes of side $w_0 \sim b^{2/3}$. Delete line portions with weighted duration below w_0/K_1 in such a cube. A bounded quadratic trajectory meets at most K/w_0 cubes, so this removes mass at most K/K_1. Remove bad cells with excessive fractional loss and cubes retaining too little of their preceding bad mass. In a remaining cube of preceding mass $M > 0$, the retained cell masses m_E sum to at least a fixed fraction of M, determinant success still has probability bounded below, and the contributing prior weight is at most KM/w_0.

Approximate each trajectory there by its affine tangent at the cube's central time. The position error is $O(Kw_0^2) \ll b$, and the direction error is $O(Kw_0) \ll b^\theta$. Radius-$O(b)$ tubes about these full tangent lines contain every successful cell and retain its determinant threshold. A trajectory contributes at most Kb weighted duration to a cell. Therefore the successful tuples have prior product weight at least $(m_E/b)^{n'}/K$. Apply (6.10) with $n = n'$ and cancel the cell-volume factor $b^{n'}$:

$$\sum_E \Big(b^\theta (m_E/b)^{n'}/K \Big)^{1/(n'-1)} \le K (M/w_0)^{n'/(n'-1)}.$$

The power-mean inequality gives a lower bound of $K^{-1} b^\theta (w_0/b)^{n'}$ for the cell count. The upper bound from (6.5) is $K(w_0/b)^{m+d}$. The ratio of the asserted lower and upper bounds is at least

$$K^{-1} b^{\theta-(1-d)/3} \longrightarrow \infty,$$

a contradiction.

From a plane to a graph. Thus the lifted directions concentrate near an m-plane on almost all the relevant cells. Even within the captured fraction, breadth supplies m base directions with determinant at least K^{-1}. Inverting this basis and using bounded X' shows that a unit normal to the plane has final component at least K^{-1}. The plane is therefore a graph with row bounded by K. Restricting to its captured incidences proves (6.11), for instance with $\kappa = \tau/2$.

Keep G fixed and renew spanning and domination by the velocity law at W_b on the captured family. Name the resulting incidence submeasure in parent γ by $\lambda_{\mathrm{ref},\gamma}$, and retain the prepared parent-mixture weights p_γ. Thus

$$\Lambda_{\mathrm{ref}} = \sum_\gamma p_\gamma \lambda_{\mathrm{ref},\gamma}, \qquad L_{\mathrm{ref}} = \Lambda_{\mathrm{ref}}(\mathrm{all}). \tag{6.12}$$

This reference includes trajectory indices and true packet labels. It is substantial: the bad-cell mass tends to zero, good cells retain their fixed captured fraction, and the renewal has vanishing loss. Hence L_{ref} is bounded below by a positive constant on the working subsequence. Freeze this post-renewal reference together with G.

Coarse lists for the graph row. For each fine cell, the tuples of width-s_0 velocity bins supporting actual spanning witnesses have mass under the product velocity law at least K^{-1}. For a fixed tuple, (6.9) and (6.11) allow at most K row bins. When s_0 is sufficiently small relative to the reciprocal

conditioning bounds, invert at the bin centers; otherwise the bound follows directly from $|G| \le K$. Choose one fine-cell witness for each row bin occurring in a coarse W_{s_0}. Sum its mass under the product velocity law: every tuple contributes to at most K row bins and the product velocity law has total mass one. There are at most K row bins. This is a support statement on the reference family, and survives every later restriction. □

6.4 Polynomial fits along reference trajectories

Use the scales a, h, M, ζ, ρ, b fixed before the graph construction. The next assertion concerns its frozen incidence law Λ_{ref}. Its relative exceptional-mass estimate will be used before the sparse selections in interpolation.

Proposition 6.3 (Polynomial qualification)**.** *Fix the scales of Section 6.3, the field G of Proposition 6.2, and its frozen reference law and total mass* (6.12)*, with the packet blocks supplied by Proposition 4.7. There are a subpower $K_{\mathrm{qual}} = N^{o(1)}$ and sets U_N of graph-reference incidences such that*

$$e_N := \frac{\Lambda_{\mathrm{ref}}(U_N)}{L_{\mathrm{ref}}} \longrightarrow 0. \tag{6.13}$$

Every reference incidence outside U_N admits a row-valued polynomial in t, of degree at most M and coefficients bounded by K_{qual}, which approximates G within h^M on reference-labeled times of its indexed trajectory of ordinary duration at least h/K_{qual} in the same Q_s-block, including that incidence.

The coarse lists do not yet compare the rows at nearby points: two points in the same cell may use different members of its list. The following selection makes a common choice in every retained cell. We will use it first for the physical row and then for a scalar residual inside a parameter block.

Lemma 6.4 (A common choice from local lists)**.** *Let μ be an unnormalized finite measure on a finite collection of components ω. In each component let $Z(e) \in \mathbb{R}^D$ and $Y(e) \in \mathbb{R}^q$ be measurable functions of an incidence e, with $|Y(e)| \le B$. Fix an integer $L \ge 2$, bounds $K_1, \dots, K_L \ge 1$, and tested widths*

$$1 = s_1 > s_2 > \cdots > s_L = r_*, \qquad A = \max_{i<L} s_i/s_{i+1}.$$

At width s_i, suppose that in every half-open s_i-cell of Z in each component, the occurring values of Y meet at most K_i half-open s_i-cells. There is a restriction $\mu' \le \mu$ such that, in every component,

$$\mu'(\omega) \ge \mu(\omega) \prod_{i=1}^{L} (4^D K_i)^{-1},$$

and any two retained incidences in the same component satisfy

$$|Y(e) - Y(\widehat{e})| \le C_{D,q}(A+B)\big(|Z(e) - Z(\widehat{e})| + r_*\big).$$

If, within each component, both Z and Y are determined by a discrete state, the restriction keeps or removes each whole state of that component. No measure is renormalized.

Proof. Order the scales as displayed. At scale s_i, color the coordinate cell $s_i(n+[0,1)^D)$ by n mod 4. In each component retain a color of largest current mass. In each retained cell choose a Y-cell of largest current mass among its at most K_i possibilities, and retain its incidences. Fix a lexicographic rule for ties. The color depends only on the component and the scale; the value-cell choice depends

only on the component, the scale, and the coordinate cell. In particular two incidences in the same retained coordinate cell use the same value-cell choice.

These two operations keep at least $(4^D K_i)^{-1}$ of the current mass in each component. Their successive application proves the mass bound. All lists may be frozen before selection, since restrictions preserve upper lists even when they remove the original witnesses.

Distinct coordinate cells of the same color are separated by at least $3s_i$ in one coordinate. Thus $|Z(e) - Z(\widehat{e})|_\infty \le s_i$ forces the same cell, and the selected value-cell gives $|Y(e) - Y(\widehat{e})| \le \sqrt{q}\, s_i$. For coordinate distance at most one, take the tested width just above the larger of that distance and r_*. This width is at most A times that larger quantity. For larger distance use $|Y(e) - Y(\widehat{e})| \le 2B$. Norm equivalence proves the stated comparison. When the two maps are state functions, every test is a test of the whole state. □

Remark (Finite-grid diagonal). Suppose all entering masses are inverse-subpower and $B, K_i = N^{o(1)}$. Starting with fixed finite grids, a sufficiently slow increase of their sizes gives

$$DL \log 4 + \sum_i \log K_i = o(\log N), \qquad A = N^{o(1)}.$$

The selected mass in Lemma 6.4 is then inverse-subpower, and the comparison loses only a factor K. At each finite-grid stage, the logarithmic loss includes the entering density and every preceding preparation cost. The diagonal reaches stage r only after this combined loss is at most $(\log N)/r$, while the maximal gap between tested exponents tends to zero. There is one selection loss per scale, irrespective of the number of cells or components. The lemma's comparison then supplies all intermediate widths.

Proof of Proposition 6.3. A substantial set of failures would still support the geometric and mass preparations needed by Proposition 5.6. We construct its scalar-normal input by counting strips, conditioning on coefficient bins, and making consistent choices from residual lists. The resulting fits contradict failure at a fixed tolerance; a slow diagonal then gives the stated subpower bounds.

Fix $\epsilon > 0$. Let $U_{N,\epsilon}$ be the reference incidences for which no such polynomial has coefficient bound N^ϵ, error at most h^M, and witness duration at least $hN^{-\epsilon}$. Use finite coefficient grids with total evaluation error at most $h^M/4$ on the bounded parent-time interval. The discovery below has an extra factor $h^{1/4}$, leaving room for this rounding and for the coefficient bound.

If $\Lambda_{\mathrm{ref}}(U_{N,\epsilon})/L_{\mathrm{ref}}$ does not tend to zero, pass to a subsequence where it is bounded below. Since L_{ref} is bounded below, this is a substantial residual of the original reference path with its unchanged parent weights. We shall find qualifying occurrences inside this residual. Keep G and all its mesh choices fixed, and renew breadth on W_b in the residual. Until the strip-filling step below, selections keep whole W_b's or remove only sufficiently small fixed fractions of their laws.

We will apply Proposition 5.6 inside a full coefficient block B_ρ. After subtracting its central curves and dividing positions by ρ, a comparison of nearby values of G_y will supply a time-only row. To obtain scalar-normal data for the fixed prior $\nu^\circ(\cdot \mid B_\rho)$, we must also control the scalar support, the spatial masses conditional on smaller coefficient bins, and the residual velocity. The construction below establishes these bounds before restricting to one such component.

Physical row comparison and strip counts. Lemma 6.4 applies on the grid between ρ and one, with parent as component, Z the center of W_b, and $Y = G(W_b)$. The coarse lists in Proposition 6.2 provide the hypotheses; cells meeting a coarse-cell boundary require only bounded enlargements. This retains dense mass and keeps whole fine states, so their conditional velocity distributions are unchanged. The displacement from an incidence to its state center is $O(Kb)$, which is negligible

compared with ρ. Hence the retained incidences obey

$$|G(W_b) - G(\widehat{W}_b)| \le K(|W - \widehat{W}| + \rho). \tag{6.14}$$

In a physical cube $C = W_\rho$, choose a row G_C within $K\rho$ of all its active rows. For each required $1 \ge s' \ge \zeta$, set $p = \rho s'$ and assign whole W_b's by their centers to width-p strips in $X - G_C \cdot z$. At active times the normal derivative is $O(K\rho)$. It is affine along a trajectory and remains so bounded between any two of those times. The normal range inside C is at most $K\rho^2 + Kb \ll p$, so a trajectory meets at most K strips there, and at most K/ρ cube/strip pairs in total.

Delete line/strip portions of duration below ρ/K_1; remove whole W_b's with excessive fractional loss, then strips with excessive total loss. Each surviving strip of preceding mass M retains comparable mass and has contributing prior weight at most KM/ρ. Only the first deletion removes fractions of the fine states. Its loss thresholds can be made small enough, in sum over the finite grid, to retain the spanning probability there. Since a fine state determines its strip and base z_p-cell, mixtures in a base cell also have a fixed positive probability of a determinant at least K^{-1}.

Apply (6.10) in $\mathbb{R}^m$ to the full affine base lines in a strip. If m_E are the retained masses in its base cells, the duration in a cell is at most Kp, and hence

$$\sum_E \big((m_E/p)^m/K\big)^{1/(m-1)} \le K(M/\rho)^{m/(m-1)}.$$

At least $K^{-1}(\rho/p)^m$ base cells are occupied. Each corresponds to a physical W_p-cell, and each physical cell meets at most K strips, since $|G_C| \le K$. A region in C with normal coordinate in an interval of length $O(Kr)$, $p \le r \le \rho$, is covered by $K(\rho/r)^m$ cubes of side r. Its physical support count is at most $K(\rho/r)^m(r/p)^{m+d}$. Division by the filling per strip proves

$$\#\{\text{occupied } p\text{-strips meeting a normal } r\text{-interval}\} \le K(r/p)^d. \tag{6.15}$$

Use each filling only to certify its upper support list. Those lists persist on further restrictions, and intermediate resolutions follow by subdivision and enlargement.

The parameter zoom. The physical cubes C were used to count scalar-normal strips. To obtain the conditional spatial mass bounds, we now condition on a full coefficient bin B_ρ and rescale its trajectories. Run the following construction in the family of components $\mathbf{q} = (\gamma, B_\rho)$, where γ is a parent and B_ρ is a full parameter block of positive ν°_γ-mass. Coordinates, cells and rows in this construction always include $\mathbf{q}$ in their labels. Within one such component subtract its central coefficient curves $X_c^{\mathbf{q}}(t), y_c^{\mathbf{q}}(t)$ and divide positions by ρ:

$$x = \frac{X - X_c^{\mathbf{q}}}{\rho}, \qquad \widetilde{y} = \frac{y - y_c^{\mathbf{q}}}{\rho}.$$

Time is unchanged. All new trajectory coefficients are bounded; aligned width-r' parameter bins are precisely the subbins $B_{\rho r'}$ of B_ρ. Choose a time-only row $F_{\mathbf{q}}(t)$ representing G_y at active points in each sufficiently fine time bin of length at most ρ, and extend it boundedly at inactive times. For the within-component calculations we suppress the superscript on the central curves and write $F = F_{\mathbf{q}}$. From (6.14),

$$|F(t) - G_y(W_b)| \le K\rho, \qquad |F(t) - F(u)| \le K(|t-u| + \rho) \tag{6.16}$$

at active occurrences. The physical positions lie within $K\rho$ of a central curve of speed K. At fixed time, $\rho(x - F(t)\widetilde{y})$, up to a common translation in each of the at most K nearby cubes C, differs

from the strip coordinate by at most $K(\rho^2+b)$. This is negligible compared with $\rho\zeta$. Thus (6.15) supplies the scalar-normal local d-support bound down to ζ, also on enlarged upper intervals by at most K subdivisions.

Take the fixed prior $\nu^\circ(\cdot \mid B_\rho)$, retaining all original trajectory tags. At the required resolutions $\zeta \le q' \le r' \le 1$, for a participating subbin $B_{\rho r'}$, the fixed-time numerator in a zoomed spatial q'-cube is at most

$$K(\rho q')^{j+d}\pi([V]_{\rho r'}).$$

Its prior denominator is at least $(\rho r')^{j+d}\pi([V]_{\rho r'})/K$, by (6.6). Consequently the conditional spatial cap is

$$K(q'/r')^{j+d}. \tag{6.17}$$

The case $r' = 1$ gives the unconditional spatial cap. These are caps for the original parameter-conditioned prior. They are not renormalized when labels are subsequently restricted.

The residual velocity. Put $R = x' - F(t)\widetilde{y}'$. Within a fixed W_b, B_ρ, its range is much smaller than ζ. Indeed, writing $v_c = y_c'$, one has

$$\rho R = G_0 + G_y v_c - X_c'(t) + (G_y - F(t))(V - v_c) + O(b^\kappa).$$

Here $|V - v_c| \le K\rho$, $|X_c''| \le K$, and the time variation within W_b is at most b. The choices of b, ρ, ζ give the asserted range.

Fix $s' \ge \zeta$ and put $\beta = \rho s'$. A β-parameter block uses at most K/s' zoomed s'-cells in its unique ρ-zoom. Delete block/cell entries of incidence mass below $s'\nu^\circ(B_\beta)/K_1$. The summed loss is at most K/K_1, before any normalization inside B_ρ. In one zoomed cell the time interval has length s', whereas the physical spatial cross-section at each time has side $K\beta$ and moves with the central curve. For a fixed velocity bin $J = [V]_\beta$, integrating (6.8) along that moving cross-section gives

$$\lambda(\text{zoomed cell}, J) \le Ks'\beta^{j+d}\pi(J).$$

Every surviving block has incidence mass at least

$$\frac{s'\nu^\circ(B_\beta)}{K_1} \ge K^{-1}s'\beta^{j+d}\pi(J).$$

Both estimates contain s', the length of the zoomed cell's time interval, so there are at most K blocks. Within one block the zoom coefficients have diameter $O(s')$; their absolute sizes are K. Across the cell's time interval, x' varies by Ks', and (6.16) varies by at most $K(s'+\rho) \le Ks'$. Thus one block contributes at most K width-s' residual bins for R.

We next remove the velocity condition with a summed denominator argument. Write $K_{\mathrm{pre}} = N^{o(1)}$ for the accumulated bounds, and let $m_0(W)$ be the earlier unnormalized fine-state weights, including mixture weights, and retain

$$\lambda(W, J) \le K_{\mathrm{pre}} m_0(W)\pi(J), \qquad \sum_W m_0(W) \le K_{\mathrm{pre}}.$$

For a coarse velocity bin $I = [V]_\rho$, the parameter list allows K_{pre} blocks B_ρ given W, I, and each has K_{pre} zoom cells meeting W, since $Kb/\rho \ll \zeta$. There are at most K_{pre}^2 refined entries $E = (W, B_\rho, s'\text{-cell})$ over W, I. Deleting those of mass below $m_0(W)\pi(I)/K_{\mathrm{cut}}$ loses at most

$$\frac{K_{\mathrm{pre}}^2}{K_{\mathrm{cut}}} \sum_W m_0(W) \sum_I \pi(I) \le \frac{K_{\mathrm{pre}}^3}{K_{\mathrm{cut}}}.$$

Choose the subpower K_{cut} larger than all accumulated inverse-subpower density losses. On each survivor,

$$\lambda(J \mid E) \le K\frac{\pi(J)}{\pi(I)}. \tag{6.18}$$

The denominator here is the retained-entry floor against the earlier fine-state weight.

Every occurring residual bin has a retained entry witnessing it. Near constancy within that entry and (6.18) make its supporting velocity bins have total mass under the velocity law at least $\pi(I)/K$, within boundedly many neighboring residual bins. A fixed fine velocity bin supports at most K residual bins by the preceding block/cell count. Summing against the common velocity law in I gives at most K residual bins in the zoomed cell.

Residual comparison and the scalar-normal input. It remains to make a common residual choice in each zoomed cell. Carry out the preceding deletions on a fixed finite grid $\{s_1, \dots, s_L\} \subset [\zeta, 1]$, including its endpoints up to dyadic rounding. Choose the thresholds successively so that the summed loss is at most half the entering dense mass. The two losses at a tested width have the forms K/K_1 and $K_{\mathrm{pre}}^3/K_{\mathrm{cut}}$ obtained above, so on a fixed grid the required thresholds remain subpower. Denote the remaining unnormalized measure, with the original parent weights, by μ_0. Its mass δ_0 is at least K^{-1}.

In component $\mathbf{q} = (\gamma, B_\rho)$ use

$$Z_{\mathbf{q}}(e) = (t, \widetilde{y}_l(t), x_l(t)), \qquad R_{\mathbf{q}}(e) = x_l'(t) - F_{\mathbf{q}}(t)\widetilde{y}_l'.$$

The dimension of $Z_{\mathbf{q}}$ is $D = j + 2$, and time retains its parent units. Freeze the width-s_i residual lists in every s_i-cell of these coordinates. The preceding argument bounds their sizes by $K_i = N^{o(1)}$. The velocity law used to count each list is common to the cell because B_ρ fixes $I = [V]_\rho$. The finer entry (W, B_ρ, C) supplied a denominator and a witness for this list; it is not an additional key for the residual choice.

Apply Lemma 6.4 with $Y = R_{\mathbf{q}}$. Choose the slowly enlarging grid together with the preceding deletions and density losses as specified in the finite-grid diagonal remark. We obtain a restriction μ_L of mass $\delta \ge K^{-1}$ such that

$$|R_{\mathbf{q}}(e) - R_{\mathbf{q}}(\widehat{e})| \le K\big(|Z_{\mathbf{q}}(e) - Z_{\mathbf{q}}(\widehat{e})| + \zeta\big) \tag{6.19}$$

for every pair of retained incidences in the same component. Coordinates and rows need not agree across components. This time the selection may remove individual edges: the last use of breadth in this construction was strip filling, and its remaining steps use only the inherited upper bounds.

Put $a_{\mathbf{q}} = p_\gamma \nu_\gamma^\circ(B_\rho)$. These weights sum to at most one. Hence some positive-weight component satisfies $\mu_L(\mathbf{q})/a_{\mathbf{q}} \ge \delta$. Divide by this parent weight and parameter-block mass. Since $\lambda \le K\nu^\circ\, dt$, the retained labels have product-prior mass at least K^{-1} for the fixed probability $\nu_\gamma^\circ(\,\cdot \mid B_\rho)\, dt$.

Polynomial fitting in one coefficient component. The preceding estimates provide the scalar-normal data of Section 5 in this component. The trajectories are $(x, \widetilde{y})$, the row is $F_{\mathbf{q}}$, and the prior remains $\nu_\gamma^\circ(\,\cdot \mid B_\rho)$. The strip count gives the scalar-normal support bound; (6.17) gives both spatial caps, including conditioning on a parameter bin; (6.16) gives row comparison; and (6.19) gives residual comparison. These are exactly the hypotheses of Definition 5.2, with the floor already chosen for Proposition 5.6. Apply Proposition 5.6. It produces a degree-M polynomial with coefficients at most K fitting F, hence G_y, to $Kh^{M+1/4}$ along labeled portions of an indexed line of duration at least h/K, including the counted occurrences. Use (6.11) and $G_0 = X_l' - G_y V_l + O(b^\kappa)$ to fit G_0 too. All witness times on such a line in the block have the same true Q_s, by its block-invariant trajectory assignment. Write K' for the combined subpower in this discovery. For the fixed $\epsilon > 0$

and all sufficiently large N, the coefficient bound is smaller than N^{ϵ}, the duration is at least $hN^{-\epsilon}$, and $K'h^{M+1/4} \le h^M/4$. Coefficient rounding still leaves error below h^M and coefficients below N^{ϵ}. These reference-labeled witnesses qualify occurrences lying in $U_{N,\epsilon}$, a contradiction. The discovered mass can be inverse-subpower: any positive mass contradicts the definition of this failure set. Therefore, for every fixed $\epsilon > 0$,

$$\frac{\Lambda_{\mathrm{ref}}(U_{N,\epsilon})}{L_{\mathrm{ref}}} \longrightarrow 0.$$

A common subpower bound. For $\epsilon_r = 1/r$, choose cutoffs beyond which this fraction is at most $1/r$, and let $r(N) \to \infty$ slowly enough to respect those cutoffs and the preceding finite-grid preparations. Set $K_{\mathrm{qual}} = N^{1/r(N)}$ and $U_N = U_{N,1/r(N)}$. Then K_{qual} is subpower and (6.13) follows. For any required fixed finite collection of applications, divide the allowed failure mass among its members before enlarging the collection. This proves a relative $o(1)$ exception in the fixed substantial mixture; it gives neither finite-N nullness nor a rate that can be passed to an arbitrary later sparse restriction. □

6.5 Cartesian interpolation of the physical field

Put $p_* = 1/k$, so $a \sim_{\log,N} h^{p_*}$, and choose

$$\begin{gathered} 0 < \alpha < \tfrac{1}{2}, \qquad \varepsilon_T = h^{p_*-1+\alpha}, \qquad a_1 = ah^{e_1}, \qquad e_1 = \alpha/2, \\ c_0 = e_1(1-d)/2, \qquad a_2 = ah^{e_2}, \qquad e_2 = c_0/3. \end{gathered} \tag{6.20}$$

In particular $0 < e_2 < 1/24$. Take a dyadic mesh ξ in block-normalized base coordinates so fine that all later evaluations are accurate, in particular $Kh\xi \ll a_1$. Velocity bins are still finer. The physical mesh b, the polynomial-qualification meshes, and the graph preparations at depths of thickness a_1, a_2 were chosen with these fixed power margins before constructing G. The displayed exponents determine the requirements in advance; none depends on the magnitude of the hidden coordinate.

Use the states

$$\mathcal{D} = (\text{parent component}, Q_s, W_b).$$

They have actual list knowledge and the required deep-history ambiguity. Return to the substantial graph-reference measure Λ_{ref}. Prepare conditional domination by the velocity law and breadth here, with strip probabilities as small a fixed constant as needed for all subsequent fixed-size tuples and fractional losses. Denote this substantial submeasure by $\Lambda_{\mathrm{prep}} \le \Lambda_{\mathrm{ref}}$. For every state of positive prepared mass, put

$$m_{\mathcal{D}} = \Lambda_{\mathrm{prep}}(\mathcal{D}), \qquad q_{\mathcal{D}} = \frac{\Lambda_{\mathrm{prep}}(U_N \cap \mathcal{D})}{m_{\mathcal{D}}}.$$

For any fixed $\eta > 0$, (6.13) gives

$$\sum_{q_{\mathcal{D}} > \eta} m_{\mathcal{D}} \le \eta^{-1}\Lambda_{\mathrm{prep}}(U_N) \le (e_N/\eta)L_{\mathrm{ref}} = o(1). \tag{6.21}$$

Thus the states with $q_{\mathcal{D}} \le \eta$ retain substantial mass. If a spanning m-tuple has probability at least $p_0 > 0$ in each prepared state, choose the fixed threshold $\eta \le p_0/(2m)$. A union bound leaves probability at least $p_0 - m\eta \ge p_0/2$ for a spanning tuple all of whose occurrences are qualified. These are the good states used below. All their prepared reference edges remain available. Qualification will supply the interpolation witnesses; after proving a statement about a state, we may restore the other prepared edges of that state without claiming they too are qualified.

Choose a parent and $Q = Q_s$ with good-state mass at least $h\nu_Q/K$. Such a choice exists since the sum of $h\nu_Q$ over packets, blocks and parent mixture weights is at most one. Write

$$u = (z - z_c)/h$$

with a fixed origin such that $|u| \le K$ on the whole block; there is no velocity shear. Dividing incidence by $h\nu_Q$ gives a base density at most K in u-space.

Over a specified sufficiently fine base cell, G to width ε_T has only K possible bins on the preceding reference used for fit times. Indeed the Q-graph lists predict X to Ka with physical gradient at most K. Since $\varepsilon_T \gg Ka$, a sufficiently small base cell meets at most K physical cells at resolution ε_T. Apply the coarse field lists in Proposition 6.2.

Conditional product sampling at a good state gives tuples of m very fine velocity bins with qualifying witnesses and determinant at least K^{-1} with probability bounded below. Domination by the velocity law makes their product velocity measure at least K^{-1}. Averaging fixes one tuple with common witnesses on states of mass at least $h\nu_Q/K$. Let A_0 have columns $\mathbf{v}_i = (1, v_i)$, the bin centers. Then $\|A_0\| + \|A_0^{-1}\| \le K$. Use a product grid of step ξ in $A_0^{-1}u$, assigning states by their fine-cell centers.

For each direction i and grid fiber, consider the polynomial fits from Proposition 6.3 on witnessing lines in that velocity bin. Their normalized-time coefficient vectors, to width ε_T, have at most K possible bins. To prove this, choose one fit and one witnessing line for each represented bin. Such a line stays within $K\xi$ of the fiber throughout the block, because its direction bin is finer than ξ. At each time its base position is known to $K\xi$; the time component of every basis direction equals one. Sample $M + 1$ independent uniform block times. For each represented bin, the probability that all these times lie in its witness set and have pairwise gaps at least h/K_1 is at least K^{-1}, for sufficiently large subpower K_1. At any successful tuple, the base-only field lists and polynomial interpolation permit at most K coefficient bins, since $h^M \ll \varepsilon_T$. Summing this probability proves the claim.

For each state fix one qualifying witness, with its polynomial fit, in each of the m selected velocity bins. Independently choose a polynomial bin on every fiber and one field bin at every grid cell, uniformly from their respective lists. Retain a state when its row $G(W_b)$ lies in the chosen field bin and all its m witness fits lie in the chosen polynomial bins on the corresponding fibers. These $m + 1$ tests involve lists of size at most K, so some choices retain mass at least $h\nu_Q/K$. Use the center of the selected field bin as the chosen cell value. It is within $K\varepsilon_T$, on every chosen fiber, of one polynomial of degree at most M in that fiber coordinate. Indeed, use representative coefficients from the selected polynomial bin: normalized time is affine in that coordinate, and its discrepancy from witness time is at most $K\xi$.

Delete cells of mass below $h\nu_Q\xi^m/K_1$. There are at most $K\xi^{-m}$ possible cells, so the loss is small. The base ceiling shows that the remaining subset occupies a fraction $\eta_0 \ge K^{-1}$ of the product grid. Choose $M + 1$ anchor coordinates and one target coordinate independently on each axis. The probability that all their product vertices are retained is at least $\eta_0^{(M+2)^m}$. For completeness, condition on the coordinates on all but one axis. Replacing one coordinate by several independent coordinates raises its conditional success probability to a fixed power. Jensen's inequality bounds the average of that power below by the same power of the preceding average. Iterating over the axes yields the stated bound. The probability of any two anchors on an axis lying within K_1^{-1} is smaller than this bound when K_1 is a sufficiently large subpower; the mesh is a fine fixed power. Fix separated anchors with an inverse-subpower fraction of successful targets.

Tensor Lagrange interpolation from the pure anchor vertices gives a polynomial row $T(u)$, of degree at most M separately in each product coordinate and bounded total degree, with coefficients and norms at most K. At a successful target, interpolate one coordinate at a time using the chosen

fiber polynomial. All necessary intermediate vertices are present, and inverse-subpower anchor gaps bound the Lagrange weights by K. The error is at most $K\varepsilon_T$. The per-cell mass floors convert the target count back to mass at least $h\nu_Q/K$. Transferring from centers to events adds a negligible error:

$$|G(W_b) - T(u)| \le K\varepsilon_T. \tag{6.22}$$

This statement depends only on the state. We now restore all prepared reference edges of the selected good states, preserving their breadth and bounds for the velocity law.

For a line of weighted duration at least h/K in this family, (6.11) and (6.22) and polynomial Remez in normalized block time give

$$|X_l'(t) - T(u_l(t)) \cdot \mathbf{v}_l| \le K\varepsilon_T \quad \text{throughout its } Q\text{-block}. \tag{6.23}$$

The comparison is polynomial in time with a fixed degree.

We will also need the joint support count

$$\#\{(\xi\text{-bin of } u, a_i\text{-bin of } X)\} \le K\xi^{-m}(a/a_i)^d, \qquad i = 1, 2. \tag{6.24}$$

Use true descendants at thickness a_i, with horizons $h_i \sim_{\log,N} a_i^k$, rounded with K loss. They have trajectory mass at least $a_i^d h_i^j/K$ and are disjoint subsets of Q's trajectories per h_i-block. Each uses at most $K(h_i/(h\xi))^m$ joint cells by its bounded gradient graph list; ξ is chosen sufficiently fine. There are h/h_i such time blocks. Thus their total count is

$$K\frac{h}{h_i}\frac{\nu_Q}{a_i^d h_i^j}\left(\frac{h_i}{h\xi}\right)^m = K\xi^{-m}\frac{\nu_Q}{a_i^d h^j} \le K\xi^{-m}(a/a_i)^d.$$

6.6 Switched lines force approximate conservativity

A conservative field has a simple geometric property: its integral along any broken base path depends only on the two endpoints. If the skew part of DT is nonzero, two orders of travel can instead reach the same base point with different accumulated values of X/h; Figure 2 illustrates this local obstruction. The proof below uses two bases of directions to turn it into a nonzero Jacobian minor. Its lower bound makes the probability of landing in the support from (6.24) tend to zero when $d < 1$, contradicting the positive probability of the switched paths.

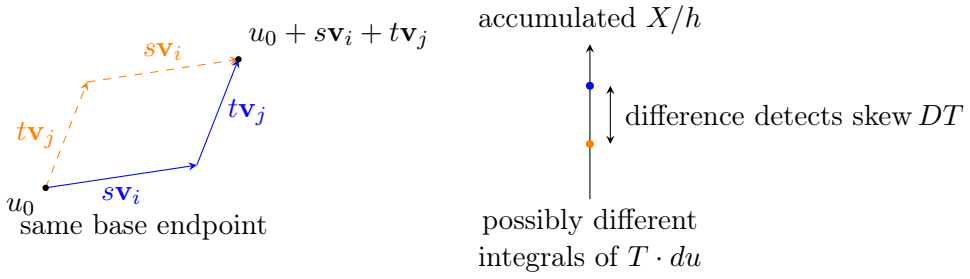


Figure 2: Two orders of the same two base displacements. The difference between their line integrals has mixed term $st\big(DT(u_0)[\mathbf{v}_i]\cdot\mathbf{v}_j - DT(u_0)[\mathbf{v}_j]\cdot\mathbf{v}_i\big)$. This is a schematic illustration of the obstruction; the proof uses $2m$ successive legs, with the first and last m directions forming bases.

Proposition 6.5 (Conservativity of the interpolated field)**.** *For the polynomial field T and enlarged box constructed in Section 6.5, with $p_* = 1/k$ and c_0 as in* (6.20)*, in coefficient norm, or equivalently in sup norm on that box up to K,*

$$\|\operatorname{skew} DT\| \le K h^{p_*-1+c_0}.$$

Proof. We prove that a large skew derivative would make switched paths fill too many endpoint cells. The transition bound lets us estimate their probabilities by Lebesgue measure in the time increments; a nonzero Jacobian minor then contradicts the support count.

If the bound fails, restrict to a subsequence where the skew coefficient norm is at least $h^{p_*-1+c_0}$. Normalize the restored state-selected incidence measure in Q to a probability $\mathbb{P}$. Its mass denominator is at least $h\nu_Q/K$, so its base density is at most K. After discarding an arbitrarily small fraction of starting events, the polynomial sublevel estimate gives skew at least $h^{p_*-1+c_0}/K$ at every retained start. A sufficiently large subpower threshold makes this discarded mass as small as needed.

Paths with two spanning sets of directions. Starting from $e_0 = (l_0, t_0)$, perform $2m$ switches. From e_{i-1}, draw $e^\diamond = (l_i, t^\diamond)$ afresh from $\mathbb{P}(\cdot \mid \mathcal{D}(e_{i-1}))$. Then retain its line and draw t_i from the conditional $\mathbb{P}$-law given l_i, producing $e_i = (l_i, t_i)$. Both operations preserve the marginal $\mathbb{P}$. A line with pre-normalization weighted duration below h/K_1 has $\mathbb{P}$-probability at most K/K_1, by the full assigned prior weight ν_Q and the preceding mass denominator. A union bound therefore removes short choices with arbitrarily small loss of path probability.

Before these longness restrictions, the velocity at each switch, conditional on its whole past, is sampled from a prepared state law. Its probability in any specified thin pencil is as small as needed. Sequential avoidance therefore gives, with probability bounded below, a well-conditioned basis from the first m velocities and another from the last m. Take sufficiently fine bins so their centers retain the determinant bounds. Consequently paths with a high-skew start, these two bases, and only long choices have probability bounded below.

A bound for the transition measure. For every current edge and past history, the *unnormalized* next-step measure, restricted to long choices, obeys

$$d\mathbb{P}\{[V_{l_i}] = J,\ t_i \in dt_i,\ l_i \text{ long} \mid \text{current edge and past}\} \le K\pi(J)\frac{dt_i}{h}. \tag{6.25}$$

The state law supplies $K\pi(J)$. On each long line, (6.1) divided by its duration bounds the endpoint-time density by K/h. The good transition kernel remains unnormalized. Iterating conditional integration backwards therefore bounds every nonnegative function of the start, control bins and endpoint times by K times the product of the starting $\mathbb{P}$-law, the velocity laws and dt_i/h. This product bound applies to the unnormalized measure of successful paths. For fixed start, the change to signed increments

$$\delta_i = (t_i - t_{i-1})/h$$

is triangular with Jacobian one in normalized times; they range in $[-1, 1]^{2m}$.

The polynomial endpoint map. For binned control directions $\widetilde{\mathbf{v}}_i$, simulate base legs $\widetilde{\mathbf{v}}_i\delta_i$ in u, and simulate increments in X/h by integrating $T\cdot\widetilde{\mathbf{v}}_i$ on these signed legs. Switching within W_b introduces only Kb physical error when the new line is evaluated at the previous time. By (6.23) and the bounded coefficients of T, the final simulation errors are

$$K(b/h + p_v) \ll \xi \quad \text{in } u, \qquad K(b/h + p_v + \varepsilon_T) \ll a_1/h \quad \text{in } X/h,$$

where p_v is the velocity-bin width. The second comparison has a fixed power margin:

$$\varepsilon_T/(a_1/h) \sim_{\log,N} h^{\alpha/2}.$$

Write the endpoint map as

$$\delta \longmapsto (u_0 + A\delta, \mathcal{R}(\delta)),$$

where A has columns $\widetilde{\mathbf{v}}_i$ and $\mathcal{R}$ is a polynomial of fixed degree. Let I, J be the first and last m indices. The columns in I determine every base displacement. Consequently the derivative in a direction $j' \in J$, minus the corresponding combination of derivatives in I, leaves the base endpoint fixed. Its effect on the accumulated height is measured by a maximal Jacobian minor. For $j' \in J$, the maximal Jacobian minor using $I \cup \{j'\}$, divided up to sign by $\det A_I$, equals

$$\partial_{j'}\mathcal{R} - (\nabla_I \mathcal{R})^T A_I^{-1} A_{j'}.$$

A Jacobian minor detects the skew derivative. Differentiating this identity at zero in each coordinate of I relates the endpoint map to the skew derivative at the start. If every minor had polynomial norm at most η, the cross Hessian would differ by at most $K\eta$ from

$$A_I^T H_{\mathrm{sym}} A_J, \qquad H_{\mathrm{sym}} = A_I^{-T}(D^2\mathcal{R}(0))_{I,I} A_I^{-1},$$

a symmetric matrix transported between the two bases. Here fixed-degree polynomial derivative bounds control a derivative at zero by the minor norm. On the other hand, directly differentiating the successive integrals gives, for $i \in I, j' \in J$,

$$(D^2\mathcal{R}(0))_{i,j'} = DT(u_0)[\widetilde{\mathbf{v}}_i] \cdot \widetilde{\mathbf{v}}_{j'}.$$

Later legs have zero duration at zero, so there are no additional cross terms. Inverting both bases bounds the skew of $DT(u_0)$ by $K\eta$. Hence for the high-skew starts at least one maximal minor has polynomial norm at least $h^{p_*-1+c_0}/K$.

The endpoint support contradiction. For every start and control tuple, choose such a minor and remove the region where its absolute value is below $h^{p_*-1+c_0}/K_1$. Fixed-degree polynomial sublevels in Lebesgue increments, together with (6.25), make the removed path mass arbitrarily small. On the passing region, fix the complementary $m-1$ increment coordinates. The regular fibers of the remaining polynomial map have bounded multiplicity, by the fixed-degree semialgebraic fiber bound. The change-of-variables formula thus bounds the Lebesgue measure mapped into one endpoint cell by

$$K\xi^m(a_1/h)h^{-(p_*-1+c_0)}.$$

The complementary coordinates have bounded volume. The resulting cell estimate has the full $m+1$-dimensional volume factor $\xi^m(a_1/h)$. It remains to sum it over the smaller support available to the actual endpoints. Every actual long path has its simulated endpoint in a bounded enlargement of the union counted by (6.24). Summing and integrating the start and velocity-bin weights bounds the successful-path probability by

$$\begin{aligned} K(a/a_1)^d(a_1/h)h^{-(p_*-1+c_0)} &\le Kh^{e_1(1-d)-c_0} \\ &= Kh^{e_1(1-d)/2} \longrightarrow 0. \end{aligned}$$

This contradicts the positive lower probability and proves the proposition. □

6.7 A potential and concentrated intercept groups

Define the polynomial

$$P(u) = h \int_0^1 T(tu) \cdot u \, dt.$$

Differentiating under the integral and comparing with $hT(u)$ leaves only the skew derivative integrated against a bounded multiple of u. Thus Proposition 6.5 gives

$$|\nabla P - hT| \le Kah^{c_0}. \tag{6.26}$$

The norms of all fixed positive derivatives of P are at most Kh on the needed enlarged box. For a long line in (6.23), the variation of $X - P(u)$ across the block is at most Kah^{c_0}: integrate the directional error over time h, using $h\varepsilon_T \le Kah^{\alpha}$ and $c_0 < \alpha$.

Bin the midpoint value of $X - P(u)$ to width a_2. Denote the bin's numeric center by C and index the intercept groups by these centers. This midpoint assignment is invariant along the assigned trajectory. At a state $\mathcal{D}$ of long labels, only boundedly many neighboring C's occur: the variation Kah^{c_0} is much smaller than a_2, the state knows X, u much more accurately, and P has controlled gradient. Within one intercept group, the normalized base density, after division by $h\nu_Q$, is at most

$$K(a_2/a)^d. \tag{6.27}$$

Indeed apply the exact-base X-cap at width a_2, whose center follows $P(u) + C$, then use $\nu_Q \sim_{\log,N} a^d h^j$.

We next bound the number of groups, retaining breadth by controlling the fractional losses in each prepared state. First remove short lines globally in Q, and states with excessive fractional loss. In each remaining state remove the fractions belonging to C's of mass below a sufficiently small fixed fraction. There are only boundedly many C's there, so the lost fraction is small. Remove short lines again in this family, still with only one C per line. Remove $(\mathcal{D}, C)$-states with excessive fractional loss, and groups with excessive total loss compared with before the last short-line deletion. Mass at least $h\nu_Q/K$ remains. If a group has remaining mass M, its contributing prior weight is at most KM/h, using the immediately preceding duration test. At each surviving $(\mathcal{D}, C)$, the law is at least a fixed fraction of the original good state law, up to the chosen small losses. The pencil probabilities were chosen smaller than these fractions. All needed breadth and bounds for the velocity law therefore survive.

Apply (6.10) in dimension m to the affine base lines of one group, in u-coordinates. Assign ξ-cells by state centers. A line spends at most $Kh\xi$ weighted physical time in one such cell, and its radius-$O(\xi)$ tube contains the assigned cell. If m_E are the group cell masses, determinant success and the prior-weight bound imply

$$\sum_E \big((m_E/(h\xi))^m/K\big)^{1/(m-1)} \le K(M/h)^{m/(m-1)}.$$

Power mean gives at least $K^{-1}\xi^{-m}$ occupied base cells per group. Each joint cell at X-width a_2 is counted by at most K intercept groups, since the variation of P on a base cell is negligible compared with a_2. The count in (6.24) with $i = 2$ now yields at most $K(a/a_2)^d$ groups.

Delete groups below

$$K^{-1}h\nu_Q(a_2/a)^d \tag{6.28}$$

with a sufficiently large denominator. The summed loss is small. On every retained long line in any group,

$$|X_l(t) - P(u_l(t)) - C| \le Ka_2 \quad \text{throughout the entire } Q\text{-block.} \tag{6.29}$$

All subsequent counts use these heavy groups and their immediately preceding long witnesses.

6.8 Seven directions and the exceptional conic

The remaining task is to replace the potential fit by an allowed quadratic test. In three base variables, even directions with every triple independent can all lie on a nonsingular conic. Third directional tests along those directions therefore need not force every higher-order term to vanish. We must identify the cubic and quartic terms that survive, and later express them using the hidden coordinate so that the final test is quadratic.

For a symmetric rank-r tensor T on $\mathbb{R}^3$, associate the homogeneous form $F_T(x) = T(x, \dots, x)$. Testing $T(p_i, p_i, p_i, \cdot, \dots, \cdot) = 0$ is equivalent to the vanishing at p_i of all partial derivatives of F_T of order $r - 3$, up to nonzero constants depending only on r. We call these the three-copy tests. For seven directions with every triple independent, the following lemma identifies their kernels. If the directions lie on a conic $q = 0$, the rank-three and rank-four kernels are q times a linear form and a multiple of q^2, while the rank-five kernel vanishes. The cubic two-copy test $T(p_i, p_i, \cdot)$ has a different role: it is injective even in the conic case and will control the third derivatives of a branch from its differentiated second-directional tests. The following lemmas also supply the conditioning bounds and the passage from approximate conic directions to exact real ones needed for those applications.

Lemma 6.6 (Seven-direction tensor tests)**.** *Let $p_i = (1, v_i) \in \mathbb{R}^3$, $1 \le i \le 7$, with every triple linearly independent.*

1. *The space of homogeneous quadratics vanishing at all p_i has dimension at most one; every nonzero member is nonsingular.*

2. *The map on symmetric cubic tensors*

$$T \longmapsto \big(T(p_i, p_i, \cdot)\big)_{i=1}^{7}$$

   is injective.

3. *If a nonzero quadratic q vanishes at all seven points, the kernels of the three-copy tests at ranks $3, 4, 5$ are, under the homogeneous-form correspondence,*

$$q\,(\mathbb{R}^3)^*, \qquad \mathbb{R}q^2, \qquad \{0\},$$

   respectively.

These assertions have polynomial conditioning. More precisely, if

$$|p_i| \le A, \qquad |\det(p_i, p_j, p_k)| \ge A^{-1}, \qquad A \ge 2,$$

the inverse of the injective map in part 2 has norm at most CA^C. On the exact-conic family, with the norm, inverse norm and inverse determinant of q also bounded by fixed powers of A, distance to each kernel in part 3 is at most CA^C times the norm of the corresponding tests. The constants depend only on these fixed finite-dimensional spaces and on the fixed conditioning powers.

Proof. Divisibility by the conic determines the algebraic kernels: first the quadratic vanishing space, then cubic injectivity and the three-copy kernels. A separate compact semialgebraic minimum argument supplies the polynomial conditioning bounds.

The algebraic kernels. All divisibility arguments can be made over $\mathbb{C}$. A complex line contains at most two of the seven points, since three real linearly independent vectors are also complex linearly independent. A singular projective conic, being a union of two lines with multiplicity allowed, cannot contain five of them.

There is exactly one conic through any five, up to scalar. Indeed, use the first three points as a projective coordinate basis. Quadratics vanishing there have form $ax_0x_1 + bx_0x_2 + cx_1x_2$. The remaining two points have all coordinates nonzero. Their vectors (x_0x_1, x_0x_2, x_1x_2) are not proportional, since their ratios recover the projective point. They give two independent conditions on (a, b, c). Thus the five-point conic space is one-dimensional, and its nonzero member is nonsingular by the preceding paragraph. Part 1 follows.

If $T(p_i, p_i, \cdot) = 0$ for a cubic form F, all its first partial derivatives belong to that quadratic space. If the space is zero, $F = 0$. Otherwise write $\partial_j F = c_j q$. Euler's identity yields $F = qL$ for a linear form L. At every tested point,

$$0 = \nabla F(p_i) = L(p_i)\nabla q(p_i).$$

Nonsingularity gives $L(p_i) = 0$, and three independent points force $L = 0$. This proves part 2.

A cubic vanishing at seven distinct points of a nonsingular conic is divisible by its equation. Here is an elementary justification of this intersection fact. Over $\mathbb{C}$, change coordinates to write the conic as $x_0x_2 - x_1^2 = 0$, with parametrization $[s : t] \mapsto [s^2 : st : t^2]$. The restriction of a cubic is a binary sextic; seven distinct projective zeros make it identically zero. For any homogeneous degree, reduction modulo $x_0x_2 - x_1^2$ leaves monomials using no pair x_0, x_2 simultaneously. Their substitutions are distinct binary monomials, with the common monomial x_1^r counted once. Thus a zero restriction has zero remainder, proving divisibility. The rank-three kernel is therefore exactly q times the linear forms.

For a quartic F in the rank-four kernel, each first partial derivative is such a cubic and is divisible by q. Euler gives $F = qH$, with H quadratic. Then

$$\partial_j F = (\partial_j q)H + q\partial_j H.$$

Choose a nonzero linear partial derivative of q. The irreducible quadratic q is relatively prime to it, so divisibility of $\partial_j F$ by q forces $q \mid H$. Hence F is a multiple of q^2. Conversely, q^2 and its first derivatives vanish at the seven points.

For a quintic in the rank-five kernel, every second derivative is a cubic divisible by q. Each first derivative consequently satisfies the preceding quartic test and is a multiple of q^2. Euler gives $F = q^2L$ with L linear. Differentiation and reduction modulo q^2 force $q \mid (\partial_j q)L$, and hence $q \mid L$ for a nonzero partial derivative. Therefore $L = 0$. This proves all kernel equalities.

Polynomial conditioning. For fixed A, the point configurations form a compact semialgebraic set. The matrix of the cubic two-copy test has polynomial entries and is injective everywhere on this set, so its least singular value has a positive minimum. For the three-copy tests restrict to the compact exact-conic family, imposing the stated coefficient and determinant bounds. Their kernel dimensions are constantly $3, 1, 0$, so their least positive singular values also have positive minima. The minima are semialgebraic functions of A, by real quantifier elimination; orthogonality to a kernel and unit norm are fixed-format polynomial conditions.

A positive semialgebraic function of one real parameter has a reciprocal polynomial lower bound at infinity. To see the rate relevant here, decompose its graph into finitely many algebraic branches for large A. On each, a nonzero relation $P(A, s) = 0$ holds. Remove powers of s from P. Its constant coefficient in s is then a nonzero polynomial in A, bounded below by a power of A for large A; all other coefficients are bounded above by powers of A. For $0 < s \le 1$, the relation forces $s \ge cA^{-C}$. Absorb bounded parameter intervals into the constant. The claimed inverse estimates follow. The rank-four estimate is applied on the exact-conic family: its rank can increase when the points move off that family. □

Lemma 6.7 (Restoration of an approximate real conic)**.** *Suppose the points satisfy the conditioning bounds of Lemma 6.6, and a real quadratic $q(x) = x^T \mathsf{Q} x$ has $A^{-1} \le \|q\| \le A$ and $|q(p_i)| \le \epsilon$. If ϵ is smaller than a sufficiently large fixed reciprocal power of A, then*

$$|\det \mathsf{Q}| \ge cA^{-C},$$

and there are exact real zeros $p_i' = (1, v_i')$ of q with

$$|p_i' - p_i| \le CA^C \epsilon.$$

For sufficiently small ϵ, every triple determinant of the shifted points has absolute value at least $(2A)^{-1}$.

Proof. For coefficient-normalized quadratics the function

$$|\det \mathsf{Q}| + \sum_{i=1}^{7} |q(p_i)|$$

has a positive minimum on the compact point family: a zero would be a nonzero singular quadratic through all seven points, contrary to Lemma 6.6. The semialgebraic argument in that lemma gives a lower bound cA^{-C}. Normalize the given q, use the smallness of its evaluations, then rescale to obtain the determinant bound.

It bounds the smallest singular value of Q below by cA^{-C}, and hence the homogeneous gradient at each p_i below by such a bound. Write $g_i = \nabla_v q(1, v_i)$. Euler's identity reads

$$\partial_0 q(1, v_i) + v_i \cdot g_i = 2q(1, v_i).$$

Since $|v_i| \le A$ and $q(1, v_i)$ is sufficiently small, the full-gradient bound implies $|g_i| \ge cA^{-C}$. With $e_i = (0, g_i/|g_i|)$,

$$q(p_i + te_i) = q(p_i) + |g_i|t + b_i t^2, \qquad |b_i| \le CA.$$

If $q(p_i) \ne 0$, take $t = -2q(p_i)/|g_i|$. For sufficiently small ϵ, the quadratic term has absolute value below $|q(p_i)|/2$, so this value has the opposite sign from $q(p_i)$. The intermediate value theorem gives a real zero at distance at most $2|q(p_i)|/|g_i|$. If $q(p_i) = 0$, no shift is needed. The first coordinate remains one. Multilinearity shows that triple determinants change by at most $CA^2 \max_i |p_i' - p_i|$, proving the final assertion. □

6.9 Restoring quadratic tests

Proposition 6.8 (Degree restoration)**.** *The potential fits in* (6.29) *and the heavy intercept groups satisfying* (6.28) *yield a candidate with all the properties in Theorem 6.1, without reducing the positive exponent e_2 in $a_2/a = h^{e_2}$.*

Proof. In version 1, four directional tests directly bound the third derivatives of the potential. In version 2, we first recover a stable hidden-coordinate branch. The tensor tests either approximate that branch by a quadratic or identify an exceptional conic; substituting the normalized hidden coordinate for the conic then restores quadratic degree.

Version 1. Choose a heavy intercept group. In normalized block time, (6.29) compares P with the quadratic $X - C$ along each of its lines. Polynomial Remez gives

$$|D_{\mathbf{v}}^3 P| \le K a_2$$

there. Conditional product sampling and averaging over the velocity law fix four separated very fine velocity bins with witnesses on common states carrying an inverse-subpower fraction of the group. Use their centers as directions. The locations in a state differ by at most Kb/h in u, and fixed derivatives of P are bounded by Kh. Finer velocity bins and $b/h \ll a_2/h$ make the transferred errors negligible. By (6.27) and (6.28), these common tests hold on inverse-subpower base volume. Polynomial Remez extends them to the enclosing box. A binary cubic is determined by its values in four separated directions $(1, V_i)$; the inverse of this Vandermonde system has norm at most K. Hence $\|\nabla^3 P\| \le Ka_2$. The quadratic Taylor polynomial of $P + C$ fits to Ka_2 on the block box. The test w minus this polynomial has hidden derivative one and zero hidden curvature, and its heavy-group mass is the required one.

Branches in version 2. The native test supplies controlled signal and derivative bounds even when w is unbounded in tangent units. In each heavy group consider

$$P_*(z, w) = P(u) + C, \qquad z = z_c + hu. \tag{6.30}$$

At path observations the residual is $O(Ka_2)$, the absolute hidden derivative lies between K^{-1} and K, and the hidden curvature is at most K. The discriminant expression

$$\Delta(u) = (P_{*,w})^2 - 2P_{*,ww}\big(P_*(z, w) - P(u) - C\big)$$

is independent of w, because the equation is quadratic in w with constant quadratic coefficient. It lies between K^{-1} and K on the observations. The heavy-group floor divided by (6.27) gives inverse-subpower base volume. Polynomial Remez bounds the coefficients and derivatives of Δ on an enlarged complex box by K. Thus, about every used point, Δ is zero-free on a complex ball of inverse-subpower radius with a fixed margin.

Each observed w_l is within Ka_2 of a real simple root of (6.30). For a quadratic, choose the root with the same derivative sign as at w_l; the derivative on the intervening segment has size at least K^{-1}, after enlarging K, by the positive discriminant and small residual. For a linear equation, divide by its nonzero coefficient. The roots extend holomorphically to the stable balls, by a holomorphic square root of Δ, or by linear division.

Choose a lattice of real comparison cores, each inside a fixed several-fold complex enlargement, with radius common up to K among the groups. Assign a core by the $(\mathcal{D}, C)$-center. Since Kb/h is negligible, it contains the state's used points with an interior margin. Boundary points can be assigned in a bounded overlapping cover. Add the sign of the nearby root. Discard sign fractions smaller than a sufficiently small fixed fraction; the core choice itself removes no fraction at a state. Delete line/intercept/core/sign portions with weighted duration below h/K_1. Each line has one intercept and at most K core/sign choices, so the total loss is at most $Kh\nu_Q/K_1$. Remove states with excessive fractional loss. The fixed-fraction comparisons to the earlier good state laws, with sufficiently small initial pencil probabilities, retain general position and domination by the velocity law.

There are at most $K(a/a_2)^d$ group/core/sign choices. One choice therefore has mass at least

$$K^{-1}h\nu_Q(a_2/a)^d. \tag{6.31}$$

Let U be its real inner ball and f its branch. Every contributing line has an immediately preceding witness set, in this same choice, of ordinary length at least h/K, including its currently counted incidences. All witnesses lie in U, with f holomorphic on a sufficient complex enlargement. The convex cores and that margin ensure that the line chords used below stay in the controlled domain.

Analytic estimates before tensor inversion. Set

$$R(u) = P_s(z_c + hu, f(u))/a. \tag{6.32}$$

We will apply the two-copy cubic test to $\nabla^3 f$ and the three-copy tests to the higher derivatives of R. First we need analytic bounds on R that hold before moving any directions to an exact conic. At the earlier witnesses, the true packet test obeys

$$|P_s(z, w_l)| \le Ka, \qquad |P_{s,w}(z, w_l)| \le K, \qquad |P_{s,ww}| \le K/a, \qquad |w_l - f(u)| \le Ka_2.$$

Its exact quadratic Taylor formula in w gives

$$|R(u)| \le K\big(1 + a_2/a + (a_2/a)^2\big) \le K.$$

The base density of this choice is bounded by the original group ceiling $Kh\nu_Q(a_2/a)^d$. Together with (6.31), this gives real projection volume at least K^{-1}. The function R is holomorphic algebraic of bounded relation degree. Lemma 3.1 bounds its norm and every required fixed derivative by K on the used interiors, before any tensor inversion.

Along a contributing line, compare f with the affine hidden trajectory w_l, and compare R with the quadratic $P_s(z, w_l)/a$, on its earlier long witness set. Normalized time on its chord has length at least K^{-1}; the witness set has inverse-subpower relative length and the chord has the stated complex margin. The one-variable algebraic norm bound yields, at the incidences,

$$|D^2_{\mathbf{v}} f| \le Ka_2, \qquad |D^3_{\mathbf{v}} R| \le Ka_2/a. \tag{6.33}$$

Let

$$S_0 = \inf_{L \text{ affine}} \|f - L\|_{C^0(U)}.$$

For $S_0 > 0$, apply Lemma 3.1 to $f - L$ with $\|f - L\|_{C^0(U)} \le 2S_0$. All fixed derivatives of order at least two are bounded by KS_0 on a slightly larger interior. If $S_0 = 0$, f is affine and these derivatives vanish.

Sample seven velocities in each retained state, sequentially avoiding all preceding points and joining lines. The prepared breadth gives a positive probability that all triple determinants of $\mathbf{v}_i = (1, v_i)$ have size at least K^{-1}. Domination by the velocity law turns this into a mass under the product velocity law at least K^{-1} of eligible bin tuples. Averaging fixes one tuple with common directional witnesses on states carrying an inverse-subpower fraction of the current mass. Using fine bin centers transfers their tests to actual base points in those states. For R, its preceding derivative bounds make all mesh errors negligible relative to a_2/a. For f, the Hessian and higher derivative errors are bounded by KS_0 times the normalized position and velocity errors. Choose those errors at most h^2/K, independently of S_0. Common-state mass and the base ceiling again give base volume at least K^{-1}. Apply Lemma 3.1 to the directional derivative functions themselves; differentiation of the simple root preserves bounded algebraic relation degree on the stable domain. We obtain throughout U, also after the fixed further differentiations needed below,

$$|D^2_{\mathbf{v}_i} f| \le K(a_2 + S_0 h), \qquad |D^3_{\mathbf{v}_i} R| \le Ka_2/a. \tag{6.34}$$

The weaker $S_0 h$ allowance absorbs the $S_0 h^2$ transfer error and subpower losses. All inverse ball radii and inverse margins here are bounded by K.

Differentiate the first tests once and apply the injective cubic two-copy test in Lemma 6.6. Its polynomial conditioning is subpower, so

$$\|\nabla^3 f\| \le K(a_2 + S_0 h). \tag{6.35}$$

If $S_0 \le a_2/h$, a quadratic Taylor polynomial f_2 approximates f on U within Ka_2. The test $w - f_2((z-z_c)/h)$ is quadratic in parent variables, has derivative one and curvature zero, and fits the long witnesses. This includes $S_0 = 0$.

The conic case. Suppose $S_0 > a_2/h$. Let u_c be the ball center and L' the affine Taylor polynomial of f there. Define

$$q(x) = \frac{1}{2S_0}\nabla^2 f(u_c)[x,x].$$

Taylor's theorem and (6.35) give

$$\frac{f(u) - L'(u)}{S_0} = q(u-u_c) + O(Kh). \tag{6.36}$$

The relative derivative bound gives $\|q\| \le K$. Conversely the definition of S_0 yields

$$1 \le \|q\|_{C^0(U-u_c)} + Kh \le K\|q\| + Kh,$$

so $\|q\| \ge K^{-1}$. The Hessian tests imply $|q(\mathbf{v}_i)| \le Kh$. Choose one parameter $A = N^{o(1)}$ controlling all coefficient norms, direction sizes, and inverse triple determinants. Since h is a fixed negative power, Lemma 6.7 applies. It gives a determinant lower bound K^{-1} for q, and exact real zeros $\mathbf{v}_i' = (1, v_i')$ with

$$|\mathbf{v}_i' - \mathbf{v}_i| \le Kh, \qquad |\det(\mathbf{v}_i', \mathbf{v}_j', \mathbf{v}_k')| \ge K^{-1}.$$

Thus even a nearly singular or nearly definite proposed quadratic is quantitatively controlled before inversion.

The independent derivative bounds for R from (6.32) show that these shifts alter its rank-three, rank-four and rank-five contractions by at most Kh. This is negligible compared with $a_2/a = h^{e_2}$, because $e_2 < 1$. The differentiated tests in (6.34) therefore remain Ka_2/a in the shifted directions. Apply the exact-conic kernels of Lemma 6.6. The fifth derivatives of R are bounded by Ka_2/a; at u_c, the cubic and quartic Taylor tensors are within Ka_2/a of q times a linear form and a multiple of q^2, respectively. Taylor expansion through degree four gives

$$R(u) = R_2(u) + q(u-u_c)L_1(u) + \lambda_0 q(u-u_c)^2 + O(Ka_2/a), \tag{6.37}$$

where R_2 is quadratic, L_1 is affine, and all displayed coefficients are bounded by K. To justify division in these factors, the coefficient norm of a product of polynomials of fixed degrees is bounded below by a positive constant times the product of their coefficient norms. Normalize the factors and use compactness and the absence of zero divisors. Since $\|q\| \ge K^{-1}$, division by q or q^2 increases coefficient norms by at most K. A constant term of L_1 can be absorbed into R_2.

On the earlier long witnesses, put

$$H'(u,w) = \frac{w - L'(u)}{S_0}.$$

Because $a_2/S_0 < h$, (6.36) and $|w - f(u)| \le Ka_2$ imply $H' = q(u-u_c) + O(Kh)$ and $|H'| \le K$. Define

$$P_{\mathrm{new}}(z,w) = P_s(z,w) - a\Big(R_2(u) + H'(u,w)L_1(u) + \lambda_0 H'(u,w)^2\Big), \qquad u = (z-z_c)/h. \tag{6.38}$$

Every term has total degree at most two in (z,w): u, H', L', and L_1 are affine in the indicated variables. Substituting H' for q introduces an error Kah, which is below a_2 by h^{1-e_2}. Replacing f by the actual w in P_s introduces an error at most

$$K|w-f| + (K/a)|w-f|^2 \le Ka_2 + Ka_2^2/a \le Ka_2.$$

Together with (6.37), this proves the Ka_2 value bound on the long witnesses.

The exact derivative perturbations are

$$\partial_w(P_s - P_{\text{new}}) = \frac{a}{S_0}(L_1 + 2\lambda_0 H'), \qquad \partial_{ww}(P_s - P_{\text{new}}) = \frac{2a\lambda_0}{S_0^2}.$$

Consequently

$$|\partial_w(P_s - P_{\text{new}})| \le Ka/S_0 < Kh^{1-e_2} \ll K^{-1}, \tag{6.39}$$

$$|\partial_{ww} P_{\text{new}}| \le K/a + Ka/S_0^2 \le K/a_2 + (K/a_2)h^{2-e_2} \le K/a_2. \tag{6.40}$$

The old derivative lower bound at the counted observations survives (6.39); the upper bound holds on the earlier witnesses. The perturbation estimates depend on the normalized value H' and the displayed lower bound on S_0; the magnitudes of w, L', and S_0 may be unbounded.

The same time block. In both cases the test is an allowed quadratic. On each contributing version 2 trajectory, $z(t)$ and $w(t)$ are affine, so its value is a polynomial of degree at most two in time and its hidden derivative has degree at most one. The earlier witness set has physical length at least h/K inside the same h-length Q_s-block. After normalizing that block to unit length, polynomial Remez loses only a factor K. Thus the value and derivative upper bounds hold throughout this block. The lower derivative is needed only on the counted incidences and was proved there. Curvature is constant in w for these quadratic tests. The version 1 comparison has the same fixed-degree time property. Extrapolation is never made from an h/K-set to a unit *parent* time interval.

The retained choice has mass $h\nu_Q(a_2/a)^d/K$, by (6.31) or the heavy intercept floor in version 1. Degree restoration makes no further discard. Dividing by its time length h gives $\nu_Q(a_2/a)^d/K$, exactly the block-normalized candidate mass requirement in Definition 4.10. □

Conclusion of Theorem 6.1. Start with any substantial residual of the reference path. The physical estimates and Proposition 6.2 retain a substantial reference family. Proposition 6.3 is an assertion of relative $o(1)$ failure under its frozen post-renewal incidence law. It follows by finding qualifying occurrences in every hypothetical substantial fixed-tolerance failure set and then taking the slow diagonal. The fixed-threshold deletion in (6.21) therefore leaves substantial good-state mass. Cartesian interpolation selects a packet and states of mass at least $h\nu_Q/K$, then restores their prepared reference edges. The conservativity estimate yields the potential and heavy intercept groups constructed above; Proposition 6.8 provides the required quadratic candidate. By (6.20), its gain has positive limit e_2ks. All meshes and degree bounds used to discover it depend on fixed exponents, while the actual output construction and its original-scale exclusion thresholds are those of Proposition 4.11. The criterion therefore gives the isotropic contradiction on every substantial residual. □

Corollary 6.9 (Zero horizon in the scalar model)**.** *For every fixed admissible choice of angular parameters and every $d < 1$, a version 1 exact problem in $C(d)$ has $H(E) = 0$. In particular there is no bad problem in version 1.*

Proof. If a bad version 1 problem existed, the critical-path construction of Proposition 4.3 would give a tangent problem with positive k and precisely the version 1 data above. Theorem 6.1 and Proposition 4.11 would contradict its maximizing-path exclusion. □

This scalar conclusion is established before the finite-narrowness analysis. It supplies the version 1 input to Theorem 8.2, whose projection labels are then used in Sections 8 to 10. The unbounded-narrowness case retains its separate limiting construction.

7 Planar rigidity and affine–skew rows

We prove that a row whose evaluation is nearly constant along many lines agrees, on many incidences, with a constant row plus a multiple of Jz, where $J(t, Z) = (-Z, t)$. This is Lemma 7.7, used in both the finite-narrowness reduction and the critical-rate argument.

The main step is planar rigidity for graphs with small restricted sumsets: many original edges have their endpoints in two parallel strips, of width equal to the mesh up to a subpower factor and with subpower-bounded common slope. Keeping original edges allows this geometric conclusion to be lifted back to incidence mass. We first prove the planar statement, then apply it to the line projections that arise from the row.

Throughout this section $h = N^{-a}$, where $a > 0$ is fixed, and $K = h^{-o(1)}$ may increase by a fixed power from line to line. For a bounded planar set U, $N_h(U)$ is its covering number by squares of side h. Fixed-grid and separated-set versions differ only by dimensional constants. Constants indexed by a fixed expression length may depend on that length, but remain subpower in h^{-1}. The final passage to strip widths $h^{1-o(1)}$ uses a slow diagonal and may introduce a new subpower factor.

7.1 Graph refinement and common endpoint sets

We begin with the graph form of Balog–Szemerédi–Gowers that records the edges surviving the sumset reduction. The three-path argument follows Sudakov et al. [28, Lemma 4.2 and the proof of Theorem 4.1]; compare also Tao and Vu [29, Theorem 2.29 and Exercise 6.4.10] for the graph form and retention of original edges. We give the proof with the polynomial dependence needed after mesh rounding.

Lemma 7.1 (Three paths retain graph edges)**.** *Let A, B be nonempty finite subsets of an abelian group, with $a_0 = |A|$, $b_0 = |B|$, and let $G \subset A \times B$ have at least $\theta a_0 b_0$ edges, $0 < \theta \le 1$. Put $C = \{x + y : (x, y) \in G\}$. There are $A' \subset A$, $B' \subset B$ such that*

$$|A'| \ge c\theta a_0, \qquad |B'| \ge c\theta b_0,$$

$$|G \cap (A' \times B')| \ge c\theta^2 a_0 b_0, \qquad |A' + B'| \le C_0 \theta^{-5} \frac{|C|^3}{a_0 b_0},$$

where $c, C_0 > 0$ are absolute constants.

Proof. We first choose endpoint sets with many three-edge walks between every pair. The sums along those walks will bound the full sumset, while degree counting will retain many edges of the original graph. Remove vertices of A of degree below $\theta b_0/2$, retaining a set A_0 and at least $\theta a_0 b_0/2$ edges. An ordered pair in A_0^2 is bad if its common neighborhood has fewer than τb_0 vertices, where $\tau = \theta^3/512$. For uniform $v_0 \in B$ let $U = N(v_0) \cap A_0$. Degree counting and Jensen give

$$\mathbb{E}|U| \ge \theta a_0/2, \qquad \mathbb{E}|U|^2 \ge \theta^2 a_0^2/4, \qquad \mathbb{E}\,\mathrm{Bad}(U) \le \tau a_0^2,$$

the last bound because a bad pair has probability below τ of lying in U^2. For one U, therefore,

$$|U|^2 - (32/\theta)\,\mathrm{Bad}(U) \ge 3\theta^2 a_0^2/16.$$

It follows that $|U| \ge \theta a_0/(2\sqrt{2})$ and $\mathrm{Bad}(U) \le \theta |U|^2/32$. Delete vertices having more than $\theta|U|/16$ bad partners. At most half of U is deleted; call the remainder A'. Define

$$B' = \{v \in B : |N(v) \cap A'| \ge \theta |A'|/4\}.$$

Every member of A' retains degree at least $\theta b_0/2$ in the original graph. Edges to $B \setminus B'$ number at most $\theta|A'|b_0/4$, so

$$|G \cap (A' \times B')| \geq \theta|A'|b_0/4, \qquad |B'| \geq \theta b_0/4.$$

Fix $u \in A'$ and $v \in B'$. Of the at least $\theta|A'|/4 \geq \theta|U|/8$ vertices in $N(v) \cap A'$, at most $\theta|U|/16$ form a bad pair with u. Each remaining u_1 has at least τb_0 common neighbors v_1 with u. Thus u, v have at least $c\theta^5 a_0 b_0$ original-graph walks u, v_1, u_1, v. The walk determines

$$(u + v_1, u_1 + v_1, u_1 + v) \in C^3,$$

and, for fixed u, v, this map is injective. Its alternating sum is $u + v$. Choose one representing pair for every distinct element of $A' + B'$; the corresponding triples are disjoint. Hence $c\theta^5 a_0 b_0 |A' + B'| \leq |C|^3$, as required. □

Here is the common finite-mesh input for the next three lemmas. The parameter law may be a law on indices: distinct direction bins with the same rounded parameter are not silently identified. The graph bounds refer to the corresponding original edge sets.

Assumption 7.2 (Planar graph data)**.** *Let $A, B \subset h\mathbb{Z}^2 \cap [-K, K]^2$, with $h^{-1}/K \leq |A|, |B| \leq Kh^{-1}$. A probability law ν on represented indices κ obeys, for a fixed $S > 0$,*

$$K^{-1} \leq |\kappa| \leq K, \qquad \nu(I) \leq K|I|^S \quad (h \leq |I| \leq 1). \tag{7.1}$$

For each index there is $G_\kappa \subset A \times B$ with

$$|G_\kappa| \geq h^{-2}/K, \qquad N_h\{x + \kappa y : (x, y) \in G_\kappa\} \leq Kh^{-1}. \tag{7.2}$$

The uniform law on G_κ obeys, for intervals I, I' of length h,

$$\mathbb{P}(x_1 \in I, y_1 \in I') \leq Kh^2. \tag{7.3}$$

Set $L = h^{-1}$ in this section. For each κ, round A and κB to $h\mathbb{Z}^2$. Since $|\kappa|$ and its reciprocal are subpower, a rounded vertex has at most K preimages, after increasing K; an image edge has at most K^2 preimages. The image graph therefore has subpower density, while the set of its edge sums has at most KL rounded values. Apply Lemma 7.1 to this image graph and pull back all preimages of the selected vertices. Every retained image edge has an original preimage, so the selected endpoint sets retain original graph edges. We obtain $A_\kappa \subset A$, $B_\kappa \subset B$ such that

$$|A_\kappa|, |B_\kappa| \geq L/K, \quad N_h(A_\kappa + \kappa B_\kappa) \leq KL, \quad |G_\kappa \cap (A_\kappa \times B_\kappa)| \geq L^2/K. \tag{7.4}$$

We use the covering-number forms of the Ruzsa triangle and fixed-iterate Plünnecke–Ruzsa inequalities [29]:

$$N_h(U - W)N_h(V) \leq CN_h(U - V)N_h(V - W), \tag{7.5}$$

$$N_h(mU - nU) \leq C_{m,n}K^{C_{m,n}}N_h(U) \quad \text{if } N_h(U - U) \leq KN_h(U). \tag{7.6}$$

These apply to h-cell unions after rounding, with only fixed dimensional changes of mesh. For the first inequality, choose one representing pair for each separated output $u - w$ and let v range over a separated subset of V; the rounded pair $(u - v, v - w)$ determines both the output and v, up to bounded choices. We also use the elementary covering lemma

$$N_h(U + V) \leq KN_h(V) \quad \Longrightarrow \quad U \subset \bigcup_{i=1}^{CK} (u_i + V - V + B(0, Ch)). \tag{7.7}$$

To see this, take a maximal disjoint collection of translates of $V + B(0,h)$ inside $U + V + B(0,h)$; volume comparison bounds their number and maximality gives the inclusion.

The sets in (7.4) depend on the parameter. We next choose one pair of endpoint sets that has large overlap with the sets for many parameters. The Ruzsa inequalities then control sums of fixed numbers of dilates of this one pair, including the scalar expressions needed later in the collision argument.

Lemma 7.3 (A common pair and scalar closure)**.** *Under Assumption 7.2, after restricting ν to a subpower fraction of its mass, there are a represented κ_0, fixed $A_* \subset A$, $B_* \subset B$, and $\mathcal{R} = \{\kappa/\kappa_0\}$ with the following properties. At least L^2/K original edges of G_{κ_0} lie in $A_* \times B_*$. Put $E = A_*$, $F = -\kappa_0 B_*$. For every fixed m, H, the set*

$$c_1 E + \cdots + c_m E + d_1 F + \cdots + d_m F \tag{7.8}$$

has at most $K_{m,H} L$ h-cells whenever each coefficient is a sum of at most H signed products of at most H elements of $\mathcal{R}$. A common subpower dilation of all coefficients is permitted. The restricted probability law on $\mathcal{R}$ still satisfies a bound of the form (7.1)*, with a new subpower constant and the same fixed exponent S.*

Proof. A common pair of endpoint sets. The rectangles $A_\kappa \times B_\kappa$ from (7.4) sit in the one universe $A \times B$, which has at most $K^2 L^2$ elements. Independent indices satisfy

$$\mathbb{E}_{\kappa,\kappa'} |(A_\kappa \times B_\kappa) \cap (A_{\kappa'} \times B_{\kappa'})| \geq \frac{(\mathbb{E}_\kappa |A_\kappa \times B_\kappa|)^2}{|A \times B|} \geq L^2/K^6.$$

Choose κ_0 with at least this average intersection, then retain indices with intersection at least $L^2/(2K^6)$. Their probability is at least $1/(2K^8)$. Set $A_* = A_{\kappa_0}$, $B_* = B_{\kappa_0}$. Because each individual set has at most KL points, the retained indices have

$$|A_\kappa \cap A_*|, |B_\kappa \cap B_*| \geq L/(2K^7).$$

The edge assertion follows from (7.4) at κ_0. Restricting and renormalizing the index law increases its interval bound by only a subpower factor; division by κ_0, whose norm and reciprocal are subpower, does likewise.

Products of scalar dilations. To compare the common sets with their dilates, put

$$d_h(U,V) = \log \frac{N_h(U-V)}{\sqrt{N_h(U) N_h(V)}}.$$

The triangle inequality (7.5) is the triangle inequality for d_h, up to $O(1)$. Changing mesh from h to h/K^C, or dilating both sets by a scalar with norm and reciprocal at most K^C, changes the bounds below by $O_C(\log K)$: a planar cell refines into at most $O(K^{2C})$ cells. The small cross sum in (7.4) gives $d_h(A_\kappa, -\kappa B_\kappa) = O(\log K)$. Applying the triangle inequality in both directions gives small self-differences for A_κ and B_κ. To use the large intersections, put $H_\kappa = A_\kappa \cap A_*$. Both $A_\kappa - H_\kappa$ and $H_\kappa - A_*$ are contained in the corresponding self-difference sets. Since $|H_\kappa| \geq L/(2K^7)$, the triangle inequality through H_κ gives $d_h(A_\kappa, A_*) = O(\log K)$. The same argument for the intersections of the B-sets gives $d_h(B_\kappa, B_*) = O(\log K)$. Consequently

$$d_h(E,F) = O(\log K), \qquad d_h(E, tF) = O(\log K) \quad (t \in \mathcal{R}). \tag{7.9}$$

Here the second relation uses $tF = -\kappa B_*$. To transfer this relation from F to E, use the triangle inequality through tF and then undo the common dilation:

$$\begin{aligned} d_h(E, tE) &\le d_h(E, tF) + d_h(tF, tE) + O(1) \\ &= d_h(E, tF) + d_h(F, E) + O(\log K) = O(\log K). \end{aligned}$$

The dilation error is uniform because both $|t|$ and $|t|^{-1}$ are subpower bounded. For $p_j = t_1 \cdots t_j$, $t_i \in \mathcal{R}$, put $p_0 = 1$. For each fixed j, the same argument gives the successive bounds

$$\begin{aligned} d_h(E, p_jE) &\le d_h(E, p_{j-1}E) + d_h(p_{j-1}E, p_{j-1}t_jE) + O(1) \\ &\le d_h(E, p_{j-1}E) + d_h(E, t_jE) + O_j(\log K) = O_j(\log K), \\ d_h(E, p_jF) &\le d_h(E, p_jE) + d_h(p_jE, p_jF) + O(1) \\ &\le d_h(E, p_jE) + d_h(E, F) + O_j(\log K) = O_j(\log K). \end{aligned}$$

Here the first bound is inductive, starting with the small self-difference of E. The norm and reciprocal of every p_j remain subpower bounded when j is fixed.

Signed sums of products. To include signed products, (7.6) with $m = 2, n = 0$ first gives $N_h(E + E) \le KL$, and hence $d_h(E, -E) = O(\log K)$. For $X = E$ or F,

$$\begin{aligned} d_h(E, -p_jX) &\le d_h(E, -E) + d_h(-E, -p_jX) + O(1) \\ &= d_h(E, -E) + d_h(E, p_jX) + O(1) = O_j(\log K). \end{aligned}$$

More concretely, for every signed product dilate U of E or F, of bounded length, $N_h(U - E) \le K_H L$, while $N_h(E) \ge L/K$. By (7.7) with $-E$, U lies in at most K_H translates of $E - E + B(0, K_H h)$. A fixed sum of such dilates is therefore covered by $K_{m,H}$ translates of a fixed iterate of $E - E$, with at most $K_{m,H}L$ cells by (7.6). For a coefficient $c = \sum_{\ell=1}^{q} \epsilon_\ell \alpha_\ell$, where $q \le H$, $\epsilon_\ell \in \{-1, 1\}$, and each α_ℓ is a product of at most H elements of $\mathcal{R}$, we use only the containment

$$cX \subseteq \epsilon_1\alpha_1X + \cdots + \epsilon_q\alpha_qX, \qquad X = E \text{ or } F.$$

Indeed the left side uses the same point of X in every summand on the right. Thus each of the $2m$ coefficient dilates in (7.8) is contained in a sum of at most H signed product dilates. The argument uses sums of products without taking their reciprocals, and therefore also covers cancellation, including $c = 0$. This proves (7.8). A common subpower dilation changes the cover by only a subpower factor. □

7.2 Flat scalar coefficients

We now seek random coefficients with two properties: their values belong to the scalar class just constructed, and their probabilities on intervals of length h are nearly as small as h. The first property keeps every fixed-coefficient endpoint sum in only $h^{-1-o(1)}$ mesh cells, forcing two independent samples to collide with probability at least $h^{1+o(1)}$. The second will give a smaller collision probability if two endpoint-difference vectors are sufficiently transverse. Their comparison in Lemma 7.5 will force the parallel-strip conclusion.

The analytic input is a finite-scale Fourier-decay theorem. For a probability measure λ, write

$$I_s(\lambda) = \iint |x - y|^{-s}\, d\lambda(x)\, d\lambda(y), \qquad I_s^r(\lambda) = I_s(\lambda * P_r),$$

where P_r is a fixed smooth approximate identity at scale r. Theorem 1.5 of Orponen et al. [25] says: for fixed $n \geq 2$ and $s_j \in (0,1]$ with $\sum_j s_j > 1$, there are $r_0, \epsilon_0, \tau > 0$, depending only on these fixed data, such that Borel probability measures λ_j on $[-1,1]$ satisfying $I^r_{s_j}(\lambda_j) \leq r^{-\epsilon_0}$, $0 < r < r_0$, obey

$$|(\lambda_1 \boxtimes \cdots \boxtimes \lambda_n)^\wedge(\xi)| \leq |\xi|^{-\tau} \quad (r^{-1} \leq |\xi| \leq 2r^{-1}), \tag{7.10}$$

where $\boxtimes$ is multiplication of independent real variables. The application below uses only exponents strictly below 1.

Lemma 7.4 (Flat generated coefficients)**.** *Let $\nu_{\mathcal{R}}$ be the restricted ratio law in Lemma 7.3. For every fixed $e > 0$, there are fixed positive integers n, k and a subpower $R \geq 1$ such that, with independent $T_{ij} \sim \nu_{\mathcal{R}}$,*

$$C = \sum_{i=1}^{k} \prod_{j=1}^{n} (T_{ij}/R) \tag{7.11}$$

satisfies

$$\sup_x \mathbb{P}(C \in [x-h, x+h]) \leq C_e h^{1-e}. \tag{7.12}$$

Every value in its support is an expression allowed in (7.8)*, with the common subpower dilation R^{-n}.*

Proof. Energy of the rescaled law. Take $R \geq \sup|T|$, with R and its reciprocal subpower, and let λ be the law of T/R on $[-1,1]$. Rescaling (7.1), also for intervals of length above $1/R$ by the trivial bound, gives $\lambda(B(x,u)) \leq K' u^S$ for $h \leq u \leq 1$, with $K' = h^{-o(1)}$. Fix $0 < s < \min(S,1)$ and $n \geq 2$ with $ns > 1$. For $h \leq r \leq 1$, smoothing at scale r bounds the energy kernel by $C_s \max(r, |x-y|)^{-s}$. The r-ball and dyadic annuli, using the interval cap, give

$$I^r_s(\lambda) \leq C_s K' \left(r^{S-s} + \sum_{j : 2^j r \leq 1} (2^j r)^{S-s} + 1 \right) \leq C'_s K'. \tag{7.13}$$

Fourier decay and interval probabilities. Fix $0 < e_0 < \min(e/2, 1)$. The constants r_0, ϵ_0, τ from (7.10) are now fixed. For small enough h, all $r \in [h, h^{e_0}]$ lie below r_0, and subpower K' makes (7.13) at most $r^{-\epsilon_0}$ uniformly over that interval. Applying (7.10) with $r = |\xi|^{-1}$ on successive annuli gives, for $\omega = \lambda^{\boxtimes n}$,

$$|\widehat{\omega}(\xi)| \leq |\xi|^{-\tau} \qquad (h^{-e_0} \leq |\xi| \leq h^{-1}),$$

up to an inessential fixed adjustment at the lower endpoint. Choose fixed k with $k\tau > 2$, and put $\eta = \omega^{*k}$, the law of (7.11). Choose a nonnegative integrable $\psi \geq 1$ on $[-1,1]$ whose Fourier transform is supported in $[-1,1]$, for example a suitably dilated squared-sinc majorant. Fourier inversion yields, uniformly in x,

$$\begin{aligned} \eta([x-h, x+h]) &\leq \int \psi((u-x)/h)\, d\eta(u) \\ &\leq Ch \int_{|\xi| \leq h^{-1}} |\widehat{\omega}(\xi)|^k \, d\xi \\ &\leq Ch \left(h^{-e_0} + \int_{h^{-e_0}}^{h^{-1}} t^{-k\tau}\, dt \right) \leq Ch^{1-e_0} \leq C_e h^{1-e}. \end{aligned}$$

The integers n, k are fixed for each e, before $h \to 0$, and the same R^{-n} multiplies every generated coefficient. □

7.3 Parallel strips from collision probabilities

Lemma 7.5 (Weighted parallel strips)**.** *Let E, F be the fixed sets in Lemma 7.3 and let μ be the uniform law on a subset of at least h^{-2}/K original edges of $G_{\kappa_0} \cap (A_* \times B_*)$, after the change $y \mapsto -\kappa_0 y$. Suppose the first-coordinate rectangle cap* (7.3) *holds for μ, with a possibly enlarged subpower K. There are two parallel strips, one in each endpoint plane, of width $h^{1-o(1)}$, containing $h^{o(1)}$ of the μ-mass. Their common slope is $h^{-o(1)}$-bounded.*

Proof. We prove that a fixed-power failure of strip concentration would force most sampled endpoint differences to include a transverse pair. The generated coefficients then give incompatible lower and upper bounds for the probability that two endpoint sums occupy the same mesh cell.

Transverse differences. Let $D \le K$ bound the endpoint norms and define

$$p(w) = \sup\{\mu(S_E \times S_F) : S_E, S_F \text{ parallel strips of width } w\}.$$

Fix $0 < \varepsilon < 1$ and $c > 0$. Suppose along a sequence that $p(w) \le h^c$ for $w = h^{1-\varepsilon}$. Draw m independent pairs of independent edges $(a_i, b_i), (a_i', b_i') \sim \mu$, and put

$$v_i = a_i - a_i', \qquad u_i = b_i - b_i', \qquad q = h^{\varepsilon/2}, \qquad \Delta = qw/10.$$

Summing the rectangle cap over the $O(D/h)$ possible cells for the other first coordinate gives a marginal cap $\mathbb{P}((a_i)_1 \in I) \le CKDh$ for an h-interval I. Independence of the two sampled edges therefore gives $\mathbb{P}(|v_i| < q) \le CKD(q+h)$. We claim

$$\mathbb{P}\left(\max_{r,s \in \{v_i, u_i : 1 \le i \le m\}} |r \wedge s| \le \Delta\right) \le [CKD(q+h)]^m + mp(w)^{m-1}. \tag{7.14}$$

The first term covers the event that every $|v_i| < q$. Otherwise choose an index i with $|v_i| \ge q$. Small wedges put every other v_j, u_j within width $w/2$ of the line $\mathbb{R}v_i$. Fix the anchor edges and all primed edges. For each $j \ne i$, its unprimed edge must then lie in one pair of parallel strips of width w, an event of probability at most $p(w)$. These unprimed edges are independent. A union bound over the anchor proves (7.14). Choose m fixed so large that $m\varepsilon/2 > 3$ and $c(m-1) > 3$; subpower K, D then make the right side at most h^3 for sufficiently small h.

The collision bounds. Fix $e < \varepsilon/8$. Independently draw C_i, D_i, $1 \le i \le m$, from the law in Lemma 7.4. Use the same coefficient tuple in two independent endpoint sums:

$$Z = \sum_{i=1}^m (C_i a_i + D_i b_i), \qquad Z' = \sum_{i=1}^m (C_i a_i' + D_i b_i').$$

The coefficient law in Lemma 7.4 uses the same common dilation for every coefficient. Thus, for every fixed coefficient tuple, Lemma 7.3 bounds the support of each sum by $K_m h^{-1}$ cells. Conditional on the tuple, Z, Z' are independent samples from the same distribution. Cauchy–Schwarz on those cells therefore gives

$$\mathbb{P}(|Z - Z'|_\infty \le h \mid (C_i, D_i)_{i=1}^m) \ge h/K_m. \tag{7.15}$$

Averaging over the coefficient tuple gives the same unconditional lower bound. For the upper bound, first fix the sampled endpoints. Outside the event in (7.14), choose two difference vectors r, s from the list $\{v_i, u_i\}$ with $|r \wedge s| > \Delta$, making the choice from endpoints alone. Condition on every coefficient except the two attached to r, s. They are still independent copies of the flat coefficient law, even if the two vectors have the same index i. The condition $|Z - Z'|_\infty \le h$ restricts these two real coefficients to a parallelogram of area $O(h^2/\Delta)$ and perimeter $O(Dh/\Delta)$. It meets at

most $C((1+D)/\Delta+1)$ squares of side h: count interior squares by area and boundary squares by perimeter. Independence and (7.12) give probability at most $C_e^2 h^{2-2e}$ for each square. Hence

$$\mathbb{P}(|Z-Z'|_\infty \le h) \le h^3 + K_m \frac{h^{2-2e}}{\Delta} = h^3 + K_m h^{1+\varepsilon/2-2e}. \tag{7.16}$$

This contradicts (7.15), since $\varepsilon/2-2e>0$ and K_m is subpower.

Strip width and slope. Thus for every fixed $\varepsilon, c>0$, eventually $p(h^{1-\varepsilon}) > h^c$. Take a sufficiently slow diagonal $\varepsilon, c \downarrow 0$, after the fixed integers and their thresholds at each stage have been chosen. This gives width $w = h^{1-o(1)}$ and mass $m_0 = h^{o(1)}$; no expression of unbounded length is used at any fixed tolerance. If q_0 is a unit vector along the common strip direction, the first-coordinate projection of either strip within $[-D,D]^2$ has length at most $C(D|(q_0)_1| + w)$. Summing (7.3) over the resulting pairs of h-intervals gives

$$m_0 \le CK(D|(q_0)_1| + w + h)^2.$$

Since m_0 is subpower below and K, D subpower above, while $w \to 0$ by a fixed power at each diagonal stage, $|(q_0)_1| \ge h^{o(1)}$. The common slope $(q_0)_2/(q_0)_1$ is therefore subpower bounded. □

Proposition 7.6 (Retained-edge planar rigidity)**.** *Under Assumption 7.2, after increasing to a new subpower K_*, there are a represented κ_0, at least h^{-2}/K_* original edges of G_{κ_0}, and scalars λ, c_A, c_B satisfying $|\lambda| + |c_A| + |c_B| \le K_*$ such that on those edges*

$$|x_2 - \lambda x_1 - c_A| + |y_2 - \lambda y_1 - c_B| \le K_* h.$$

Proof. The rounding and Lemma 7.1 give (7.4). Lemma 7.3 fixes κ_0, A_*, B_* while retaining h^{-2}/K edges of G_{κ_0}; its scalar closure and Lemma 7.4 meet the inputs of Lemma 7.5. Give those retained original edges their uniform law. Conditioning on a subset of size at least h^{-2}/K inflates (7.3) only by a subpower factor, since $|G_{\kappa_0}| \le K^2 h^{-2}$. The invertible dilation $y \mapsto -\kappa_0 y$ likewise changes an h-interval into at most K h-intervals, preserving the cap with another subpower factor. Lemma 7.5 selects subpower mass of these same edges. Undoing that dilation preserves the common slope and edge identities; it changes strip width and intercept by at most a subpower factor. Endpoint norms bound both intercepts once the slope is subpower bounded. Absorb all losses, including the slow diagonal, into K_*. The diagonal in Lemma 7.5 enters this new subpower function; its growth in h^{-1} is the quantitative bound used here. □

7.4 An affine–skew row from line incidences

We apply the planar rigidity theorem to a row evaluated along moving lines. In a fixed direction, the line intercept and the row's evaluation are nearly constant. Long line segments therefore give small planar projection sets. Two directions provide the endpoints of a graph, and a third direction provides its small restricted sumset.

Lemma 7.7 (Affine–skew row fit)**.** *Let $z(t) = (t, Z(t))$, $0 \le t \le 1$, be affine trajectories with velocity $(1, v)$, and suppose positions and velocities are bounded by a subpower factor K. Let μ be an incidence measure of mass at least $1/K$, dominated by K times a trajectory prior times Lebesgue measure in time. At a fine fixed-power mesh ξ, suppose a probability law π on velocities satisfies $\pi(I) \le K|I|^S$ for some fixed $S>0$ and all intervals with $\xi \le |I| \le 1$. Suppose also that every pair of position and velocity cells obeys*

$$\mu(z_\xi, v_\xi) \le K\xi^2 \pi(v_\xi).$$

Let $T \in [-K,K]^2$ be a row on the incidences. Suppose it has at most K bins of width $E \geq \xi$ per position cell z_ξ, and $T(1,v)$ lies within KE of a constant depending only on the trajectory. Then a restriction of mass at least $1/K$ satisfies

$$T = B + \lambda J z + O(KE), \qquad J(t,Z) = (-Z,t), \quad |B| + |\lambda| \leq K.$$

The subpower factor K may increase in the conclusion.

Proof. Long line segments will give small projection sets, and Proposition 7.6 will place many original projection edges in parallel strips. The first stored incidence law supplies the projection covers; a second one supplies the point-mass lower bounds needed to lift the selected edges back to incidence mass.

We may assume $\xi \leq E \leq 1$, since the conclusion is immediate for $E > 1$. Put $V_0 = (\xi/E)T$, and consider the occupied cells of $(z, V_0)_\xi$. There are at most K per position cell and at most $K\xi^{-2}$ in all. For a slope bin d of width ξ, with representative v_d, put $\mathbf{u}_d = (1, v_d)$. On point-cell centers use the linear projection

$$P_{\mathbf{u}_d}(z, V_0) = (z \cdot J\mathbf{u}_d, V_0 \cdot \mathbf{u}_d).$$

For the true direction, its first coordinate is constant on a line; the second is constant to error $K\xi$ by the row hypothesis.

Projection covers and point-mass witnesses. Let μ_{cov} denote the present incidence law, which will provide the projection covers. Delete lines with too little labeled time or incidence weight. Prior-times-time domination leaves each used trajectory with a μ_{cov}-witness time set of length at least $1/K$. It visits at least $1/(K\xi)$ witness point cells, throughout which its projection varies by at most $K\xi$. Replacing its true velocity by the bin representative changes each projection coordinate by at most $K\xi$.

For each direction bin, projected values separated by a sufficiently large multiple of $K\xi$ consequently use disjoint witness sets. Since the total number of point cells is at most $K\xi^{-2}$, its currently occurring projection can be covered by at most $K\xi^{-1}$ mesh cells. The law μ_{cov} will continue to supply these covers, even after the incidence law is restricted further.

Discard slope bins of velocity-bin weight below ξ^2. The joint cap, summed over at most K/ξ such bins and the position cells, makes their total mass negligible. Pigeonhole comparable velocity-bin weights on dense incidence mass, leaving M bins. Their total velocity-bin weight is at least $1/K$. The uniform law on them therefore has an interval bound $K\rho^{S'}$, for some fixed $S' > 0$ and $\xi \leq \rho \leq 1$.

Normalize this current incidence law to probability and freeze it as μ_{pt}. Every joint point-direction cell has mass at most $K\xi^2/M$. Delete point cells of marginal mass below ξ^2/K_1, and point-direction edges of mass below $\xi^2/(K_1 M)$, with a sufficiently large subpower K_1. The cell and edge counts bound the total mass removed by these deletions. At least $\xi^{-2}M/K$ unweighted edges remain. Deleting low-degree points leaves at least ξ^{-2}/K point cells of degree at least M/K.

Every surviving point has the stated lower mass bound in the stored μ_{pt}-marginal. Thus a later selection of ξ^{-2}/K such point cells will lift to dense mass by restoring all their μ_{pt}-incidences, regardless of which graph directions were selected. This second frozen law supplies point-mass floors; the first supplies the long-line projection covers.

Two projections and the third-direction sumset. At each surviving point there are at least M^3/K ordered triples of incident slopes separated pairwise by at least $1/K$. Indeed the interval bound makes the fraction within a sufficiently small reciprocal-subpower neighborhood negligible compared with the degree fraction. Pigeonhole the first two directions v_1, v_2. They have at least $\xi^{-2}M/K$ common-neighbor incidences with third directions also separated from them. Retain third directions

with at least ξ^{-2}/K such neighbors. There are at least M/K retained directions; use their uniform law.

Let $\mathcal{A}$ and $\mathcal{B}$ be the projections of the common-neighbor point cells under $P_{\mathbf{u}_1}$ and $P_{\mathbf{u}_2}$, rounded to the ξ-lattice. The earlier projection covers give $|\mathcal{A}|, |\mathcal{B}| \le K\xi^{-1}$. Each endpoint pair has at most K point preimages, since the basis determinant is at least $1/K$ and its entries are bounded by K. For each retained third direction, its common-neighbor points therefore define a graph $G_\kappa \subset \mathcal{A} \times \mathcal{B}$ with at least ξ^{-2}/K distinct edges.

For that direction write

$$\mathbf{u}_d = c_1\mathbf{u}_1 + c_2\mathbf{u}_2, \quad c_1 = \frac{v_2 - v_d}{v_2 - v_1}, \quad c_2 = \frac{v_d - v_1}{v_2 - v_1}, \quad \kappa = \frac{c_2}{c_1} = \frac{v_d - v_1}{v_2 - v_d}.$$

The map $P_{\mathbf{u}}$ is linear in $\mathbf{u}$. Thus, for the unrounded projections a, b of a common point,

$$a + \kappa b = c_1^{-1} P_{\mathbf{u}_d}(z, V_0).$$

Both c_i and their reciprocals are subpower bounded. Rounding and this dilation increase the covering number by at most a factor K, so the third-direction projection cover gives

$$N_\xi\{a + \kappa b : (a, b) \in G_\kappa\} \le K\xi^{-1}.$$

We verify the remaining planar hypotheses. The graph sizes imply $|\mathcal{A}|, |\mathcal{B}| \ge \xi^{-1}/K$, and both endpoint sets are bounded by K. Separation from the basis slopes gives $K^{-1} \le |\kappa| \le K$. The ratio law has the same positive-exponent interval bound up to K, since

$$|v_d - v_{d'}| \le K|\kappa_d - \kappa_{d'}|$$

on the retained choices. Finally, prescribing length-ξ intervals for the first endpoint coordinates (a_1, b_1) restricts z to at most K cells by the invertible basis map. There are only K point options above those cells. Each graph therefore has only K possible edges at such a first-coordinate rectangle. Its uniform edge law obeys the required cap, as does the uniform law on any subset of at least ξ^{-2}/K edges. Every such subset uses at least ξ^{-2}/K distinct surviving point cells.

Recovering the row and restoring incidence mass. Proposition 7.6 with $h = \xi$ gives a represented κ_0 and retains at least ξ^{-2}/K_* original edges of G_{κ_0} in parallel strips of width $K_*\xi$, with common slope and intercepts bounded by a new subpower K_*. The subsequent estimates require only that K_* be subpower, with no fixed-power dependence on the earlier K. Absorb K_* into that subpower factor.

On the corresponding point cells,

$$V_0 \cdot \mathbf{u}_i = b_i + \lambda_0 z \cdot J\mathbf{u}_i + O(K\xi), \qquad i = 1, 2.$$

The estimate depends only on (z, V_0) and remains valid throughout each selected cell. Inverting the basis and undoing $V_0 = (\xi/E)T$ gives $T = B + \lambda Jz + O(KE)$ there.

Bounded endpoint multiplicity leaves at least ξ^{-2}/K distinct selected point cells. Each has frozen μ_{pt}-mass at least ξ^2/K_1, so their whole-cell mass is dense. Restore those incidences. The row estimate applies even when their directions were not among the selected graph edges, and no incidence removed before freezing μ_{pt} is restored.

It remains to bound the coefficients after undoing the scaling. If $|\lambda|$ were power-large, the bound $|T| \le K$ would confine this dense selected z-mass to a ball of radius $K/|\lambda|$. Summing the joint cell cap over velocities bounds its mass by

$$K(K/|\lambda| + \xi)^2,$$

a contradiction. Hence $|\lambda| \le K$, and an occupied point then bounds $|B|$ by K as well. □

8 Finite narrowness and scalar projections

We treat a tangent parent with positive finite narrowness. Scalar projection first supplies full-time graphs. Wide projected intervals produce a time improvement; the other case forces a horizontal affine model, whose critical rates are treated in Section 9. Throughout this section, $K = N^{o(1)}$ is a tangent subpower factor which may increase, and dense means mass at least K^{-1} in the stated normalization. Slowly increasing lists of requirements are handled only after each fixed finite list has been established. Conversion to original-resolution outputs is still supplied by Proposition 4.11.

Remark 8.1 (Reference weights and current masses)**.** We use the reference-weight pruning of Lemma 4.8 in the following scaled form. Suppose reference weights R_e satisfy $\sum_e R_e \le K^{(0)}M$, while a current restricted law has mass at least $M/K^{(2)}$. If an appended datum has at most $K^{(3)}$ possibilities for each e, deleting entries of current mass less than $R_e/K^{(1)}$ loses at most

$$\frac{K^{(3)}}{K^{(1)}}\sum_e R_e \le \frac{K^{(0)}K^{(3)}}{K^{(1)}}M.$$

Choose the subpower $K^{(1)}$ after $K^{(0)}, K^{(2)}, K^{(3)}$ to make this negligible. If, before normalization, the numerator at such an entry and a velocity bin b is at most $K\pi(b)R_e$, the retained entry's conditional velocity probability is at most $KK^{(1)}\pi(b)$. Summing over entries gives the same type of bound at their common appended state. These are current floors obtained at a specified pruning stage. Older witnesses may separately be stored for geometric counts. Further restriction preserves an unnormalized upper bound, but does not by itself preserve a normalized conditional bound or a current lower floor. A new use of either is justified by the displayed summable calculation.

8.1 A scalar projection theorem at finite meshes

Theorem 8.2 (Scalar mesh projection)**.** *Let ν be a probability law on trajectories $(t, y(t), x(t))$, where y is scalar affine, x is scalar quadratic, and all coefficients are bounded by $K = N^{o(1)}$. Additional trajectory coordinates are permitted. Let $\mu_{\mathrm{act}} \le K\nu \otimes dt$ have mass at least $1/K$. Fix $a = N^{-s}$, $s > 0$, and a dyadic mesh $\sigma \sim_{\log,N} N^{-M}$ in (t, y), with fixed $M > s$. Suppose $c > 0$ and its reciprocal are polynomially bounded and, for some $d < 1$,*

- *each (t, y)-mesh cell and x-bin of width $p = N^{-r}$, $0 \le r \le s$, have active mass at most $K\sigma^2 cp^d$;*
- *the number of occupied (σ, σ, a)-cells, multiplied by σ^2, is at most $Kc^{-1}a^{-d}$.*

The depth hypotheses use fixed powers, with the rounding and interpolation conventions of Section 2. In particular, $d \ge 0$.

After a dense restriction and passage to a subsequence, there are labels with disjoint trajectory assignments and a common scale $\sigma/K \le \beta \le K$, up to subpower factors, such that each label has

- *a moving affine interval of width $K\beta$ containing its assigned y-trajectories throughout unit time;*
- *a quadratic $F(t, y)$ with $|x - F(t, y)| \le Ka$ throughout unit time on its assigned trajectories;*
- *assigned active mass at least $ca^d\beta/K$.*

The quadratic norm is at most K in coordinates adapted to the moving interval of scale β.

If, at $r = s$, appending a y'-bin of width N^{-u} gives the mass ceiling a further factor $KN^{-S_1 u}$ for $0 < u \le u_1$, with fixed $S_1, u_1 > 0$, then $\beta \sim_{\log,N} 1$.

Proof. Smoothing converts the finite-mesh bounds into an exact-base problem to which Corollary 6.9 applies. Its packets yield full-time fits by extrapolation, and a common choice of coefficient jitter returns these fits to the original incidence law. If $d < 0$, summing the unit-width cap over terminal support contradicts the dense total mass, proving the stated consequence $d \geq 0$.

Smoothing the mesh law. Let ν be the given trajectory prior and $\mu_{\mathrm{act}} \leq K\nu \otimes dt$ the original active incidence measure. Round the five coefficients of y, x on a mesh $\sigma' \ll \sigma a^2$ by a fixed power. Index the positive-weight rounded vectors by tags j, and let ω_j be the ν-weight of tag j. Mark a dyadic time σ-bin I for this tag when

$$\mu_{\mathrm{act}}(j \times I) \geq \omega_j \sigma / K_1.$$

Write M_j for the union of its marked bins. The unmarked aggregate mass is negligible when the subpower K_1 is large enough.

Let ρ_σ be the product of five independent uniform noise laws of width σ. The new trajectory index is (j, u), and its coefficients are the rounded vector of tag j plus u. Define the noisy prior and its marked incidence measure by

$$\nu_{\mathrm{jit}} = \sum_j \omega_j \delta_j \otimes \rho_\sigma, \qquad d\mu_{\mathrm{mark}}(j, u, t) = \mathbf{1}_{M_j}(t)\, d\nu_{\mathrm{jit}}(j, u)\, dt.$$

Thus the noise distribution is the same for every tag; the independence is between its five coordinates. The domination

$$\mu_{\mathrm{act}}\{(j, t) : t \in M_j\} \leq K \sum_j \omega_j |M_j| = K\mu_{\mathrm{mark}}(\mathrm{all})$$

shows that μ_{mark} has dense mass. This is a tempered single-component exact problem in base N, version 1 with hidden $w = x$; the mark and tag tables have polynomial sizes.

For the marked noisy law, the exact-base density in an x_p-bin is at most Kcp^d. Fix (t, y). The marked tags that can contribute have total prior weight at most $K\sigma cp^d$: their original incidences satisfy the prescribed position constraints enlarged by $K\sigma$, because motion over a time bin and the coefficient perturbation both introduce errors at most $K\sigma$. Summing the original cell caps and using the mark threshold gives this weight bound. The noisy intercept of y contributes density at most K/σ, proving the exact-base ceiling.

The independent slope noise gives a further factor $K \min(1, \rho/\sigma)$ when y' is confined to an interval of width ρ. If the optional angular estimate is assumed, the same mark calculation at terminal x-width gives

$$Kca^d N^{-S_1 u}$$

for the joint exact-base density with a velocity bin of width N^{-u}, for every tested $u < M$ in its range. The slope enlargement is much smaller than that bin. All these unnormalized ceilings survive further restrictions.

The noisy terminal support, measured by base Lebesgue measure and counting x_a-bins, is at most $Kc^{-1}a^{-d}$. Every marked noisy cell is among K neighbors of a cell carrying original active incidences. Homogenize the profiles by the counting rules of Section 2, and pass to a subsequence on which $C_0 = \lim \log_N c$. Dense mass, the terminal ceiling, and the support bound give

$$n(s) \sim_{\log, N} ca^d, \qquad f(r) \geq dr - C_0, \qquad f(s) - f(r) \leq d(s - r).$$

If the angular estimate was assumed, the resulting problem belongs to $C(d)$; choose its positive angular depth range below M. Then Corollary 6.9 gives true packets with arbitrarily small horizon

exponents on subsequences. Without that estimate, we first obtain it by cropping the slope and intercept.

Obtaining an angular gain. When no angular gain is assumed, also homogenize the joint profiles $f(r,u)$ with velocity bins through $U = 4(M+s+1)$. Put $R = (1-d)/2 > 0$, fix a sufficiently small $\epsilon > 0$, and maximize

$$(1-R)r - f(r,u) + \epsilon u, \qquad (r,u) \in [0,s] \times [0,U].$$

At $(s,0)$ its value is $C_0 + Rs$. The slope-noise ceiling bounds the objective above by $C_0 + Rr + \epsilon u - (u-M)_+$. Continuity of the profiles therefore gives a maximizer (r_0,u_0) with

$$s - r_0 \le \epsilon U/R, \qquad u_0 \le M/(1-\epsilon), \qquad f(r_0,u_0) \le f(s,0) + \epsilon U.$$

In particular $r_0 > 0$ and $u_0 + r_0 \le U$. Optimality with $u = u_0$ fixed gives a terminal slope cap $1-R$ ending at r_0. Optimality with $r = r_0$ fixed gives

$$f(r_0, u_0+b) - f(r_0,u_0) \ge \epsilon b, \qquad 0 \le b \le r_0.$$

Crop the noisy trajectories by the intercept and velocity of y at dyadic width G of order N^{-u_0}. Make each crop $\mathcal{C}$ a component with its conditional prior and component weight $p_{\mathcal{C}}$, its prior probability. Subtract the central affine motion and divide y by G. There are polynomially many crops. Discard extremely light crops, renormalize their component weights if necessary, and homogenize $p_{\mathcal{C}} = N^{-\alpha+o(1)}$ on the used components. The resulting data remain tempered and have dense mass: the reciprocal crop probabilities and the affine coordinate changes have polynomial bounds.

The new pure density exponent is

$$f'(r) = f(r,u_0) + u_0 - \alpha.$$

Indeed, given exact (t,y) and the old velocity bin, the crop label has only boundedly many possibilities, including for its intercept since $y(0) = y(t) - ty'$. Refinement by that label preserves the typical old exponent, while the within-component density is multiplied by $G/p_{\mathcal{C}}$. Summing exceptional masses with the new component weights increases their bound by only a subpower factor. The old and new velocity grids at corresponding joint precisions boundedly overlap once the crop is fixed, as in Lemma 2.13. Thus the joint formula at r_0 and relative velocity depth b replaces $f(r,u_0)$ by $f(r_0,u_0+b)$. The cropped problem belongs to $C(1-R)$ with angular exponent ϵ. Apply Corollary 6.9 again to obtain arbitrarily small time exponents.

Lifting such packets to original noisy coordinates, write β for their spatial width scale including the G factor. Their mass per block time length within the single-component noisy law is at least

$$K^{-1} p_{\mathcal{C}} N^{-f'(r_0)} (\beta/G) \ge \beta c a^d N^{-\epsilon U - o(1)}$$

by the floor in Definition 2.8 and the profile formulas. They have an explicit quadratic fit over that time block at thickness $N^{-r_0+o(1)}$ (the spatial frame change was affine in t,y). Disjointness over components/labels lifts to disjointness within each block using the crop assignments. Total selected success lifts to dense total noisy mass by the choice of component weights (any overall renormalization for discarded components changes mass by at most a subpower factor).

Full-time fits and return to the original law. In the cropped construction, let ϵ and the time exponents tend sufficiently slowly to zero. In the optional angular case, use the terminal systems obtained without cropping and let only the time exponents decrease. At each fixed tolerance, first take samples sufficiently late along the corresponding successful subsequence. The resulting diagonal

has dense noisy incidence mass in true assignments with block length $q \geq 1/K$, interval scale $\beta \leq K$, per-label mass at least $qca^d\beta/K$, and quadratic error Ka throughout the assigned block. These are tangent subpower conclusions.

The interval scale satisfies $\beta \geq \sigma/K$. The block bound confines y' to width $K\beta/q \leq K\beta$ about the moving-center slope. At exact base the graph test uses only K bins of x_a. Integrate the exact-base ceiling, including its additional slope-noise factor, over base area $Kq\beta$, and compare with the mass floor. This forces the asserted lower bound. In the optional angular case the same comparison with the augmented ceiling, at a sufficiently small positive u, excludes every fixed-power deficiency of β from one.

The quadratic predictor also has norm at most K in moving interval coordinates. Normalize time to the block and use $(y - y_c(t))/\beta$ for the spatial variable. Since $|x| \leq K$, the mass floor and base ceiling show that $|F| \leq K$ on a set of Lebesgue measure at least $1/K$ in a K-bounded box. Polynomial Remez bounds its norm. As $q \geq 1/K$, extrapolation extends this bound to unit time with a factor K. The same extrapolation extends the affine position bounds and quadratic fit errors along each assigned trajectory. The moving center is bounded by K as well, because an occupied chart contains a bounded assigned trajectory.

Keep one block's assignments carrying dense selected μ_{mark}-mass; there are only K blocks since $q \geq 1/K$. Their floors give $\sum_{\mathrm{labels}} ca^d\beta \leq K$. Averaging over the noise coordinate u in the displayed product prior fixes *one common coefficient jitter vector* for which the assigned marked-time mass, measured by $\sum_j \omega_j \delta_j \otimes dt$, is still dense. Assign every original trajectory of a selected tag to that tag's label in this block.

Undo this common jitter in every label. If its affine and quadratic components are $\Delta y(t), \Delta x(t)$, use center $y_c - \Delta y$ and predictor $F(t, y + \Delta y(t)) - \Delta x(t)$. The bounded norms and full-time fits persist for the original trajectories: their remaining rounding errors are $O(\sigma')$, negligible in moving coordinates compared with a because $\beta \geq \sigma/K$.

It remains to return from marked-time mass to μ_{act}. Every marked tag-bin that contributes at the fixed jitter has original active mass at least $\omega_j\sigma/K_1$. Restore its entire original active part. Full-time extrapolation permits this also for bins meeting block edges. The mark threshold therefore retains at least K_1^{-1} times the successful marked-time mass, so the restored original mass is dense. Finally discard labels whose restored mass is too small compared with $ca^d\beta$, using an enlarged subpower denominator and the summed numerator bound above. The discarded mass is negligible. Further dyadic pigeonholing makes β common up to subpower factors and proves the assertion.

Subpower conclusions over an ensemble of eligible models with common exponent bounds can be taken uniformly: a fixed-power failure along a sequence of individually chosen models would contradict existence with subpower losses on a further subsequence. Equivalently use the worst infimum log-loss over models at each stage then diagonalize. Theorem 8.2 can thus be used anew on dense restrictions meeting its hypotheses. □

8.2 Projection of a finite-narrowness parent

Assume now $0 < \ell < \infty$, $y = (Z, Y)$. Use tangent units and

$$v = Z', \quad V_1 = (1, v), \qquad a_s = N^{-s}, \quad h_s = N^{-ks}, \quad g_s = N^{-\ell s}, \quad b_s = h_s g_s.$$

The true label Q_s specifies a block of length $q_s \sim_{\log,N} h_s$ and a tilted box

$$|Y - Y_Q(t, Z)| \leq Kb_s, \qquad |Y' - A_Q V_1| \leq Kg_s,$$

where Y_Q is affine with derivative row A_Q and coefficients bounded by K. By the matrix description in Proposition 4.7, unit packet coordinates are interchangeable, up to K, with translated time and

Z divided by q_s, and $U = (Y - Y_Q(t, Z))/b_s$. These changes also preserve the stable local graph comparisons. Occupancy bounds the speed of the major center by K, so constant major centering within a block suffices.

Start with any substantial residual and choose a typical parent with dense success and the prepared upper bounds. A candidate in such a parent suffices. Use its restricted incidence measure, dominated by $K\nu \otimes dt$, and retain the parent velocity law π. The joint cap for $(t, Z, Y, X_p, [v]_\chi)$ is $Kp^d\pi([v]_\chi)$ per unit base volume at the required finite meshes.

We will integrate these caps over moving triangular-coordinate boxes whose inverse widths and distortions have fixed-power bounds. Do so on much finer base meshes, holding the velocity bin fixed. A scalar-bin offset may vary with the base, provided its variation in one such cell is at most K bin widths; use the prepared lists for those tests. Conditional velocity domination by $K\pi$ at deep-label and appended-point states is renewed by deleting states too light relative to their earlier reference weights. The lower masses below use the same parent normalization.

The projected groups. Choose a sufficiently fine fixed-power χ, and record

$$\alpha = ([v]_\chi, [Z(0)]_\chi), \qquad Z = Z_\alpha(t) + O(K\chi), \qquad w_\alpha = \pi([v]_\chi)\chi.$$

Here Z_α uses representative affine coefficients and w_α is a weight. At depth s, also crop $(Y(0), Y')$ at width g_s into bins γ. Let $p_{\alpha\gamma}$ be the prior mass of the joint group. Condition its prior, divide its active measure by $p_{\alpha\gamma}$, and normalize Y by subtracting the representative affine motion and dividing by g_s. We shall apply Theorem 8.2 with

$$c = \frac{w_\alpha g_s}{p_{\alpha\gamma}}, \qquad a = a_s.$$

Use a time and normalized-Y mesh $\sigma \sim N^{-M}$, with $M > s$ sufficiently large in terms of s, k, ℓ, and take χ much finer. Before conditioning, integration of the joint cap while Z tracks its specified motion bounds a projected cell and an X_p-bin by

$$Kp^d\sigma^2 g_s\chi\pi([v]_\chi) = K\sigma^2 w_\alpha g_s p^d.$$

This is the required projected density bound.

The projected support. Fix a true Q_s and a velocity bin. Its major width and time extent allow at most Kh_s/χ intercept bins for $Z(0)$. For each there are at most K compatible γ's: the row A_Q determines the slope to Kg_s, and $Y_Q(t, Z_\alpha(t))$ determines position to Kb_s at an occurrence. In one group the contributed normalized (t, Y)-support has mesh-enlarged area at most Kq_sb_s/g_s. Its strip slope is bounded by K.

The stable graph lists of Q_s allow only K terminal X_{a_s}-bins per mesh cell. Indeed the base uncertainty in packet axes is at most $K(\sigma + \chi)/b_s$, which is negligible even compared with a_s; derivatives are bounded by K on each stable ball. Weighting the support measures, including these scalar bins, by $w_\alpha g_s$ and summing therefore gives at most $Kq_sh_sb_s$ per true label. Since

$$\nu_Q \sim_{\log,N} a_s^d h_s b_s, \qquad \sum_Q q_s\nu_Q \le 1,$$

the total weighted support is at most Ka_s^{-d}.

Discard groups whose weighted support exceeds $p_{\alpha\gamma}a_s^{-d}$ by a sufficiently large subpower factor; their total prior mass is small by this sum. Also discard groups with poor conditional success. For the remaining nonempty groups, the support inequality and dense conditional mass with the unit cap give fixed-power bounds on c and c^{-1}. Both hypotheses of the scalar mesh theorem now hold.

We obtain labels Φ_s with disjoint trajectory assignments within each (α,γ)-group and across those groups, carrying unconditioned active mass at least $w_\alpha a_s^d \beta_s/K$. Here β_s is the interval scale in parent Y-coordinates, and throughout unit time

$$g_s\sigma/K \le \beta_s \le K g_s, \qquad |X - F_{\Phi_s}(t,Y)| \le K a_s.$$

The positions and slopes lie in the corresponding affine packets, and F_{Φ_s} has norm at most K in their coordinates. The scalar theorem supplies uniform subpower losses over the eligible groups. Homogenize the scale exponents by dense selection on subsequences. At finitely many depths, prepare these outputs successively on dense restrictions, renewing light marginal floors as needed; the earlier bound $\sum_{\Phi_s} w_\alpha a_s^d \beta_s \le K$ keeps the label counts available. For s in a compact positive range, the exponent M may be common and does not depend on the finer choice of χ.

To compare the two kinds of label, subdivide the true time blocks into dyadic intervals of length

$$q \sim_{\log,N} \min(h_s, \beta_s/g_s), \qquad m \sim_{\log,N} q g_s.$$

A domain D specifies one such interval, a Y'-bin of width g_s, and a Y-bin of width m at its midpoint; its moving-center slope is the binned value of Y'.

Lemma 8.3 (Comparison on common stable domains)**.** *Let Q_s, Φ_s, and D be the true labels, projected labels, and common comparison domains constructed in Section 8.2. After a dense restriction their graph predictions agree to Ka_s on every retained common comparison ball. Given (α, D, Q_s) there are at most K retained Φ_s, and conversely at most K true labels occur given (α, D, Φ_s).*

Proof. We put both lists on common domains before pairing their graphs. This allows each graph to keep its own marginal witnesses for the later list count, even when its mass in the current pair law is small.

Common domains and marginal witnesses. Fix α. For entries in either list use the reference weight

$$T_D = q w_\alpha a_s^d m.$$

The sum of these reference weights is subpower bounded. For a true label Q_s, sum over its subblocks, the possible α's, the K slope options per α, and the Kb_s/m position slices. The result is at most $Kq_s\nu_Q$. For a projected label Φ_s, there are $K\beta_s/m$ position slices per subblock and K slope choices, giving a total at most $Kw_\alpha a_s^d \beta_s$. Summing over either list gives $\sum T_D \le K$. Thus Remark 8.1 lets us discard light entries and retain marginal witnesses of mass at least T_D/K, on any dense restriction.

Evaluate the local graphs for Q_s at $Z_\alpha(t)$. They still predict X to Ka_s at the active points. The affine map from normalized D-coordinates to the true packet axes has linear part bounded by K: in the transverse row,

$$|Y'_D - A_Q(1, v_\alpha)| \le K g_s, \qquad q g_s, m \le K b_s.$$

Replacing Z by $Z_\alpha(t)$ changes packet coordinates by at most $K\chi/b_s$, so the evaluation error is negligible at width a_s.

Choose, before pairing any graphs, a grid in normalized D with step much smaller than the common inverse-subpower stable radii and their fixed holomorphic margins. The enlargement of a cell containing an occurrence then lies in that occurrence's stable neighborhood. Index each sheet by the cell and its stable ball/branch choice; there are K subentries. All witnesses for a subentry lie in its common comparison ball. Delete light marginal subentries with threshold T_D/K_1, enlarging

K_1 so that the deletion is negligible. The reference sum remains subpower, including on a putative dense set of bad pairs. This construction does not form pairwise intersections of separately chosen balls and hence does not multiply witnesses by the number of partners.

On these domains both kinds of graph are algebraic of bounded degree and have norm at most K. For Φ_s, this uses polynomial extrapolation over transverse variation at most $K\beta_s$. The original joint cap, integrated across the α-range of Z and using the graph test at width Ka_s, bounds each index-base marginal on sufficiently fine comparison cells by KT_D times normalized cell area. The Jacobian to physical (t, Y) is at most Kqm. At still finer meshes these bounds follow from the true-base caps already prepared above. Jitter the common paired event point uniformly within its tiny comparison cell. Stable margins and bounded derivatives preserve agreement of values to Ka_s, and the resulting index-base marginals satisfy the same exact density ceilings.

Matching with separate reference weights. Take norms and scaled center jets on the common comparison balls, with their fixed holomorphic margins. Suppose that graph discrepancies exceed a_s by a fixed power on dense pair mass. Pigeonhole their norm at a dyadic scale M_0, and group pairs within each (α, D)-comparison subbox by neighboring center-jet bins at that scale, through a sufficiently large fixed order. By Lemma 3.1, partner bins are boundedly many neighbors, and all graphs in a group lie within $O(M_0)$ of a common graph. An index enters boundedly many groups at a fixed discrepancy scale, since it has only one jet bin. There are $O(\log N)$ discrepancy scales and K stable subentries, so the sum of reference weights is still at most K.

Each color has at most $K(M_0/a_s)^d$ indices in the group. Indeed, all stored witnesses for one index lie in the comparison subbox and within KM_0 in X of the common graph. Sum their floors T_D/K under the cap at that width, using disjointness within each list, or its subpower multiplicity. For subpower-large M_0, the unit cap gives the same bound.

Let B be a group's current bad-pair mass, and let $W_c = \sum_{i \in \mathcal{I}_c} T_i$ be its reference total in color $c = 1, 2$. Since the total bad mass is dense and the reference total is subpower, one group satisfies

$$B \geq (W_1 + W_2)/K.$$

Restriction preserves the index-base ceiling $f_i(z) \leq KT_i$. With the separate color priors $p_i = T_i/W_c$, normalization of the pair law gives

$$\frac{f_i(z)}{B} \leq K \frac{W_c}{B} \frac{T_i}{W_c} \leq Kp_i.$$

Integrating the same ceiling also gives $B \leq KW_c$. Thus the separate reference totals are comparable up to subpower factors. With these separate reference priors and normalization by the total bad-pair mass, Corollary 3.4 applies with $R = M_0/a_s$ to exclude this group. Letting the fixed power tolerance decrease slowly gives discrepancy at most Ka_s on the retained pairings.

For fixed (α, D, Q_s), every retained projected test is now within Ka_s of one of the K graphs of Q_s on the appropriate comparison subbox. Sum its stored marginal floor under the cap at width Ka_s. There are at most K possible Φ_s. The same argument with the colors reversed gives the converse list bound. Summing over stable subentries and subboxes introduces only a factor K and uses the separate velocity-bin weights throughout. □

8.3 Three wide moving intervals

Proposition 8.4 (Wide-interval improvement)**.** *Use the finite-narrowness parent data of Section 8.2: the joint caps, conditional domination by the velocity law at fine point and path states,*

and the true Q_s graph lists. The observable X may be the native signal or any other bounded quadratic-along-lines signal with these data. At scales $s, s+u, s+c_2$, put

$$a = a_s, \qquad a_1 = aN^{-u}, \qquad a_2 = aN^{-c_2}.$$

Suppose that simultaneous projected labels Φ, Φ_1, Φ_2 of the type constructed above, including their mass bounds, occur on dense mass and use the same α's. Let $\beta_{\min}, \beta_{\max}$ be their smallest and largest interval scales. Assume $0 < c_2 < (1-d)u/4$ and

$$\beta_{\max} \le Kg_s, \qquad b_s/\beta_{\min} + h_s\beta_{\max}/\beta_{\min} \le N^{-(C_d+3)u}, \qquad C_d = 10/(1-d),$$

up to subpower factors, with common scale exponents. Take all comparison meshes, including the intercept mesh χ, sufficiently fine in fixed powers relative to these widths and h_s, b_s, a_1, so the transverse evaluation errors are negligible even at width a_1.

Then some true $Q = Q_s$ admits a quadratic predictor in the full base (t, Z, Y) for X, of error Ka_2 throughout its block on assigned trajectories, carrying original active mass at least

$$q_s\nu_Q(a_2/a)^d/K.$$

Here ν_Q is the full trajectory assignment weight.

Proof. We first show that the finer projected fits differ from a coarse fit by nearly constant offsets on a true block. We then find common nonconstant transverse coefficients, subtract them, and apply scalar projection in (t, Z).

Use first the short domains in Lemma 8.3, now at the Q_s block length and width b_s, slope bins g_s. Given $Q = Q_s, \alpha$ there are only K relevant short domains, and only K possible retained Φ's across this interval. The reverse count given α, D, Φ is at most K as well. The Φ_i sliced entries have marginal floors $K^{-1}q_s w_\alpha a_i^d b_s$, by the same width slicing since their slope uncertainties are $\le Kg_s$.

Comparing the projected fits at two domain sizes. Compare Φ_i and Φ on long domains of time length $\rho \sim_{\log,N} \beta_{\min}/\beta_{\max}$, slope-bin width $\beta_{\max}$, and moving spatial width $\beta_{\min}$. Let A be the common time-times-width Jacobian. An entry of color i has reference weight

$$T_i = Aw_\alpha a_i^d.$$

As in Lemma 8.3, slicing gives a subpower bound for their sum and retained marginal floors T_i/K. The individual index-base ceilings are KT_i.

The polynomial discrepancies on these long domains are at most $KaN^{C_d u}$ after negligible deletion. To prove this, use neighboring-jet groups at discrepancy scale M_0 and match width a, as in Lemma 8.3. The list in color i has at most $K(M_0/a_i)^d$ members. Since $a_1 = aN^{-u}$ is the smallest thickness, whenever $M_0/a \ge N^{C_d u}$,

$$(M_0/a_i)^d \le (M_0/a)^d N^{du} \le (M_0/a)^{d+d/C_d} \le (M_0/a)^{(1+d)/2}.$$

The strict inequality $d + d/C_d < (1+d)/2 < 1$ absorbs the subpower losses. Normalize the current pair law with the separate color priors T_i/W_i, using its bad mass B and the reference totals W_i exactly as in the earlier proof. Thus Corollary 3.4 excludes any fixed-power excess over $aN^{C_d u}$. This argument keeps each color's factor a_i^d in its own prior.

On the *whole* short domain of a retained pairing this difference $F_{\Phi_i} - F_\Phi$ now varies by $\ll a_i$: the normalized diameter needed in the long coordinates measured from the pairing is at most $K(h_s/\rho + b_s/\beta_{\min})$, allowing slope error Kg_s over the short time. Polynomial bounds on fixed

enlargements suffice. Likewise on the whole fiber box $Z = Z_\alpha(t) + O(K\chi)$, $Y = Y_Q(t, Z) + O(Kb_s)$ in this time interval the same estimate applies. Thus at retained events

$$X = F_\Phi(t, Y) + D_i + O(Ka_i)$$

where D_i can be a constant determined by Q, α, Φ, Φ_i (evaluate the difference at the block center on the specified central fiber); $|D_i| \le Ka$.

There are at most $K(a/a_i)^d$ possible Φ_i's and hence constants given Q, α, Φ: on each relevant short domain the participating Φ_i have their tests within Ka of F_Φ, so their marginal floors sum by the cap. Conversely given a short domain, α, Φ_i, there are only K possible paired Φ.

The reverse bound and the true-label comparison list give only K refinements of each sliced Φ_1 entry. After pruning light source entries, typical source entries consisting of $(\alpha, D, \Phi_1, \Phi, Q)$ have active weight at least $K^{-1}q_s w_\alpha a_1^d b_s$. Their normalized (t, Y) density on the short domain is $\le K$ times uniform by the finest label test and cap (at sufficiently fine meshes).

Use horizontal coordinates $z = (t - t_*, Z - Z_*)/q_s$ (constant centering per Q) and $U = (Y - Y_Q(t, Z))/b_s$. Expand

$$F_\Phi(t, Y) = f_0^\Phi(z) + f_1^\Phi(z)U + f_2^\Phi U^2.$$

Here f_1^Φ is affine and f_2^Φ constant. We seek one polynomial

$$P_{\mathrm{nc}}(z, U) = C_1(z)U + C_2U^2, \qquad e = aN^{-(1-d)u/2},$$

with C_1 affine and C_2 constant, such that on dense mass $(f_1^\Phi(z), f_2^\Phi) = (C_1(z), C_2) + O(Ke)$. Subtracting it will leave, to error $Ke \ll a_2$, only the z-dependent coefficient and the already counted offsets D_2. We can then count scalar bins without retaining U and project in (t, Z).

The coefficients f_1^Φ, f_2^Φ are bounded by K near their active fibers by the moving-interval norm, since Kb_s fits inside the interval scale. Evaluation throughout bounded Q-coordinates, or sub-power enlargements, requires only fixed-power accuracies: the original coefficients are bounded by $K\beta_{\min}^{-O(1)}$, and occupied interval centers are bounded by K. Taking χ sufficiently small therefore makes replacement of Z by $Z_\alpha(t)$ negligible.

Coefficient agreement at a fine state. Condition on a fine state consisting of Q and (z, U, X) in tiny cells. All comparison precisions, including the direction grid used below, may be prepared at arbitrarily fine fixed powers. After light-state deletion, conditional v-bin weights are bounded by $K\pi$. This follows from the K fine-state options per end label. Within the conditioning state, the projected labels may still depend on velocity.

Normalize only the whole dense current law to sample a source event, then sample a target independently conditional on the same state. Source entries $(\alpha, D, \Phi_1, \Phi, Q)$ retain their *unnormalized* Φ_1 ceilings after state deletion. Their reference weights $q_s w_\alpha a_1^d b_s$ sum to at most K. For the target velocity we instead use the conditional kernel bound $K\pi(d_v)$. We claim that the two nonconstant coefficient pairs agree to Ke except with negligible sampling probability.

Fix a source entry and a target velocity bin d_v of width χ. Their joint law is bounded by the source law times $K\pi(d_v)$. We now choose a containing list of target polynomials before considering their offsets.

First restrict source time to a sufficiently short fixed-power bin. The fine state ties the source and target base positions much more closely than χ, so the target intercept lies within $K\chi$ of $Z_\alpha(t) - tv_{d_v}$. Throughout this time bin there are only K possible target α's, uniformly in U, X. For each, the whole tilted Q-box meets only K slope/midpoint short domains, and the comparison list gives K coarse predictors Φ on their union. A finer target label chooses one of at most KN^{du} offsets D_1 for such a predictor. It does not enlarge the coarse polynomial list, and its offset does not change the derivative polynomial. The source entry already fixes its own polynomial and offset.

Use source variables t, U_α, $U_\alpha = (Y - Y_Q(t, Z_\alpha(t)))/b_s$; here $|U_\alpha| \le K$. Expand *both* candidate polynomials for these variables at source $Z_\alpha(t)$. Their values with offsets (using $i = 1$ in the offset formula) have to agree within Ka_1: fine-state agreement controls X, t, Z, Y between the events, and in the new expression only (t, Y) are used as arguments of either polynomial. A power-negligible error also covers transport of the two pairs of nonconstant coefficients to this source evaluation from their respective z's. All these errors, including those for then freezing the polynomial coefficients on the indicated tiny source time bin, can fit inside Ka_1 by the fixed power bounds, χ accuracy and still finer bins.

Put $\tau = (1 - d)u/8$. For each frozen polynomial pair with nonconstant coefficient gap at least e, the derivative gap exceeds $eN^{-\tau}$ outside a set of U_α-length $KN^{-\tau}$. If the quadratic coefficient difference is much smaller than e/K, the linear term gives this uniformly; otherwise apply the sublevel estimate for the affine derivative. Sum this exceptional length over the K coarse polynomial choices. Offsets introduce no further exceptional sets.

On the complement, monotonicity on a bounded number of intervals for each polynomial and offset bounds the total length where values can match to Ka_1 by $KN^{du}a_1N^\tau/e$. The two bounds are therefore

$$KN^{-\tau}, \qquad KN^{du}\frac{a_1N^\tau}{e} = KN^{-3(1-d)u/8}.$$

Both save a fixed power. The errors in freezing the coefficients are already within Ka_1, so a coefficient gap of Ke at the actual events is covered by these estimates. Thickening the at most KN^{du} intervals by sufficiently fine fixed-power mesh steps preserves the saving.

Integrating over the fine time bins weights each by its length, so their number contributes no additional factor. State deletion only decreases the source entry's unnormalized ceiling, and normalization of the whole dense sampling law introduces a factor K, with every individual entry kept unnormalized.

At such fine source meshes in normalized time and U_α, the source upper weight is $\le Kq_sw_\alpha a_1^d b_s$ times mesh area per entry, by the joint cap and the Φ_1 test. This holds with normalization to our overall sampling probability as well. Thus the length tests integrate against this upper source measure and the velocity probability to give a power-small multiple of each reference weight. Their sum has already been bounded. This proves the claimed coefficient agreement.

A common coefficient field. Conditional averaging now assigns a deterministic coefficient pair to each fine state, agreeing to Ke with the source pairs on dense mass. Prune light states to retain velocity domination, then choose Q with dense mass relative to $q_s\nu_Q$, using $\sum_Q q_s\nu_Q \le 1$. Its normalized z-marginal is bounded by K times area. We next show that these statewise pairs agree on dense mass with $(C_1(z), C_2)$, for one affine C_1 and constant C_2, both bounded by K.

Choose first a fixed-power grid step $\omega \ll \chi$, small enough that all coefficient evaluation errors below are $\ll e$. This choice is independent of the eventual basis, since the admissible bases and their inverses are bounded by K. Prepare a direction grid and the common conditioning-state mesh much finer than ω. These direction bins refine the width-χ bins stored in α; each meets only boundedly many of those older bins.

Now sample from π. Each typical state's compatible direction bins have total velocity-bin weight at least $1/K$; the ball cap gives separated pairs weight at least $1/K$. Averaging selects two fine bins with representatives $(1, v^1), (1, v^2)$, separated to reciprocal-subpower order, such that dense state mass has a witness from each. Use these vectors as the axes of the step-ω product grid in z, and assign each state to a nearby vertex.

Along a grid fiber parallel to either axis, its witnesses have only K possible α's over the entire fiber. Indeed the fiber prescribes an affine motion of that slope in (t, Z), with contact errors at most $Kq_s\omega$. The witness slope bin is much finer than ω, so the full-time intercept has only K bins

at width χ. The earlier comparison then gives only K possible Φ's for this fiber and Q. Their coefficient pairs restrict to an affine function and a constant on the fiber. At each compatible state one option on each fiber agrees with the assigned pair to Ke. Select one polynomial pair per fiber by averaging; dense state mass satisfies both selected fits. Keep all matching occurrences, including those with directions outside the two witness bins.

There are now at least $K^{-1}\omega^{-2}$ vertices each carrying weight at least ω^2/K with that agreement (by the z upper bound and discarding light vertices), out of a product of side counts $\le K/\omega$. A dense fraction up to powers of K of 3-by-3 product grids have all vertices present: the expected proportion of common neighbors of three independently chosen first coordinates is at least the cube of the overall density, by convexity; cube again for three independent opposite coordinates. Each side count is also $\ge 1/(K\omega)$, so deleting near coincidences (on step-ω grids) permits demanding reciprocal-subpower coordinate gaps by taking those cutoffs sufficiently small compared with this density, still with many grids. Fix the first two anchor coordinates on each side so there are $\ge \omega^{-2}/K$ target pairs completing such grids.

Bilinear interpolation from the four anchors fits the first entry at the target vertices to Ke; a constant from one anchor fits the second. Choose the four first-entry values from the selected affine polynomials on the two anchor rows. The interpolant agrees identically with those row polynomials. On a target column it is therefore within Ke of the selected column polynomial at both anchor rows. Affine interpolation along that column, using the inverse-subpower gaps, gives the target fit. The fiber constants compare through their crossings with one anchor row. Occupied anchor bounds and the gaps bound both coefficient sets by K. Grid accuracy transfers the fits to events, and the vertex floors and target count give dense fitted mass.

Removing the mixed term. It remains to remove the mixed bilinear term. Let μ_Q be the unnormalized law in Q before the fiber and anchor selections. At that stage Q belongs to the fine state with conditional velocity domination, and $\mu_Q(\text{all}) \le Km_Q$, where $m_Q = q_s\nu_Q$. For either selected axis v^j, this earlier law satisfies

$$\mu_Q\{|v - v^1| < r \text{ or } |v - v^2| < r\} \le Km_Q \sum_{j=1}^{2} \pi(B(v^j, 2r)) \le Km_Q r^{S'}.$$

This upper bound survives every subsequent restriction. If the final fitted law has mass m_Q/K_{ret}, choose the reciprocal-subpower radius r slowly enough that $KK_{\text{ret}}r^{S'} = o(1)$, first proving the assertion at fixed power radii. A dense non-axis residual remains. The law is dominated by K times the original trajectory prior restricted to the full assignment of Q, times Lebesgue measure in time. That restricted prior has total mass ν_Q, so some indexed line has retained duration at least q_s/K. Its fixed α and Q permit only K coarse predictors, so one Φ holds for duration q_s/K. On those retained times the bilinear interpolant agrees to Ke with the affine function f_1^{Φ}. In the chosen basis the components of $(1, v)$ are

$$\frac{v^2 - v}{v^2 - v^1}, \qquad \frac{v - v^1}{v^2 - v^1}.$$

Both are bounded below in absolute value by a reciprocal subpower. The quadratic coefficient of the bilinear interpolant on this line is its mixed coefficient times their product. Remez on the retained time set therefore bounds the mixed coefficient by Ke. Removing that term gives the claimed affine C_1, while the fitted second entry gives the constant C_2. The argument used the earlier unnormalized direction bound throughout.

Scalar support after subtraction. The common field just proved supplies the proposed $P_{\text{nc}} = C_1(z)U + C_2U^2$. Subtract it from X in Q. The resulting quadratic signal is subpower bounded on assigned trajectories. On a sufficiently fine z-mesh, with U no longer recorded, each cell has

at most $K(a/a_2)^d$ residual bins of width a_2 after pruning. Indeed, at fixed $[v]_\chi$, the values are $f_0^\Phi(z) + D_2 + O(Ka_2)$: there are K choices for α, Φ and $K(a/a_2)^d$ for D_2. The error Ke fits within this width because $c_2 < (1-d)u/4$.

To sum this list count over velocities, use the output state consisting of the z-cell and residual scalar bin in the chosen Q. It has only K options per sufficiently deep end label, since P_{nc} has fixed-power coefficient bounds. Delete light appended entries against the earlier end-history weights, whose sum inside Q is at most $Kq_s\nu_Q$. The retained output states then have conditional velocity domination, hence compatible velocity bins of mass under the velocity law at least $1/K$. Integrating the preceding count against π proves the claimed residual-bin count. The fine z-mesh determines the intercept to width χ up to K choices and freezes $f_0^\Phi(z)$ to Ka_2; all these meshes are prepared simultaneously.

Normalize the current mass by $q_s\nu_Q$. In unit Q-time and scaled Z, the mass per unit projected base volume in a scalar bin of width $a_2 \le p \le 1$ is at most $K(p/a)^d$, also with the angular factor from π. For a z-cell of side σ_*, integrating over $|U| \le K$ uses old base volume at most $Kq_s^2 b_s \sigma_*^2$. On still finer meshes the X-bin can follow the offset P_{nc}. Now use $\nu_Q \sim_{\log,N} a^d h_s b_s$ and the velocity law ball bound at sufficiently small positive exponents. The scaled velocity remains v.

The prior for the next projection is the true assignment of Q. Restrict it, if necessary, to trajectories with bounded coefficients on the normalized full block. Every trajectory in the final dense events has these bounds by construction. Since Q was selected heavy relative to its full trajectory weight, this conditioning changes the normalization by only a factor K. Theorem 8.2 applies to $(X - P_{\mathrm{nc}}, Z)$, with $c = a^{-d}$, terminal width a_2, and its positive angular cap. The terminal density is then $ca_2^d = (a_2/a)^d$. The support count just proved is its second hypothesis. It gives labels of full horizontal width up to K, hence a quadratic prediction in (t, Z) throughout the block, with normalized mass at least $(a_2/a)^d/K$. Restoring P_{nc} gives the asserted quadratic in the full base and the required mass.

These last comparison meshes may have arbitrarily fine fixed-power widths. The equations with errors $O(a_i)$ are evaluated on true base variables at their existing accuracy, while the prepared pure and joint caps and domination by the velocity law apply to the finer native point states. If this construction is used inside an earlier true parent after affine changes, select heavy descendants using their full assignments nested in that parent's prior, whose weights sum to at most that prior in each block. □

For the original parent observable X, subtracting this quadratic from its native polynomial gives a time-improvement candidate. Fix, for example, $s = 1$. Choose $u > 0$ sufficiently small depending on s, k, ℓ, d, followed by $0 < c_2 < (1-d)u/4$. The three projections and their mutual comparison lists with the corresponding true packets are then prepared as above. If the widths satisfy $\beta_{s'} \sim_{\log,N} g_{s'}$ at all three scales, Proposition 8.4 applies. Otherwise, after passing to a subsequence, one of the three depths, relabeled s, satisfies

$$\beta_s = N^{-B_0 + o(1)}, \qquad \ell s < B_0 \le \ell s + M.$$

We now deal with this remaining case. In choosing χ it suffices to anticipate the fixed upper bounds here and the comparison errors through these scales.

8.4 The horizontal model

The horizontal model has equation $Y' = -Z + tv$. The approximation satisfied by the normalized trajectories on the horizontal-model branch is recorded in (8.1); the corresponding horizontal and transverse box scales appear in the proposition below.

Proposition 8.5 (Reduction to the horizontal model)**.** *In the finite-narrowness setting of Section 8.2, suppose $\beta_s = N^{-B_0+o(1)}$ with $\ell s < B_0 \le \ell s + M$. On every substantial residual either there is a strip-improvement candidate, or a true-parent normalization gives $k = \ell$ and* (8.1)*, with fixed $e_0 > 0$. At sufficiently small prescribed relative depths the whole true boxes have horizontal scale H and transverse scale H^2 in* (8.2)*. Mass is normalized by the chosen parent's full assignment times its time length.*

Proof. Narrow projected intervals give short lists of transverse velocities and hence short lists of rows. Lemma 7.7 fits these rows by a constant row plus a skew row. A small skew coefficient gives a strip candidate; otherwise comparison with the true boxes either gives another strip candidate or forces $k = \ell$ and the horizontal box scales.

From projected intervals to a row list. Choose $r > s$ with $(k + \ell)r > B_0 > \ell r$. Given Q_r, its Q_s-history, and sufficiently fine horizontal position and velocity bins, there are only K possible β_s-bins for Y'. Indeed α is known up to neighbors. The Q_s, Φ_s comparison domains then have K possibilities: Q_r determines Y to Kb_r, while Q_s determines the g_s-slope bin. Translating from the occurrence time to a comparison-subblock midpoint changes the position by at most Kg_sq, its prescribed width, and $b_r \lesssim_{\log,N} \min(b_s, \beta_s)$. Thus only K midpoint bins, and hence K projected labels, are possible. Each label's full-time moving interval confines Y' to width $K\beta_s$. Take the horizontal and velocity meshes sufficiently fine also relative to χ.

Normalize the true parent $Q = Q_r$. Translate time and Z by constants and divide by q_r, keeping horizontal velocity $V_1 = (1, v)$. Subtract Y_Q and divide the transverse coordinate by $q_r g_r$. The normalized transverse velocity is

$$\frac{Y' - A_Q V_1}{g_r},$$

bounded by K, and the preceding list has accuracy $E = N^{-(B_0-\ell r)}$. A sufficiently deep true label has row

$$T = (A_{\text{deep}} - A_Q)/g_r$$

whose evaluation TV_1 predicts this velocity to $\ll E$. Fix base and velocity meshes $\xi \ll E$, fine enough for these lists.

We verify the remaining row-fit hypotheses. At a state consisting of the deep row label and fine horizontal location, use sufficiently deep histories, append this state, and delete light states to obtain conditional velocity domination by $K\pi$. Its supporting velocity bins have mass under the velocity law at least $1/K$. The angular cap therefore supplies pairs separated by a reciprocal subpower, with weight at least $1/K$ under the product velocity law. The two evaluations of the fixed row at actual witnesses bound T by K.

For any fixed such pair at a horizontal cell, the two lists for the normalized transverse velocity constrain compatible E-row bins to K possibilities by interpolation. Bounded rows and velocities make the binned-velocity error negligible. Choose one state per row-bin option and sum with the product velocity law. Since every option has witness-pair weight at least $1/K$, there are at most K row options per horizontal cell.

The horizontal joint cell cap, before dividing mass by $q_r\nu_Q$, is

$$K\xi^2 q_r \nu_Q \pi(v_\xi).$$

It follows by summing the joint state bounds with their earlier weights on which the Q-base cap holds. The velocity law interval bounds through ξ follow from the power-scale caps and interpolation. Choose a Q_r heavy relative to $q_r\nu_Q$, and use its full conditioned trajectory prior. All hypotheses of Lemma 7.7 now hold in the normalized chart.

The horizontal equation and its scales. The row fit gives normalized transverse velocity $(B + \lambda Jz) \cdot V_1 + O(KE)$ on dense mass. If $|\lambda|$ is power-small on a subsequence, this is a strip candidate: multiply by g_r and restore $A_Q V_1$. Otherwise $|\lambda| \sim_{\log,N} 1$. Subtract $B \cdot z$ from the transverse coordinate and divide by λ. Reusing (t, Z, Y) for these coordinates, we obtain

$$|Y' - Jz \cdot V_1| \le N^{-e_0} \tag{8.1}$$

for some fixed $e_0 > 0$, decreased with room if needed. The expression on the left before absolute values is constant along each trajectory.

At a small relative depth $s' > 0$, a child $Q_{r+s'}$ has block length $H = q_{r+s'}/q_r \sim_{\log,N} h_{s'}$, horizontal diameter at most KH, and a fixed row predicting transverse velocity to $KN^{-\ell s'}$. Compare this row with Jz using separated velocities in a typical fine-location state. Interpolation gives row error at most $K(N^{-\ell s'} + N^{-e_0})$. These comparisons use unnormalized domination and earlier weights summing inside Q_r to at most $Kq_r\nu_Q$; delete light refinements in that normalization.

Choose s' small enough that the N^{-e_0} error is negligible at both relative scales. After summable deletion, each used child carries normalized mass at least $H\nu_{Q_{r+s'}}/(K\nu_Q)$. Prior-times-time domination then forces its active time extent to be at least H/K. But its fixed row predicts the second component of Jz, namely t, to $KN^{-\ell s'}$. Therefore

$$N^{-ks'} \le KN^{-\ell s'},$$

so $k \ge \ell$. If $k > \ell$, replace Jz in (8.1) by its value at an active child center. The error from the horizontal diameter is smaller by a power than the nominal velocity width, giving a strip candidate at $Q_{r+s'}$. Restoring earlier coordinates multiplies velocity error by $|\lambda| g_r$; dividing the displayed child mass floor by H and restoring the ν_Q denominator gives the required candidate mass. Thus it remains only to consider $k = \ell$.

Whole horizontal boxes. Choose an occupied center (z_c, Y_c) of a used child and set

$$\widehat{z} = (z - z_c)/H, \qquad \widehat{Y} = (Y - Y_c - Jz_c \cdot (z - z_c))/H^2. \tag{8.2}$$

The true strip row is within KH of Jz_c, by the preceding two-velocity comparison at a point within horizontal distance KH of the center. With $k = \ell$, its true transverse spatial width is KH^2 up to subpower factors. The coordinate changes used to obtain the horizontal equation alter these comparisons only by K. A slope discrepancy of KH over horizontal distance KH gives a positional discrepancy of at most KH^2, so these centered coordinates describe the entire true tilted box, up to K.

The two-velocity comparison can use common-location meshes much finer than the box widths: the row is fixed, and the variation of Jz between the nearby true-velocity witnesses is absorbed by the stated row error. Prepare finitely many requested depths simultaneously, then let the grids grow slowly, discarding labels and states too light relative to their earlier weights. Time is unit time in Q_r, and the prior is its full true line assignment conditioned to probability. Descendant mass floors and deletions continue to use the full nested assignment weights even if only some assigned lines participate in the later restrictions. □

8.5 Affine approximation and residual normalization

The horizontal equation first forces the local hidden sheets to be affine to their fitted accuracy. We then subtract one parent affine fit. This gives a bounded residual with caps and state lists at arbitrarily fine prescribed meshes; those preparations will not require the horizontal equation at the same accuracy. Finally, comparison of nested affine fits gives the slope bounds used in the next section.

Lemma 8.6 (Affine approximation of the hidden sheets)**.** *Let Q_r be the selected true parent at initial tangent depth r on the horizontal-model branch of Proposition 8.5, with normalized base* (8.1) *and centered child coordinates* (8.2)*. Keep the hidden signal w in its initial tangent units of derivative scale one. For finitely many sufficiently small depths s' relative to Q_r, including $s' = 0$, there are K affine entries per true label fitting w to width $Ka_ra_{s'}$ throughout the corresponding block.*

Proof. The long line segments bound the horizontal second derivatives of each sheet. The horizontal commutators then control its full Hessian, and an absorption argument bounds its distance from an affine function.

Stable sheets and long line segments. Fix one of the required depths, including depth zero. Until we return to the parent at the end of the proof, (t, Z, Y) denotes the current label's centered coordinates from (8.2), and $z = (t, Z)$. Keep the hidden variable w in the initial tangent units of derivative scale one. Its fitted width is $\Delta = a_ra_{s'}$, up to K. In this centered frame the actual event direction is

$$\mathbf{W} = (1, v, Jz \cdot V_1 + O(KN^{-e_0}/h_{s'})).$$

Indeed subtracting the central plane and rescaling changes the horizontal row to $J(z_{\text{old}} - z_c)/H$. For Q_r itself the unit horizontal coordinates of (8.1) suffice. Choose the required depths small enough that the displayed velocity error is at most N^{-e}, with fixed $e > 0$.

Use the stable-ball preparation of Proposition 4.7 for the hidden test sheet f, choosing the branch on which the observed w lies within $K\Delta$. Above the high-curvature cutoff, the affine quadratic center already gives the required fit. Below that cutoff, the discriminant bounds and simple-root stability give balls with fixed holomorphic margins and inverse-subpower radii. Passing from packet axes to the centered boxes changes norms by at most a factor K, and the ball and branch choices number at most K per true label.

In each ball/branch entry, remove trajectories whose associated time set has relative length below a sufficiently small reciprocal subpower. Charge this deletion to the full true trajectory weights using prior-times-time domination. Also discard entries too light relative to block length times their true assignment weight. These thresholds sum within Q_r, so the deletions are negligible. The true-label base cap then gives inverse-subpower Lebesgue filling in every retained entry's centered box. Store the long ball/branch witnesses for subsequent restrictions. After the fine-state prunings below, repeat the light-entry deletion when current base filling is needed; the earlier long witnesses still give the chord estimates.

Horizontal second derivatives. Let M_f be the best sup-norm affine approximation error on the real comparison ball. By Lemma 3.1, applied with margin after subtracting an affine function, all derivatives of order at least two through any fixed required order are bounded by KM_f. Along a long witnessing chord, the Hessian evaluated twice on the actual direction $\mathbf{W}$ is at most $K\Delta$ at its events. To see this, take a complex parameter disk around the event of radius comparable to the ball radius divided by the direction length, with the fixed margin supplied above. It contains the earlier same-ball/branch times of relative parameter measure at least $1/K$. On those times, the sheet differs from the affine line signal by $K\Delta$. The one-variable form of Lemma 3.1 bounds its second derivative at the event; the parameter rescalings introduce only a factor K. The hidden values and affine coefficients may be arbitrarily large: these derivative estimates involve the fitted width Δ and the affine approximation error M_f.

Use fine states consisting only of the path label, fine base position, and ball/branch choice. They have K ambiguity given sufficiently deep labels. After the usual light-state deletion, conditional domination by the velocity law supplies three actual velocity witnesses in one state, separated by reciprocal- subpower distances and carrying the long-branch tests. Transfer their Hessian estimates to a common base point and to the horizontal directions $(1, v, Jz \cdot V_1)$. The velocity error and

sufficiently fine base meshes give an error KM_fN^{-e}, because only second and higher derivatives enter.

Put

$$\mathcal{X} = \partial_t - Z\partial_Y, \qquad \mathcal{Z} = \partial_Z + t\partial_Y.$$

For fixed v, the square $(\mathcal{X} + v\mathcal{Z})^2 f$ is exactly the frozen horizontal second directional derivative: the coefficient of ∂_Y is $-Z + tv$, and

$$(\mathcal{X} + v\mathcal{Z})(-Z + tv) = -v + v = 0.$$

In particular an affine function contributes zero before the actual direction is perturbed. Consequently the error above is controlled by M_f even when the affine part is large.

Interpolate the quadratic in v at the three separated witnesses. On the base projection of retained events in a well-filled entry,

$$\mathcal{X}^2 f, \quad \mathcal{Z}^2 f, \quad (\mathcal{X}\mathcal{Z} + \mathcal{Z}\mathcal{X})f = O\big(K(\Delta + M_f N^{-e})\big).$$

These expressions are algebraic analytic functions of bounded relation degree on the stable enlargement. The Remez and derivative bounds in Lemma 3.1 extend the estimates to the real comparison ball and to every fixed further derivative needed below.

From horizontal derivatives to the Hessian. The central commutator $\mathcal{T} = [\mathcal{X}, \mathcal{Z}] = 2\partial_Y$ recovers the remaining second derivatives. Since $\mathcal{T}$ is central,

$$\mathcal{X}^2\mathcal{Z} = \mathcal{Z}\mathcal{X}^2 + 2\mathcal{T}\mathcal{X}, \qquad \mathcal{X}\mathcal{Z}\mathcal{X} = \mathcal{Z}\mathcal{X}^2 + \mathcal{T}\mathcal{X}.$$

Consequently

$$\mathcal{X}(\mathcal{X}\mathcal{Z} + \mathcal{Z}\mathcal{X}) - 2\mathcal{Z}\mathcal{X}^2 = 3\mathcal{T}\mathcal{X},$$
$$\mathcal{Z}(\mathcal{X}\mathcal{Z} + \mathcal{Z}\mathcal{X}) - 2\mathcal{X}\mathcal{Z}^2 = -3\mathcal{T}\mathcal{Z}.$$

The preceding differentiated bounds therefore control $\mathcal{T}\mathcal{X}f$ and $\mathcal{T}\mathcal{Z}f$, and then

$$\mathcal{T}^2 f = \mathcal{X}\mathcal{T}\mathcal{Z}f - \mathcal{Z}\mathcal{T}\mathcal{X}f.$$

With $E_f = K(\Delta + M_f N^{-e})$, all the following expressions are bounded by E_f, after increasing K:

$$\begin{aligned}
\mathcal{T}^2 f &= 4f_{YY}, \\
\mathcal{T}\mathcal{X}f &= 2(f_{tY} - Zf_{YY}), & \mathcal{T}\mathcal{Z}f &= 2(f_{ZY} + tf_{YY}), \\
\mathcal{X}^2 f &= f_{tt} - 2Zf_{tY} + Z^2 f_{YY}, & \mathcal{Z}^2 f &= f_{ZZ} + 2tf_{ZY} + t^2 f_{YY}, \\
\tfrac{1}{2}(\mathcal{X}\mathcal{Z} + \mathcal{Z}\mathcal{X})f &= f_{tZ} + tf_{tY} - Zf_{ZY} - tZf_{YY}.
\end{aligned}$$

Since the comparison coordinates are bounded by K, these identities bound the entire ordinary Hessian by $K(\Delta + M_f N^{-e})$. Taylor's affine polynomial on the ball gives

$$M_f \le K\Delta + KM_f N^{-e}.$$

For sufficiently large N, $KN^{-e} < 1$, so the second term is absorbed for every finite M_f.

Extension along assigned trajectories. The resulting affine fit agrees with w to $K\Delta$ on the stored long ball/branch times of each contributing line. Their difference is affine on that line, so extrapolation gives the same bound, up to K, throughout the corresponding block. There are at most K fits per true label. Make these preparations for each fixed finite family of depths, then let the family grow slowly, keeping the earlier entry and long-line witnesses when restricting further. □

Lemma 8.7 (Residual normalization and fine-state bounds)**.** *Choose one successful parent affine fit L from Lemma 8.6 and set $x = (w - L)/a_r$. On the resulting dense restriction, x is affine along trajectories and $|x| + |x'| \le K$.*

Let μ be the restricted incidence measure in normalized Q_r coordinates, with mass divided by $q_r \nu_{Q_r}$, without further normalization by its retained mass. For any prescribed finite family of fixed-power base, scalar and velocity meshes, the preparations give

$$\begin{aligned} \mu\{(t, Z, Y) \in C,\ x \in J_p\} &\le K|C|p^d, \\ \mu\{(t, Z, Y) \in C,\ x \in J_p,\ v \in b_\chi\} &\le K|C|p^d \pi(b_\chi). \end{aligned} \tag{8.3}$$

Here C is a cell in the normalized three-dimensional base, $|C|$ is its volume, J_p is a scalar bin of fixed-power width $0 < p \le 1$, and b_χ is a velocity bin at the prescribed fixed-power width χ. The law π is the earlier parent velocity law. The bounds persist under restriction; K may depend on the finite mesh preparation, with increasing families handled by slow diagonalization.

Sufficiently deep true labels determine these base and residual bins with K choices, also beyond the accuracy range of the approximate horizontal equation.

Proof. Return to the Q_r coordinates of (8.1). The full-block parent fit makes $x = (w - L(z, Y))/a_r$ affine and bounded along the used lines, and therefore also gives $|x'| \le K$. We first recover fine residual states from the bounded analytic residuals. The reference-weight pruning then transfers the velocity bounds to those states.

Recovering fine residual states. Freeze the dense law after choosing L, on which the actual residual x is bounded. At a sufficiently deep true level R, prune light ball/branch entries against block length times their full true assignment weights. In the initial tangent mass units these references sum within Q_r to at most $Kq_r\nu_{Q_r}$. The resulting deletion bound is available before any fine x-state is constructed. The base ceiling gives inverse-subpower filling in each retained branch entry's own true axes. On those filled base points its analytic residual

$$g_R = (f_R - L)/a_r$$

is bounded by K. By Lemma 3.1, its norm and fixed derivatives are bounded by K on the stable comparison interior, even if L has enormous coefficients.

Choose R so that its graph error is below the desired residual mesh. A still deeper label supplies the relative location accuracy required by the derivative bounds. It then determines at most K residual and base bins in the true packet axes, including beyond the accuracy range of the approximate horizontal equation. This recovery precedes the reference-weight pruning that transfers conditional velocity bounds to the new bins.

Transferring the velocity bounds. Append the recovered state to each earlier end history e. In initial tangent mass units its reference weight R_e obeys $\sum_e R_e \le Kq_r\nu_{Q_r}$, and the retained velocity numerator is at most $K\pi(b)R_e$. There are at most K appended choices per history. Delete current appended entries of mass below R_e/K_1, choosing K_1 after the inverse success fractions for Q_r and L. By Remark 8.1, the reference sum bounds the mass removed by this deletion. Division on a retained entry gives conditional velocity domination; summing with its pre-deletion weight gives the corresponding joint bound at the appended state.

The normalized joint cap. The pure residual cap is Kp^d per unit normalized base volume. Indeed integrate the hidden w-cap at width pa_r about the exact offset L, with the Q_r Jacobian and prior denominator. This bound also holds for the current pre-deletion weights. Combining it with the transferred numerator gives $Kp^d\pi(b_\chi)$ at the required meshes, proving (8.3).

If a probability prior on bounded participating trajectories is needed, conditioning away the other trajectories changes the normalization by at most a factor K. The full true descendant assignment weights relative to Q_r remain the reference weights for counts and deletions. In particular, the child target mass per unit block time is $a_{s'}^d h_{s'}^3$, up to K. □

For subsequent counts let $\nu^\flat$ be the full trajectory prior conditioned on the true assignment of Q_r. It is not conditioned on the success of L. The current measure μ is dominated by $K\nu^\flat \otimes dt$. Write Q_s for a true child at depth s relative to Q_r, and $H_s = q_{r+s}/q_r \sim_{\log,N} h_s$ for its block length. The H/H^2 geometry and $k = \ell$ give

$$\nu^\flat(Q_s) \sim_{\log,N} a_s^d h_s^3, \qquad m_{Q_s} = H_s \nu^\flat(Q_s), \qquad \sum_{Q_s} m_{Q_s} \le 1. \tag{8.4}$$

These full assignment weights remain the reference weights after restrictions of the current incidence law.

Lemma 8.8 (Slopes of nested affine fits)**.** *Use the residual x from Lemma 8.7, and let s denote depth relative to Q_r. At the required small depths, the affine predictions $L_s(t, Z, Y)$ for x may be prepared on a dense restriction so that, on active boxes,*

$$|\nabla_z L_s + (\partial_Y L_s) Jz| \le K(1 + a_s/h_s), \qquad |\partial_Y L_s| \le K(1 + a_s/h_s^2). \tag{8.5}$$

The predictions have error Ka_s throughout their assigned blocks, with K subentries per true label. The coordinates (t, Z, Y) in this statement are those of the normalized parent Q_r.

Proof. The affine predictions follow by subtracting L and dividing the fits of Lemma 8.6 by a_r. We compare them along a nested depth grid and keep full-domain witnesses for each pair. For consecutive depths s_{i-1}, s_i, the paired ancestor and child affine subentries have only K options per true child label, whose history and time block are already specified. Discard pairs whose mass is below a sufficiently small subpower fraction of child block length times the child's full trajectory weight. These thresholds are summable. More explicitly, if the number of allowed pairs is K_{pair}, deletion below $\epsilon m_{\text{child}}/K_{\text{pair}}$ loses at most

$$\epsilon \sum_{\text{child}} m_{\text{child}} \le \epsilon.$$

Choose ϵ negligible relative to the current dense mass. Each fixed depth grid is pruned first, and the grid is then allowed to grow slowly.

The child base cap gives inverse-subpower filling for every retained pair's full-domain witnesses. Both affine fits are close to x there, so affine Remez bounds their difference on the whole child box by $Ka_{s_{i-1}}$. Its slope along the child's centered horizontal plane is at most $Ka_{s_{i-1}}/h_{s_i}$, and its transverse slope is at most $Ka_{s_{i-1}}/h_{s_i}^2$. The same bounds apply at an active descendant location: that location lies in the enlarged ancestor box, and Jz varies by at most Kh_{s_i} within the child box.

Start at relative depth zero with the zero prediction and the bound $|x| \le K$. Telescope over a grid with depth gaps tending sufficiently slowly to zero. For $p = 1, 2$ and $s_i \le s$,

$$a_{s_{i-1}}/h_{s_i}^p \le K(1 + a_s/h_s^p).$$

The sum of the slope increments therefore gives (8.5). All losses remain subpower on a slowly growing grid. The same witness deletion also prepares nonconsecutive ancestor/child pairs when needed. □

For later counts, one may replace Y by $Y^0 = q_0 - tp$, where

$$Z = p + tv, \qquad P = (p, q_0) = (Z(0), Y(0)),$$

with error at most N^{-e_0}, while keeping x unchanged. The bounds transfer by bounded enlargements at meshes and evaluation accuracies safely coarser than that error, including all slope amplifications. Conversely, a fit with controlled coefficients can then use the true Y in a candidate. A time improvement in these units lifts by restoring $w = L + a_r x$, with hidden derivative one.

8.6 Full-time coefficient counts in the horizontal model

For the critical and subcritical rates, the next lemma supplies one last consequence of the prepared model: when $k \le 1/2$, small scalar spacetime support concentrates the full-time affine coefficients.

Lemma 8.9 (Horizontal coefficient count)**.** *Suppose $k = \ell \le 1/2$, with the bounded residual x of Lemma 8.7. Fix sufficiently small $s > 0$ and $D = a_s^u$, with bounded $u \ge 1$. Group by a sufficiently fine velocity bin d_0 and a D-bin of $P = (Z(0), Y(0))$, and put $W_G = \pi(d_0)D^2$. On a dense restriction, typical groups have active and prior mass comparable to W_G up to K, and their (t, x)-support requires at most Ka_s^{-1-d} squares of side a_s. Their full-time affine coefficient bins can be restricted to at most Ka_s^{-d} bins of width a_s, each of active mass at least $W_G a_s^d/K$. After subdividing time at a fixed-power length $\eta \ll a_s$ and summably pruning, the group, time subbin and x_{a_s} determine a list of at most K retained coefficient bins.*

Proof. Stored full-domain witnesses bound the number of true boxes meeting each group, and hence the scalar spacetime support of the current incidences. Lemma 5.1 then clusters the affine coefficients on each dense residual; a duration lower bound transfers the prior-mass bound to current active mass. Finally, lower mass bounds in short time subbins give the list bound at a prescribed time and scalar value.

Work with the full prior $\nu^\flat$, current measure μ, and true-child reference weights in (8.4). Restrictions may be taken anew on dense mass in this normalization. The depths are sufficiently small for the horizontal equation, centered boxes, and affine preparations, simultaneously at the finitely many required scales. When $k \le 1/2$, (8.5) and the bounded parent positions bound every ordinary slope of L_s by K. We retain the joint finite-mesh caps for (t, Z, Y, x, v).

Fix small $s > 0$, $D = a_s$ (or $D = a_s^u$, $u \ge 1$ bounded, depths small enough), and group by a very fine v-bin d_0 and a P_D-bin, both trajectory data. Write $W_G = \pi(d_0)D^2$ for a group G. The direction width here and the error in (8.1) are taken smaller than D by a power. For a parameter center (p_c, q_c) and direction representative v_{d_0}, positions are within KD of the moving center

$$z_c(t) = (t, p_c + tv_{d_0}), \qquad Y_c(t) = q_c - tp_c$$

on occurrences of the group. Thus their integrated mass over any required time interval I of fixed power length (also scale one), in an x-bin of width ϱ, $a_s \le \varrho \le 1$, is at most

$$K|I|W_G\varrho^d.$$

Sum the joint caps at fixed d_0 on much finer base meshes to integrate over the moving spatial domain of area $\le KD^2$ per unit time. A band of width $K\varrho$ with t-dependent polynomial center of bounded degree and derivative $\le K$ works too. Grid changes, enlargements, and interpolation introduce a factor K. In particular $\mu(G) \le KW_G$, since x is bounded by K. Conversely one can discard groups of mass too small compared with W_G/K, and groups of too small mass fraction

relative to $\nu^{\flat}(G)$, at negligible loss of mass by $\sum_G W_G \le K$ over occurring groups (P bounded by K), and disjointness. On remaining groups active mass and prior mass therefore compare up to K with W_G.

Counting support by whole true boxes. Before the group restrictions, discard true Q_s's of current mass less than m_{Q_s}/K_1, and freeze the retained witnesses. The reference-weight bound in (8.4) makes the loss at most $1/K_1$. These are witnesses from the bounded-residual law after choosing L, so its pure base cap applies to them. The full assignment supplies the reference weights; the stored bounded-residual law supplies the witnesses. Later contact with a group will locate the entire box containing these witnesses.

Fix a time bin of length a_s. Each contributing true box meets the group's reference motion to KD. In a true time block of length H_s, all these boxes lie in one enlarged horizontal box. To verify this, note that the reference motion satisfies $Y_c' = Jz_c \cdot (1, v_{d_0})$ exactly and that $D \le h_s^2$. At contact the error therefore fits both horizontal and transverse widths. Use the plane with row $Jz_c(t_*)$ at the reference block midpoint. Each meeting box has horizontal extent Kh_s; changing its centered plane to this reference plane changes transverse positions by at most Kh_s^2. Thus its whole domain lies in the common enlargement, whose base volume is at most Kh_s^4.

The pure base cap on the frozen bounded-residual law is K per unit volume, by its unit scalar cap and $|x| \le K$. Each contributing true label has disjoint whole-domain witnesses of mass at least $h_s^4 a_s^d/K$. Their number in a meeting time block is consequently at most Ka_s^{-d}. Only boundedly many true time blocks meet the a_s-bin, up to rounding losses. Each label has K affine entries. Its ordinary slope bounds, $D \le a_s$, and the bounded speed of the reference motion imply that each entry supplies only K scalar bins on the group's occurrences during this time bin. Over all time bins, the (t, x)-support therefore needs at most Ka_s^{-1-d} squares of side a_s.

From support to heavy coefficient bins. Normalize one typical group's weights by W_G. The band cap bounds incidence mass with affine x-coefficients in a ball of radius $\varrho \ge a_s$ by $K\varrho^d$; for large ϱ, use the total mass bound. On any dense residual, discard trajectories carrying too little integrated incidence mass relative to their prior. The total prior is at most K, so a sufficiently small reciprocal-subpower threshold loses negligible mass. On the remaining trajectories the prior itself satisfies the coefficient-ball cap $K\varrho^d$.

Apply Lemma 5.1 with $p = a_s$, zero quadratic coefficient, and the support count just proved. It finds a prior mass of at least a_s^d/K in an ordinary coefficient bin, after splitting a possible subpower enlargement. The duration threshold gives the same lower bound, up to K, for the residual active mass in that bin.

This conclusion holds on every dense residual, so all but negligible mass can be assigned to heavy bins. Explicitly, if for a fixed $\delta > 0$ the bins of mass below $W_G a_s^d N^{-\delta}$ carried a nonvanishing fraction of the dense total success, summation would select one group where this bad-bin residual is dense relative to W_G. Applying the preceding clustering argument only to those bins is a contradiction. Let δ decrease slowly. We may therefore discard whole bins of mass below $W_G a_s^d/K$, retain dense mass, and bound the number of remaining coefficient bins per group by Ka_s^{-d}. The assignments are full-time coefficient assignments within the group, so they predict x to Ka_s throughout unit time.

Lists at a prescribed time and value. Finally subdivide time into bins of fixed-power length $\eta \ll a_s$. Delete coefficient entries of mass less than $\eta W_G a_s^d/K'$ in a subbin. The number of coefficient bins is at most Ka_s^{-d}, so summing these thresholds over the time partition and the groups gives negligible loss for sufficiently large subpower K'. Store the retained subbin masses for the following count.

Fix a group, a time subbin, and an x_{a_s}-bin. If a trajectory ℓ_0 in a coefficient entry meets that

scalar bin at t_0, every trajectory ℓ in the same coefficient entry and every time t in the subbin satisfy

$$|x_\ell(t) - x_{\ell_0}(t_0)| \le Ca_s + K|t - t_0| \le Ca_s + K\eta \le Ka_s.$$

Thus the entry's entire stored subbin mass lies in the enlarged scalar band. The time/band cap is $K\eta W_G a_s^d$; summing the subbin floors beneath it gives at most K compatible coefficient entries. □

9 The critical horizontal rate and the return to wide intervals

We complete the finite-narrowness argument in the horizontal model of Proposition 8.5. All depths in this section are relative to the selected true parent Q_r. Thus

$$a_s = N^{-s}, \qquad h_s = N^{-ks}, \qquad 0 < k = \ell \le 1, \qquad m_{Q_s} = H_s \nu^\flat(Q_s) \sim_{\log,N} h_s^4 a_s^d,$$

where $0 \le d < 1$, H_s is the true block length in the normalized parent, and $\nu^\flat$ is its trajectory prior. The constants denoted by K are subpower in these tangent units. We retain the full true assignment weights and the earlier witnesses when restricting the current incidence law. Restoring a counted mass to the initial tangent parent multiplies it by $q_r \nu(Q_r)$, up to the stated subpower factors.

Recall the horizontal notation $z = (t, Z)$, $V_1 = (1, v)$, and $J(t, Z) = (-Z, t)$. The normalized scalar $x = (w - L)/a_r$ is affine on each participating trajectory, with $|x| + |x'| \le K$. The derivative x' in this section always uses the full normalized Q_r-time variable. We use the horizontal equation, affine slope estimates, and coefficient counts of (8.1) and (8.5) and Lemma 8.9. In particular, with $P = (p, q_0) = (Z(0), Y(0))$,

$$Z = p + tv, \qquad Y = q_0 - tp + O(N^{-e_0}) \tag{9.1}$$

holds throughout the participating lines for a fixed $e_0 > 0$. The pure and joint caps, together with the end-history bounds for the velocity law of Proposition 4.7 and Lemma 4.8, are upper bounds on the unnormalized restricted measures. The velocity law π of v has its interval nonconcentration bound. These conventions will be used even after an individual descendant or predictor is selected.

At the critical rate $2k = 1$, we fit the derivative of x and then count its remaining trajectory-dependent intercepts. Three successive changes of trajectory show that intercept bins carrying most of the mass each occupy a substantial part of the three-dimensional base; the true descendant counts then force one heavy bin. For $2k \ne 1$, we instead construct wide projected intervals and apply Proposition 8.4. Below the critical rate this requires full-time predictors on parameter boxes of width h_s^2; Proposition 9.5 reaches that scale from the coefficient counts of the preceding section.

Here is the elementary deletion principle used to make several finite families of labels available at once.

Lemma 9.1 (Simultaneous marginal floors)**.** *Let a finite measure of mass $M \ge K^{-1}$ carry finitely many finite label families. Suppose that each family partitions its incidences, up to a subpower multiplicity, and that its proposed positive floor numerators T_j have total at most K, summed over all families. For a sufficiently large subpower K_1, one may retain at least $9M/10$ so that every remaining label has current mass at least T_j/K_1. Earlier label witnesses and trajectory priors may be kept without restriction as reference objects.*

Proof. Use the fixed-threshold deletion argument of Lemma 2.11, with thresholds T_j/K_1 and K_1 chosen so that $\sum_j T_j/K_1 < M/10$, including the subpower multiplicities. Each deleted label becomes permanently empty, so the process terminates after finitely many deletions and charges each

label at most once; the total charge is less than $M/10$. Only the current measure is restricted; earlier witnesses and trajectory priors remain unchanged. For increasing finite grids, choose their sizes slowly enough that the multiplicities and K_1 remain subpower. □

9.1 The critical rate

Proposition 9.2 (Critical-rate improvement)**.** *In the horizontal model of Proposition 8.5, with the residual bounds and coefficient counts of Lemmas 8.7–8.9, suppose $2k = 1$. Then every substantial residual yields a time-improvement candidate in the sense of Proposition 4.11. More precisely, for a sufficiently small fixed $s > 0$ and any fixed $0 < c < ks/2$, one can choose a true Q_s and an affine base predictor whose hidden-variable test has width Ka_ra_{s+c}, hidden derivative one, and counted mass at least*

$$K^{-1}q_r\nu(Q_r)m_{Q_s}N^{-dc}$$

in the initial tangent parent. Its fit holds throughout the selected block on the assigned trajectories.

Proof. An affine primitive for the trajectory derivatives. We first find an affine base function F_* whose derivative follows x' on a dense family in one Q_s. Integrating leaves an intercept for each trajectory. At the finer width $a' = a_sN^{-c}$, choose a sufficiently fine base mesh of side ω. True descendant counts bound the number of occupied pairs of base cells and intercept bins by $K\omega^{-3}(a_s/a')^d$. We will show that bins carrying almost all the fitted mass each occupy at least $K^{-1}\omega^{-3}$ base cells. Their number is therefore at most $K(a_s/a')^d$, forcing the desired heavy intercept.

Fix $s > 0$ sufficiently small that all required multiples of s lie in the horizontal preparation range and $100s < e_0$, put $a = a_s = h_s^2$, and apply Lemma 8.9 with $D = a$. Direction and time meshes d_0, η can be e.g. of sizes N^{-100s}; use finer fixed-power meshes for summations/conditionings as needed. Fix also $0 < c < ks/2$; include depths $s + c, 3s$ in the preparations before zooming in a Q_s.

At an occurrence in Q_s, use $\widehat{z} = (z - z_c)/H_s$ with a box center as in Proposition 8.5. Given $Q_s, \widehat{z}_\xi, v_\xi$ for sufficiently fine fixed power $\xi \ll a$ (finer also than needed for d_0, η), there are only K options for the x'-bin of width a, with prime still in full Q_r-time units. Indeed p is recovered using $Z = p + tv$; Y is bounded to KH_s^2 accuracy using Q_s, z; hence q_0 is predicted using $q_0 = Y + tp + O(N^{-e_0})$, with only K options for P_D altogether, since $D = a = h_s^2$. Likewise the L_s-lists restrict x_a with only K options. The time subbin and d_0 are known up to neighbors. Thus the group, time subbin and possible x_a-bins are available, and Lemma 8.9 gives the derivative list.

We next turn the derivative lists into a list of possible rows. On this law,

$$x' = G_* \cdot V_1 + O(Ka), \qquad G_* = \nabla_z L_{3s} + (\partial_Y L_{3s})Jz, \qquad |G_*| \le K,$$

by (8.1) and (8.5), and differentiating the affine fits along lines (error $\le Ka_{3s}/h_{3s}$ by the fitted-block bounds).

Use typical states of $(Q_s, \widehat{z}_\xi, [G_*]_a)$, with conditional velocity-bin cap $K\pi$ after light-entry and light-conditioning-state deletion. Indeed this appended state has only K options given sufficiently deep histories: L_{3s} has only K entries for its label, and (8.5) here controls row variation. The end-label domination with earlier masses summing inside Q_r to $\le Kq_r\nu_{Q_r}$ thus applies by Lemma 4.8.

Each surviving such state has supporting pairs of velocity bins, separated by at least $1/K$, of weight under the product velocity law $\ge 1/K$. Use conditional domination and the angular interval bound, choosing the inverse-subpower separation threshold sufficiently small relative to prior subpower losses. For any fixed such pair over $Q_s, \widehat{z}_\xi$, the two x' evaluation lists constrain the compatible row-bin centers to K options by linear interpolation (evaluation errors Ka using bounded rows and velocities). Summing this multiplicity bound with the weights under the product velocity law gives only K occurring a-bins for G_* per $Q_s, \widehat{z}_\xi$.

The remaining input for Lemma 7.7 is the joint cell cap. Pick one Q_s with dense success relative to m_{Q_s}. Integrate the original joint caps on much finer base meshes, at v_ξ, over the transverse range KH_s^2 and the K affine predictions for x at accuracy Ka. Before division by m_{Q_s}, the bound is $K(H_s\xi)^2 H_s^2 a^d \pi(v_\xi)$; afterwards it is $K\xi^2\pi(v_\xi)$.

In centered coordinates horizontal velocity remains V_1 and time has length one; the constant translations leave the velocities unchanged. Apply Lemma 7.7 to $T = G_*$, $z = \widehat{z}$, $E = a$, with line constant x' and the block-conditioned prior. On dense mass relative to m_{Q_s}, $G_* = B + \lambda J\widehat{z} + O(Ka)$, $|B| + |\lambda| \le K$. To realize this fitted row along the trajectories, observe that the affine function

$$F_*(z, Y) = (B - \lambda J z_c/H_s) \cdot z + \lambda Y/H_s$$

has slopes at most K/H_s and satisfies

$$\left| x' - \frac{d}{dt} F_*(z, Y) \right| \le Ka.$$

Indeed use (8.1) with $N^{-e_0}/H_s \ll a$; these last two trajectory derivatives are constant.

Intercepts and the upper cell count. Take $a' = aN^{-c}$. For each fitted indexed trajectory l, let $c_l = (x - F_*)(t_{\mathrm{mid}})$ be its exact midpoint value, and let $C = [c_l]_{a'}$ denote its bin. The exact value obeys

$$x(t) - F_*(z(t), Y(t)) = c_l + O(Kah_s), \qquad Kah_s \ll a',$$

throughout the block, since $c < ks/2$. Thus a sufficiently precise current base position and x-value determine only boundedly many possible C's. Replacing c_l by its bin center incurs $O(a')$, which is included in the eventual fit error.

In centered coordinates $(\widehat{z}, \widehat{Y})$, with plane Jz_c subtracted and transverse scaling by H_s^2, use a base mesh $\omega \ll a'$, say N^{-5s}. The total number of (base cell, intercept bin) pairs active here is at most

$$K\omega^{-3}(a/a')^d.$$

Indeed the number of true descendants Q_{s+c}, over all their subblocks, is at most

$$K \left(\frac{a}{a'}\right)^d \left(\frac{h_s}{h_{s+c}}\right)^4,$$

by the time-length ratios and disjoint nested trajectory assignments. Each descendant's domain has volume at most $K(h_{s+c}/h_s)^4$ in centered Q_s coordinates, including mesh enlargement. To see this, use its horizontal scale and transverse square scale at a meeting center. Its centered plane has tilt at most K in Q_s units, and ω is finer by a fixed power than its relative widths. Multiplying this volume bound by $O(\omega^{-3})$ bounds the number of cells in each box. Summing over boxes cancels the two fourth-power scale ratios. By (8.5) and the slopes of F_*, its L_{s+c}-entries specify only K intercept bins per such cell.

Three horizontal traversals and the lower cell count. We now prove the complementary estimate: intercept bins carrying almost all the fitted mass each occupy at least $K^{-1}\omega^{-3}$ base cells. Together with the preceding upper count, this will bound the number of such bins by $K(a/a')^d$.

The switching state records C, centered base position, and x, the latter two on meshes much finer than ω. It has only K possibilities per sufficiently deep true history. Indeed Lemma 8.7 gives this conclusion for the fine base and residual bins, even beyond the accuracy of the horizontal equation. Those bins determine only K possible C's, because F_* has the stated slope bounds and $x - F_* = c_l + O(Kah_s)$, with $Kah_s \ll a'$.

Apply the reference-weight pruning of Remark 8.1 to these appended states inside the selected Q_s. The earlier end-history weights satisfy $\sum_e R_e \le K m_{Q_s}$, and the velocity numerator for history e is bounded by $K\pi(b)R_e$. Deleting light history–state entries, and then any light aggregate states needed for the conditioning, retains a law μ_{last} with

$$m_{\mathrm{last}} := \mu_{\mathrm{last}}(\mathrm{all}) \ge m_{Q_s}/K$$

and conditional velocity-bin weights at most $K\pi(b)$. All reference weights and earlier full-domain witnesses are unchanged. Set $\widetilde{\mu} = \mu_{\mathrm{last}}/m_{\mathrm{last}}$. This probability is bounded by K times the true trajectory prior in Q_s times uniform centered time.

Call a trajectory good when its $\widetilde{\mu}$-marginal density relative to that prior is at least $1/K_1$, where K_1 is a sufficiently large subpower. Bad trajectories carry negligible mass, and the conditional time density on every good trajectory is at most K, after enlarging K.

Starting with an event of law $\widetilde{\mu}$, perform the following two draws three times. First draw a fresh event conditional on the current switching state. Then, conditional on its entire indexed trajectory, draw a new event on that trajectory. Both conditional resampling operations preserve $\widetilde{\mu}$. They also preserve C, which is both trajectory data and part of the switching state. Denote the rounded velocity on the ith fresh trajectory by c_i, and its new centered time by τ_i.

Retain for the estimate only paths whose three fresh trajectories are good and for which

$$|c_2 - c_1|, |c_3 - c_2|, |\tau_2 - \tau_1| \ge 1/K'.$$

Here K' can be chosen as a sufficiently large subpower so that the total failure probability is $o(1)$. This follows from stationarity, the conditional velocity bound and the velocity law's interval bound, and the time-density bound on good trajectories. We restrict paths without renormalizing any transition kernel. In particular, conditional on the full past, the subkernel of (c_i, τ_i) on good fresh trajectories is bounded by $K\pi \otimes d\tau_i$. Successive use of this bound therefore dominates the joint control–time law by $K\pi^{\otimes 3} \otimes d\tau_1 d\tau_2 d\tau_3$. The bound holds before imposing endpoint or later regularity tests.

Up to error $o(\omega)$, the three traversals follow

$$\frac{d\widehat{Z}}{d\widehat{t}} = c_i, \qquad \frac{d\widehat{Y}}{d\widehat{t}} = -\widehat{Z} + \widehat{t}c_i.$$

The refresh changes the start of each traversal only within the tiny base mesh. The horizontal equation contributes error $O(KN^{-e_0}/H_s)$, and velocity rounding contributes still less. Take the meshes sufficiently fine and s sufficiently small that the accumulated error over three traversals is $o(\omega)$. Signed time increments are allowed.

Fix the starting event (τ_0, Z_0, Y_0) and the control triple, and put $D_j = c_{j+1} - c_j$. Exact integration gives

$$\begin{aligned} Z_f &= Z_0 + c_1(\tau_3 - \tau_0) + \sum_{j=1}^{2} D_j(\tau_3 - \tau_j), \\ Y_f &= Y_0 + (-Z_0 + \tau_0 c_1)(\tau_3 - \tau_0) + \sum_{j=1}^{2} D_j\tau_j(\tau_3 - \tau_j). \end{aligned} \tag{9.2}$$

A switch at τ_j changes the transverse velocity by $\tau_j D_j$. Consequently

$$\begin{aligned} \det \frac{\partial(\tau_3, Z_f, Y_f)}{\partial(\tau_1, \tau_2, \tau_3)} &= \det \begin{pmatrix} -D_1 & -D_2 \\ D_1(\tau_3 - 2\tau_1) & D_2(\tau_3 - 2\tau_2) \end{pmatrix} \\ &= 2D_1D_2(\tau_2 - \tau_1). \end{aligned} \tag{9.3}$$

This determinant is bounded below by an inverse subpower on the retained paths. At fixed τ_3, Z_f, the internal times satisfy one nontrivial linear equation. On that line the equation for Y_f has degree at most two and cannot be constant at a regular solution. Thus there are at most two regular preimages. The change-of-variables formula bounds the time-triple measure mapping into an enlarged base cell by $K\omega^3$, uniformly in the fixed starting event and control triple. Integrating against $\pi^{\otimes 3}$ and using the preceding subkernel bound gives

$$\mathbb{P}(\text{regular endpoint in a specified base cell} \mid \text{start}) \le K\omega^3. \tag{9.4}$$

The simulation error requires only a bounded enlargement of the cell.

Write $q_C = \widetilde{\mu}(C)$, and let e_C be the failure probability averaged over the starting law conditional on C. Since the total failure probability is $\sum_C q_C e_C = \eta = o(1)$, the bins $\mathcal{B} = \{C : e_C \le 1/2\}$ have total mass at least $1 - 2\eta$. For each such bin, integrate (9.4) over its conditional starting law. Every endpoint still has intercept bin C, while regular paths have probability at least $1/2$. Hence

$$\#\{\text{base cells occupied by } C\} \ge \frac{1}{2K\omega^3}, \qquad C \in \mathcal{B}.$$

This is the lower cell count on bins carrying almost all the fitted mass.

The heavy candidate in the original mass units. The upper count of (base cell, C) pairs and the lower count for $C \in \mathcal{B}$ give $\#\mathcal{B} \le K(a/a')^d$. Since $\sum_{C\in\mathcal{B}} q_C \ge 1 - 2\eta$, one of them has

$$\mu_{\text{last}}(C) = m_{\text{last}} q_C \ge K^{-1} m_{Q_s} (a'/a)^d = K^{-1} m_{Q_s} N^{-dc}.$$

This is mass on the original residual in horizontal-problem units, before probability normalization. Its $x - F_*$ offset test matches throughout the fitted block at width Ka'. Restoring w gives the affine test $w - L - a_r(F_* + \text{offset})$ in the initial tangent parent, with hidden derivative one and width $Ka_r a'$. Restoring mass multiplies, up to K, by $q_r \nu_{Q_r}$, yielding the required counted mass $K^{-1} q_r \nu_{Q_r} m_{Q_s} N^{-dc}$. This is the time-improvement candidate at initial depth $r + s$. □

9.2 Full-time parameter estimates below the critical rate

The coefficient count gives predictors at thickness a_s on parameter boxes of width a_s. For the return to wide intervals we need the larger width $h_s^2 = a_s^{2k}$. We record the intermediate parameter widths as follows.

Definition 9.3 (Full-time parameter estimates)**.** Suppose $2k < 1$. Choose $s_{\max} > 0$ sufficiently small compared with e_0, allowing true-chart preparations up to the required fixed multiples of $s_{\max}$. Fix a common velocity grid d_0 of width $\chi \le N^{-e_0}$, a fixed power up to dyadic rounding.

For $2k \le u \le 1$, property $\mathcal{G}(u)$ means the following. For every fixed $0 < s < s_{\max}$ and every dense input, a further dense restriction and subsequence have full-time labels in the groups (d_0, P_D), where $D = a_s^u$. Their trajectory assignments are disjoint, their individual active masses are at least

$$K^{-1}\pi(d_0) D^2 a_s^d,$$

and each label has a predictor

$$x = H(t, P) + O(Ka_s)$$

throughout unit time on its assigned trajectories. The function H is affine in P, with coefficients affine in time, and is bounded by K on the group's scaled parameter box over unit time. The counted incidences may be restricted in time. All masses and depths use the fixed horizontal-model conventions; dyadic rounding and subpower factors are allowed as before.

Lemma 9.4 (Comparison with full-time predictors)**.** *Suppose $2k < 1$, $2k \le u \le 1$, and $\mathcal{G}(u)$ holds. Fix a sufficiently small depth s_0, and put $a_0 = a_{s_0}$, $D_0 = a_0^u$, and $G = (d_0, P_{D_0})$. For any prescribed finite set of nearby deeper depths in the sufficiently small depth regime of Lemma 8.8, there is a dense restriction with true labels and affine entries prepared as in that lemma at these depths and at s_0, and with their paired full-domain witnesses retained by Lemma 9.1. On this prepared input, $\mathcal{G}(u)$ as defined in Definition 9.3 supplies full-time labels H_0 at s_0. A further dense restriction satisfies the following conclusions.*

For each Q_{s_0} time block I, write

$$\mathcal{L}_{s_0}(t, P) = L_{s_0}(t, p + tv_{d_0}, q_0 - tp).$$

Every retained pairing of H_0 with (Q_{s_0}, L_{s_0}) satisfies

$$\|\mathcal{L}_{s_0} - H_0\| \le Ka_0$$

on the whole product of I and its parameter box. Given H_0, I, at most K such true-label entries occur. For each prescribed deeper depth s, the paired true affine entries also have stored full-domain witnesses of mass at least m_{Q_s}/K, and $\|L_s - L_{s_0}\| \le Ka_0$ on the whole child box. The witnesses and full trajectory assignment priors are kept when the current incidence law is further restricted.

Proof. We first compare each child affine fit with its ancestor on the whole child box. We then compare the ancestor with the full-time predictor on their common time and parameter domain; the marginal mass bounds turn this norm comparison into the claimed list bound.

Nested affine fits. For nested true labels we use Lemma 9.1 to retain full-domain witnesses for paired affine entries at s_0 and any nearby required deeper relative depths s. Prepare as in (8.5) before the small slices or full-time selections: discard light pairs compared with $m_{Q_s} = H_s\nu^\flat(Q_s)$, using the $\le K$ choices of matched subentries per true child label including its ancestor. Each remaining pair has earlier witnesses together near both fits, weighing $\ge m_{Q_s}/K$. Thus $L_s - L_{s_0}$ has norm at most Ka_0 throughout the child box by affine Remez, using inverse-subpower base coverage from the child mass and density cap there. We can prepare all such data first on sufficiently fine fixed finite grids, increasing sufficiently slowly.

Marginal masses on the common domain. Slice by $G = (d_0, P_{D_0})$ and Q_{s_0}'s time block I; put $W_G = \pi(d_0)D_0^2$. Both lists H_0 and (Q_{s_0}, L_{s_0}) have typical individual marginal floor T_{GI}/K, $T_{GI} = |I|W_G a_0^d$ there. Indeed the thresholds summed for H_0 use the earlier full-time label floors and $\sum_I |I| = 1$.

For each Q_{s_0}, d_0 there are at most $Kh_{s_0}^3/D_0^2$ parameter cells of P to test. Use true assigned positions of contributing trajectories at the time midpoint, within horizontal scale Kh_{s_0} and tilted transverse scale $Kh_{s_0}^2$. The horizontal equation error on these lines is constant in velocity through the block by (8.1). Thus, up to $K(\chi + N^{-e_0}) \ll D_0$, the spatial positions come from $P \mapsto (p + tv_{d_0}, q_0 - tp)$, where v_{d_0} is the binned velocity; this has determinant one and distortion bounded by K. A meeting parameter bin fits the K-enlarged range since $D_0 \le Kh_{s_0}^2$.

Summing the thresholds with this count, the K affine subentries, $\sum \pi(d_0) = 1$ and $\sum m_{Q_{s_0}} \le 1$ gives a total of $\le K$ for the thresholds T_{GI} in each list. The light individual entries can therefore be deleted while the pre-deletion counterparts are kept as witnesses for the heavy entries. The assignments counted within each list on these domains are disjoint up to the K-subentry losses.

On this product domain, H_0 and $\mathcal{L}_{s_0}$ both match x to Ka_0 at paired points, by (8.1) and (8.5). We use Corollary 3.4 to upgrade this agreement to the whole domain. The same calligraphic notation will be used for true affine entries at other depths.

For either index on sufficiently fine (t, P) meshes, index-base marginals obey the upper bound KT_{GI} times box-uniform, by the joint cap at d_0, x_{a_0} near its graph. Indeed use the map $P \mapsto$

$(p + tv_{d_0}, q_0 - tp)$ to true spatial positions up to $K(\chi + N^{-e_0})$, summing over much finer true base cells. Partition the time and parameter domains internally with absolute side steps on the order of $N^{-Cs_{\max}}$, with sufficiently large fixed C and still $Cs_{\max}$ well below e_0. These dominate the reconstruction error, and errors inside such cells in both graph evaluations are negligible compared with a_0. The true L_{s_0} slopes are bounded by K here; H_0 has polynomial norm $\le K$ on the full-time D_0-scaled group. Thus jitter the paired base point together within its cell to realize the density ceilings.

Norm comparison and the list bound. If a power-excess norm discrepancy occurs on dense mass, pigeonhole a comparable discrepancy M_0, larger than a_0 by a power (those bounds and an active value bound the occurring graph norms by K). On each domain group the two lists into combinations of neighboring polynomial jet bins at size M_0 in domain-scaled variables; paired discrepancies of that size permit only boundedly many neighbors (up to K). Within such groups both list counts are $\le K(M_0/a_0)^d$: sum their individual earlier T_{GI}/K floors under the cap near a common graph representative at KM_0-thickness, integrating the joint cap over that domain. Including repetitions in neighboring groups, the reference weights T_{GI} still sum to $\le K$ over all domains. Hence some group has dense bad-pair mass relative to that sum. After subtracting a common representative and jittering as above, normalization gives each index-base marginal the bound required by its normalized index weight, up to K. Corollary 3.4 with $R = M_0/a_0$ therefore excludes this group on the unit-scaled box.

Letting the power tolerance decrease slowly and pruning failures leaves discrepancy at most Ka_0. For fixed H_0, I, every other retained index has its earlier mass lower bound near the graph of H_0 at Ka_0-thickness. The same cap bounds the sum of those masses and gives the asserted K-list. □

Proposition 9.5 (Parameter bootstrap below the critical rate)**.** *If $2k < 1$, there is a fixed $s_{\max} > 0$ for which $\mathcal{G}(2k)$ holds. It can be applied anew to any dense input sequence and any subsequence. Any prescribed finite set of depths below $s_{\max}$ can be used simultaneously on a further dense restriction, with current individual floors, unchanged earlier full-domain witnesses, and the original true assignment priors.*

Proof. At $u = 1$, Lemma 8.9 gives the property by binned affine coefficients, with no P-dependence in the predictors. For each $2k < u \le 1$, we subtract a full-time predictor and use the first-jet conclusion of Proposition 5.4 to obtain short lists of residual values and slopes. Affine evolution then gives a finer full-time predictor on the same parameter box. Iterating this improvement and taking a slow diagonal produces the endpoint labels.

Fix $2k < u \le 1$, and suppose $\mathcal{G}(u)$ holds. For a target depth $s_1 < s_{\max}$, choose $s_0 = s_1/(1+\theta)$, where the fixed positive θ will be specified below. Write

$$a_0 = a_{s_0}, \qquad D_0 = a_0^u, \qquad W_G = \pi(d_0)D_0^2.$$

Prepare the nearby true depths, and then take the full-time predictors H_0 supplied by $\mathcal{G}(u)$ at s_0. Lemma 9.4 compares these predictors with the true affine entries. We will improve the fitted width from a_0 to a_{s_1} on the same parameter box.

The conditional chart and the target predictor. Choose one such paired L_{s_0}, Q_{s_0} per H_0, I losing only subpower in aggregate success. Thus L_{s_0} is time-block determined within H_0. Use typical H_0 as charts with dense conditional success and conditional prior mass denominator

$$\nu^\flat(H_0) \sim_{\log,N} W_G a_0^d.$$

Indeed restore floors or discard now-light H_0 by the threshold sums; the joint cap near its graph gives the matching incidence upper bound. Prior-times-time domination and deletion of poor success

fractions, charged to the disjoint full-time assignments, give the stated prior denominator. Fix one such chart and retain the trajectory probability

$$\nu_H = \nu^\flat(\cdot \mid H_0)$$

throughout this bootstrap step. Let λ_H be the current incidence measure in this chart divided by $\nu^\flat(H_0)$. This dense submeasure is measured relative to the fixed trajectory prior ν_H. Write

$$d\lambda_H = f_H \, d\nu_H \, dt, \qquad 0 \le f_H \le K_{\mathrm{pre}}, \qquad \delta_H = \lambda_H(\mathrm{all}) \ge 1/K,$$

where K_{pre} is subpower. In this chart put

$$y = (P - P_c)/D_0, \qquad \bar{x} = (x - H_0(t,P))/a_0.$$

Thus $y \in \mathbb{R}^2$ is static, $\bar{x}$ affine on lines with slope and range $\le K$ by the full-time prediction. On active time blocks put $F(t) = \nabla_y(\mathcal{L}_{s_0} - H_0)/a_0$, a bounded-by-$K$ time-only row by the comparison (it can be set to zero on unused blocks). Put $S(t) = \bar{x}(t) - F(t)y$.

The immediate target is a short list of residual values and slopes at one time. With $w = N^{-\theta s_0}$, where θ is chosen below, we seek a bounded time-only row $A(t)$ and a chart-dependent time t_* such that on dense labels only Kw^{-d} bin pairs

$$\big(S_l(t_*)_w, (\bar{x}_l' - A(t_*)y_l)_w\big)$$

are needed. Since y_l is static and $\bar{x}_l$ is affine in time, each pair determines a full-time prediction affine in y, with affine time coefficients and error Kw. Restoring H_0 gives thickness $Ka_0w = Ka_{s_1}$ on the unchanged parameter box:

$$D_0 = a_0^u = a_{s_1}^{u/(1+\theta)}.$$

This is how the step improves $\mathcal{G}(u)$ to $\mathcal{G}(u/(1+\theta))$. The following preparations verify the hypotheses of Proposition 5.4, whose first-jet conclusion supplies this value-and-slope list.

Mixed caps with the conditional prior fixed. Take the accuracy exponent $E = (u-2k)s_0/10$, $\zeta = N^{-E}$; in particular

$$D_0/h_{s_0+E}^2 \ll \zeta.$$

We first obtain integrated mixed caps for λ_H, then prepare time labels to obtain instantaneous caps under ν_H. For widths $r, w \in [\zeta, 1]$, integrate the joint d_0-cap at thickness $a_0 w$ about the H_0 offset. On P's moving spatial domain the result, before division by the chart denominator, is at most $K\pi(d_0)(D_0 r)^2(a_0 w)^d$ per unit time. Subdivide P as finely as needed to evaluate that offset. The absolute time and parameter meshes may have depths that are large fixed multiples of $s_{\max}$: the coefficient amplification is at most $K/(D_0 a_0)$, and the reconstruction buffer controls these errors. Still finer true meshes may be used for the integration. Consequently the normalized bound over a tested time bin I is $K|I|r^2 w^d$.

From integrated to instantaneous bounds. Set $E_l^0 = \{t : f_H(l,t) \ge \delta_H/2\}$, and let $d\rho_H^0 = \mathbf{1}_{E_l^0}(t) d\nu_H(l) dt$. The omitted λ_H-mass is at most $\delta_H/2$, and on the retained event

$$(\delta_H/2)\, d\rho_H^0 \le d(\lambda_H|_{E^0}) \le K_{\mathrm{pre}}\, d\rho_H^0.$$

In particular ρ_H^0 has mass at least $\delta_H/(2K_{\mathrm{pre}})$.

Choose one tiny time partition, fine enough for the current finite width grid that $K|I| \ll \zeta$. For each indexed line l and time bin I, retain $E_l^0 \cap I$ only if $|E_l^0 \cap I| \ge |I|/K_{\mathrm{dur}}$. Let E_l be the union of these retained sets and $d\rho_H = \mathbf{1}_{E_l}(t) d\nu_H(l) dt$. The deletion removes at most $1/K_{\mathrm{dur}}$ in indicator

mass and $K_{\mathrm{pre}}/K_{\mathrm{dur}}$ in weighted mass. Choose K_{dur} sufficiently large that $1/K_{\mathrm{dur}}$ is negligible compared with $\rho_H^0(\mathrm{all})$, and $K_{\mathrm{pre}}/K_{\mathrm{dur}}$ is negligible compared with $\lambda_H(E^0)$.

For fixed $t \in I$, let A_t be the trajectories currently labeled at t, with y in a specified r-square and $\bar{x}(t)$ in a specified w-interval. Bounded speed puts their entire retained portion of I in the same enlarged constraints. The duration floor and integrated cap therefore give

$$\frac{|I|}{K_{\mathrm{dur}}}\nu_H(A_t) \le \rho_H^0(I \times \text{enlarged constraints}) \le \frac{2}{\delta_H}\lambda_H(I \times \text{enlarged constraints}) \le K|I|r^2w^d.$$

After division by $|I|$, and absorption of subpower factors,

$$\nu_H\{l : t \in E_l,\ y_r, \bar{x}_l(t)_w \text{ in specified bins}\} \le Kr^2w^d.$$

These are line–time-bin deletions. All subsequent selections restrict E_l, keeping ν_H fixed, so this instantaneous upper bound persists. Null exceptional times and trajectories may be omitted. The displayed density comparison also returns any retained indicator mass to the weighted incidence law with a subpower factor.

Scalar support and local counts. We also have at each active t, writing $w = N^{-s'}$, $0 \le s' \le E$,

$$N_w\{S(t)\} \le Kw^{-d}.$$

This support count uses the earlier full-domain pair witnesses in the horizontal law, frozen before conditioning to H_0. Current ρ_H-labels locate the contributing boxes; the witness floors and their common-box ceiling are both measured in that earlier law. Use the true Q_s, $s = s_0 + s'$, paired with the shared Q_{s_0}, L_{s_0} on this time block. Their domains meeting the occurrences at t are all in a common enlarged horizontal box per child time interval (as in the full-time count): contact error along the G reference motion is bounded by KD_0, now much smaller than h_s^2 up to K. Sum their earlier full-domain pair witnesses of individual mass $\ge K^{-1}h_s^4a_s^d$ in this common box, satisfying the same L_{s_0}-test. The pure cap near that fit gives a total ceiling $Kh_s^4a_0^d$, hence $\le K(a_0/a_s)^d$ possibilities, counting the affine subentries as well. On a pairing, S differs within Kw from a common time-only term plus $(\mathcal{L}_s - \mathcal{L}_{s_0})/a_0$ by affine parameter dependence and the definition of F. At fixed t this last function has parameter slopes bounded by K/h_s^2: use $\|L_s - L_{s_0}\| \le Ka_0$ on the whole centered child box from the pair witnesses. Thus its variation across the D_0-box is $\le Kw$. Evaluation errors using (8.1) fit this estimate as well. This proves the support count.

At each tested width w, discard scalar bins of instantaneous ν_H-labeled mass less than w^d/K, with K sufficiently enlarged. The support count, integrated in time, bounds the mass removed by this deletion. To bound mass in an r-interval of S, $w \le r \le 1$, apply the mixed cap with scalar width r and parameter squares y_r. On each such square the bounded row $F(t)$ turns the S-constraint into at most K enlarged $\bar{x}$-bins of width r. Summing the Kr^{2+d} cap over at most Kr^{-2} parameter squares gives Kr^d. Division by the retained scalar-bin floor gives the local count $K(r/w)^d$.

Prepare the true-chart and pair witnesses on fixed finite exponent grids before applying the baseline property to that dense input. After obtaining H_0, the preceding counts hold on these grids, and the scalar-bin deletion removes a sum of arbitrarily small reciprocal-subpower masses. Let the grids densify slowly enough that their cardinalities and total losses remain subpower. Rounding and interpolation then give the local counts and caps on the whole width range. The next two preparations further restrict these labels, first on finite grids and then on a slower diagonal.

Regularity of the time-dependent row. First we require the Lipschitz comparison for F on active times, with ζ-slack. At a tested dyadic width r, sample uniform time pairs from common r-intervals (averaged with length weights). The mean prior line weight jointly labeled there is at least an

inverse-subpower by Jensen on the labeled time fractions in these intervals. But for $|t - t'| \le r$ active with $M = |F(t) - F(t')| \gg r$, this shared-line weight is $\le K(r/M)^{1-d}$.

To prove this bound, given a scalar source $S(t)_r$-bin, target $S(t')$'s for common lines lie within $K(M+r)$ of it since $\bar{x}$ has bounded speed up to K. There are $\le K(M/r)^d$ target bin choices by local scalar counts. The signal equation therefore confines the projection of static y onto the difference axis to that many intervals of length Kr/M. Each interval meets at most $K/(Mr)$ squares of side r for y ($M \le K$). With the given scalar source bin, each such square has $\le Kr^{2+d}$ prior weight at t. Summing over the $\le Kr^{-d}$ source bins proves the estimate.

Consequently the pairs with M/r larger than a sufficiently large subpower contribute negligibly to the shared-label lower bound. By averaging choose one anchor time per interval; retaining labels at times with $|F - F_{\text{anchor}}| \le Kr$ leaves dense total mass (one can count even just shared lines for this lower bound). Retain also a separated residue class of intervals with dense total mass, so that active times within distance r compare in the same interval. Iterate at all tested scales, upper bounds persisting under restriction, to obtain the comparison $K(|t - t'| + \zeta)$ using slow grids.

Slope comparison and coefficient-bin masses. Next prepare the slope comparison on spacetime $(t, y, \bar{x})$ and floors for coefficient bins under the fixed prior ν_H. At width r, a typical active y_r-group has prior and label mass of order r^2, up to K, by the cap, summable thresholds and deletion of poor success fractions. At fixed time it has at most Kr^{-d} bins of $\bar{x}_r$, by the S-count and boundedness of F.

For each line and r-time interval I_r, omit the pair if its current labeled duration is less than r/K, with K sufficiently large. Every now-occurring spacetime bin has witnesses within its Kr-enlargement for duration at least r/K. Integrating the fixed-time support count over these witnesses gives at most Kr^{-1-d} spacetime bins per y_r-group. Also retain lines with total labeled duration at least $1/K$, and denote this set by $\mathcal{L}_r$. For clustering in a typical group, use the auxiliary finite prior obtained by restricting ν_H to that group and $\mathcal{L}_r$, and dividing by r^2. Its mass is at most K. Testing a coefficient ball on those long labeled times and using the integrated spatial caps bounds this auxiliary prior's coefficient-ball masses by Kw^d, for $w \ge r$. The reference probability ν_H itself remains fixed.

Lemma 5.1 now permits retaining joint bins $B_r = (y_r, [\bar{x} \text{ coefficients}]_r)$, each of current ρ_H-mass at least r^{2+d}/K. Indeed apply clustering anew to any dense putative residual of lighter bins in typical groups, with the same support and long-duration preparations. The cluster's auxiliary prior floor, multiplied by the long-duration threshold, gives a heavy ordinary active bin, a contradiction. At reciprocal-subpower widths all parameter counts are already subpower.

Next prune the distinct entries (B_r, I_r), where I_r is an r-time interval, below the current mass floor $|I_r| r^{2+d}/K$. The heavy-bin count makes these thresholds summable. If such an entry meets a given spacetime r-cube, its entire earlier subentry has $(y, \bar{x})$ within Kr of that cube: coefficients differ by $O(r)$ inside B_r, and motion over I_r is bounded by Kr. Summing its full subentry floor under the integrated spatial cap gives at most K possible coefficient bins, hence slope bins, for the cube.

In each cube select one slope bin with dense aggregate success, then retain a separated residue class of cubes so that selected points within distance r lie in the same cube. For example, color all coordinate indices modulo a fixed integer greater than two and keep a dense class. Iterating over the tested scales gives the residual velocity comparison with ζ-slack, since $y' = 0$.

For each participating full parameter bin B_r, the active floor just obtained implies $\nu_H(B_r) \ge r^{2+d}/K$, since time has length one. This fixed-prior floor survives later label restrictions. The

absolute instantaneous cap therefore gives

$$\frac{\nu_H\{l \in B_r : t \in E_l,\ (y_l, \bar{x}_l(t)) \text{ in a } q\text{-cube}\}}{\nu_H(B_r)} \le K\frac{q^{2+d}}{r^{2+d}}, \qquad \zeta \le q < r.$$

Only static y and the affine $\bar{x}$-coefficients are binned; hidden trajectory indices remain in ν_H. Nested dyadic grids allow interpolation using a participating finer bin's prior floor inside its containing bin. The finite iterations and slow diagonal retain dense mass, with the finest-width comparisons also controlling smaller distances. These are all the hypotheses of Proposition 5.4, under the unchanged prior ν_H and the current labels E_l.

One bootstrap step. The first-jet conclusion of Proposition 5.4, applied to the normalized signals $\bar{x}, y$, gives on dense labels in the chart a slope row $A(t)$ bounded by K and

$$H\big((\bar{x}' - A(t)y)_w \mid t, S(t)_w\big) = o(\log N)$$

for $w = N^{-\theta s_0}$, taking $\theta = (1-d)(u-2k)/10000$; here sample trajectory and time independently before imposing the retained labels. The finest affine time-fit exponent can be $E/4$, with dyadic rounding. Indeed, for $H_* = N^{-E/4}$, the floor satisfies $\zeta = N^{-E} \ll H_*^3$, and

$$\theta s_0 = \frac{(1-d)E}{1000} < \frac{(1-d)E}{256}.$$

These inequalities verify the floor and reading conditions of the first-jet conclusion. Moreover, $D_0/h_{s_0+E}^2 = N^{-(10-2k)E} \ll \zeta$, so the preceding support comparison has a fixed power saving. All dense successes and subpower bounds can be taken uniform across the typical charts with the stated common inputs (or test against sequences of such charts). The negligible log entropy permits retaining, given time and scalar bin, subpower lists of the indicated residual bin (discard very small conditional probabilities). The scalar support bound therefore gives at most Kw^{-d} total bin pairs at each fixed active time. Discard also lines with too short total labeled duration, then pick by averaging a single time t_* within the chart with dense line weight thus covered. These trajectories account for dense labeled incidence mass using their inverse-subpower labeled durations, also restoring earlier weights with a subpower factor.

Assign these lines disjointly by the bin pair at t_*, fixed within their earlier full-time H_0 label. For bin centers b, b', predict throughout by

$$\bar{x}(t) = b + F(t_*)y + (t - t_*)\big(b' + A(t_*)y\big) + O(Kw)$$

using exact affine evolution. Restoring H_0 gives the prescribed predictor type bounded by K on the unchanged full-time scaled parameter box, at thickness $a_0 w = a_{s_1}$. Typical heavy entries among the new labels have the required active mass lower $\pi(d_0)D_0^2 a_{s_1}^d/K$: per earlier chart the prior denominator compares with $W_G a_0^d$, and there are only Kw^{-d} new labels, so drop light ones by summing. This proves $\mathcal{G}(u/(1+\theta))$ at the arbitrary target $s_1 < s_{\max}$. All parameter and time accuracies needing reconstruction here used only fixed multiples of $s_{\max}$ in absolute depth, independently of how small $u - 2k > 0$ is; each fixed application has positive fixed jet exponents before any endpoint diagonal.

A fixed reconstruction buffer. The accuracy requirements in the preceding step are uniform in the distance $u - 2k$ in the following precise sense. The normalized parameter and scalar widths are at least $\zeta = N^{-E}$, with $E \le s_{\max}/10$. The first-jet argument uses the fixed-order fitting scales $H^{1/4}$, $H^{1/8}$, $H^{(1-d)/32}$, and interpolation widths above this floor. As E decreases, these meshes become

coarser. The largest coefficient amplification when returning the baseline normalized predictor to physical parameter coordinates is

$$\frac{K}{D_0 a_0} \le K N^{2 s_{\max}}, \qquad 2k < u \le 1.$$

All further differentiated tests have fixed degree and order. Thus one absolute exponent $C_0 s_{\max}$, for a fixed sufficiently large C_0, supplies the internal time and parameter meshes for every finite bootstrap step. Choose $s_{\max}$ at the outset so that $C_0 s_{\max} < e_0/2$, enlarging C_0 first to include the fixed reconstruction and evaluation amplifications. The N^{-e_0} reconstruction error is then below the meshes and their required scalar accuracies by a fixed power. The number of tested widths, coloring losses, and finite-iteration constants may depend on the step; they do not require an exponent proportional to $1/E$. Finer true-base meshes used only to integrate an already established cap do not invoke reconstruction at those meshes.

Predictors at the endpoint parameter width. Put $\alpha = (1-d)/10000$. The recurrence is

$$u_{j+1} = \frac{u_j}{1 + \alpha(u_j - 2k)}, \qquad u_0 = 1.$$

It satisfies

$$u_{j+1} - 2k = \frac{(u_j - 2k)(1 - 2k\alpha)}{1 + \alpha(u_j - 2k)} > 0,$$
$$u_j - u_{j+1} = \frac{\alpha u_j (u_j - 2k)}{1 + \alpha(u_j - 2k)} > 0.$$

The limit must therefore be $2k$. Fix a desired target depth $s < s_{\max}$ and a dense input sequence. For each finite j, perform its finite sequence of bootstrap steps on a further subsequence. Absorb every loss, grid count and norm bound in $K_j(N)$. Take N sufficiently late on that subsequence that $\log K_j(N)/\log N < 1/j$, that the retained relative mass is at least $N^{-1/j}$, and that every required finite accuracy buffer holds. A diagonal with $j = j(N) \to \infty$ chosen only after these thresholds has subpower total losses. This construction is made for the given input sequence; it asserts no uniform diagonal over all possible dense inputs.

Let $D_j = a_s^{u_j}$, $D_* = a_s^{2k}$, and $R_j = D_*/D_j$. Then $\log_N R_j = s(u_j - 2k) \to 0$. Keep the old predictor labels separate, even if several lie in the same new D_*-square, and split each assignment by that new grid. An old square meets only boundedly many new squares. Since the predictor is affine in the two parameter coordinates, its norm on a meeting new square over unit time is at most $CK_j(1 + R_j)$. Predictions on its assigned trajectories are unchanged.

At the fixed bootstrap step j, index the old predictor labels by ι, and let $d_0(\iota)$ be the velocity bin of label ι. Set $T_{j,\iota} = \pi(d_0(\iota)) D_j^2 a_s^d$. The earlier active floors give $\sum_\iota T_{j,\iota} \le K_j$. The required endpoint numerator for each split of label ι is $R_j^2 T_{j,\iota}$; hence the sum of these new numerators is at most $CK_j R_j^2$. This is subpower. Choose the new floor constant larger than that sum divided by a small inverse-subpower fraction of the current total mass. Removing light split labels keeps dense mass and yields precisely the $\pi(d_0) D_*^2 a_s^d / K$ floor. Both its norm bound and its floor are therefore actual endpoint objects, even if R_j itself tends to infinity.

For several prescribed depths, apply this construction successively to the current dense input, preserving earlier assignments, predictors and witnesses. Each resulting family's floor numerators have subpower sum. Include also the full-time chart labels, with floor numerators $\nu^\flat(H)$ from their unchanged trajectory assignments. These numerators sum to at most one per family by disjointness. Apply Lemma 9.1 to all the families together. The final law therefore has both the required predictor floors and current chart mass at least $\nu^\flat(H)/K$, despite cascading deletions. Prior-times-time

domination supplies the reverse comparison. No further deletion is needed to obtain the conditional prior denominators, and no earlier full-domain witness or true prior is replaced by the final restricted law. This proves $\mathcal{G}(2k)$ with the stated simultaneous-use assertion. □

9.3 Returning to wide intervals away from the critical rate

Proposition 9.6 (Return to wide intervals)**.** *In the horizontal model of Proposition 8.5, with the residual bounds of Lemmas 8.7–8.8, suppose* $2k \neq 1$*. Then every substantial residual supplies the hypotheses of Proposition 8.4 at three fixed nearby depths. Consequently it yields a time-improvement candidate in the sense of Proposition 4.11, with a fixed positive depth gain, a fit throughout the full assigned block, and hidden derivative one. The assertion holds both for* $2k < 1$ *and for* $2k > 1$*.*

Proof. Parameter charts and their unchanged prior denominator. Keep the horizontal-model coordinates, relative depths and true assignment priors fixed. We will construct projected labels at three nearby depths whose common transverse width is $D_0 = h_{s_0}^2$. Put

$$D_0 = N^{-2ks_0}, \qquad \widetilde{P} = (P - P_c)/D_0$$

using groups at P_{D_0}, $d_0 = [v]_\chi$, $P_c = (p_c, q_c)$ the parameter-bin center (dyadic roundings as usual), common fine direction width $\chi \le N^{-e_0}$. Take $s_0 > 0$ sufficiently small compared with e_0 for the fixed multiples below, also $\frac{3}{2}s_0 < s_{\max}$ when using $\mathcal{G}(2k)$. Depth choices below can be fixed in advance including sufficiently fine χ for the wide comparisons.

Prepare the earlier affine data (including (8.5)) for the needed depths first. In particular for pairs used between s_0 and a deeper s_j we can keep joint whole-domain witnesses of mass at least $m_{Q_{s_j}}/K$ for the true labels and paired entries L_{s_j}, L_{s_0}. Use Lemma 9.1 to discard light such pairs before smaller slices/full-time selections, using the true child reference sums and subentry counts to bound the deleted mass; $L_{s_j} - L_{s_0}$ then has norm at most Ka_{s_0} throughout the child box. Whole-domain witnesses of this mass order for just the deeper labels, in the bounded x-problem here, suffice where no ancestor comparison is needed.

If $2k > 1$, use the groups (d_0, P_{D_0}) directly as charts with baseline predictor $H_0 = 0$, thickness unit $A_0 = 1$. If $2k < 1$, use Proposition 9.5 and Lemma 9.1 to take simultaneous $\mathcal{G}(2k)$ labels/predictors H_0, H_b at $s_0, s_b = \frac{3}{2}s_0$, with the same d_0's. Compare each with L_{s_j}, Q_{s_j} at its own depth by Lemma 9.4 (in both cases write $\mathcal{L}_{s_j}(t, P) = L_{s_j}(t, p + tv_{d_0}, q_0 - tp)$ at either depth). Thus for $2k < 1$ the discrepancy of $\mathcal{L}_{s_j}$ to its paired full-time predictor at $s_j = s_0, s_b$ is bounded by Ka_{s_j} on the product of that time block and its parameter domain of width N^{-2ks_j} up to K. In this case take one paired L_{s_0}, Q_{s_0} per H_0 and time block using the list bound, and use H_0 labels as charts with $A_0 = a_{s_0}$.

In either case keep typical such charts with active mass and prior mass denominator both of order up to K

$$W_G A_0^d, \qquad W_G = \pi(d_0) D_0^2.$$

Indeed the thresholds sum to subpower by the earlier label floors (or $\sum W_G \le K$ over occurring groups in the trivial case). The joint d_0, x-cap near H_0 at A_0, integrated over the moving parameter domain, bounds the incidence mass per chart by the same quantity, as in the comparisons ($\chi + N^{-e_0}$ errors negligible here). And one can discard poor success fractions relative to trajectory assignment weights in $\nu^\flat$, by disjointness; we have prior-times-time domination in the upper direction as well. Condition the prior to each chart and divide active masses there by the prior denominator. If trimming other lines of the assignments is needed, e.g. to use bounded x, the resulting normalization changes by only a subpower factor since all our incidences already meet the trajectory bounds.

A scalar parameter and its fiber direction. The scalar projection will use

$$R(t) = \widetilde{P}_2 - 2t\widetilde{P}_1, \qquad W(t) = (1, 2t).$$

At fixed time the fibers of R in the parameter plane have direction $W(t)$. We seek a correction after which the affine parameter predictors vary by no more than their fitted width along these fibers. They can then be evaluated at $\widetilde{P} = (0, R)$ and counted using only (t, R).

Suppose first that $2k < 1$. The bounded-by-K time signal on paired events in a chart

$$G_0(t) = (D_0/A_0)\nabla_P(\mathcal{L}_{s_0} - H_0) \cdot W(t)$$

agrees with $(D_0/A_0)\nabla_P(H_b - H_0) \cdot W(t)$ there to error

$$K\big(D_0/h_{s_b} + (D_0/A_0)a_{s_b}/h_{s_b}^2\big) \le N^{-\kappa}$$

with, for example, $\kappa = \frac{s_0}{4}\min\{k, 1-2k\} > 0$ after absorbing subpower losses. Indeed $\nabla_P\mathcal{L}_j \cdot W = \partial_Z L_j + t\partial_Y L_j$. The first error compares L_{s_b} and L_{s_0} by the whole-child-box bound KA_0: the slopes of the difference are bounded by KA_0/h_{s_b} along the central horizontal plane and by $KA_0/h_{s_b}^2$ transversely, and the horizontal spatial displacement $(1, t)$ at the occurrence differs from $(1, t_c)$ of that plane by at most Kh_{s_b}. The second error compares the finer calligraphic fit with H_b using norm closeness throughout its paired parameter domain. The savings before subpowers are $N^{-ks_0/2}$ and $N^{-(1-2k)s_0/2}$ respectively.

We now choose one such quadratic time polynomial while retaining mass from all paired trajectories. In a typical chart let $\lambda_{\mathrm{pair},H}$ be the incidence measure after the pairings just used, divided by the chart's prior denominator, and let ν_H be that fixed conditional trajectory probability. Thus $\lambda_{\mathrm{pair},H} \le K\nu_H \otimes dt$. Denote its time marginal by $(\lambda_{\mathrm{pair},H})_t$, and write

$$m_H(t) = \frac{d(\lambda_{\mathrm{pair},H})_t}{dt}, \qquad M_H = \int_0^1 m_H(t)\,dt \ge 1/K.$$

Retain $T_0 = \{t : m_H(t) \ge M_H/2\}$. This discards at most $M_H/2$ of the paired incidence mass. Domination by $K\nu_H \otimes dt$ then supplies an indexed trajectory whose paired time set $T_b \subset T_0$ has $|T_b| \ge M_H/(2K)$. Its label H_b is fixed along the whole trajectory.

For this fixed label, $(D_0/A_0)\nabla_P(H_b - H_0) \cdot W(t)$ is a quadratic in time. It approximates the shared signal $G_0(t)$ to $KN^{-\kappa}$ on T_b, so boundedness of G_0 and Remez give coefficient bounds K. Restore every paired line at those same times. The s_0 entry was chosen per chart and time block, so G_0 is shared by all of them, and the restored mass is

$$\lambda_{\mathrm{pair},H}\{t \in T_b\} = \int_{T_b} m_H(t)\,dt \ge (M_H/2)|T_b| \ge M_H^2/(4K).$$

It is therefore dense. This argument uses the selected line to find the time polynomial; the retained mass is that of all paired lines on its time set.

Choose $H_{\mathrm{corr}}(t, \widetilde{P})$, affine in parameters with affine time coefficients and bounded by K on the scaled domain, whose derivative along $W(t)$ is this polynomial. For $b_0 + b_1 t + b_2 t^2$, take $(b_0 + b_1 t)\widetilde{P}_1 + (b_2 t/2)\widetilde{P}_2$. When $2k > 1$, set $H_{\mathrm{corr}} = 0$.

Choosing all three depths with positive margins. Let $C_d = 10/(1-d)$, as in Proposition 8.4, and put $C = C_d + 3$. Reduce s_0 so that $100s_0 < e_0$, all preceding fixed reconstruction buffers hold, and $3s_0/2 < s_{\max}$ in the subcritical case. Set

$$B_* = \begin{cases} \min\{s_0/2, \kappa, ks_0/(1+k)\}, & 2k < 1, \\ \min\{s_0/2, (2k-1)s_0\}, & 2k > 1. \end{cases}$$

Choose

$$\delta = B_*/4, \quad s = s_0 + \delta, \quad u = \min\{\delta/4, k\delta/(4C)\}, \quad c_2 = (1-d)u/8, \quad s_i \in \{s, s+u, s+c_2\}.$$

All these are fixed positive exponents. Writing $\Delta_i = s_i - s_0$, we have $0 < \Delta_i \le 5B_*/16 < B_*$, so $s_i < 3s_0/2$ and $D_0/h_{s_i} = N^{-k(s_0-\Delta_i)} \ll 1$. For $2k < 1$,

$$\frac{D_0/h_{s_i}}{a_{s_i}/A_0} = N^{-[ks_0-(k+1)\Delta_i]} \ll 1, \qquad \frac{N^{-\kappa}}{a_{s_i}/A_0} = N^{-(\kappa-\Delta_i)} \ll 1.$$

For $2k > 1$,

$$\frac{D_0}{a_{s_i}} = N^{-[(2k-1)s_0-\Delta_i]} \ll 1.$$

These inequalities supply all the directional error margins. They also give $0 < c_2 < (1-d)u/4$, as required for the wide-interval argument. On the paired events

$$\left|\nabla_{\widetilde{P}}\big((\mathcal{L}_{s_i} - H_0)/A_0 - H_{\mathrm{corr}}\big) \cdot W(t)\right| \le K a_{s_i}/A_0. \tag{9.5}$$

For $2k > 1$ use the horizontal bound (8.5) times D_0. For $2k < 1$ compare the true affine fits L_{s_i}, L_{s_0} on the child box in the same horizontal direction as just used (with error at most KD_0/h_{s_i} here), then use the time-signal approximation. These derivatives only concern affine dependence on parameters.

The scalar projection: measure ceilings and support. Apply Theorem 8.2 to the scalar parameter R defined above and the signal

$$x^\circ = (x - H_0(t,P))/A_0 - H_{\mathrm{corr}}(t,\widetilde{P}), \qquad w_i = a_{s_i}/A_0.$$

Both signals are affine on trajectories with coefficients $\le K$ in the conditional charts, by the full-time bounds. We check the scalar mesh hypotheses there at terminal width w_i, with density prefactor one and angular gain, say using (t,R) side σ_R of depth $20s_0$. At fixed time the map from $\widetilde{P}$ to (R,R') has determinant of magnitude 2. Before division by the chart denominator, the mass with (t,R) in a cell, x° in a bin of width $1 \ge \varrho \ge w_i$, and optionally R' in an interval of width N^{-v_0}, $0 < v_0 \le s_0$, is at most

$$K\pi(d_0)\sigma_R^2 D_0^2 (A_0\varrho)^d$$

times N^{-v_0} if using that interval. Indeed reconstruct P from (Z,Y) with error $K(N^{-e_0}+\chi)$, using the determinant-one map $P \mapsto (p + tv_{d_0}, q_0 - tp)$. Errors even after division by D_0 are negligible compared with the cell precisions and a_{s_i}. Integrate the true joint cap in much finer (t,Z,Y) cells at fixed d_0, over spatial area at most $KD_0^2\sigma_R$ at each such time (times the further width factor if using R'). Test x to width $KA_0\varrho$ with offset using reconstructed $H_0 + A_0 H_{\mathrm{corr}}$; coefficients in the indicated scaled parameters are bounded by K, so this uses only subpower lists per fine base cell. Thickening by sufficiently fine fixed-power true meshes changes the volume bound at most by K. This proves the mass ceilings after division by $\sim_{\log,N} W_G A_0^d$.

At each (t,R)-cell the relevant Q_{s_i} domains lie in a common enlarged horizontal box per time block (the cell meets only K blocks). Specifically for the bin representative R_* use reference motion

$$Z_{\mathrm{ref}} = p_c + tv_{d_0}, \qquad Y_{\mathrm{ref}} = q_c + D_0 R_* - tp_c.$$

At contact their spatial discrepancies are $D_0\widetilde{P}_1(1,t)$ up to $K(D_0\sigma_R + N^{-e_0} + \chi)$ errors. The reference motion is exactly horizontal. Use its point/plane at block midpoint: $D_0 \ll h_{s_i}$ and the contact errors fit inside $h_{s_i}^2$, with slope change Kh_{s_i} for $(1,t)$. The aligned true boxes extend from these contacts

with horizontal size Kh_{s_i}, transverse size $Kh_{s_i}^2$ relative to planes differing by at most Kh_{s_i} in slope there. Thus the common enlarged box of base volume $Kh_{s_i}^4$ suffices for their whole domains.

The common box bounds the number of contributing true labels, including their affine entries, by Kw_i^{-d}. Indeed for $2k > 1$ sum the disjoint whole-domain witness floors $m_{Q_{s_i}}/K$ by the base cap alone (bounded x and unit cap). For $2k < 1$ these earlier paired witnesses lie near the shared L_{s_0} of the chart/time-block selection at s_0, so their summed weight is $\le KA_0^d h_{s_i}^4$ by the pure x-band cap. These counts use witnesses and caps in the full horizontal problem before the parameter-chart conditionings.

Each affine entry yields only K bins for x° at w_i on the cell. It predicts using $(\mathcal{L}_{s_i} - H_0)/A_0 - H_{\mathrm{corr}}$ with error Kw_i; replacing the true spatial variables as in this formula is allowed by (8.1) and (8.5). Moreover one can evaluate this expression at $\widetilde{P} = (0, R(t))$ instead by (9.5) and bounded $\widetilde{P}_1$. To control its variation within the cell, use (8.5) and an active position: the ordinary slopes of L_{s_i} are at most $K(1 + a_{s_i}/h_{s_i}^2) \le KN^{(2k-1)+s_i}$. The functions H_0, H_{corr} have norm K on the scaled parameter domain and its bounded enlargements. After division by A_0, the evaluated entry's time and R derivatives are bounded by $KN^{5s_0/2}$, using $s_i < 3s_0/2$ and $0 < k \le 1$. Its variation across a σ_R-cell is therefore at most $KN^{-35s_0/2} \ll w_i$, because $w_i \ge N^{-3s_0/2}$. Furthermore,

$$D_0\sigma_R \ll h_{s_i}^2, \qquad \frac{N^{-e_0} + \chi}{D_0} \ll \sigma_R$$

when $100s_0 < e_0$. Enlarging the fixed preliminary buffer also covers division by A_0 and all scalar evaluation errors. Thus the projection uses reconstruction only at accuracies strictly coarser than its known error. Each of at most $K\sigma_R^{-2}$ cells has at most Kw_i^{-d} terminal scalar bins, as required.

Theorem 8.2, with its angular gain, gives disjoint full-time trajectory labels of active mass $\ge K^{-1}w_i^d$ in conditional units, predicting x° throughout to Kw_i by quadratics $F_i(t, R)$ with coefficients $\le K$. Indeed their interval width in that theorem and its reciprocal are bounded by K; the interval center and slope are then bounded by K on occupied labels, so its moving-interval norm bound suffices. Retain simultaneous data at the three depths by applying successively to dense restrictions with the same ceilings/support bounds. Keep the ensemble of typical charts using the uniform subpower conclusions for the conditional charts, not just one frozen d_0; thus there is still dense total mass in the horizontal problem.

Returning the predictors and their masses to true coordinates. Split $p = Z(0)$ to width χ to return to α for wide intervals, with representative p_α. Use $P^\# = (p_\alpha, Y + tp_\alpha)$ instead of P, and corresponding $\widetilde{P}^\#, R^\#$, in

$$H_0 + A_0H_{\mathrm{corr}} + A_0F_i(t, R).$$

The resulting predictor for x is quadratic in t, Y since $P^\#, \widetilde{P}^\#, R^\#$ now depend affinely on those variables (the first scaled-parameter component is constant). It has norm K on the relevant moving interval of width KD_0 about $q_c - tp_\alpha$. Throughout the assigned lines we can use error Ka_{s_i}, since (8.1) holds throughout participating lines and $\|\widetilde{P}^\# - \widetilde{P}\| \le K(N^{-e_0} + \chi)/D_0$; trim lines to the trajectory bounds on counted incidences if needed. Thus the packet width scale for Y can be D_0.

Their typical sliced weights are $\ge K^{-1}a_{s_i}^d w_\alpha D_0$, $w_\alpha = \pi(d_0)\chi$. Before this split the floor per projected label (on obtaining that projection) is $\pi(d_0)D_0^2A_0^d w_i^d/K$; assignments at a depth in and between charts are disjoint. Remove light slices by summing thresholds over all those labels and the $\le KD_0/\chi$ options each. Explicitly, an unsliced floor numerator is

$$T_i = \pi(d_0)D_0^2A_0^d w_i^d = \pi(d_0)D_0^2a_{s_i}^d.$$

The numerator required for one slice is $t_i = \pi(d_0)\chi D_0 a_{s_i}^d$. Since there are at most KD_0/χ slices, their total numerator is at most KT_i. The old floors and disjoint assignments give a subpower sum

of the T_i's. A larger common floor constant therefore makes the mass removed by the light-slice deletion negligible.

Before the final common pruning, prepare the conditioning states needed for Proposition 8.4. Each state consists of a true Q_s and sufficiently fine true-base and x-bins; no projected predictor label is appended. By Lemma 8.7, there are K choices per sufficiently deep end history e. Its earlier reference weight R_e bounds the velocity numerator by $K\pi(b)R_e$, and the references sum inside Q_s to at most Km_{Q_s}. Thus the numerators for all appended history–state entries have total at most $K\sum_{Q_s} m_{Q_s} \le K$.

Apply Lemma 9.1 to these entries and all three sliced projected layers together. This restores the projected floors while giving each retained history–state entry mass at least R_e/K. Division of its persistent velocity numerator, then summation over histories, gives conditional domination by $K\pi$ on the final law. All earlier whole-domain witnesses remain available.

Width D_0 fits the wide condition since $b_s/D_0 \sim_{\log,N} N^{-2k(s-s_0)}$, $D_0 \ll h_s = g_s$; the chosen parameters satisfy

$$(C_d+3)u \le k\delta/4 < \min\{2k(s-s_0), ks\}.$$

Consequently both wide inequalities hold with a fixed power of slack: b_s/D_0 decays with exponent $2k(s-s_0)$, and $h_s\beta_{\max}/\beta_{\min}$ decays with exponent ks, since all three widths β_i equal D_0 up to subpowers.

We finish by checking the wide-interval input and returning its output to the initial tangent parent.

First, use H_s as the block length called q_s in Proposition 8.4. The velocity v is unchanged by the horizontal normalizations, so the same velocity law and interval caps apply. The current dense law has the pure and joint caps in (8.3). The common pruning just performed gives conditional velocity domination at the fine true-base and x-states, also for offsets of fixed-power variation in the true base. Selecting a heavy descendant does not change these conditional laws, since Q_s is already part of the state.

Second, each true label has only K affine entries L_s predicting x to Ka_s. Their norms in the centered Q_s boxes and the needed enlargements are at most K: use an active value of x, the horizontal slope bound in (8.5) multiplied by h_s, and the transverse slope bound multiplied by the transverse width h_s^2. The aligned planes therefore give the required true tilted domains. The wide argument applies with these lists and the Φ-labels on true base positions, including its final fine projection mesh.

Third, Proposition 8.4 gives a quadratic prediction on a true relative descendant Q_s, with horizontal-problem mass at least $H_s\nu^\flat(Q_s)N^{-dc_2}/K$. Let q_{desc} and ν_{desc} be that descendant's block length and full assignment weight in the initial tangent parent. The time change and conditional full-assignment prior in Lemma 2.13 give

$$q_{\mathrm{desc}} = q_r H_s, \qquad \nu_{\mathrm{desc}} = \nu_{Q_r}\nu^\flat(Q_s).$$

The additional restrictions and bounded-trajectory normalizations already used affect comparisons of retained incidence mass by only K. Restoring the factor $q_r\nu_{Q_r}$ therefore gives counted mass at least

$$\frac{q_r\nu_{Q_r}H_s\nu^\flat(Q_s)N^{-dc_2}}{K} = \frac{q_{\mathrm{desc}}\nu_{\mathrm{desc}}N^{-dc_2}}{K}.$$

Finally restore $w = L + a_r x$. The resulting test has error $Ka_r a_{s+c_2}$ throughout the full assigned block. Its hidden derivative is one, and its base terms are affine or quadratic as required.

This is a candidate of the precise kind used in Proposition 4.11. Its mass is charged to the original residual and its fit is on the full assigned block. The argument works anew on each substantial residual, so the candidate-upgrade criterion applies. □

Together Propositions 9.2 and 9.6 finish the tangent contradiction for $0 < \ell < \infty$.

10 Unbounded optimized narrowness and completion of the proof

The remaining case of the second model is excluded by the following proposition. Here ℓ is the infimum, over maximizing sequences, of the terminal narrowness supremum defined in Section 4.

Proposition 10.1 (Exclusion of unbounded optimized narrowness)**.** *In the second model's extremal setup of Proposition 4.3, the global optimized narrowness ℓ is finite.*

Assume, for a contradiction, that $\ell = \infty$. We will construct terminal-known atlases at a fixed positive depth whose time exponents tend to zero. Completing these atlases to the terminal depth contradicts the positive critical time rate k. Two nested true systems first give thin charts valid for the whole time interval (Proposition 10.2). Normalization then makes the logarithmic exponents of their mass-to-area ratios tend to zero (Lemma 10.3). Lemmas 10.4 and 10.5 prepare these packets for the scalar projection theorem. Its quadratic fits yield a known atlas with the required depth gain and coverage; the final step establishes actual subpower bounds for this atlas.

Throughout this section, $K_0 = N_0^{o(1)}$ denotes a subpower factor within a fixed exact problem; it may depend on that problem. After taking the outer diagonal, $K = N_0^{o(1)}$ denotes a subpower factor along that diagonal. The latter controls the scalar analysis, whereas the output atlas must retain the former type of bound.

10.1 Nested true systems

Normalize a true intermediate system on a maximizing critical path by Lemma 2.13. Rescale the remaining depth interval to length one. Denote the resulting maximizing sequence by E_i. In this section, *original coordinates* means these coordinates of E_i, before the further localizations below. The own-depth knowledge and saturation of the initial true system give

$$n_i(0) \sim_{\log} 1.$$

Consequently the relative-profile conclusion of Proposition 4.3 gives

$$f_i(r) \longrightarrow dr \quad \text{uniformly for } 0 \le r \le 1, \qquad d_i \longrightarrow d = d_* < 1, \qquad H(E_i) \longrightarrow k > 0. \tag{10.1}$$

Here $n_i(r) = N_0^{-f_i(r)}$ in logarithmic order, and $E_i \in C(d_i)$. The fixed angular parameters are denoted by $S, \eta > 0$.

The original trajectories satisfy $|y| + |V| \le K_0$, and their native signal $X = P_*(t, y, w)$ has hidden derivative scale B. At an exact base (t, y), the conditions

$$|P_*| \le K_0, \qquad |(P_*)_w| \ge B/K_0$$

on incidences require only K_0 hidden bins of width $1/B$. Indeed a quadratic has boundedly many monotonicity intervals, and on the tested part its inverse image of an interval of length $O(K_0)$ has total length $O(K_0^2/B)$. Together with the zero-depth density bound in (10.1), this gives a base-density ceiling K_0 in each component after removal of atypical entries.

Choose parameters in the order

$$0 < u < \eta, \qquad 0 < s_1 < s_2 < 1, \qquad (d + 2k)s_2 < \frac{Su}{4}. \tag{10.2}$$

The segment construction in Proposition 4.3 supplies nested true nodes Q_1, Q_2, with depths $r_{j,i} \to s_j$ and common homogenized time lengths q_j, such that

$$q_j = N_0^{-h_{j,i}+o(1)}, \qquad h_{j,i} \to k s_j, \qquad g_j = N_0^{-\alpha_{j,i}}, \qquad \alpha_{2,i} - \alpha_{1,i} \longrightarrow \infty. \tag{10.3}$$

Here $q_j^2 g_j$ is the spatial determinant in original coordinates, up to actual subpower factors. To obtain the last limit, first normalize at the earlier node and choose the later endpoint as in the critical-path construction. The intervening problem, divided by its actual depth length, is again maximizing. The global hypothesis $\ell = \infty$ therefore gives, by part 4 of Proposition 4.3, true systems whose horizon tends to k and whose narrowness tends to infinity. Multiplication by the intervening depth length, which tends to $s_2 - s_1 > 0$, preserves divergence when these systems are lifted.

All labels include their required path histories. We retain their full-layer trajectory assignments, so that, within each original component γ,

$$\nu_\gamma(Q_j) \sim_{\log} n_i(r_{j,i}) q_j^2 g_j. \tag{10.4}$$

The subsequent path incidences are subsets of these assignments. Assignments at one node and in one time block are disjoint. The mass relation (10.4) concerns the full-layer assignments; for subsequent restrictions, it supplies upper mass bounds.

Both labels are known by $p_L = p_1$. The angular fullness statement therefore makes their largest spatial widths q_j up to actual subpowers. Write D_j for the spatial matrix, $y_j(t)$ for its affine center, and n_j for a unit left minor axis. The position bounds and their endpoint consequences give

$$\begin{aligned} |n_j \cdot (y - y_j(t))| &\le K_0 q_j g_j \quad \text{throughout the } Q_j\text{-block}, \\ |n_j \cdot (V - y_j')| &\le K_0 g_j. \end{aligned} \tag{10.5}$$

The full center velocities are bounded by K_0, since the trajectories have bounded velocities and fit the chart throughout a nondegenerate block.

At a common path incidence,

$$|\sin \angle(n_1, n_2)| \le K_0 g_1. \tag{10.6}$$

In fact $D_2 = D_1 R$, where the relative matrix satisfies $\|R\| \le K_0 q_2/q_1$. Thus $\|n_1^T D_2\| \le K_0 q_2 g_1$. The major axis of D_2 has length at least q_2/K_0, which proves (10.6).

10.2 Localization for the whole time interval

The true charts control each trajectory only on their own blocks. We first compare two occurrences of the later chart on the same trajectory. Their nearly parallel normal strips will determine one thin chart valid for the entire original time interval.

Proposition 10.2 (Full-time localization)**.** *Assume $\ell = \infty$ in the second-model extremal setup of Proposition 4.3. Let Q_1, Q_2 be the nested true systems of Section 10.1, with parameters satisfying* (10.2) *and* (10.3)*. A permitted further selection admits full-time charts C with tangential width one and normal width*

$$g_C = g_2 N_0^{2u}. \tag{10.7}$$

Their labels are known by the original p_1 with actual subpower lists. The native test remains P_*. If n_C and $y_C(t)$ are the normal and affine center of C, then every retained Q_1-incidence satisfies, after a consistent choice of signs,

$$|n_1 - n_C| \le K_0 g_1, \qquad |n_C \cdot (y_1' - y_C')| \le K_0 g_1. \tag{10.8}$$

All assignments, geometric tests and coordinate changes have tempered bounds within each fixed E_i.

Proof. Outside negligible pair mass, two occurrences of the later chart have nearly parallel normals. Binning their normal directions and affine center motions then gives the full-time charts, and the terminal lists recover these bins.

Concentration of the normals. Sample two independent times t_0, t on the same trajectory, using the original trajectory and component priors, and require a current path incidence at both times. If the path has mass p, the pair mass is at least p^2 by Jensen's inequality, and hence at least $1/K_0$.

We first prove that, outside negligible pair mass,

$$|\sin \angle(n_2(t_0), n_2(t))| \le g_2 N_0^u. \tag{10.9}$$

Before counting labels, remove from the t-incidences the atypical sampled entries for the terminal joint angular cap at depth $u/2$, the pure terminal lower density, and the zero-depth upper density. The removed pair mass is negligible: the additional time has length at most one, and the density tolerances can be relaxed slowly relative to the subpower path mass.

For fixed t_0, γ, saturation and disjointness give at most

$$\frac{K_0}{n_i(r_{2,i}) q_2^2 g_2} \tag{10.10}$$

possible first labels $Q_2(t_0)$. Fix one. If (10.9) fails, the two velocity strips in (10.5) intersect in a ball of radius at most $K_0 N_0^{-u}$. At each $p_1(t)$, its actual list contains only K_0 possible second labels. Thus only K_0 angular bins of side comparable to $N_0^{-u/2}$ must be considered. On the retained entries their total numerator is at most $K_0 N_0^{-Su/2}$ times the original pure p_1 density.

We sum the denominator only over its permitted good zero-depth ancestors before integrating. At each exact base only K_0 such hidden bins occur, and their total density is at most K_0. Moreover, a trajectory assigned to the fixed first label stays, over the whole unit interval, in its extrapolated transverse strip of width $K_0 g_2$. Its tangential coordinate remains bounded. The area available at time t is therefore at most $K_0 g_2$. Multiplying by (10.10), and then integrating the times and weighting the components, bounds the failure mass by

$$\frac{K_0 N_0^{-Su/2}}{q_2^2 n_i(r_{2,i})} = N_0^{-Su/2 + (d+2k)s_2 + o_i(1) + o_{N_0}(1)}. \tag{10.11}$$

This is power-small by (10.2). The exceptional density entries were removed before the sum, so their mass is not multiplied by the number of first labels.

Construction of the charts. Fix t_0 by averaging, retaining subpower mass of paired successes. The original tempered path and the angle test define the retained selection; the density masks serve only to estimate its mass. Assign each successful trajectory at t_0 according to the following binned data of $Q_2(t_0)$: its unit normal, and the intercept and velocity coefficients of its affine normal center. Use bins of width g_C, treating simultaneous sign reversal by finitely many fixed orientation charts. These bins, without the identity of $Q_2(t_0)$, are the C-label.

For each bin choose a representative unit normal and a representative affine normal motion. The transverse bound in (10.5) extrapolates to $K_0 g_2$ over unit time. Rounding the normal and the two coefficients adds at most $K_0 g_C$, because positions and velocities are bounded by K_0. A tangential column of length one and a normal column of length g_C therefore give the required full-time geometry. The native P_* already has the full-block bounds for a depth-zero chart.

Terminal lists and alignment. Terminal knowledge must recover both the normal and the affine center motion. A later $Q_2(t)$ in the list at $p_1(t)$ has normal within $K_0 g_2 N_0^u = o(g_C)$ of the first normal, modulo sign. Both labels constrain the same bounded affine trajectory. Their normal

intercepts and normal velocities consequently differ by at most $K_0 g_2 N_0^u$ as well: compare each coefficient to the corresponding coefficient of that trajectory, using (10.5), and then account for the change of normal. Each later label thus determines boundedly many neighboring bins for all the C-data. Composing with the actual $Q_2(t)$-list proves that C is known by $p_1(t)$ with actual subpower lists.

Since

$$\frac{g_C}{g_1} = N_0^{-(\alpha_{2,i}-\alpha_{1,i})+2u},$$

the binning error is much smaller than g_1 for large i. Combining with (10.6) gives the first inequality in (10.8). At a retained occurrence, comparison of both center velocities with V gives a Q_1 normal error of $K_0 g_1$ and a C normal error of $K_0 g_C$. The change of normal contributes at most $K_0 g_1$ because $V - y_1'$ is bounded. This proves the second inequality. The relevant normals and center velocities are constant parameters of the labels, so this comparison holds throughout their blocks.

Slicing at a fixed real t_0, the finite bin tests, the original path predicates and the geometric angle test all have tempered complexity. Although their bounds can grow with i, they remain fixed polynomial bounds along each exact sequence E_i. □

10.3 Normalization of mass and spatial area

The full-time charts give a common thin direction. The next normalization makes their masses comparable to their spatial areas in the two-limit sense needed for scalar projection. The resulting time exponent tends to zero, while the depth gain constructed below stays positive.

Lemma 10.3 (Normalization and relative density cancellation)**.** *Assume $\ell = \infty$ in the second-model extremal setup of Proposition 4.3, and let C be the full-time charts of Proposition 10.2. After a further permitted selection and passage to subsequences, these charts contain a system of charts A, known by original p_1, with common time length*

$$q_A = N_0^{-h_{A,i}+o(1)}, \qquad h_{A,i} \longrightarrow 0.$$

Write D_A for their original spatial matrices and $J_A = |\det D_A|$. Their original assigned trajectory masses satisfy

$$\frac{\nu_\gamma(A)}{J_A} = N_0^{-db_i+\varepsilon_i+o_{N_0}(1)} = N_0^{o_{i\to\infty}(1)+o_{N_0\to\infty}(1)}, \qquad b_i \longrightarrow 0, \quad \varepsilon_i \longrightarrow 0. \tag{10.12}$$

If e_Z is a unit tangent to C, then the systems may also be prepared so that

$$\|D_A^T e_Z\| \geq q_A/K_0, \qquad \|D_A^T n_C\| \leq K_0 g_C q_A. \tag{10.13}$$

Actual lists, native derivative bounds and geometric slacks remain subpower within each fixed exact problem.

Remark. The second equality in (10.12) is a two-limit statement. It does not assert an actual subpower bound for fixed i.

Proof. Normalization inside C changes absolute densities. Their ratio at two depths cancels this change and gives the required mass-to-area estimate. We then check the original mass bounds, terminal lists and spatial axes.

Normalization inside the full-time chart. Normalize spatially inside C, keeping the time interval, hidden coordinate, thickness and native P_* unchanged. The depth-zero case of Lemma 2.13 applies

to this known atlas without requiring saturation at depth zero. Its new component (γ, C) has conditional trajectory prior $\nu_\gamma(\cdot \mid C)$. Put

$$W = \sum_{\gamma, C} \omega_\gamma \nu_\gamma(C).$$

Then $K_0^{-1} \le W \le 1$, and the normalized new component weights are $\omega_\gamma \nu_\gamma(C)/W$. Multiplying the conditional law by its component weight restores the original restricted trajectory law, divided only by W. The terminal density formula, within each component, is

$$n_i'(1) \sim_{\log} n_i(1) \frac{g_C}{\nu_\gamma(C)}. \tag{10.14}$$

At intermediate depths only the one-sided comparison of Lemma 2.13 is needed. To see the cap inheritance explicitly, homogenize $g_C/\nu_\gamma(C) = N_0^{t_i + o(1)}$. Then $f_i'(1) = f_i(1) - t_i$ and $f_i'(r) \ge f_i(r) - t_i$. Hence

$$f_i'(1) - f_i'(r) \le f_i(1) - f_i(r) \le d_i(1 - r).$$

A primed velocity bin maps into an old velocity ball of its radius times K_0, since the norm of the C-matrix is bounded by K_0. The old good terminal angular numerators therefore give the same angular cap after the density change. Thus the homogeneous new problem E_i' belongs to $C(d_i)$.

Every true terminal system in E_i' lifts through C, so

$$H(E_i) \le H(E_i') \le k(d_i).$$

Since the outer sequence is maximizing, both endpoints tend to k. Consequently E_i' is also maximizing. Its relative profiles have the linear limits from Proposition 4.3.

Short packets and density cancellation. Choose small positive depths $b_i \to 0$ slowly. More explicitly, fix $b = 1/m$, first take i late enough for the relative-profile comparison at $b, 1$ with error at most $1/m$, and choose true packets in the length-b coarsening with horizon at most $b + 1/m$. The universal initial bound $H \le L$ provides these packets for the coarsening, regardless of its membership in the same cap class. Let $m = m(i)$ tend to infinity slowly enough to satisfy these requirements together with the finite preparations already imposed. Then

$$f_i'(1) - f_i'(b_i) = d(1 - b_i) + o_i(1), \tag{10.15}$$

and the packet horizon tends to zero. Saturate those primed true packets and compose them with C, including C in the resulting label A.

The primed spatial determinant of A is J_A/g_C, up to actual subpowers. Saturation gives

$$\frac{\nu_\gamma(A)}{\nu_\gamma(C)} \sim_{\log} n_i'(b_i) \frac{J_A}{g_C}.$$

Combining with (10.14) yields

$$\frac{\nu_\gamma(A)}{J_A} \sim_{\log} n_i(1) \frac{n_i'(b_i)}{n_i'(1)}. \tag{10.16}$$

The logarithmic exponent of the right-hand side is

$$-f_i(1) + f_i'(1) - f_i'(b_i) = -db_i + o_i(1),$$

by (10.1) and (10.15). This proves (10.12). The primed profile enters this exponent only through the difference $f_i'(1) - f_i'(b_i)$. An arbitrarily large absolute primed density shift therefore cancels in the quotient in (10.16).

Original masses and terminal lists. At the C-stage the factors $\nu_\gamma(C)$ cancel conditional normalization globally; only the actual subpower W^{-1} remains. At the A-stage, disjointness gives the original mass bound

$$\sum_{\gamma,I,A} \omega_\gamma q_A \nu_\gamma(A) \le 1. \tag{10.17}$$

Here the sum includes the time blocks, and the inequality follows from disjointness within each block. Every later homogeneous selection has subpower total success in its fixed exact problem. An original exception of fixed power-small mass is therefore still negligible relative to that success. Components or labels with too small a conditional success fraction may be discarded by summing (10.17). This controls exceptional mass after summing over the labels, even when conditioning on individual C or A would magnify that mass.

The terminal lists retain actual subpower sizes. At p_1, the actual list for C identifies the possible full-time charts. Given one such C, its stored affine spatial map determines the primed exact base. The hidden coordinate and B have not changed, so the old hidden bin of width N_0^{-1}/B determines the coarser primed b_i-bin with at most two choices. Composition with the actual own-depth list of the primed true packet therefore gives the required labels. If the two list sizes are $L_C, L_{A'}$, the resulting A-list has size at most $2L_C L_{A'} = N_0^{o(1)}$ within fixed i, irrespective of the density shift in (10.14).

The spatial axes. Composition gives the second inequality of (10.13) and also an original largest-axis upper bound $K_0 q_A$. Suppose on subpower retained mass that the first inequality fails by a fixed power $N_0^{-\varepsilon}$. The endpoint position tests then place the original velocity, for each A, in a ball whose radius is bounded by $K_0(N_0^{-\varepsilon} + g_C)$. Choose a fixed angular depth $v > 0$ smaller than ε, the allowed angular range, and a positive power of smallness for g_C. The actual A-list at p_1 requires only K_0 corresponding velocity bins. On good terminal angular entries these bins carry at most $K_0 N_0^{-Sv}$ times the original pure denominator. After bad entries are removed, integration over the old observation bounds the total mass by the same angular estimate. The resulting power-small mass contradicts the retained subpower mass. Homogenization and slowly vanishing fixed tolerances therefore give the first inequality of (10.13). □

10.4 The depth gain and the diagonal

Retain the final tempered path in original components as the reference incidence measure μ_{ref}. Write $\mu_{\mathrm{ref},\gamma}$ for its within-component measure, without the component weight ω_γ, so that

$$\mu_{\mathrm{ref}} = \sum_\gamma \omega_\gamma \mu_{\mathrm{ref},\gamma}, \qquad \|\mu_{\mathrm{ref}}\| = \mathfrak{p}_{\mathrm{path}} \ge K_0^{-1}.$$

Here $\|\mu\|$ denotes total mass. The earlier full-layer assignments and their prior masses remain fixed. For later restricted measures, a subscript γ likewise excludes the component weight.

We fix the depth gain and the losses allowed in coefficient matching before selecting diagonal samples. Fix $d < d_0 < 1$. Let ε_* be the tolerance in Corollary 3.4 for quadratic graphs on two base variables, fixed comparison domains and dimension bound d_0. Choose fixed parameters

$$0 < \theta < \min\{1, \varepsilon_*/4\}, \qquad 2\theta < \varepsilon < \varepsilon_*.$$

Let $C_{\mathrm{match}} \ge 1$ dominate the fixed exponents in the list, coefficient, fit and marginal losses in the matching application below. These depend only on the fixed degrees and comparison dimensions. Choose

$$0 < \delta < \min\left\{\frac{\theta s_1}{4}, \frac{\varepsilon s_1}{2C_{\mathrm{match}}}\right\},$$

and put

$$c' = \frac{\theta s_1}{2} \in (0,1). \tag{10.18}$$

For sufficiently large i, an actual known atlas in E_i at depth c' and time length q_A is impossible on an exact subsequence. Indeed $h_{A,i} < kc'/2$, and normalization followed by a true terminal system in the remaining problem would give

$$H(E_i) \le \frac{kc'}{2} + k(d_i)(1-c'). \tag{10.19}$$

The right-hand side tends to $k - kc'/2 < k$, contrary to (10.1).

The finite-sample exclusion argument of Lemma 4.6 gives the following constraint on the sampled problems. Within fixed i, fix a sufficiently large polynomial complexity bound for the proposed outputs, and allow their list sizes and geometric slacks to be bounded by a sufficiently large fixed power of $K_{\mathrm{path}} \log N_0$. Here K_{path} collects actual subpower bounds for the old problem and prepared path, including the reciprocal reference mass. Exclude outputs covering at least

$$\mathfrak{p}_{\mathrm{path}} / \log N_0. \tag{10.20}$$

If infinitely many samples in this fixed exact problem admitted such outputs, their bounded complexity and actual subpower controls would supply an exact known atlas on a subsequence, contradicting (10.19). Thus one may sample N_0 arbitrarily late in this excluded range.

We take a diagonal $N_0 \to \infty$, $i \to \infty$, sampling each exact sequence late enough that the actual path factors and the two-limit errors in (10.12) are absorbed in $K = N_0^{o(1)}$. The constants $u, s_1, s_2, \theta, \delta$ remain fixed. All later finite geometric operations have polynomial bounds depending only on the old problem/path bounds and these fixed constants; the output complexity bound above is chosen large enough for those operations before the sample is taken. The output lists and slacks will have subpower bounds within each fixed exact problem, as required by the finite-sample exclusion.

Before the sparse searches, prepare analytical caps by removing atypical pure densities at depths through s_1, including $0, r_{1,i}$, and bad pure lower or joint angular upper entries at p_1. The loss is $o(\mathfrak{p}_{\mathrm{path}})$. One may use slowly increasing finite depth grids, shrinking fixed exponent tolerances and interpolation between neighboring widths. At every fixed i and tolerance the exceptional mass is power-small, so late sampling makes all these preparations simultaneous. In particular the pure ceilings through s_1 are KN_0^{-rd}, and the ceiling at $r_{1,i}$ is $Kn_i(r_{1,i})$. These caps concern restricted measures and their specified good old ancestors. The analytical masks are used only in the restricted measures; the reference path and final assignments retain their geometric predicates. Denote the resulting within-component submeasures by $\mu_{\mathrm{prep},\gamma} \le \mu_{\mathrm{ref},\gamma}$, and put

$$\mu_{\mathrm{prep}} = \sum_{\gamma} \omega_\gamma \mu_{\mathrm{prep},\gamma}.$$

The next lemma makes the remaining light-label and light-entry deletions and then fixes μ_{prep}. The projected laws introduced afterward will all be restrictions of this one measure.

10.5 Stable graphs on the earlier charts

The scalar projection keeps time and the tangential coordinate $e_Z \cdot y$, together with X. Its support count will use the earlier Q_1-charts. We first give their native signal a finite graph description

with derivative bounds. The lower masses needed here are obtained before any conditioning on an A-chart or a projected cell. For an earlier chart with block left endpoint t_1, write

$$\widehat{u} = ((t - t_1)/q_1, D_1^{-1}(y - y_1(t)))$$

for its normalized base coordinates, and put $a = N_0^{-s_1}$.

Lemma 10.4 (Graph descriptions on the earlier charts)**.** *Assume $\ell = \infty$ in the second-model extremal setup of Proposition 4.3, with the reference measure and diagonal preparations of Section 10.4. After analytical deletions of mass $o(\mathfrak{p}_{\mathrm{path}})$, each Q_1-label still meeting μ_{prep} has the following properties.*

1. *It has an earlier witness measure W_{γ,I,Q_1}, supported in its original time block and full spatial domain, with mass at least $q_1\nu_\gamma(Q_1)/K$. These witnesses are disjoint within each original block and satisfy the prepared zero-depth cap.*

2. *In its normalized base coordinates $\widehat{u}$, its retained incidences are assigned to at most K pairs (U, G). Here U is a real box and G is a bounded-degree algebraic function, holomorphic on a fixed-factor complex enlargement of U. Every assigned incidence has $\widehat{u} \in U$ and satisfies*

$$|X - G(\widehat{u})| \le Ka, \qquad |G| + |\nabla G| \le K,$$

where the value and derivative bounds for G hold on a neighborhood of U.

The final prepared measure has mass $\|\mu_{\mathrm{prep}}\| = (1 - o(1))\mathfrak{p}_{\mathrm{path}}$. The earlier witness measures are retained before the later box/graph deletions and may therefore extend beyond this final measure.

Proof. We first retain witnesses of sufficient mass for counting whole chart domains. The quadratic tests then give local holomorphic branches or affine centers, and a second mass selection supplies the derivative bounds for their compositions with the native signal.

Earlier witness measures. Delete Q_1-labels whose incidence mass under $\mu_{\mathrm{prep},\gamma}$, in their original block, is less than

$$q_1\nu_\gamma(Q_1)/K. \tag{10.21}$$

Choose the reciprocal-subpower threshold small enough relative to the reference mass. The loss is negligible because

$$\sum_{\gamma,I,Q_1} \omega_\gamma q_1 \nu_\gamma(Q_1) \le 1.$$

Write W_{γ,I,Q_1} for each surviving restriction to its Q_1-label and block at this stage. Keep these measures as earlier witnesses when subsequent restrictions decrease μ_{prep}. The assigned prior $\nu_\gamma(Q_1)$ in (10.21) is the original full-layer prior in (10.4). The floor measures the whole Q_1-restriction before conditioning on a particular A.

Branches of the quadratic tests. The scalar signal will be described by graphs in the normalized coordinates of each surviving earlier chart. Write $z = Bw$, let $P(\widehat{u}, z)$ be the true Q_1-test, and set $a_1^{\mathrm{fit}} = N_0^{-r_{1,i}} \sim_{\log,N_0} a$. On the earlier retained incidences,

$$|P| \le K_0 a_1^{\mathrm{fit}}, \qquad K_0^{-1} \le |P_z| \le K_0, \qquad |P_{zz}| \le K_0/a_1^{\mathrm{fit}}. \tag{10.22}$$

The depth-$r_{1,i}$ pure ceiling, the K_0 compatible hidden bins and the chart Jacobian give a base-density ceiling $Kq_1\nu_\gamma(Q_1)$ in $\widehat{u}$-coordinates. The floor (10.21) therefore implies a real base projection of volume at least $1/K$ in a box of size at most K_0.

The discriminant

$$\mathcal{D}(\widehat{u}) = P_z^2 - 2P_{zz}P$$

is independent of z and is a fixed-degree polynomial in $\widehat{u}$. Its values on that projection are bounded by (10.22). Polynomial Remez and fixed-degree coefficient bounds consequently bound $\mathcal{D}$ and its first derivatives by K on the needed complex enlargements of the normalized box.

Choose a sufficiently large subpower cutoff K'. If $|P_{zz}| \geq (a_1^{\mathrm{fit}}K')^{-1}$, the affine quadratic center

$$z_c(\widehat{u}) = -\frac{\text{coefficient of } z \text{ in } P}{P_{zz}}$$

approximates the observed z to Ka_1^{fit}, because $P_z = P_{zz}(z - z_c)$. For the remaining tests choose K' large enough that $2|P_{zz}P| \ll |P_z|^2$ on the incidences. Their discriminants are positive and bounded below by $1/K$ there.

Partition the normalized box into real subboxes of inverse-subpower radius. Use fixed-factor complex enlargements with an additional margin as comparison domains. For tests in the small-curvature case, the derivative bound on $\mathcal{D}$ permits a radius such that its variation on the enlargement of every occupied box is much smaller than its incidence lower bound. The discriminant therefore never vanishes there, and the two simple roots are holomorphic. For a linear equation the same construction keeps its linear coefficient nonzero. In the large-curvature case use the affine center z_c on these same boxes. At each occurrence the chosen center or root branch approximates z to Ka_1^{fit}, by (10.22) and the preceding center estimate.

Mass selection and graph bounds. There are K box/branch entries per label. Delete light entries again, using the same $q_1\nu_\gamma(Q_1)$ mass bounds and a smaller reciprocal-subpower threshold. The total loss remains negligible, and each retained entry has an earlier real projection of volume at least $1/K$. Composition of its center or root branch with the native P_*, using $w = z/B$, gives the graph of the scalar signal. The native derivative and curvature bounds imply an error at most Ka_1^{fit} for X at these incidences. The resulting function is holomorphic and algebraic of bounded relation degree; on the tested real projection its size is at most K, since $|X| \leq K_0$. Lemma 3.1, applied on the comparison domains with margin, bounds its value and first derivatives on the used interiors by K.

The real boxes lie strictly inside those interiors. Boundary points may be assigned to a neighboring box, or to a bounded overlapping cover, without changing the K count. Two points using the same entry are thus compared within a derivative-controlled neighborhood; different entries contribute separate graphs to the list. All floors used to obtain these graph bounds precede conditioning on a particular projected cell. The bounds belong to the functions and remain valid after further restrictions.

The box/branch norm bounds use the incidence witnesses retained after the second deletion. The earlier witnesses W_{γ,I,Q_1}, kept before it, remain available for counting whole compatible domains. All deletions have negligible total mass by the original assignment mass bounds. This completes the definition of μ_{prep}, whose support remains fixed in the subsequent argument. □

10.6 The scalar projection: caps and support

Scalar mesh projection requires both a mass ceiling for each scalar bin and a bound on the number of occupied bins. The graph description just prepared will supply the support bound: we count the earlier charts compatible with a projected cell, and then the scalar bins met by their graphs.

Fix a component γ, an A-block I, and its rescaled time $\tau \in [0,1]$. In this subsection the index A includes its component and block whenever they are suppressed. Put

$$Z = e_Z \cdot y, \qquad \zeta = \frac{Z - Z_*}{q_A}, \tag{10.23}$$

where Z_* is the tangential component of its center at the block midpoint. Thus ζ is affine with $d\zeta/d\tau = e_Z \cdot V$, while $X = P_*$ is quadratic along each trajectory. On the unit block, both ranges and coefficients have subpower bounds within each fixed exact problem. Let T_A be this coordinate change on incidences, retaining the trajectory data, and define

$$\lambda_A = \frac{(T_A)_*(\mu_{\mathrm{prep},\gamma}|_{I,A})}{q_A \nu_\gamma(A)}, \qquad R_A = \omega_\gamma q_A \nu_\gamma(A). \tag{10.24}$$

The denominator is the original assigned prior mass multiplied by the block length. Thus λ_A is a subprobability dominated by the rescaled conditional trajectory prior $\nu_\gamma(\cdot \mid A)$ times $d\tau$. Every later restriction keeps this denominator, independently of its current success mass. Disjointness gives the global identities and bound

$$\|\mu_{\mathrm{prep}}\| = \sum_A R_A \|\lambda_A\|, \qquad \sum_A R_A \le 1. \tag{10.25}$$

The projected law means the pushforward of λ_A to (τ, ζ, X, ζ').

Lemma 10.5 (Projected cap and support bounds)**.** *Assume $\ell = \infty$ in the second-model extremal setup of Proposition 4.3. For the prepared measure of Lemma 10.4 and the laws λ_A in* (10.24)*, the projection in every used A has the following properties, with $a = N_0^{-s_1}$ as above.*

1. *An X-bin of width $p = N_0^{-r}$, $0 \le r \le s_1$, has density at most Kp^d per unit Lebesgue area in (τ, ζ). At $p = a$, adjoining a ζ'-bin of width N_0^{-v}, $0 < v \le u_1$, gains a factor KN_0^{-Sv}, for some fixed $0 < u_1 < u$.*

2. *For every fixed sufficiently large M, there exists a dyadic $\sigma \asymp N_0^{-M}$ such that each (τ, ζ)-mesh cell of width σ meets at most Ka^{-d} occupied X-bins of width a. In particular the total number of occupied (σ, σ, a)-cells is at most $Ka^{-d}\sigma^{-2}$.*

Proof. Projected densities. At fixed original time, an interval of length b in ζ cuts area at most $K_0 J_A b$ in the A-domain. In unit spatial coordinates of A, it tests a linear form whose norm is $\|D_A^T e_Z\|/q_A \ge 1/K_0$, by (10.13). Thus this estimate is also a density bound for the area pushforward.

At each exact original base, an X-bin of width p requires at most K_0 hidden bins of width p/B, by the native derivative lower bound and quadratic monotonicity. Integration of the prepared pure cap across the A-domain, with time change $dt = q_A\, d\tau$ and normalization by $q_A \nu_\gamma(A)$, bounds the projected density by

$$Kp^d \frac{J_A}{\nu_\gamma(A)} \le Kp^d,$$

where the last K uses (10.12) only on the chosen diagonal.

For the angular refinement, the normal component of original V is specified by C to accuracy $K_0 g_C$. Once a ζ'-bin of width N_0^{-v} is specified, the full original V therefore fits in K_0 old bins of that width, for any fixed $0 < v \le u_1 < u$ and sufficiently large i. On the good sampled terminal entries, their joint numerator is at most KN_0^{-Sv} times the old pure terminal density. Summing these denominators within permitted good hidden-bin ancestors at depth s_1 and integrating over

the base gives the angular cap. The restriction to good entries precedes this integration, so the bound applies entirely to the prepared law.

Counting whole compatible domains. Fix A, one projected cell, and a Q_1-time block meeting that cell. There are at most two such blocks, because $q_A\sigma \ll q_1$. We show that every compatible Q_1-domain in this component and block lies in a common enlarged moving box of area $K_0 q_1^2 g_1$.

At a compatible occurrence, its center has tangential distance at most $K_0 q_1$ from the cell's Z-location. Its normal distance from the affine C-center is at most $K_0 q_1 g_1$, using (10.8), the true position bounds, and $g_C \ll q_1 g_1$. The normal center difference changes at rate at most $K_0 g_1$ by (10.8). The tangential center difference changes at rate at most K_0. The same center bounds therefore hold throughout the entire Q_1-block.

The frames obey the same constant angle comparison throughout that block. The projection of a major axis of length $K_0 q_1$ onto n_C has length at most $K_0 q_1 g_1$, and the minor axis already has that size. Thus the whole domain of each compatible label, not only its intersection with A, lies in the claimed common box. The entire witness measure W_{γ,I,Q_1}, whose mass satisfies (10.21), is consequently supported there.

Those witnesses are disjoint in the original block and obey the original zero-depth base cap. If there are L_1 compatible labels, (10.4) and (10.21) give

$$\frac{L_1}{K} q_1 n_i(r_{1,i}) q_1^2 g_1 \le K q_1^3 g_1.$$

It follows that

$$L_1 \le \frac{K}{n_i(r_{1,i})} \le K N_0^{ds_1}. \tag{10.26}$$

The bound holds for each fixed A and cell, using the original zero-depth cap and witnesses summed within one component and block.

Normalized coordinate precision. Fix one compatible Q_1, with the normalized base coordinates $\widehat{u}$ of Lemma 10.4. For two occurrences in the projected cell, write $\Delta t = t' - t$ and $\Delta y = y' - y$, allowing the two occurrences to be on different trajectories. Their projected coordinates give

$$|\Delta t| + |\Delta Z| \le K_0 q_A \sigma.$$

Since both are in the same C-domain,

$$n_C \cdot \Delta y = n_C \cdot y_C' \Delta t + O(K_0 g_C).$$

Decompose $n_1 = c_1 n_C + c_2 e_Z$, with $|c_2| \le K_0 g_1$, and subtract the same Q_1-center at the two times. The constant velocity comparison in (10.8) gives

$$|n_1 \cdot (\Delta y - y_1' \Delta t)| \le K_0 (g_1 q_A \sigma + g_C), \tag{10.27}$$

$$|\text{major component of } \Delta y - y_1' \Delta t| \le K_0 (q_A \sigma + g_C).$$

Divide by the respective widths $q_1 g_1, q_1$, and include the normalized time difference. Thus

$$|\Delta \widehat{u}| \le K_0 \left(\frac{q_A \sigma}{q_1} + \frac{g_C}{q_1 g_1} \right). \tag{10.28}$$

The factor g_1 in the first term of (10.27) cancels before the narrow width is inverted.

For clarity, write $\Delta_i = \alpha_{2,i} - \alpha_{1,i}$. The two terms in (10.28), including subpower factors, are bounded on the diagonal by

$$N_0^{-M+ks_1+o(1)} + N_0^{-\Delta_i+2u+ks_1+o(1)}.$$

Choose $M > 1 + s_1 + ks_1$, and then use $\Delta_i \to \infty$. Both terms are smaller than $a = N_0^{-s_1}$ by a fixed power. This estimate depends on the ratio g_C/g_1 and holds for every absolute size of g_1 allowed by the construction.

The relation (10.28) and the derivative and approximation bounds in Lemma 10.4 show that points of one projected cell using one graph vary in X by at most Ka. Each graph therefore meets only K X_a-bins. There are K graphs per label and at most the number of labels in (10.26), which proves the cellwise support bound. Finally the (τ, ζ)-range has area at most K, so the total cell count is at most $Ka^{-d}\sigma^{-2}$. Both conclusions concern the same fixed law λ_A. They persist under every subsequent restriction. □

10.7 Quadratic fits and the improved atlas

We now construct a known atlas at the fixed positive depth c' of (10.18), with time length q_A, actual subpower lists and geometric slacks, and the coverage and tempered complexity required by the finite-sample exclusion.

Proof of Proposition 10.1. The scalar theorem produces quadratic fits on every substantial residual. Greedy selection gives positive coverage, and graph matching makes the fitted coefficients recoverable from the terminal observation. Subtracting a binned fit then gives the forbidden atlas.

Let μ_{prep} be the fixed prepared measure of Lemma 10.4. Its projections satisfy Lemma 10.5. Residuals and covered pieces below are submeasures of it; their A-normalizations always use the denominator in (10.24).

A candidate on every substantial residual. Consider any subsequence and any residual $\mu_{\mathrm{res}} \le \mu_{\mathrm{prep}}$ of mass at least a fixed positive fraction of $\mathfrak{p}_{\mathrm{path}}$. Set

$$\lambda_{A,\mathrm{res}} = \frac{(T_A)_*(\mu_{\mathrm{res},\gamma}|_{I,A})}{q_A \nu_\gamma(A)} \le \lambda_A.$$

The identity in (10.25), applied to this residual, shows that some A satisfies $\|\lambda_{A,\mathrm{res}}\| \ge 1/K$. Theorem 8.2 applies with fixed conditional prior $\nu_\gamma(\cdot \mid A)$ and active law $\lambda_{A,\mathrm{res}}$. Its two mesh hypotheses, with $c = 1$, follow from Lemma 10.5:

$$\sigma^2 \#\{\text{occupied } (\sigma,\sigma,a)\text{-cells}\} \le Ka^{-d},$$

together with the pure caps and a positive terminal angular gain. The upper caps and occupied-cell bound persist under this restriction, and the displayed mass floor supplies the required density relative to the fixed prior.

On a further subsequence the theorem gives a quadratic $F(\tau, \zeta)$ fitting X to Ka on assigned trajectories throughout the unit block, with residual incidence mass in original units at least

$$\frac{q_A \nu_\gamma(A) a^d}{K}. \tag{10.29}$$

Indeed the angular conclusion makes its moving interval scale $\beta \sim_{\log, N_0} 1$. The theorem's adapted norm bound then bounds the ordinary coefficients of F by K: the interval center motion and all occupied coordinate ranges are bounded by K.

Greedy coverage in original mass. In every A, greedily insert quadratic tests whose coefficients are at most N_0^δ, whose whole-block fit is at most aN_0^δ, and whose residual masked incidence mass is at least

$$q_A \nu_\gamma(A) a^d N_0^{-\delta}. \tag{10.30}$$

Let $\mu_c \le \mu_{\mathrm{prep}}$ denote the incidences covered by these assignments, measured with their original component weights; at any stage the residual is $\mu_{\mathrm{prep}} - \mu_c$. On insertion assign all still-unassigned A-trajectories meeting the geometric fit. These assignments are invariant over the block. Each insertion removes at least $a^d N_0^{-\delta} \nu_\gamma(A)$ trajectory mass, since its incidence duration is at most q_A. Therefore there are at most $N_0^\delta a^{-d}$ inserted tests in each A.

Maximality forces substantial coverage on a subsequence. If the covered fraction $\|\mu_c\|/\mathfrak{p}_{\mathrm{path}}$ tended to zero, the uncovered incidences would be a substantial residual of the prepared path. The preceding application of Theorem 8.2 would produce another candidate on a further subsequence. Since $K \le N_0^\delta$ eventually, its coefficient, fit and mass bounds would make it eligible in (10.30), contradicting termination. Thus for some fixed $\kappa > 0$, along a subsequence,

$$\|\mu_c\| \ge \kappa \mathfrak{p}_{\mathrm{path}}.$$

Thus the insertion count is controlled by each candidate's reciprocal-subpower floor, while maximality gives the positive total coverage.

Posterior comparison. Keep μ_c unnormalized, including its analytical masks. Let $O(\xi) = (p_1(\xi), A(\xi))$ be the observation of an incidence ξ, with its component and block included, and put $m_c = O_* \mu_c$. For m_c-almost every observation o, let π_o be the conditional probability on incidences. Define the posterior-pair measure by

$$\Pi = \int (\pi_o \otimes \pi_o)\, dm_c(o). \tag{10.31}$$

The base measure in this integral is the unnormalized observation pushforward. Consequently $\|\Pi\| = \|\mu_c\|$, and both incidence marginals of Π equal μ_c. The two assigned polynomials F, G are evaluated at the same (τ, ζ). At the common original base their hidden coordinates lie in one terminal bin; the native derivative and curvature bounds give

$$|X_F - X_G| \le K_0 N_0^{-1}.$$

Since $s_1 < 1$, the two fits consequently imply

$$|F(\tau,\zeta) - G(\tau,\zeta)| \le a N_0^{O(\delta)}. \tag{10.32}$$

Use the maximum norm on the six quadratic coefficients in the original (τ, ζ)-coordinates, and define

$$\mathcal{B} = \{(\xi_1, \xi_2) : \|\operatorname{coeff} F - \operatorname{coeff} G\|_\infty > a^{2\theta}\}, \qquad \Pi_{\mathrm{bad}} = \mathbf{1}_{\mathcal{B}} \Pi.$$

Coefficient comparison. We claim that $\|\Pi_{\mathrm{bad}}\| = o(\|\Pi\|)$. Otherwise it has a fixed positive relative mass on a subsequence. The mass bound in (10.17) then gives a component and an A-block for which the within-component restriction $\Pi_{\mathrm{bad},\gamma,A} = \omega_\gamma^{-1} \Pi_{\mathrm{bad}}|_{\gamma,I,A}$ has

$$b_A := \|\Pi_{\mathrm{bad},\gamma,A}\| \ge \frac{q_A \nu_\gamma(A)}{K}.$$

Normalize only these selected pairs, after the coordinate change:

$$\widehat{\Pi}_A = \frac{(T_A \times T_A)_* \Pi_{\mathrm{bad},\gamma,A}}{b_A}, \qquad (\widehat{\Pi}_A)_j \le \frac{q_A \nu_\gamma(A)}{b_A} \lambda_A \le K \lambda_A \quad (j = 1, 2). \tag{10.33}$$

Here $(\widehat{\Pi}_A)_j$ is the j-th incidence marginal. The domination follows because each marginal before normalization is bounded by $\mu_{c,\gamma}|_{I,A}$, hence by $\mu_{\mathrm{prep},\gamma}|_{I,A}$.

For this probability law each individual polynomial index has marginal base density at most

$$KN_0^{O(\delta)}a^d \quad \text{in } (\tau,\zeta). \tag{10.34}$$

Indeed, at a fixed base the fit of one assigned polynomial permits only $N_0^{O(\delta)}$ X_a-bins. Their caps under λ_A, followed by (10.33), give this ceiling. Each color has a list of at most $N_0^{\delta}a^{-d}$ indices. If $\rho_{j,f}$ is uniform on that list, then $a^d \le N_0^{\delta}\rho_{j,f}$; hence the index-base marginal is bounded by $KN_0^{O(\delta)}\rho_{j,f}\,d\tau\,d\zeta$, as required for local graph matching.

After enlarging and rescaling the ζ-range by a factor between one and K, it lies in a fixed bounded comparison domain. The polynomial norm upper bounds become KN_0^{δ}, while a coefficient discrepancy larger than $a^{2\theta}$ gives a norm gap bounded below by a constant times that quantity, allowing subpower comparison factors. These two-variable quadratic graphs are holomorphic on any fixed larger comparison domain. We apply Corollary 3.4 with common reference zero, norm scale $M = 1$, matching scale $\Delta = a$, and the tolerance ε fixed before (10.18); thus $R = a^{-1}$. For every fixed loss exponent $C \le C_{\mathrm{match}}$,

$$N_0^{C\delta} = R^{C\delta/s_1}, \qquad C\delta/s_1 < \varepsilon/2.$$

This accounts for the list count, polynomial norm, value discrepancy and index-base density. The range rescaling and the bad-pair normalization contribute only factors $K = R^{o(1)}$, so all these upper bounds fit the tolerance ε at late samples. The norm gap is at least $R^{-2\theta}/K$, which exceeds $R^{-\varepsilon}$ because $2\theta < \varepsilon$. Finally $\widehat{\Pi}_A$ has mass one and $d < d_0 < 1$. Thus all hypotheses of Corollary 3.4 hold, contradicting its conclusion. This proves the claim about Π_{bad}.

Deterministic coefficient lists. Bin the six coefficients of a quadratic in (τ,ζ), in these original A-coordinates before the range rescaling, to width a^{θ}. The successful posterior pairs lie in bounded neighboring bin lists, because $a^{2\theta} \ll a^{\theta}$. For an observation $o = (p_1, A)$, let β_o be the coefficient-bin pushforward of the incidence posterior π_o, and let $L(b)$ be the bounded neighboring list around a bin b. For independent bins B_1, B_2 with law β_o, the matching event is $B_2 \in L(B_1)$, and

$$\int \beta_o(L(b))\,d\beta_o(b) = \mathbb{P}\{B_2 \in L(B_1) \mid o\}.$$

Choose a maximizing center b for each o. Integration against m_c then gives deterministic bounded lists retaining a fixed positive fraction of the covered mass. Composing with the actual A-lists at p_1 multiplies list cardinality only by an actual subpower factor.

The new test and its actual bounds. Within each A, merge all assigned tests with the same coefficient bin. For a bin b, let $\widetilde{F}_{A,b}$ be the quadratic with its binned coefficients. The final chart label is (A, b). In original coordinates use

$$P_{\mathrm{new}}(t,y,w) = P_*(t,y,w) - \widetilde{F}_{A,b}\left(\frac{t-t_A}{q_A}, \frac{e_Z\cdot y - Z_*}{q_A}\right), \tag{10.35}$$

where t_A is the block's left endpoint. The error on every assigned trajectory throughout the block is at most

$$aN_0^{\delta} + Ka^{\theta}.$$

Our fixed choices give $\delta < \theta s_1/4$, $\theta < 1$ and $c' = \theta s_1/2$. Thus both exponents $s_1 - \delta$ and θs_1 are strictly larger than c', so at late diagonal samples the error is at most $N_0^{-c'}$. The positive power buffer absorbs the merely diagonal subpower factor K.

The polynomial subtracted in (10.35) does not involve w. Thus

$$(P_{\mathrm{new}})_w = (P_*)_w, \qquad (P_{\mathrm{new}})_{ww} = (P_*)_{ww}.$$

The original native bounds give the actual full-block derivative upper bound and, on retained original path incidences, the actual lower bound. The native curvature bound K_0B^2 is stronger than the required $K_0B^2N_0^{c'}$. The chart retains the actual frame and time block of A. Its geometric slacks and derivative bounds are therefore bounded by fixed powers of actual path factors. The time exponent still tends to zero.

Assignments and lists on the original law. Store the ordered quadratic tests for each A, assign trajectories by the first geometric whole-block fit, and merge tests with the same coefficient bin. Each whole-block fit is a fixed-degree semialgebraic condition on the trajectory coefficients, obtained by eliminating its single time quantifier. The at most $N_0^{\delta}a^{-d}$ tests in each A, their priorities and merger table therefore have polynomial complexity. The coordinate changes, coefficients and inverse lengths also have polynomial bounds within each fixed i.

Attach this geometric label map to the original tempered path μ_{ref}, using a failure label on unassigned pieces. It agrees with the posterior construction on μ_c. The latter measure has the bounded coefficient-bin lists just proved, composed with the actual A-lists at p_1, and

$$\mu_c \le \mu_{\mathrm{prep}} \le \mu_{\mathrm{ref}}.$$

Thus Lemma 4.12 applies with original law μ_{ref}, analytical submeasure μ_c, output label (A, b), and original observation p_1. The failure label is omitted from every list. The unnormalized success mass already obtained is exactly the mass used in this application of the lemma.

All predicates entering this application are the original path predicates and the finitely many geometric fit tests. The analytical cap masks and residual searches are absent. The polynomial bounds for these geometric data give a fixed mesh exponent and list table of the complexity allowed by the prechosen finite-sample exclusion. The table loses only $O(N_0^{-1})$ mass, and its list sizes remain fixed powers of actual path factors and logarithms.

A fixed fraction of $\kappa\mathfrak{p}_{\mathrm{path}}$ therefore survives. Since $\mathfrak{p}_{\mathrm{path}} \ge N_0^{-o(1)}$, the flattening error is negligible and the resulting coverage exceeds (10.20) for all sufficiently late samples. Together with the geometric and derivative bounds above, this is precisely the excluded known atlas at depth c', with time q_A. It contradicts (10.19) and proves Proposition 10.1. □

10.8 Completion of the proof

Proof of Theorem 1.1. Let $K \subset \mathbb{R}^4$ contain a unit line segment in every direction. Suppose that $\dim_H K < 4$, and choose $\sigma > 0$ with $\dim_H K < 4 - \sigma$. By Theorem 2.17, the Hausdorff covers of this arbitrary set produce an exact problem of the second model in $C(d_0)$, for some $d_0 < 1$ and fixed positive angular parameters, with $H > 0$. That reduction uses outer slope measure and finite selections of witnesses, so it applies to nonmeasurable K and nonmeasurable witnessing line families.

Optimize cap dimension and horizon as in Proposition 4.3. The resulting critical dimension is less than one and the maximizing time rate satisfies $k > 0$. The auxiliary first model has already been excluded by Corollary 6.9; this supplies the scalar results used in the subsequent second-model branches.

For the second model, consider the global optimized narrowness ℓ. If $\ell = 0$, Theorem 6.1 gives a forbidden improvement. For $0 < \ell < \infty$, the projection and comparison arguments in Propositions 8.4 and 8.5 either give such an improvement immediately or reduce to the horizontal case with narrowness rate equal to k. At $2k = 1$, Proposition 9.2 excludes that case. Above the critical rate, Proposition 9.6 supplies the wide intervals required by Proposition 8.4. Below the critical rate, Proposition 9.5 first reaches the limiting parameter scale, and Proposition 9.6 then supplies those intervals. In each instance the original-unit construction of Proposition 4.11 yields

the actual atlas forbidden by the critical-path optimization. Finally, the global case $\ell = \infty$ is excluded by Proposition 10.1.

Every possible optimized narrowness has therefore been excluded, contradicting the positive-horizon problem supplied by Theorem 2.17. It follows that $\dim_H K \geq 4$. The ambient upper bound is $\dim_H K \leq 4$, so $\dim_H K = 4$. The argument imposes no compactness, measurability or stickiness condition on K or on its witnessing line family. □

11 Geometric consequences

Theorem 1.1 has consequences for other notions of dimension, for unions of segments with prescribed directions, and for several curved incidence problems. The direction-set and curved conclusions use the transfer theorems of Keleti and Máthé, Gao, Liu, and Xi, and Nadjimzadah.

11.1 Packing and Minkowski dimensions

Every Kakeya set $K \subset \mathbb{R}^4$ also has packing dimension four, since Hausdorff dimension is at most packing dimension, which is at most the ambient dimension. If the set is bounded, both its lower and upper Minkowski dimensions are four: Hausdorff dimension is at most lower Minkowski dimension, and lower Minkowski dimension is at most upper Minkowski dimension, which is at most four. Here the lower and upper Minkowski dimensions are respectively the lower and upper limits of $\log N_\delta(K)/\log(1/\delta)$, where $N_\delta(K)$ is the least number of balls of radius δ covering K, as $\delta \downarrow 0$. Thus the theorem also controls the common-scale covering growth of each bounded Kakeya set.

11.2 Projection to four dimensions

Theorem 1.1 also implies that every Kakeya set $E \subset \mathbb{R}^d$, $d \geq 4$, has $\dim_H E \geq 4$; in particular, this holds in dimension five. Fix a four-dimensional linear subspace V, and let P_V be orthogonal projection onto V. For each unit $v \in V$, a witnessing segment $a + [0,1]v \subset E$ projects to the unit segment $P_V a + [0,1]v$. Hence $P_V E$ is a Kakeya set in $V \simeq \mathbb{R}^4$, so $4 = \dim_H(P_V E) \leq \dim_H E$. The final inequality follows because projection does not increase covering diameters, with no compactness or measurability assumption. The resulting lower bound is four; the full conjecture in $\mathbb{R}^d$ asks for dimension d.

11.3 Arbitrary directions and extension of segments

Keleti and Máthé transfer the full-direction assertion to arbitrary direction sets. We identify unoriented directions in $\mathbb{R}^4$ with the real projective space $\mathbb{RP}^3$.

Corollary 11.1 (General direction-set bound)**.** *Let $D \subset \mathbb{RP}^3$ be nonempty and let $E \subset \mathbb{R}^4$. If E contains a nondegenerate line segment in every direction in D, then*

$$\dim_H E \geq 1 + \dim_H D.$$

Proof. By Keleti and Máthé [20, Theorem 1.4], there is a compact Besicovitch set $B \subset \mathbb{R}^4$ such that $\dim_H E \geq \dim_H B - (3 - \dim_H D)$. Their Besicovitch convention requires a unit segment in every direction, so Theorem 1.1 gives $\dim_H B = 4$. Substitution proves the bound. □

In particular, a direction set of Hausdorff dimension three already forces full spatial dimension. We will use this below when the directions form a nonempty open patch. Neither E nor D is required to be measurable, and the positive segment lengths may vary.

The same conclusion gives a line-segment extension principle by the implication proved in Keleti and Máthé [20, Section 1.2]. For any nonempty family $\mathcal{S}$ of nondegenerate segments in $\mathbb{R}^4$, let $\mathcal{L}$ be the family of their supporting full lines. Then

$$\dim_H \left(\bigcup_{S \in \mathcal{S}} S \right) = \dim_H \left(\bigcup_{L \in \mathcal{L}} L \right).$$

Thus extending every segment in a fixed family to its whole line preserves the Hausdorff dimension of the union. This application uses the authors' same-dimensional implication from the general direction-set bound.

11.4 Nikodym sets on constant-curvature manifolds

The Nikodym formulation selects geodesics through a positive-volume family of base points. The Kakeya formulation selects a segment for every direction. Theorem 1.1 controls this different incidence requirement through the established Kakeya-to-Nikodym implication of Gao, Liu, and Xi. We use their normalized local geodesic formulation [11, Definition 1.3 and Theorem 2.1]: the metric is normalized so that the injectivity radius is at least 10, and the geodesic segments below have unit length. For a Borel set $\Omega \subset M$ and $0 < \lambda < 1$, put

$$\Omega_\lambda^\star = \left\{ x \in M : \begin{array}{l} \text{there is a unit geodesic segment } \gamma_x \text{ through } x \\ \text{such that } |\gamma_x \cap \Omega|_g \geq \lambda |\gamma_x|_g \end{array} \right\},$$

where the bars denote geodesic length. The set Ω is a *Nikodym set* if $\Omega_\lambda^\star$ has positive Riemannian volume for every λ sufficiently close to 1.

Corollary 11.2 (Four-dimensional Nikodym dimension)**.** *Let (M^4, g) be a Riemannian manifold of constant sectional curvature, with the metric and geodesic lengths normalized as in Section 11.4. Every Nikodym set $\Omega \subset M$ has $\dim_H \Omega = 4$.*

Proof. Theorem 1.1 applies in particular to the compact Euclidean Kakeya sets in the convention of Gao et al. [11, Definition 1.1]. It therefore supplies their Kakeya Hausdorff lower bound with $d = \alpha = 4$. Gao et al. [11, Theorem 1.5] transfer this bound to Nikodym sets on constant-sectional-curvature d-manifolds, giving $\dim_H \Omega \geq 4$. The reverse inequality holds for every subset of a four-dimensional Riemannian manifold. □

This resolves Conjecture 1.4 of Gao et al. [11] in dimension four and includes Euclidean $\mathbb{R}^4$. The geodesic straightening and Kakeya-to-Nikodym transfer are results of those authors, in the constant-sectional-curvature setting used here.

11.5 Curved Kakeya sets for a fixed translation-invariant phase

A separate consequence concerns the structured curved-Kakeya class of Gao, Liu, and Xi: a direction-indexed family from one phase in a Euclidean chart. We use their phase and core conventions [11, Definitions 1.7–1.11], with the active support made explicit. One common local change of coordinates straightens a direction patch of this family. This lets the Euclidean dimension theorem control the original curved set through local bi-Lipschitz invariance.

Write $X = (x, t) \in \mathbb{R}^3 \times \mathbb{R}$, and fix a sufficiently small chart

$$U = B_\rho^3 \times (-\rho, \rho), \qquad Y = B_\rho^3,$$

where B^3_ρ is the open Euclidean ball about the origin. Fix one smooth phase on a neighborhood of $\overline{U} \times \overline{Y}$,

$$\phi(x,t;y) = x \cdot y + \psi(t;y), \qquad \psi(0;y) = 0.$$

For $(X;y)$ in this neighborhood, let

$$G_0(X;y) = \bigwedge_{j=1}^{3} \partial_{y_j} \nabla_X \phi(X;y), \qquad \mathcal{D}_y = G_0(X;y) \cdot \nabla_X,$$

where the wedge is identified with its normal vector in $\mathbb{R}^4$, and $\mathcal{D}_y$ differentiates in X at fixed y. Impose the mixed-rank and curvature conditions

$$\operatorname{rank} \nabla_X \nabla_y \phi(X;y) = 3,$$
$$\det \Big(\nabla_y^2 \langle \nabla_X \phi(X;y), G_0(X;y_0) \rangle \big|_{y=y_0} \Big) \neq 0,$$

and Bourgain's condition in the invariant formulation of Guo et al. [12, Theorem 2.1],

$$\mathcal{D}_y^2 \partial^2_{y_i y_j} \phi(X;y) = C(X;y) \mathcal{D}_y \partial^2_{y_i y_j} \phi(X;y) \qquad (1 \le i,j \le 3),$$

with the same scalar $C(X;y)$ for every pair i,j. All three conditions hold throughout the chosen neighborhood of $\overline{U} \times \overline{Y}$, after one fixed phase-dependent localization. The support hypotheses therefore apply to the curve portions inside the active chart. For $y \in Y$ and $z \in U$, define the full local core

$$\Gamma^\phi_y(z) = \{X \in U : \nabla_y \phi(X;y) = \nabla_y \phi(z;y)\}.$$

Corollary 11.3 (Fixed-chart curved Kakeya dimension)**.** *In the fixed translation-invariant phase and domain setting of Section 11.5, assume the mixed-rank, curvature and Bourgain conditions stated there, and let $E \subset \mathbb{R}^4$. Suppose that for every $y \in Y$ there exists $\omega_y \in B^3_\rho$ such that*

$$\Gamma^\phi_y((\omega_y, 0)) \subset E.$$

Then $\dim_H E = 4$.

Proof. A common change of coordinates turns a direction patch of the cores into straight segments. Corollary 11.1 then gives full dimension in the new coordinates, which transfers back by local bi-Lipschitz invariance.

Straightening the phase. For this phase, $G_0 = (-\partial_t \nabla_y \psi, 1)$ in the standard orientation. The mixed rank is automatic, and the remaining conditions become

$$\det \nabla_y^2 \partial_t \psi(t;y) \neq 0, \qquad \nabla_y^2 \partial_t^2 \psi(t;y) = c(t,y) \nabla_y^2 \partial_t \psi(t;y).$$

The nonsingular matrix determines this scalar smoothly and independently of x. Gao et al. [11, Corollary 4.2 and Theorem 1.12] show that c depends only on t on the connected local product and straighten the phase. More precisely, their proof gives, after shrinking to an open time interval I containing 0 and a nonempty direction ball $Y_0 \subset Y$,

$$\kappa(u,s) = (u + B(\alpha(s)), \alpha(s)), \qquad \alpha(0) = 0, \qquad \alpha'(s) \neq 0,$$

$$\phi(\kappa(u,s);y) = u \cdot y + s h(y) + q(y) + f(s), \qquad \det \nabla_y^2 h(y) \neq 0,$$

whenever $s \in I$, $y \in Y_0$, and $\kappa(u,s) \in U$. This is one smooth coordinate change for the fixed phase.

The resulting direction patch. Since $\psi(0;y)=0$, the core through $(\omega_y,0)$ is

$$\Gamma_y^{\phi}((\omega_y,0))=\{(\omega_y-\nabla_y\psi(t;y),t)\in U\}.$$

Put $V=U\cap(\mathbb{R}^3\times\alpha(I))$ and $F=\kappa^{-1}(E\cap V)$. Every anchor $(\omega_y,0)$ lies in the open set V, so continuity supplies a nontrivial closed time interval on which its core remains in V. In the new coordinates the gradient equality defining that core is exactly

$$u=(\omega_y-B(0))-s\nabla h(y).$$

Indeed, the $\nabla q(y)$ terms cancel from the two gradient values, and $f(s)$ has zero y-gradient. Thus F contains a positive-length straight segment in the direction

$$\nu(y)=\frac{(-\nabla h(y),1)}{\sqrt{1+|\nabla h(y)|^2}}\qquad\text{for every } y\in Y_0.$$

The nonsingular Hessian of h makes ν a local diffeomorphism. Its image therefore contains a nonempty open patch in the upper hemisphere of S^3. The corresponding unoriented directions form an open set $D\subset\mathbb{RP}^3$, so $\dim_H D=3$. Corollary 11.1 gives $\dim_H F=4$; it allows the positive segment lengths to depend on y.

Return to the original coordinates. A smooth local diffeomorphism preserves Hausdorff dimension of arbitrary subsets, using a countable cover by coordinate neighborhoods on which it and its inverse are Lipschitz. Consequently

$$4=\dim_H F=\dim_H(E\cap V)\le\dim_H E\le 4.$$

The argument applies to arbitrary E and arbitrary anchor choices $y\mapsto\omega_y$, with positivity as the only condition on the length of each initial segment. □

The straightening and general-dimensional set transfer are due to Gao et al. [11, Theorem 1.12 and Section 4.2]. The corollary applies their transfer to full gradient-defined cores in one fixed active chart.

The translation-invariance assumption specifies the class being straightened here. Bourgain's condition alone need not allow one coordinate change to straighten the curves, as the examples of Nadjimzadah [22, Example 1.24 and Proposition 1.25] show.

11.6 A three-dimensional quadratic curved consequence

A further application uses Nadjimzadah's lifting theorem to pass from the four-dimensional linear result to curved sets in dimension three. For a real 2×2 matrix B, let I_2 denote the identity matrix. Use the fixed compact anchor ball $W_B\subset\mathbb{R}^2$, chosen sufficiently large, and the fixed time interval $I_B=[-\tau_B,\tau_B]$, with $\tau_B>0$ sufficiently small, specified after equation (1.16) of Nadjimzadah [23].

Corollary 11.4 (Rank-one quadratic curved Kakeya sets)**.** *Let B be a real 2×2 matrix with $\operatorname{rank}B=\operatorname{rank}(B^2)=1$, and let W_B, I_B be the fixed domains specified in Section 11.6. Suppose that a compact set $K\subset\mathbb{R}^3$ contains, for every $y\in[-1,1]^2$, the full curve*

$$\{(w_y+(tI_2+t^2B)y,t):t\in I_B\}$$

for some $w_y\in W_B$. Then $\dim_H K=3$.

Proof. The linear hypothesis. Corollary 11.1 supplies the linear Kakeya hypothesis in the convention of Nadjimzadah [23, Section 1.2, equations (1.8)–(1.13)]. Indeed, a compact linear-chart set $F \subset \mathbb{R}^4$ in that convention contains a segment

$$\{(b_\eta + t\eta, t) : t \in J\}$$

for every slope η in a fixed three-dimensional parameter ball, with anchors in a fixed compact ball and one fixed nondegenerate compact interval J. Slopes in the interior give a nonempty open set of directions $[\eta : 1] \in \mathbb{RP}^3$, and every segment has positive length. The corollary therefore gives $\dim_H F = 4$.

The quadratic lift. The geometry of Nadjimzadah's lift explains the two rank hypotheses. Write $B = UV$, where $U : \mathbb{R} \to \mathbb{R}^2$ is injective and $V : \mathbb{R}^2 \to \mathbb{R}$ is surjective, and define

$$\pi : \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}^2 \times \mathbb{R}, \qquad \pi(x, z, t) = (x + tUz, t).$$

The lifted segments in Nadjimzadah [23, equations (4.1)–(4.9)] are

$$L_{y,w,r}(t) = \big((w, r) + t(y - Ur, Vy), t\big), \qquad t \in I_B,$$

with r in a fixed compact interval with interior. They map to the prescribed curves over the entire time interval:

$$\pi(L_{y,w,r}(t)) = (w + ty + t^2UVy, t) = (w + (tI_2 + t^2B)y, t).$$

Since $B^2 = U(VU)V$ has rank one, $VU \neq 0$. The slope map $\Theta(y, r) = (y - Ur, Vy)$ consequently satisfies

$$\det D\Theta = \det \begin{pmatrix} I_2 & -U \\ V & 0 \end{pmatrix} = VU \neq 0.$$

Thus varying y and r in their interiors produces an open slope patch in dimension four. Definition 2.1 of Nadjimzadah [23] retains every $t \in I_B$ in these lifted segments, each of which has length at least $2\tau_B$.

The dimension bound. With the four-dimensional linear hypothesis established, Nadjimzadah [23, Theorem 1.3] applies with $3 + \operatorname{rank}(B^2) = 4$. It gives infimal Hausdorff dimension $3 - \operatorname{rank} B + \operatorname{rank}(B^2) = 3$ for the compact curved sets in the statement. Hence $\dim_H K \geq 3$, and the ambient bound gives equality. □

This is a lower-dimensional application of the headline result: the prescribed quadratic curves lift to linear Kakeya sets in one additional dimension. Compactness and containment of the full curves over the fixed interval are part of this application. The anchor choice $y \mapsto w_y$ may be arbitrary within the prescribed compact ball.

References

[1] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, Berlin, Heidelberg, 2nd edition, 2006. ISBN 978-3-540-33098-1. doi: 10.1007/3-540-33099-2.

[2] Jonathan Bennett, Anthony Carbery, and Terence Tao. On the multilinear restriction and Kakeya conjectures. *Acta Mathematica*, 196(2):261–302, 2006. doi: 10.1007/s11511-006-0006-4.

[3] A. S. Besicovitch. On Kakeya's problem and a similar one. *Mathematische Zeitschrift*, 27:312–320, 1928. doi: 10.1007/BF01171101.

[4] Tainara Borges, Tiklung Chan, Mingfeng Chen, Diankun Liu, Yakun Xi, and Yufei Zhan. Restriction and Kakeya maximal estimates in $\mathbb{R}^4$. arXiv:2511.22824v1, 2025.

[5] Jean Bourgain. On the dimension of Kakeya sets and related maximal inequalities. *Geometric and Functional Analysis*, 9(2):256–282, 1999. doi: 10.1007/s000390050087.

[6] Jean Bourgain and Larry Guth. Bounds on oscillatory integral operators based on multilinear estimates. *Geometric and Functional Analysis*, 21(6):1239–1295, 2011. doi: 10.1007/s00039-011-0140-9.

[7] Anthony Carbery and Stefán Ingi Valdimarsson. The endpoint multilinear Kakeya theorem via the Borsuk–Ulam theorem. *Journal of Functional Analysis*, 264(7):1643–1663, 2013. doi: 10.1016/j.jfa.2013.01.012. arXiv:1205.6371v1.

[8] Michael Christ. Quasiextremals for a Radon-like transform. arXiv:1106.0722v1, 2011.

[9] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006. doi: 10.1002/047174882X.

[10] Roy O. Davies. Some remarks on the Kakeya problem. *Proceedings of the Cambridge Philosophical Society*, 69(3):417–421, 1971. doi: 10.1017/S0305004100046867.

[11] Chuanwei Gao, Diankun Liu, and Yakun Xi. Curved Kakeya sets and Nikodym problems on manifolds. arXiv:2503.11574v1, 2025.

[12] Shaoming Guo, Hong Wang, and Ruixiang Zhang. A dichotomy for Hörmander-type oscillatory integral operators. *Inventiones mathematicae*, 238(2):503–584, 2024. doi: 10.1007/s00222-024-01288-8. arXiv:2210.05851v2.

[13] Larry Guth. The endpoint case of the Bennett–Carbery–Tao multilinear Kakeya conjecture. *Acta Mathematica*, 205(2):263–286, 2010. doi: 10.1007/s11511-010-0055-6.

[14] Larry Guth and Joshua Zahl. Polynomial Wolff axioms and Kakeya-type estimates in $\mathbb{R}^4$. *Proceedings of the London Mathematical Society*, 117(1):192–220, 2018. doi: 10.1112/plms.12138. arXiv:1701.07045v3.

[15] Larry Guth, Hong Wang, and Joshua Zahl. A streamlined proof of the Kakeya set conjecture in $\mathbb{R}^3$. arXiv:2601.14411v1, 2026.

[16] Nets Hawk Katz and Keith M. Rogers. On the polynomial Wolff axioms. *Geometric and Functional Analysis*, 28(6):1706–1716, 2018. doi: 10.1007/s00039-018-0466-7. arXiv:1802.09094v3.

[17] Nets Hawk Katz and Terence Tao. New bounds for Kakeya problems. *Journal d'Analyse Mathématique*, 87:231–263, 2002. doi: 10.1007/BF02868476. arXiv:math/0102135.

[18] Nets Hawk Katz and Joshua Zahl. A Kakeya maximal function estimate in four dimensions using planebrushes. *Revista Matemática Iberoamericana*, 37(1):317–359, 2021. doi: 10.4171/RMI/1219. arXiv:1902.00989v3.

[19] Nets Hawk Katz, Izabella Łaba, and Terence Tao. An improved bound on the Minkowski dimension of Besicovitch sets in $\mathbb{R}^3$. *Annals of Mathematics*, 152(2):383–446, 2000. doi: 10.2307/2661389. arXiv:math/9903166v2.

[20] Tamás Keleti and András Máthé. Equivalences between different forms of the Kakeya conjecture and duality of Hausdorff and packing dimensions for additive complements, 2023. arXiv:2203.15731v3.

[21] Izabella Łaba and Terence Tao. An improved bound for the Minkowski dimension of Besicovitch sets in medium dimension. *Geometric and Functional Analysis*, 11:773–806, 2001. doi: 10.1007/PL00001685. arXiv:math/0004015v1.

[22] Arian Nadjimzadah. Bourgain's condition, sticky Kakeya, and new examples. *Geometric and Functional Analysis*, 36:1131–1173, 2026. doi: 10.1007/s00039-026-00750-4. arXiv:2511.10918v4.

[23] Arian Nadjimzadah. Lifting curved Kakeya sets to linear Kakeya sets, 2026. arXiv:2609.06299v1.

[24] OpenAI. *The Kakeya maximal conjecture in three dimensions*. Preprint, 23 September 2026. https://github.com/openai/math/blob/main/preprints/The-Kakeya-maximal-conjecture-in-three-dimensions-September-23-2026/paper.pdf.

[25] Tuomas Orponen, Nicolas de Saxcé, and Pablo Shmerkin. On the Fourier decay of multiplicative convolutions. arXiv:2309.03068v2, 2024.

[26] Mukul Rai Choudhuri. An improved bound on the Hausdorff dimension of sticky Kakeya sets in $\mathbb{R}^4$. *Revista Matemática Iberoamericana*, 42(4):1227–1256, 2026. doi: 10.4171/RMI/1621.

[27] N. Roytwarf and Y. Yomdin. Bernstein classes. *Annales de l'Institut Fourier*, 47(3):825–858, 1997. doi: 10.5802/aif.1582.

[28] Benny Sudakov, Endre Szemerédi, and Van H. Vu. On a question of Erdős and Moser. *Duke Mathematical Journal*, 129(1):129–155, 2005. doi: 10.1215/S0012-7094-04-12915-X.

[29] Terence Tao and Van H. Vu. *Additive Combinatorics*, volume 105 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2006. doi: 10.1017/CBO9780511755149.

[30] Hong Wang and Joshua Zahl. Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions. arXiv:2502.17655v1, 2025.

[31] Hong Wang and Joshua Zahl. Sticky Kakeya sets and the sticky Kakeya conjecture. *Journal of the American Mathematical Society*, 39(2):515–585, 2026. doi: 10.1090/jams/1067. arXiv:2210.09581v2.

[32] Hong Wang and Dmitrii Zakharov. Kakeya and Furstenberg problems for sticky sets of tubes. arXiv:2609.22035v1, 2026.

[33] Thomas H. Wolff. An improved bound for Kakeya type maximal functions. *Revista Matemática Iberoamericana*, 11(3):651–674, 1995. doi: 10.4171/RMI/188.

[34] Joshua Zahl. A discretized Severi-type theorem with applications to harmonic analysis. *Geometric and Functional Analysis*, 28:1131–1181, 2018. doi: 10.1007/s00039-018-0455-x. arXiv:1801.05106v3.